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Standard Model Effective Field Theory at d = 6 and Standard Model deviations

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Abstract

The Standard Model (SM) of particle physics stands as one of the most remarkable achievements in our understanding of the nature. It encapsulates the behavior of elementary particles and their interactions with astonishing precision, yet it is not the ultimate theory of the universe.
The realm of High Energy Physics has uncovered phenomena that defy the SM's explanatory power, from the mystery of dark matter to the unification of fundamental forces.
In this context, the Standard Model Effective Field Theory (SMEFT) emerges as a critical tool, bridging the gap between the Standard Model and new physics.

Our main goal will be to establish a solid theoretical foundation for the SMEFT which allowed us to study the possible deviations from the SM by using the randomly generated Wilson coefficients as pseudo-data.

The dissertation is organized as follows.

In the **first chapter** we will introduce the fundamental ingredients involved in the construction of a generic effective field theory. We will study how to build operators in mass dimension higher than four and how to assemble the minimal complete set of them by redefinition of fields. The fundamental role of Wilson coefficients will be also discussed.
Once derived the new effective Lagrangian, we will investigate the process of achieving the correct normalization of the kinetic terms and the Feynman rules. This is accomplished through a procedure known as canonical normalization.
We will conclude the chapter by analyzing a simple case of study: the Scalar Effective Field Theory. (SEFT).

The **second chapter** serves as a bridge to the SMEFT. After a short overview about the fields and symmetries in the SM, we will outline the fundamental building blocks necessary for the construction of the SMEFT, adopting the Warsaw basis [1] as our framework. In this regard, we will discuss the different classes of operators involved at $d = 6$ and the relative canonical normalization to obtain the final effective Lagrangian.
In the last part of the chapter our focus will be led to the technology involved in the computation of transition amplitude up to one-loop in the SMEFT. By introducing the language of counterterms, we will illustrate how the renormalization procedure works in $d = 6$.

The computational aspect in the SMEFT will be discussed in the **third chapter**.
In the first part of the chapter, we will provide an in-depth exploration of the computational challenges encountered throughout the development of this dissertation. The introduction of operators of mass dimension higher than four significantly amplifies the number of Feynman diagrams involved in a generic transition amplitude. To address this challenge, we have developed a series of dedicated codes in `FORM` [2] to semi-automate the generation of our results.
We will check our first result in the SMEFT by assembling and confirming the well known Ward-Slavnov-Taylor (WST) identities. To accomplish this, it will be necessary to update the β_h-scheme [3], the diagonalization of neutral sector and the self energies.
After the definition of the input parameter set (IPS), we will conclude the chapter by providing the calculation of Electroweak (EW) corrections for $Z \to \bar{l}l$, $H \to \bar{b}b$ and $H \to \gamma\gamma$. The integration over the full Lorentz invariant phase space (LIPS) will allow us to obtain the following pseudo-observables (POs) : $\Gamma(Z \to \bar{l}l)$, $\Gamma(H \to \bar{b}b)$ and $\Gamma(H \to \gamma\gamma)$.

We will also compute the $g - 2$ of the muon, $\sin\theta^l_{\text{eff}}$ and the mass of the W boson.

The numerical analysis of these quantities will be presented in the **fourth** and **fifth chapter**. By implementing several routines in `FORTRAN`, we will investigate the possible deviations from the SM in the SMEFT language.
Working in an equiprobable space of Wilson coefficients, the goal will be to

understand how large the deviations are from the SM once the SMEFT scale and the range of coefficients are selected. The correlations between different POs are also discussed.
In particular, the fifth chapter will focus on the precise determination of the W boson mass. Within the context of the LEP scenario, we will update the renormalization equations to account for the novel effects introduced by the inclusion of operators in $d = 6$. A finite prediction for the W mass is obtained after performing the mixing among the Wilson coefficients. Subsequently, we will delve into the analysis of potential deviations from the SM expectation.

Our conclusions are reported in the **sixth chapter** while the analytic expression of the results are presented in the **Appendix D**.

Conventions

The Pauli metric is adopted. In this way we work with four-momentum defined by

$$p_\mu = (\vec{p}, p_4) = (\vec{p}, ip_0).$$

The metric tensor in the common Bjorken-Drell convention is then replaced by the Kronekcker delta $\delta_{\mu\nu}$. It follows:

$$p^2 = (\vec{p}, ip_0) \cdot (\vec{p}, ip_0) = |\vec{p}|^2 - p_0^2 = -m^2.$$

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Chapter 1

The use of Effective Filed Theory in HEP problems

1.1 Introduction

The main priorities in High-Energy Physics are testing the SM and searching for new physics. The Run-1 of LHC has discovered a resonance which is compatible with the Higgs boson of the SM [4],[5]: the excellent experimental accordance has confirmed the validity and consistency of the theory as a gauge field theory, renormalizable at every energy scale.

Some phenomena still lack a satisfactory explanation within the SM, such as the theory of gravitation, dark matter, or the neutrino masses. The forthcoming challenge lies in the construction of a new theory as an extension of the Standard Model, marking the next crucial step in our understanding of the universe.
In the near future, the LHC may provide evidence for physics beyond the Standard Model (BSM) by producing new particles that could manifest themselves in kinematic distributions or cross sections.
We can effectively explore BSM physics through the use of Effective Field Theories (EFTs), which offer an approximate description of a high-energy theory characterized by a cutoff scale beyond which the predictability diminishes.

There are two different approaches to study EFTs: *top-down* and *bottom-*

up. In the *top-down* approach, the high energy theory is well-known at all scales and the low energy behavior is derived by integrating out the heavy degrees of freedom. The Lagrangian of the full theory is then expanded in a sum of terms of decreasing relevance, generating new operators and a new low-energy coupling. The obvious disadvantage is the large number of BSM models at our disposal.

The Fermi theory provides an example for the *top-down* approach [6]. Let us consider the lowest order (LO) amplitude for the muon decay:

$$\mathcal{M}_{\text{SM}} \sim \frac{g^2}{2} \left[\bar{\psi}_e \gamma_\mu (1+\gamma_5) \psi_{\nu_e}\right] \frac{\delta_{\mu\nu}}{p^2 + M^2} \left[\bar{\psi}_{\nu_\mu} \gamma_\nu (1+\gamma_5) \psi_\mu\right]. \tag{1.1}$$

Working in the limit $p^2 << M^2$, we can expand the propagator as

$$\frac{1}{p^2 + M^2} = \frac{1}{M^2}\left(1 + \frac{p^2}{M^2} + ...\right). \tag{1.2}$$

This expansion is equivalent to the process of integrating out the W field, substituting a boson-mediated interaction with a point-like interaction described by a four-fermion operator in mass dimension $d = 6$.
The final result of the expansion reads as follows:

$$\mathcal{M}_{\text{Fermi}} \sim 2\sqrt{2}\, G_F \left[\bar{\psi}_e \gamma_\mu (1+\gamma_5) \psi_{\nu_e}\right] \left[\bar{\psi}_{\nu_\mu} \gamma_\nu (1+\gamma_5) \psi_\mu\right], \tag{1.3}$$

where the Fermi constant G_F is introduced as new coupling constant, determined by the matching of two amplitudes:

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8\, M^2}. \tag{1.4}$$

We may also use a *bottom-up* approach, working in a model independent way, without reference to any other theory. Starting from a theory X in mass dimension $d = 4$, it is possible to construct the effective field theory XEFT by adding to the Lagrangian a basis of operators in mass dimension $d > 4$, suppressed by an unknown cut-off scale Λ.
A Wilson coefficient, denoted by a_i, is assigned to each operator to quantify our lack of knowledge about the new physics effects. The new XEFT has to

respect all the symmetries of the X theory, without introducing new degrees of freedom. This requirement leads us to formulate the most generic Lagrangian as follows:

$$\mathcal{L}_{XEFT} = \mathcal{L}_X^{(4)} + \sum_{d>4} \sum_i \frac{a_i^{(d)}}{\Lambda^{d-4}} \mathcal{O}_i^{(d)} = \mathcal{L}_X^{(4)} + \frac{\mathcal{L}^{(5)}}{\Lambda} + \frac{\mathcal{L}^{(6)}}{\Lambda^2} + \dots . \tag{1.5}$$

It is crucial at this point to address the foundational philosophy underpinning the construction of EFTs [7].
Let us start with a theory T_1 and extend it to T_2 by introducing higher-dimensional operators, with T_2 being valid up to an unknown energy scale Λ_2. The experimental data can be explained by fitting Wilson coefficients in T_2 and ideally, if a coefficient differs from zero, we have to consider the list of all possible BSM models. If a model reproduces this behavior, caution is necessary because may not represent the full theory.
In the same way we can extend T_2 to T_3 (valid within $\Lambda_3 > \Lambda_2$) that could explain something more than T_2 and so on.
We cannot be certain if this succession of effective theories has an endpoint. Nonetheless, it offers a significant advantage: the ability to compute physical quantities with remarkable precision. This enables us to concentrate our efforts on the pertinent degrees of freedom and interactions, thus 'simplifying' our calculations.

1.2 Building blocks in EFT

1.2.1 Power counting, scale Λ and renormalization

Examining Eq. (1.5), we notice that each operator in dimension d is suppressed by Λ^{d-4} to maintain the proper dimensionality of the Lagrangian. To achieve this, we introduce the concept of power counting as a crucial element in constructing operators for $d > 4$ and in establishing a coherent EFT.
Considering natural units ($c = \hbar = 1$ and $E = M = L^{-1}$), a generic action in $d = 4$ reads:

$$S = \int \mathrm{d}^4x \, \mathcal{L}(x) \to [S] = 0, \tag{1.6}$$

and $[\mathcal{L}(x)] = 4$.
By examining the kinetic terms in $\mathcal{L}$, we can determine the mass dimensions

for both fermionic and bosonic fields:

$$[\psi] = \frac{3}{2} \ , \ \ [\phi] = 1 \ , \ \ [F_{\mu\nu}] = 2 \tag{1.7}$$

with [1]:

$$[A_\mu] = 1 \ , \quad [D_\mu] = 1 \ , \quad [g] = 0. \tag{1.8}$$

For example, the dimension of a generic operator $\mathcal{O}$ defined by

$$\mathcal{O} = (D_\mu \Phi)^\dagger (D_\nu \Phi) F_{\mu\nu}, \tag{1.9}$$

will be equal to:

$$[\mathcal{O}] = 2[D_\mu] + 2[\Phi] + [F_{\mu\nu}] = 6, \tag{1.10}$$

The cut off Λ ($\sim$ TeV) represents the scale where the new physics must be solved, separating the UV physics from the low energy limit and suppressing the effect of high dimension operators.

These considerations lead us to define a generic scattering amplitude [8], at a specific momentum scale p, with a single insertion of a d-operator as

$$\mathcal{A} \sim \left(\frac{p}{\Lambda}\right)^{d-4}. \tag{1.11}$$

It logically follows that a single insertion of an operator in $d = 5$ yields terms equal to p/Λ, while an insertion of a 6-operator or two insertions of a 5-operator give terms equal to $(p/\Lambda)^2$, etc.

This construction significantly affects the renormalizability of an EFT. When we consider the Eq.(1.5), it becomes clear that a generic EFT is not assured to be renormalizable, which would necessitate an infinite number of counterterms to reabsorb the ultraviolet (UV) divergences. However, if we truncate the expansion at a certain order in $1/\Lambda$, we should be able to perform the correct procedure to renormalize our EFT using a finite number of counterterms, ensuring the predictive power of the theory.

[1] Working in dimensional regularization, all is evaluated in $4 - \epsilon$ dimensions and the coupling constant would result $[g] = \epsilon$. In order to maintain a dimensionless coupling, we factor-out a parameter μ from the coupling such that :

$$g = \mu^\epsilon g \ , \quad [\mu] = 1.$$

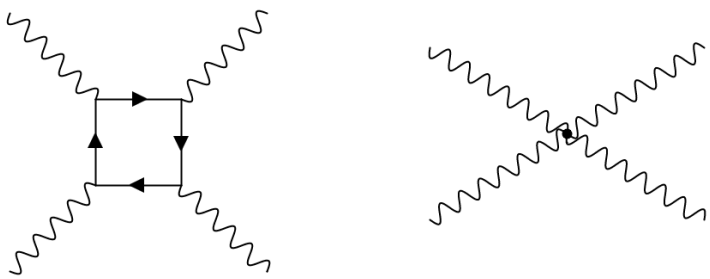


Figure 1: Full theory vs EFT for $\gamma\gamma$ scattering

1.2.2 The role played by the coefficients

We have explained how the use of EFT allows to perform calculations without knowing what happens at high energy scale.
Only the relevant degrees of freedom and the symmetries of low-energy theory are taken into account, in order to reproduce the correct matrix elements when we perform the matching procedure between the EFT and the full theory.
To give an idea, let us consider the $\gamma\gamma$ scattering.
In the low energy approximation of QED, when $E_\gamma << m_e$, we can integrate out all the fermionic component. This leads to the effective Lagrangian

$$\begin{aligned}\mathcal{L}_{QEDEFT} = &-\frac{1}{4}F_{\mu\nu}F_{\mu\nu} \\ &+\frac{\alpha^2}{m_e^4}\left[a\,(F_{\mu\nu}F_{\mu\nu})^2 + b\,F_{\mu\nu}F_{\nu\rho}F_{\rho\sigma}F_{\sigma\mu}\right] + \mathcal{O}\left(\frac{\alpha^4}{m_e^8}\right),\end{aligned} \tag{1.12}$$

where $a,\ b$ are the Wilson coefficients associated to the operators in $d = 8$.
It is clear that the new operators allow the $\gamma\gamma$ scattering without the electron mediation, as we observe in Fig.(1):

In the limit $E_\gamma << m_e$ we obtain, by the matching condition, the following coefficients:

$$a = \frac{1}{36}\ , \quad b = \frac{7}{90}. \tag{1.13}$$

Consequently, we can say that the Wilson coefficients constitute the essential link between the low energy theory and the full one. Let us suppose that an experiment reproduces a certain phenomena that can not be explained

by the low energy theory X. We decide to interpret the data by using its effective extension, XEFT, so the use of XEFT to fit experimental data is traduced by "fitting the Wilson coefficients to explain the data"; this implies to treat the coefficients as pseudo-data in our analysis.
We find that a coefficient differs from zero. Therefore, we have to analyze all the possible BSM models at our disposal that reproduce the fit, but more caution is needed.
Even if we are not sure that the chosen-model corresponds to the full theory, if we assume it to be the correct model, we should consider the low-energy limit and select the BSM parameters accordingly for the pseudo-data [9].

1.2.3 Bases and redefinition of fields

Let us give some details about the operators in a generic EFT.
All the operators involved must be allowed by the symmetry group which characterizes the low energy theory X and they must be in a minimal complete set. Given a symmetry group G and two operators $\mathcal{O}_1$, $\mathcal{O}_2$ in G with the relative Wilson coefficients a_1, a_2, we can say that $\mathcal{O}_1$, $\mathcal{O}_2$ form a complete set if all elements in G can be write as linear combination of a_1, a_2.
There is not a single way to choose a basis, all bases are equivalent and there is no problem in extracting Wilson coefficients; it could happen that a certain combination of coefficients in one basis becomes a single Wilson coefficient in another one and so on.
The interpretation of the fits requires that the experimental bounds, on Wilson coefficients in one basis, have to be translated in bounds of another one. At the moment, only at $d = 6$ level is possible to obtain a linear correspondence among different bases.

The construction of a complete set of operators requires one more step.
Let us suppose that we have constructed an XEFT in d dimensions and all possible operators have been found. It is possible to have some degeneracy, a redundancy [2] among these operators; a well defined strategy is necessary in order to reduce the number of operators and obtain a minimal basis.
The answer, up to $d = 6$, is given by using equation of motion (EoM).

[2] An operative definition of redundancy is given in [9]: considering two operators $\mathcal{O}_1$, $\mathcal{O}_2$ with coefficients a_1, a_2, redundancy means that we can only measure $a_1 + a_2$. Selecting $a_1 = 0$ or $a_2 = 0$ is equivalent to select another basis.

Let us starting by considering the following Lagrangian:

$$\mathcal{L}^{(4)} = -\frac{1}{2}\partial_\mu\phi\partial_\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4. \tag{1.14}$$

We analyze what happens when we include

$$\mathcal{L}^{(6)} = \frac{a}{\Lambda^2}\phi^3\Box\phi. \tag{1.15}$$

From the EoM

$$\frac{\delta\mathcal{L}}{\delta\phi} - \partial_\mu\frac{\delta\mathcal{L}}{\delta\partial_\mu\phi} = 0 \tag{1.16}$$

and setting $\mathcal{L} = \mathcal{L}^{(4)} + \mathcal{L}^{(6)}$, we derive:

$$\Box\phi = m^2\phi + \frac{\lambda}{6}\phi^3 - 6\,\frac{a}{\Lambda^2}\left[\phi\partial_\mu\phi\partial_\mu\phi + m^2\phi^3 + \frac{\lambda}{6}\phi^5\right]. \tag{1.17}$$

Replacing Eq(1.17) in Eq.(1.15), the box operator vanishes at $\mathcal{O}(\Lambda^{-2})$.
The use of EoM does not present issues up to $d = 6$, while when we consider the next order at $\mathcal{O}(\Lambda^{-4})$, it becomes apparent that these corrections differ from those obtained through the field redefinition

$$\phi \to \phi + \frac{1}{\Lambda^2}\mathrm{P}_1(\phi) + \frac{1}{\Lambda^4}\mathrm{P}_2(\phi). \tag{1.18}$$

It is therefore reasonable to consider the redefinition of fields as the best strategy in order to eliminate the effects of redundancy, obtaining a complete minimal-basis of operators in dimension d .
While the redefinition of fields changes the starter Lagrangian, the S-matrix elements do not and the theorem that assures this assumption is known as *Equivalence Theorem* [10],[11],[12] , reported in **Appendix A**.
From a physical perspective, the intuitive idea is that the fields are considered as integration variables in the functional path integrals, rather than as particles.

Starting from a Lagrangian $\mathcal{L}$, after the redefinition of fields, we arrive at

$$\mathcal{L}' = \mathcal{L} + \Delta\mathcal{L}, \tag{1.19}$$

where $\mathcal{L}'$ results in a different functional form that leads to a different normalization of the the kinetic terms .

The same functional form as $\mathcal{L}$ can be obtained by redefining both fields and parameters (such as couplings, mixing angles, and masses), a process known as *canonical normalization*. This ensures the correct normalization of the sources that enter the S-matrix.

1.3 Application: the scalar theory SEFT

To consolidate our findings, the next step will be analyze a simple theory containing one real scalar field [8],[9].
Starting from Eq.(1.14), we write down all the operators in $d = 6$ (the dimensional apex is omitted):

$$\begin{aligned}
&\mathcal{O}_0 = \phi^6, \\
&\mathcal{O}_1 = \phi^3 \Box \phi \ , \qquad \mathcal{O}_2 = \phi^2 \partial_\mu \phi \partial_\mu \phi \ , \qquad \mathcal{O}_3 = \phi \Box^2 \phi, \\
&\mathcal{O}_4 = \partial_\mu \phi (\partial_\mu \Box \phi) \ , \quad \mathcal{O}_5 = (\Box \phi)^2 \ , \qquad \mathcal{O}_6 = (\partial_\mu \phi \partial_\nu \phi)(\partial_\mu \phi \partial_\nu \phi).
\end{aligned} \tag{1.20}$$

At this point we identify three steps in order to write the EFT Lagrangian:

- IBP in order to perform a first reduction, obtaining

$$\mathcal{O}_2 = -\frac{1}{3}\mathcal{O}_1 \ , \qquad \mathcal{O}_4 = -\mathcal{O}_3 \ , \qquad \mathcal{O}_5 = \mathcal{O}_3 \ , \qquad \mathcal{O}_6 = -\mathcal{O}_3 - 2\mathcal{O}_4. \tag{1.21}$$

- defining linear combinations $\bar{a}_i$ such that

$$\bar{a}_0 = a_0 \ , \quad \bar{a}_1 = a_3 - a_4 + a_5 + a_6 \ , \quad \bar{a}_2 = a_1 - \frac{1}{3}a_2, \tag{1.22}$$

- Wilson coefficients rescaling

$$\bar{a}_i = g^{n-2}\frac{\bar{a}_i}{\Lambda^{k-4}}, \tag{1.23}$$

  with n= number of fields, k= mass dimension of the operators.

Assembling the various ingredients, we obtain:

$$\mathcal{L} = -\frac{1}{2}\partial_\mu \phi \partial_\mu \phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 + \frac{1}{\Lambda^2}\left(g^4 \bar{a}_0 \mathcal{O}_0 + \bar{a}_1 \mathcal{O}_3 + g^2 \bar{a}_2 \mathcal{O}_1\right). \tag{1.24}$$

The issue of redundancy is addressed by implementing a redefinition of fields similar to Eq. (1.18). When we truncate the expansion at Λ^{-2}, a useful redefinition can be expressed as follows:

$$\phi \to \phi + \frac{1}{\Lambda^2}\Big[\mathrm{f}(\phi) + \mathrm{g}(\phi)\Box\phi\Big] + \text{h.o}, \tag{1.25}$$

and all the operators containing $\Box^n\phi$ are removed.
Let us suppose that a solution for the functions f and g is given by

$$\mathrm{f}(\phi) = \alpha g^2\phi^3 \ , \quad \mathrm{g}(\phi) = \beta, \tag{1.26}$$

with:

$$\alpha = -\frac{1}{2}\left(\frac{\bar{a}_1}{6} + \bar{a}_2\right) , \quad \beta = -\bar{a}_1. \tag{1.27}$$

What we obtain is the new Lagrangian

$$\begin{aligned}\mathcal{L}' = &-\frac{1}{2}\partial_\mu\phi\partial_\mu\phi\left(1 + 2\bar{a}_1\frac{m^2}{\Lambda^2}\right) + \frac{1}{2}m^2\phi^2 - \frac{1}{24}g^2\phi^4\left[1 - 2\big(6\bar{a}_2 + \bar{a}_1\big)\frac{m^2}{\Lambda^2}\right] \\ &+ \frac{1}{72}\frac{g^2}{\Lambda^2}\phi^6\left(72\bar{a}_0 + 6\bar{a}_2 + \bar{a}_1\right) - \frac{1}{4}\frac{g^2}{\Lambda^2}\phi^2\partial_\mu\phi\partial_\mu\phi\left(6\bar{a}_2 + \bar{a}_1\right) + \text{h.o.}\end{aligned} \tag{1.28}$$

To generate the correct Feynman rules and guarantee the proper normalization of the kinetic terms, we apply the canonical normalization redefining fields and parameters as

$$\phi \to \left(1 + \frac{\bar{m}^2}{\Lambda^2}\,\zeta_\phi\right)\bar{\phi}, \quad g \to \left(1 + \frac{\bar{m}^2}{\Lambda^2}\,\zeta_g\right)\bar{g}, \quad m^2 \to \left(1 + \frac{\bar{m}^2}{\Lambda^2}\,\zeta_m\right)\bar{m}^2, \tag{1.29}$$

where the ζ_i quantities are given by

$$\zeta_\phi = -\bar{a}_1 \ , \qquad \zeta_g = 3\bar{a}_1 + 6\bar{a}_2 \ , \qquad \zeta_m = 2\bar{a}_1. \tag{1.30}$$

By introducing a new combinations $\hat{a}_i$ such that:

$$\hat{a}_0 = \bar{a}_0 - \frac{1}{12}\bar{a}_1 - \frac{1}{2}\bar{a}_2 \ , \qquad \hat{a}_1 = \bar{a}_1 \ , \qquad \hat{a}_2 = \frac{3}{2}\bar{a}_1 + 9\bar{a}_2, \tag{1.31}$$

the correct normalized Lagrangian results

$$\bar{\mathcal{L}} = -\frac{1}{2}\partial_\mu\bar{\phi}\partial_\mu\bar{\phi} - \frac{1}{2}\bar{m}^2\bar{\phi}^2 - \frac{1}{4!}\bar{g}^2\bar{\phi}^4 + \frac{\bar{g}^2}{\Lambda^2}\hat{a}_2\,\bar{\mathcal{O}}_2 + \frac{\bar{g}^4}{\Lambda^2}\hat{a}_0\,\bar{\mathcal{O}}_0, \tag{1.32}$$

where $\bar{\mathcal{O}}_2 = \bar{\phi}^2\partial_\mu\bar{\phi}\partial_\mu\bar{\phi}$, $\bar{\mathcal{O}}_0 = \bar{\phi}^6$. It is worth noting that the S-matrix elements will depend on the hat-coefficients.

For completeness, we close this section reporting the different compensations that EoM and the redefinition of fields give at $\mathcal{O}\left(\frac{1}{\Lambda^4}\right)$.
Let us consider Eq.(1.14) and its $d = 6$ extension

$$\mathcal{L}^{(6)} = \frac{1}{\Lambda^2}\bar{a}_2\mathcal{O}_1, \tag{1.33}$$

with $\mathcal{O}_1 = \phi^3\Box\phi$.
The EoM on the ϕ field lead to

$$(\Box - m^2)\phi = \frac{1}{6}g^2\phi^3 - \frac{g^2}{\Lambda^2}\bar{a}_2\left(6\phi\partial_\mu\phi\partial_\mu\phi + g^2\phi^5 + 6m^2\phi^3\right), \tag{1.34}$$

and the transformed Lagrangian reads

$$\mathcal{L} = \frac{g^2}{\Lambda^2}\bar{a}_2\left(m^2\phi^4 + \frac{1}{6}g^2\phi^6\right) - \left(\frac{g^2}{\Lambda^2}\bar{a}_2\right)^2\left(g^2\phi^8 + 6\phi^4\partial_\mu\phi\partial_\mu\phi + 6m^2\phi^6\right). \tag{1.35}$$

On the other hand, performing the following redefinition of fields

$$P_1 = -g^2\bar{a}_2\phi^3 \qquad P_2 = \alpha\,(g^2\bar{a}_2)^2\phi^5, \tag{1.36}$$

we obtain

$$\begin{aligned}\mathcal{L}' = \frac{g^2}{\Lambda^2}\bar{a}_2\left(m^2\phi^4 + \frac{1}{6}g^2\phi^6\right) + \frac{1}{2}(g^2\bar{a}_2)^2\Big[&\big(39 - 10\alpha\big)\phi^4\partial_\mu\phi\partial_\mu\phi \\ &- (1 + 2\alpha)m^2\phi^6 - 6g^2(3 + 2\alpha)\phi^8\Big],\end{aligned} \tag{1.37}$$

confirming that redefinition of fields and the use of EoM are consistent only up to $\mathcal{O}(\Lambda^{-2})$, but at higher orders, the redundant operators provide a distinct compensation.

Chapter 2

SMEFT and the Next-to-leading-order (NLO) investigation

2.1 SM overview and conventions

A comprehensive and formal description of the SM can be found in [13], [14], [15]. We focus our attention on the relevant technical aspects of the theory, fixing some useful conventions in order to describe the extension to the SMEFT.

The SM is the theory that explains three of the four forces known in nature: electromagnetic, weak and strong. The interactions and the fields are encoded in the Lagrangian, characterized by the local symmetry

$$SU(3)_C \times SU(2)_L \times U(1)_Y \tag{2.1}$$

which guarantees the invariance of Lagrangian when we perform a gauge transformation of the fields.

The fields'content in the SM Lagrangian can be summarized as follows:

- a singlet B_μ, a triplet W^I_μ (I= 1,...,3) and an octet G^A_μ (A=1,...,8) for gauge bosons;
- a doublet of complex scalar field Φ to describe the Higgs sector;

- fermion families;
- Faddeev-Popov ghost $X^{\pm}$, Y^Z, Y^A.

The neutral physical fields Z and A are related to W^3_μ and B_μ by the transformation

$$\begin{pmatrix} Z \\ A \end{pmatrix} = \begin{pmatrix} c_\theta & -s_\theta \\ s_\theta & c_\theta \end{pmatrix} \begin{pmatrix} W^3 \\ B \end{pmatrix}, \tag{2.2}$$

while the charged ones are reletad to $W^{1,2}_\mu$ by

$$W^{\pm}_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \mp iW^2_\mu). \tag{2.3}$$

Looking at the EW sector, the local gauge invariance is preserved by the covariant derivative D_μ:

$$D_\mu = \partial_\mu - \frac{i}{2}g_0 W^I_\mu \tau_I - \frac{i}{2}g g_1 B_\mu, \tag{2.4}$$

where $g_1 = -s_\theta/c_\theta$, τ_a are the Pauli matrices and θ is the weak mixing angle.

For a correct treatment of the neutral sector [3], we have to set:

$$g_0 = g\big(1 + g^2\Gamma\big), \tag{2.5}$$

where Γ is fixed by requiring that the $\gamma - Z$ transition vanishes at $p^2 = 0$.

2.1.1 The Yang-Mills sector

The dynamic of gauge bosons is described by the Yang-Mills Lagrangian $\mathcal{L}_{YM}$:

$$\mathcal{L}_{YM} = -\frac{1}{4}W^I_{\mu\nu}W_{\mu\nu,I} - \frac{1}{4}B_{\mu\nu}B_{\mu\nu} \tag{2.6}$$

where the field-strength tensors are defined by

$$\begin{aligned} W^I_{\mu\nu} &= \partial_\mu W^I_\nu - \partial_\nu W^I_\mu + g_0 \varepsilon^{IJK} W^J_\mu W^K_\nu, \\ B_{\mu\nu} &= \partial_\mu B_\nu - \partial_\nu B_\mu. \end{aligned} \tag{2.7}$$

2.1.2 The Higgs sector

The scalar sector is defined by the Lagrangian

$$\mathcal{L}_H = -(D_\mu \Phi)^\dagger D_\mu \Phi - \mu^2 \Phi^\dagger H - \frac{\lambda}{2}(\Phi^\dagger \Phi)^2, \tag{2.8}$$

which describes the Higgs dynamics and the mechanism for generating the gauge bosons and chiral fermions masses via *spontaneous symmetry breaking* (SSB), under the conditions $\lambda > 0$ and $\mu^2 < 0$.
The scalar field is a complex doublet given by

$$\Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} H + 2\frac{M}{g} + i\phi^0 \\ \\ \sqrt{2} i\phi^- \end{pmatrix}, \tag{2.9}$$

where H denotes the Higgs field, $\phi^\pm$ and ϕ^0 are the Goldstone boson associated to the EW symmetry breaking, M is the W bare mass and g is the bare coupling constant of $SU(2)$.
The scalar-potential $V(H^\dagger H)$ defined by

$$V(\Phi^\dagger \Phi) = -\mu^2 \Phi^\dagger \Phi - \frac{\lambda}{2}(\Phi^\dagger \Phi)^2, \tag{2.10}$$

induces tadpoles terms, relevant at one loop level, which have to be treated in the well-known β_h-scheme .
Setting:

$$\mu^2 = \beta_h - \frac{2\lambda M^2}{g^2}\ , \quad \lambda = \frac{g^2 M_H^2}{4M^2}, \tag{2.11}$$

and defining the Higgs VEV $v = 2M/g$, β_h is fixed order by order in perturbation theory by requiring $\langle 0| H |0\rangle = 0$.

2.1.3 Gauge fixing and Faddeev-Popov sectors

Expanded the product $(D_\mu \Phi)^\dagger D_\mu \Phi$, we find zeroth-order mixing terms as

$$M\left(\frac{1}{c_\theta} Z_\mu \partial \phi^0 + W_\mu^+ \partial_\mu \phi^- + W_\mu^- \partial_\mu \phi^+\right) \tag{2.12}$$

and these must cancel in order to ensure the development of a consistent perturbation theory.

We introduce a gauge-fixing Lagrangian $\mathcal{L}_{gf}$

$$\mathcal{L}_{gf} = -\mathcal{C}^+\mathcal{C}^- - \frac{1}{2}\Big[(\mathcal{C}^Z)^2 + (\mathcal{C}^A)^2\Big], \tag{2.13}$$

where

$$\mathcal{C}^A = -\frac{1}{\xi_A}\partial_\mu A_\mu, \quad \mathcal{C}^Z = -\frac{1}{\xi_Z}\partial_\mu Z_\mu + \xi_Z \frac{M}{c_\theta}\phi^0, \quad \mathcal{C}^\pm = -\frac{1}{\xi}\partial_\mu W^\pm_\mu + \xi M \phi^\pm. \tag{2.14}$$

and the cancellation of the mixing terms is obtained.
As consequence, $\mathcal{L}_{gf}$ breaks the gauge invariance which can be restored by the introduction of Faddeev-Popov ghost fields $X^\pm$, Y^Z, Y^A.

2.1.4 Fermion sector

We define the isodoublet

$$\psi = \begin{pmatrix} u \\ d \end{pmatrix} \tag{2.15}$$

and the chiral projection

$$\psi_{L,R} = \frac{1}{2}(1 \pm \gamma_5)\psi. \tag{2.16}$$

The fermion-gauge boson coupling is derived by the application of the covariant derivative on the L-R and fields:

$$D_\mu \psi_L = (\partial_\mu + g B^i_\mu T_i)\psi_L, \quad D_\mu \psi_R = (\partial_\mu + g B^i_\mu t_i)\psi_R, \quad (i = 0, ..., 3) \tag{2.17}$$

where we have used the generators of SU(2)×U(1)

$$T^a = -\frac{i}{2}\tau^a, \quad T^0 = -\frac{i}{2} g_2 I, \quad (a \neq 0) \tag{2.18}$$

and

$$t^a = 0, \quad t^0 = -\frac{i}{2}\begin{pmatrix} g_3 & 0 \\ 0 & g_4 \end{pmatrix}, \quad (a \neq 0). \tag{2.19}$$

Furthermore we have introduced the quantities $g_i = -s_\theta/c_\theta \lambda_i$ with:

$$\lambda_2 = 1 - 2Q_u, \quad \lambda_3 = -2Q_u, \quad \lambda_4 = -2Q_d \tag{2.20}$$

where λ_i fixed by requiring that the e.m current has the well-know structure $igs_\theta Q_f \bar{f}\gamma_\mu f$.

The problem of writing a mass term in SM Lagrangian for the fermion fields can be found in their chiral nature. It is well known that a term like $m\bar{\psi}\psi$ breaks the gauge symmetry, so we are led to write the Higgs-fermion interaction as

$$\mathcal{L}_S^{ferm} = -\sum_f m_f \bar{f} f + \mathcal{L}_S^{ferm,\mathrm{I}}, \tag{2.21}$$

where $\mathcal{L}_S^{ferm,\mathrm{I}}$ is the fermion-scalar interaction Lagrangian defined by

$$\mathcal{L}_S^{ferm} = -\alpha_f \bar{\psi}_{_L} \Phi u_{_R} - \beta_f \bar{\psi}_{_L} \Phi^c d_{_R} + \text{h.c.}. \tag{2.22}$$

Here we have introduced the conjugate scalar field $H_i^c = \varepsilon_{ij} H_j^*$ which results

$$\Phi^c = -\frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{2} i\phi^+ \\ \\ H + 2\frac{M}{g} - i\phi^0 \end{pmatrix}, \tag{2.23}$$

and the Yukawa couplings as

$$\alpha_f = \frac{1}{\sqrt{2}} g \frac{m_u}{M}, \quad \beta_f = -\frac{1}{\sqrt{2}} g \frac{m_d}{M}. \tag{2.24}$$

2.1.5 The strong sector

The QCD Lagrangian can be summarized as the sum of three fundamental pieces:

- gluons dynamic described by

$$\mathcal{L}_c = -\frac{1}{4} G^A_{\mu\nu} G^A_{\mu\nu} \tag{2.25}$$

  with $G^A_{\mu\nu} = \partial_\mu \mathrm{g}^A_\nu - \partial_\nu \mathrm{g}^A_\mu + g_s f^{ABC} \mathrm{g}^B_\mu \mathrm{g}^C_\nu$. Here f^{ABC} are the SU(3) structure constants and g_s is the strong coupling constant;

- quarks-gluon interaction given by

$$\mathcal{L}_c^{ferm} = \frac{1}{2} i g_s \big(\bar{q}_i^\sigma \gamma_\mu \lambda^A_{ij} q_j^\sigma\big) G^A_\mu, \tag{2.26}$$

  where λ^A are the Gell-Mann matrices

- FP sector of QCD

$$\mathcal{L}_c^{FP} = \bar{\kappa}^A \partial^2 \kappa^A + g_s f^{ABC} \partial_\mu \bar{\kappa}^A \kappa^B G_\mu^C, \tag{2.27}$$

  where κ^A is the ghost color field.

2.2 Construction of SMEFT

2.2.1 Introduction

Our main purpose is to construct a consistent theory of SM deviations: the basic idea is that the new physics is much heavier than the SM and a cut-off scale between that and the BSM physics is required.
Following the *bottom-up* approach, we consider the SM as an EFT.

The SMEFT Lagrangian is assembled by completing $\mathcal{L}_{SM}$ with all the possible operators in $d > 4$, preserving the Poincaré and the gauge symmetry given by the group in Eq.(2.1).
Considering the EW sector, SMEFT assumes that $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ is spontaneously broken to $\mathrm{U}(1)_{\mathrm{em}}$ by the Higgs VEV v, where the Higgs is a component of a doublet scalar field Φ in analogy with Eq.(2.9).

Form the Eq.(1.5), we can write the SMEFT Lagrangian as:

$$\mathcal{L}_{SMEFT} = \mathcal{L}_{SM}^{(4)} + \sum_{d>4} \sum_i \frac{a_i^{(d)}}{\Lambda^{d-4}} \mathcal{Q}_i^{(d)}. \tag{2.28}$$

The Wilson coefficients, denoted as $a_i^{(d)}$, are dimensionless quantities that quantify our lack of knowledge regarding high-energy physics. To extract signals or establish constraints on new physics, it becomes necessary to compute cross sections and decay amplitudes in the context of the SMEFT, then fit the coefficients to experimental data. In SM, all coefficients are set to zero, therefore any non-zero value observed may indicate the presence of a new physics event.

The operators $\mathcal{Q}_i^{(d)}$ are suppressed by the cut-off scale Λ ($\sim$ TeV) much larger than the EW symmetry breaking one; a $d = 5$ operator like $\mathcal{Q}^{(5)}$ will be suppressed by Λ^{-1}, while an operator like $\mathcal{Q}^{(6)}$ will be suppressed by Λ^{-2}

and so on.

In order to construct a theory which explain the possible deviations from the SM, a fundamental requirement is to have a minimal and non-redundant set of operators that form a basis. The selection of this basis is determined by grouping the operators into equivalence classes and then choosing the non-equivalent basis operators from each class. In the SMEFT this partitioning helps distinguish between potential tree-generated (PTG) and loop-generated (LG) operators .
The PTG operators arise from tree-diagrams of the underlying theory and they capture the effects of new physics beyond the EW symmetry breaking scale. The LG operators originate from quantum loop effects in the underlying theory, they incorporate the effect of quantum corrections at higher energy scale and they are suppressed by a factor $1/(16\pi^2)$.
The distinction between PTG and LG operators enables a systematic treatment of deviations from the SM and facilitates the analysis of Wilson coefficients.
Assuming that the high energy theory is weakly-coupled and renormalizable, the Einhorn-Wudka classification result valid [16]. Working in a different context, for example considering the next high-energy theory as an EFT that involves some strongly interacting theory, the classification changes, as reported in Eq.(A1-A2) of [17].

For our purposes, we will extend the Standard Model up to $d = 6$. When we consider $d = 5$, it can be shown that only one operator in that dimension is available, but it does not conserve lepton number [18] From the Lagrangian:

$$\mathcal{L}^{(5)} = \frac{a^{(5)}}{\Lambda} \varepsilon_{ij} \bar{L}_L^{c,i} \Phi^j \varepsilon_{kl} L_L^k \Phi^l, \tag{2.29}$$

we can get a Majorana mass term for the left-handed neutrinos in the form

$$\frac{1}{2} m_\nu \nu_L^a \nu_L^b \varepsilon_{ab} \tag{2.30}$$

where

$$m_\nu = \frac{1}{2} \frac{a^{(5)} v^2}{\Lambda}. \tag{2.31}$$

Since from experimental data results $m_\nu < 0.5$eV, for $a^{(5)} \sim 1$ we can give a constraint on the cut-off scale:

$$\Lambda > 10^{14} \text{ GeV } \ [\sim \text{GUT scale}] \tag{2.32}$$

The net result is that a Majorana mass term in the Lagrangian violates the lepton number conservation when the weak interaction is spontaneously broken.

In $d = 6$ scenario [19],[20], the standard SMEFT basis in use is the Warsaw basis, reported in **Appendix B**.
Setting the number of fermion generations $N_f = 3$, we can construct 2499 operators that conserve baryonic and leptonic number ($\Delta B = \Delta L = 0$), while for $N_f = 1$ we obtain 76 different operators [3].

The operators in $d = 7$ and $d = 8$ are treated in more detail in Ref.[21],[22],[23].

2.2.2 Operator classes in the Warsaw basis

Let us considering the extension of $\mathcal{L}_{SM}$ in $d = 6$. It is possible to construct seven classes of operators in the pure bosonic sector:

$$\left\{X^3,\ X^2\Phi,\ X^2D^2,\ X\Phi^2D^2,\ \Phi^6,\ \Phi^4D^2,\ \Phi^2D^4\right\}, \tag{2.33}$$

where X denotes a generic gauge field strength $X_{\mu\nu}$, Φ is the scalar doublet and D is the covariant derivative.
Removing the redundant operators Φ^2D^4, $X\Phi^2D^2$, and X^2D^2, we write the minimal non-redundant bosonic set as

$$\left\{X^3,\ \Phi^6,\ \Phi^4D^2,\ X^2\Phi^2\right\}. \tag{2.34}$$

In the same way we construct the classes that involve the fermionic fields:

$$\left\{\psi^2\Phi^3,\ \psi^2X\Phi,\ \psi^2\Phi^2D, \psi^4\right\}. \tag{2.35}$$

The eight classes are presented in **Appendix B**.

[3] Appendix B reports only 59 operator that satisfy $\Delta B = \Delta L = 0$ for $N_f = 1$. Hermitian operators have real coefficients in the Lagrangian, while non-Hermitian operators have complex coefficients. Considering $(\Phi^\dagger\Phi)^3$, this is an Hermitian operator and just one real coefficient is associated to it. On the other hand, the configuration $(\Phi^\dagger\Phi)(\bar{l}e\Phi)$ is not an Hermitian operator, it has a complex coefficient so it counts as two Hermitian operators.

Class 1: X^3

In this class we find four operators, which involves three gauge field strengths. These operators modify the kinetic terms of the SM gauge bosons and can be employed in the analysis of multijet events [24] or in studies related to *vector boson scattering* (VBS) [25].

Class 2: Φ^6

This class involves only the operator $\mathcal{Q}_\Phi = (\Phi^\dagger\Phi)^3$, which modify the self interactions of the higgs field as well as the scalar potential structure. The study of this class is crucial to understand the dynamics of EW symmetry breaking and the origin of the mass in the SM.

Class 3: $\Phi^4 D^2$

Here we find the extension in $d = 6$ of the Higgs-vectorial gauge bosons interactions.
The contribution of these operators is relevant in process as production of Higgs boson via *vector boson fusion* (VBF), e.g $ZZ \to H$.

Class 4: $X^2\Phi^2$

In this class we find eight operators with two Higgs doublet and two gauge field strength tensors.
These operators lead to point-like interactions $H\gamma\gamma$ or Hgg, banned by SM. To give an idea, the process $H \to \gamma\gamma$ is available in the SMEFT at LO, while in the SM it takes place at one loop level.

Class 5: $\psi^2\Phi^3$

These operators involve a fermion pair and three Higgs fields, leading to a modification of the Yukawa coupling.
Unlike in the Standard Model (SM), where Yukawa interactions are proportional to the fermion masses, the operators in Class 5 introduce off-diagonal terms in the Yukawa matrices.

Class 6: $\psi^2 X\Phi$

Here we find two fermion fields, a gauge field strength and the Higgs doublet. When the H field gets a VEV, these operators lead to dipole operators for process as $\mu \to e\gamma$ or $b \to s\gamma(g)$.

Class 7: $\psi^2 H\Phi^2 D$

This class modify the coupling of EW bosons with the fermions.

Class 8: ψ^4

The last class involves four fermion operators ψ^4 which can be grouped into different sets, depending on the chirality properties of the operators. This class is relevant in process with external fermion legs and in fermion self energies computation.

2.2.3 Canonical normalization and S-matrix

To connect the SMEFT Lagrangian to the S-matrix elements, we have to require the correct normalization of the kinetic terms.
Let us consider :

$$\mathcal{L}_{SMEFT}(\{\chi\},\{p\},\{a_i\}) = \mathcal{L}_{SM}(\{\chi\},\{p\}) + \mathcal{L}_6(\{\chi\},\{p\},\{a_i\}), \quad (2.36)$$

where $\{\chi\}$, $\{p\}$ and $\{a_i\}$ are the collections of fields, parameters and Wilson coefficients.
The canonical normalization requires the introduction of new quantities $\bar{p}$, $\bar{\chi}$ defined by:

$$p = \left(1 + \frac{M^2}{\Lambda^2} dR_p\right)\bar{p}\ , \quad \chi = \left(1 + \frac{M^2}{\Lambda^2} dR_\chi\right)\bar{\chi}\ , \quad (2.37)$$

where the corrective factors dR_i are collected in [26], Sec.2.4.
Following the β_h-scheme, we redefine the β_h parameter as

$$\beta_h = 12\frac{M^4}{g^2}\frac{a_H}{\Lambda^2} + \left(1 + \frac{M^2}{\Lambda^2} dR_{\beta_h}\right)\bar{\beta}_h \quad (2.38)$$

and $\bar{\beta}_h$ has to be fixed order-by-order to have zero vacuum expectation value for the normalized Higgs field.

In the same way, we have to redefine the gauge parameters in Eq.(2.14) (normalized to one), obtaining

$$\xi_i = 1 + dR_{\xi_i} R_\Lambda \ , \quad i = Z, A, W. \tag{2.39}$$

where we have introduced the new quantity $R_\Lambda = M^2/\Lambda^2$.

The normalization of the physical fields in the neutral sector can lead to a non-zero transition between the Z boson and the photon at $\mathcal{O}(\Lambda^{-2})$. Defining the "normalized" fields $\bar{Z}, \bar{A}$ from

$$\begin{pmatrix} Z \\ A \end{pmatrix}_\mu = \begin{pmatrix} R_{ZZ} & R_{ZA} \\ R_{AZ} & R_{AA} \end{pmatrix} \begin{pmatrix} \bar{Z} \\ \bar{A} \end{pmatrix}_\mu \tag{2.40}$$

and introducing the following combination of Wilson coefficients

$$a_{AZ} = 2\bar{s}_\theta \bar{c}_\theta (a_{\Phi W} - a_{\Phi B}) + (2\bar{c}_\theta^2 - 1) a_{\Phi WB}, \tag{2.41}$$

it is possible to construct a new vertex as

$$\sim \ \frac{a_{AZ}}{\Lambda^2}\left(p_\mu p_\nu - \delta_{\mu\nu} p^2\right) \ . \tag{2.42}$$

We can now present the normalized SMEFT Lagrangian, from which we can derive the Feynman rules and proceed with the calculation of S-matrix elements:

$$\begin{aligned} \mathcal{L}_{SMEFT}(\{\bar{\chi}\}, \{\bar{p}\}, \{a_i\}) = \mathcal{L}_{SM}(\{\bar{\chi}\}, \{\bar{p}\}) \\ + \mathcal{L}_6(\{\bar{\chi}\}, \{\bar{p}\}, \{a_i\}) + \mathcal{L}_6^{AZ}\left(\bar{Z}_\mu, \bar{A}_\mu, a_{AZ}\right), \end{aligned} \tag{2.43}$$

where $\mathcal{L}_6^{AZ}\left(\bar{Z}_\mu, \bar{A}_\mu, a_{AZ}\right)$ describes the $Z - \gamma$ transition at $\mathcal{O}(\Lambda^{-2})$.

To provide an illustration of how canonical normalization works, let us focus on the scalar sector. The SM potential is modify by the Class 2 operator $\mathcal{Q}_\Phi = (\Phi^\dagger \Phi)^3$, leading to a shift in the minimum of the scalar potential [27]. The new VEV is defined by:

$$v_T = \left(1 + \frac{3v^2}{8\lambda} a_H\right) v, \tag{2.44}$$

where v is the VEV in the SM and a_H is the rescaled Wilson coefficient associated to $\mathcal{Q}_H$.
The kinetic part is extended by the insertion of operators in class 3, which involve only Higgs field and derivative terms. We can write:

$$\begin{aligned}\mathcal{L}_{\Phi,kin} &= -(D_\mu\Phi^\dagger)(D_\mu\Phi) - a_{\Phi\Box}(\Phi^\dagger\Phi)\Box(\Phi^\dagger\Phi) - a_{\Phi D}(\Phi^\dagger D_\mu\Phi)^*(\Phi^\dagger D_\mu\Phi)\\ &= -\frac{1}{2}(\partial_\mu H)^2 - \frac{1}{2}(\partial_\mu\phi^0)^2 - \partial_\mu\phi^-\partial_\mu\phi^+\\ &\quad + a_{\Phi\Box}\, v_T^2(\partial_\mu H)^2 - a_{\Phi D}\,\frac{v_T^2}{4}(\partial_\mu H)^2 - a_{\Phi D}\frac{v_T^2}{4}(\partial_\mu\phi^0)^2.\end{aligned} \tag{2.45}$$

In order to obtain the same functional form as that found in the SM, we have to redefine the fields as

$$\begin{aligned} H &\to \left(1 + a_{\Phi\Box}v_T^2 - a_{\Phi D}\frac{v_T^2}{4}\right) H,\\ \phi^0 &\to \left(1 - a_{\Phi D}\frac{v_T^2}{4}\right)\phi^0.\end{aligned} \tag{2.46}$$

and the new Higgs doublet results:

$$\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix}\left[1 + a_{\Phi\Box}v_T^2 - a_{\Phi D}\frac{v_T^2}{4}\right]H + v_T + i\left[1 - a_{\Phi D}\frac{v_T^2}{4}\right]\phi^0\\ \sqrt{2}i\phi^-\end{pmatrix}. \tag{2.47}$$

We notice that a redefinition is not necessary for the charged Goldstone bosons since they preserve the canonical normalization even after the insertion of the $\mathcal{Q}_H$ operator; Additionally, the field redefinition does not modify the value of the minimum in the scalar potential, and the relationship in Eq.(2.44) still remains valid.

2.3 NLO calculation

2.3.1 Amplitude

The most generic amplitude that can be constructed within the SMEFT framework is written as

$$\mathcal{M} = \sum_{n=N}^{\infty} g^n \mathcal{M}_n^{(4)} + \sum_{n=N_6}^{\infty}\sum_{l=1}^{n}\sum_{k=1}^{\infty} g^n \left[\frac{1}{(\sqrt{2}G_F\Lambda^2)^k}\right]^l \mathcal{M}_{n,l,k}^{4+2k}, \tag{2.48}$$

$$
\begin{array}{lllll}
g/d & \rightarrow & & & \\
\downarrow & & & & \\
 & g\mathcal{M}_1^{(4)} & + gg_6\mathcal{M}_{1,1,1}^{(6)} & + gg_8\mathcal{M}_{1,1,2}^{(8)} & ... \\
 & g^3\mathcal{M}_3^{(4)} & + g^3g_6\mathcal{M}_{3,1,1}^{(6)} & + g^3g_6^2\mathcal{M}_{3,2,1}^{(6)} & ... \\
 & ... & ... & ... & ...
\end{array}
$$

Figure 2: The double expansion for a generic $1 \rightarrow 2$ amplitude.

where g is the SM coupling constant, G_F is the Fermi constant and Λ is the cut-off scale. The index l indicates the number of insertions for a given SMEFT operator and k indicates the inverse mass dimension of the Lagrangian terms inserted.
By setting N, we establish LO at $d = 4$. For instance, when N=1, it encompasses processes such as HVV (with $V = Z, W$), while for N=3, it covers $H \rightarrow \gamma\gamma$.
When we consider tree-level-initiated processes, the value of N_6 is equal to N, while for loop-initiated ones we have to set $N_6 = N - 2$.

Let us consider H$\gamma\gamma$. In the SM, there is no vertex like $\gamma\gamma$, so we need to start the computation of the amplitude at the one-loop order. However, in the SMEFT there is a vertex like this one, generated by the operator $\mathcal{Q}_{HB} = H^\dagger H B_{\mu\nu} B_{\mu\nu}$, which guarantee a LO contribution.

We introduce the notation:

$$\frac{1}{(\sqrt{2}G_F\Lambda^2)^k} = g_{4+2k}, \tag{2.49}$$

where the new coupling constant g_{4+2k} denotes the level of precision required: g_6 indicates a single insertion of $\mathcal{Q}^{(6)}$, g_6^2 indicates two distinct insertions of $\mathcal{Q}^{(6)}$, g_8 denotes a single insertion of an operator in $d = 8$ as $\mathcal{Q}^{(8)}$, etc.
The obvious conclusion is that the SMEFT involves a double expansion in $1/\Lambda$ and in g. as reported in Fig.(2) [27]: where the term $gg_6\ \mathcal{M}_{1,1,1}$ define the LO in the SMEFT with a single insertion of a generic operator $\mathcal{Q}^{(6)}$, while $g^3g_6\ \mathcal{M}_{3,1,1}$ denotes its natural extension to the one-loop corrections.

2.3.2 One-loop corrections and Renormalization in SMEFT

The NLO corrections are necessary to show the consistency of the SMEFT as a well defined field theory. Perform an NLO calculation for a given process means constructing the observables with the complete one-loop correction part, including all operators at a fixed order in the power counting.
The main ingredients [28] for the NLO calculation are defined by:

- SMEFT vertices in the tree-level SM diagrams;
- SMEFT tree-level diagrams with a non SM-topology;
- SMEFT vertices inserted in the one-loop SM diagrams;
- SMEFT one-loop diagrams with a non SM topology.

Considering just one insertion in $d = 6$, a generic one-loop amplitude reads as follows:

$$\mathcal{M} = g^N \left(\mathcal{M}^{(0,4)} + \frac{g_6}{\sqrt{2}} \mathcal{M}^{(0,6)} \right) + \frac{g^{N+2}}{16\pi^2} \left(\mathcal{M}^{(1,4)} + \frac{g_6}{\sqrt{2}} \mathcal{M}^{(1,6)} \right), \qquad (2.50)$$

where $\mathcal{M}^{(m,n)}$ represents the part of amplitude at the order m in perturbation theory and mass dimension n, i.e the notation (0,4) indicates the LO terms in $d = 4$ while (1,6) denotes the one-loop terms in $d = 6$.

The implementation of NLO calculations introduces UV poles. To cancel out these divergences and ensure a predictive theory, a renormalization procedure is necessary.
Once divergences are removed by introducing counterterms on parameters (and fields), we are able to predict the finite value of an observable at a some order of accuracy. If the infrared and collinear divergences are involved in the calculation, they vanish by adding the real emission in accordance with the Kinoshita-Lee-Nauenberg (KLN) theorem [29],[30].

Generally an EFT is not strictly renormalizable. As we have said in our introduction, the expansion in power of $1/\Lambda$ leads to an infinite number of counterterms and the predictive power of the theory can be restored by the truncation of the series in Eq.(1.5) at some order in Λ.

The bare-fields $\{\chi_0\}$ and the bare-parameters $\{p_0\}$ have to be generally replaced by the renormalized quantities $\chi_{\rm ren}$ and p_{ren} defined by the relations:

$$\chi_0 = \mathrm{Z}_\chi\ \chi_{\rm ren}\ , \quad p_0 = \mathrm{Z}_p\ p_{ren}. \tag{2.51}$$

The counterterms Z_i have to respect the following decomposition:

$$\mathrm{Z}_i = 1 + \frac{g^2}{16\pi^2}\left(d\mathrm{Z}_i^{(4)} + \frac{g_6}{\sqrt{2}} d\mathrm{Z}_i^{(6)}\right)\Delta_{UV}. \tag{2.52}$$

where $\mathrm{d}Z_i^{(4)}$ denotes the counterterm relative to the SM while $\mathrm{d}Z_i^{(6)}$ represents the new part in $d = 6$. Here, we have introduced the UV regulator

$$\Delta_{UV} = \frac{2}{\epsilon} - \gamma_E - \ln\pi - \ln\frac{\mu_R^2}{\mu^2} = \frac{1}{\bar{\epsilon}} - \ln\frac{\mu_R^2}{\mu^2}, \tag{2.53}$$

where $\epsilon = 4 - d$, d is the space-time dimension in *dimensional regularization* scheme, γ_E is the Euler-Mascheroni constant, μ_R is the renormalization scale and μ is an auxiliary mass which is canceled in any UV-renormalized quantity.

The presence of a $d = 4$ component implies that the renormalization procedure in $d = 6$ can be carried out similarly to the SM.
In our calculation we will adopt the LEP strategy: for a given process $|i\rangle \to |f\rangle$, we compile the self-energy corrections, perform Dyson resummation and introduce counterterms on the parameters. Then we compute the wave function renormalization (WFR) factors for each external legs and no counterterms are introduced on the fields. The introduction of three-point (or higher) Green functions leads us to obtain a finite $d = 4$ amplitude, with a residual divergence UV divergence in $d = 6$. In this regard, we have to perform a **mixing** among the Wilson coefficients, defining:

$$a_i = \sum_j \mathrm{Z}_{ij}^a\ a_j^{ren}. \tag{2.54}$$

The matrix Z_{ij} is expressed by

$$\mathrm{Z}_{ij}^a = \delta_{\mu\nu} + \frac{g^2}{16\pi^2}\ d\mathrm{Z}_{ij}^a\ \Delta_{UV}, \tag{2.55}$$

and we fix it by requiring the cancellation of the remaining UV terms [4].

[4]The renormalized Wilson coefficients show a logarithmic scale dependence which should be removed by resummation in terms of LO coefficients of anomalous dimension matrix. More details are given in [31].

2.4 Propagators and WFR factors

2.4.1 Boson field

Since a large part of the renormalization procedure involves two-point Green functions, we want analyze the case of a generic scalar field X with a bare mass m_X.
Let us consider the bare propagator Δ_X: turning to the dressed one , we sum up all the one-loop insertions performing the Dyson resummation.
What we obtain is

$$\bar{\Delta}_X^{-1}(p^2) = p^2 + m_X^2 - \frac{g^2}{16\pi^2}\Sigma_X(p^2) \tag{2.56}$$

where we have omitted an overall factor $1/(2\pi)^4 i$.
The self-energy $\Sigma_X(p^2)$ follows the decomposition

$$\Sigma_X(p^2) = \Sigma_X^{(4)}(p^2) + \frac{g_6}{\sqrt{2}}\Sigma_X^{(6)}(p^2), \tag{2.57}$$

where each term can be divided into a finite part and a divergent one:

$$\Sigma_X^{(i)}(p^2) = \Sigma_X^{(i),fin}(p^2) + \Sigma_X^{(i),UV}(p^2)\frac{1}{\bar{\epsilon}}, \quad i = 4, 6. \tag{2.58}$$

Therefore we define the renormalized mass $m_{X,ren}$ as

$$m_X = m_{X,ren}\left[1 + \frac{g}{16\pi^2}\left(dZ_{m_X}^{(4)} + \frac{g_6}{\sqrt{2}}dZ_{m_X}^{(6)}\right)\frac{1}{\bar{\epsilon}}\right]. \tag{2.59}$$

with dZ_{m_X} representing the relative counterterm.
The next step is to expand the self energy in Eq.(2.57) around $p^2 = -M_X^2$, where M_X is the physical mass of the X field:

$$\Sigma_X(p^2) = \Sigma_X(-M_X^2) + (p^2 + M_X^2)\Sigma_X'(p^2)\Big|_{p^2=-M_X^2} + \mathcal{O}\big((p^2 + M_X^2)^2\big)\ , \tag{2.60}$$

where we have used the shorthand notation $f'(x) = df(x)/dx$.
The complete equation for the propagator reads

$$\begin{aligned}\bar{\Delta}_X^{-1}(p^2) = p^2 + m_{X,ren}^2\left[1 + \frac{g}{16\pi^2}\ 2\left(dZ_{m_X}^{(4)} + \frac{g_6}{\sqrt{2}}dZ_{m_X}^{(6)}\right)\frac{1}{\bar{\epsilon}}\right] \\ - \frac{g^2}{16\pi^2}\left[\Sigma_X(-M_X^2) + (p^2 + M_X^2)\Sigma_X'(p^2)\Big|_{p^2=-M_X^2}\right] + \mathcal{O}\big((p^2 + M_X^2)^2\big).\end{aligned} \tag{2.61}$$

Defining the physical mass as

$$m_{X,ren} = M_X \left[1 + \frac{g}{16\pi^2} \left(\delta^{(4)}_{M_X} + \frac{g_6}{\sqrt{2}} \delta^{(6)}_{M_X} \right) \right], \tag{2.62}$$

where δ_{M_X} denotes the finite part of the correction, the resummed propagator results

$$\begin{aligned} \bar{\Delta}_X^{-1}(p^2) = p^2 + M_X^2 &+ \frac{g^2}{16\pi^2}\, 2M_X^2 \left(dZ^{(4)}_{M_X} + \frac{g_6}{\sqrt{2}} dZ^{(6)}_{M_X} \right) \frac{1}{\bar{\epsilon}} \\ &+ \frac{g^2}{16\pi^2}\, 2M_X^2 \left(\delta^{(4)}_{M_X} + \frac{g_6}{\sqrt{2}} \delta^{(6)}_{M_X} \right) \\ &- \frac{g^2}{16\pi^2} \left[\Sigma_X(-M_X^2) + (p^2 + M_X^2)\Sigma'_X(p^2) \Big|_{p^2=-M_X^2} \right] + \mathcal{O}\big((p^2 + M_X^2)^2\big). \end{aligned} \tag{2.63}$$

The counterterms are fixed by setting:

$$\bar{\Delta}_X^{-1}(-M_X^2) = 0 \tag{2.64}$$

and that implies:

$$\begin{aligned} dZ^{(4)}_{m_X} &= \frac{1}{2M_X^2} \Sigma_X^{(4,UV)}(-M_X^2), \\ dZ^{(6)}_{m_X} &= \frac{1}{2M_X^2} \Sigma_X^{(6,UV)}(-M_X^2). \end{aligned} \tag{2.65}$$

The new counterterms δ_{M_X} are fixed by requiring

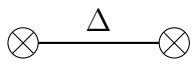
$$M_X^2 = m_X^2 - \frac{g^2}{16\pi^2} \mathrm{Re}\Sigma_X(M_X^2), \tag{2.66}$$

and the final expression can be written as

$$\begin{aligned} \Delta_X^{-1}(p^2) &= \left[1 - \frac{g^2}{16\pi^2} \Sigma'_X(p^2) \Big|_{p^2=-M^2} \right] (p^2 + M_X^2) + \mathcal{O}\big((p^2 + M_X^2)^2\big) \\ &= Z_X(p^2 + M_X^2) + \mathcal{O}\big((p^2 + M_X^2)^2\big), \end{aligned} \tag{2.67}$$

Note that a UV divergence from the derivative function Σ' is still present. Z_X is the WFR factor and its function is strictly connected with the normalization of the sources.

Let us denote by J a generic source. The propagation source-source can be represented by

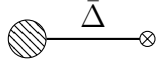


$$= J\Delta J = J\frac{1}{p^2+m_X^2}J, \tag{2.68}$$

where the source terms are figured by a crossed dot and the bare propagator is denoted by Δ.
If we take into account the renormalized propagator $\bar{\Delta}$, in order to preserve the propagation probability equal to one (we are considering a non-interacting propagation), we have to write:

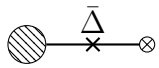
$$J\frac{1}{Z_X(p^2+M_X^2)}J = (Z_X^{1/2}J)\frac{1}{Z_X(p^2+M_X^2)}(Z_X^{1/2}J). \tag{2.69}$$

At this point it is crucial to exercise caution when dealing with the S-matrix. Indicating with $\mathcal{M}$ (the full blob) a generic S-matrix element, we find



$$= \mathcal{M}\ \frac{1}{Z_X(p^2+M_X^2)}\ Z_X^{1/2}J. \tag{2.70}$$

After that, we cut the Green function multiplying by $(p^2+M_X^2)$ and take the limit $p^2 \to M_X^2$, obtaining:

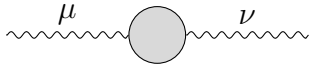


$$= \mathcal{M}\ Z_X^{-1/2}J. \tag{2.71}$$

The net effect is a factor $Z^{-1/2}$ to each external leg, where Z_X is denoted by the following equation:

$$Z_X^{-1/2} = 1 + \frac{1}{2}\ \frac{g^2}{16\pi^2}\Sigma'_X(p^2)\Big|_{p^2=-M^2}. \tag{2.72}$$

The same approach is applicable when we deal with a vector boson field V. An irreducible vector boson self energy may be expressed as



$$= \frac{g^2}{16\pi^2}\Big[A(p^2)\delta_{\mu\nu} + B(p^2)p_\mu p_\nu\Big], \tag{2.73}$$

with

$$\begin{aligned} A(p^2) &= A^{(4)}(p^2) + \frac{g_6}{\sqrt{2}}A^{(6)}(p^2) \\ B(p^2) &= B^{(4)}(p^2) + \frac{g_6}{\sqrt{2}}B^{(6)}(p^2). \end{aligned} \tag{2.74}$$

The complete propagator is obtained by repeating insertion of self energy diagrams:

$$\bar{\Delta}_{V,\mu\nu} = \frac{\delta_{\mu\nu}}{p^2 + M_V^2 - \frac{g^2}{16\pi^2}A} + \frac{Bp_\mu p_\nu}{(p^2 + M_V^2 - A)(p^2 + M^2 - A - Bp^2)}. \quad (2.75)$$

Assuming that V couples to conserved currents, we may set $B = 0$.

The pole of propagator is at some value $p^2 = s_V$, with $s_V = -M_V^2 + A(s_V)$ and we may expand

$$A(p^2) = A(s_V) + (p^2 - s_V)A_1 + (p^2 - s_V)^2 A_{rest}. \quad (2.76)$$

In going to S-matrix, we cut the Green function finding that each external leg contributes with a factor $(1 - A_1)^{-1/2}$.

2.4.2 Fermion propagator

We start from the resummed propagator of a fermion field with bare mass m_0(we omit the prefactor $1/(2\pi)^4 i$)

$$G(p) = \left[i\not{p} + m_0 - \Sigma(i\not{p}) \right]^{-1}, \quad (2.77)$$

where the self energy is defined by

$$\Sigma(\not{p}) = \frac{g^2}{16\pi^2}\left[\Delta_f + (V_f - A_f\gamma_5)i\not{p} \right]. \quad (2.78)$$

The various components respect the decomposition

$$\begin{aligned} \Delta_f &= \Delta_f^{(4)} + \frac{g_6}{\sqrt{2}}\Delta_f^{(6)} \text{ (scalar)}, \\ V_f &= V_f^{(4)} + \frac{g_6}{\sqrt{2}}V_f^{(6)} \text{ (vectorial)}, \\ A_f &= A_f^{(4)} + \frac{g_6}{\sqrt{2}}A_f^{(6)} \text{ (axial)}. \end{aligned} \quad (2.79)$$

The WFR factors are extracted by defining the factors $Z_V,\ Z_A$, such that

$$G(p) = \left[(Z_V + Z_A\gamma_5)(i\not{p} + m)(Z_V - Z_A\gamma_5) \right]^{-1}. \quad (2.80)$$

Expanding up to $\mathcal{O}(g^2)$, we obtain:

$$Z_V = 1 - \frac{g^2}{16\pi^2} W_V, \quad Z_A = \frac{g^2}{16\pi^2} W_A, \tag{2.81}$$

where it is always valid the decomposition

$$W_{V;A} = W^{(4)}_{V,A} + \frac{g_6}{\sqrt{2}} W^{(6)}_{V,A}. \tag{2.82}$$

The next step is to expand Eq.(2.77) around $i\not{p} = m$ and matching with expansion of Eq.(2.80) up to g^2. In this way, we obtain the renormalized equation for the mass and the expressions of W factors:

$$\begin{aligned} m_0 &= m\left\{1 + \frac{g^2}{16\pi^2}\left[\frac{1}{m}\Delta_f(m) - V_f(m)\right]\right\}, \\ W_V &= \frac{1}{2}V_f(m) + m\Delta'_f(m) - m^2 V'_f(m), \\ W_A &= \frac{1}{2}A_f. \end{aligned} \tag{2.83}$$

In going to S-matrix, the net effect will be given by the following replacement:

$$\begin{aligned} u(p) &\to \left[1 + \frac{g^2}{16\pi^2}\big(W_V + W_A\gamma_5\big)\right]u(p) = Z_u^{-\frac{1}{2}}u(p), \\ \bar{u}(p) &\to \bar{u}(p)\left[1 + \frac{g^2}{16\pi^2}\big(W_V - W_A\gamma_5\big)\right] = \bar{u}(p)Z_{\bar{u}}^{-\frac{1}{2}}, \end{aligned} \tag{2.84}$$

and similar for the $v-$spinors.

Chapter 3

SMEFT in act

3.1 Set-up

The introduction of operators in $d = 6$ significantly amplifies the number of Feynman diagrams involved in a generic transition amplitude.
The numerical implementation of the final results, for example cross sections or decay width amplitude, heavily depends on the chosen "symbolic manipulation" technique. Different computational approaches can be employed to handle the calculations, simplifying the evaluation of the Feynman diagrams and facilitating the extraction of the relevant quantities.

To address these challenges, specific software tools have been developed in the last years as `Package-X` [32] and `FeynCalc` [33] which are designed specifically for the software *Mathematica by Wolfram*.

For a semi-automatic and analytical computation of Feynman diagrams, we have utilized the symbolic manipulation program `FORM`, which facilitates the organization and manipulation of algebraic expressions.

An overview of the functionality of our program, saved in the main file `SMEFTO.frm`, is presented below.

The first step concerns the construction of three distinct library files `Symb.h`, `Lib.h` and `Vert.h` which have to be included in the main one:

```
#include Symb.h
#include Lib.h
#include Vert.h
```

In the first file, we define all the functions, symbols and vectors involved in the computation, in the second one we gather all the necessary procedures such as manipulation of matrix strings, solving one-loop integrals, and reducing Dirac matrix strings, among others.. The last file contains the vertices.

Given a generic one-loop diagram, we store it in a global variable, let us call it `diag1`. Then, we insert the vertices from `Vert.h` by calling:

```
#call VertSMEFT
```

The following procedure, stored in `Lib.h`,

```
#procedure Nwc
```

is called in order to introduce a more readable notation for the coefficients. The treatment of $d = 6$ expressions is simplified by introducing linear combinations of Wilson coefficients, as reported in [26],Eq.(206). This operation is executed by using the following command:

```
#call Makewc.
```

which provides a new vector-like notation (see Table 1 on the next page).

When fermions are involved, the spinorial structure is handled by implementing the Dirac equations through the following command

```
#call DiracEq(x)
```

where `x` is the index that distinguishes between different spinorial lines.

The manipulation and reduction of products of gamma matrices are performed using the Kahane's algorithm [34] and subsequently by the Chisholm one [35]. Furthermore, if a fermionic loop is involved, we need to trace. These three operations are carried out by performing

Table 1: Vector-like notation for Wilson coefficients

$a_{AA} = W_1$	$a_{uWB} = W_{11}$	$a_{HuA} = W_{21}$	$a_{uu} = W_{39}$	$a_{HG} = W_{49}$
$a_{ZZ} = W_2$	$a_{uBW} = W_{12}$	$a_{HuV} = W_{22}$	$a_{dd} = W_{40}$	$a^{(8)}_{Qd} = W_{50}$
$a_{AZ} = W_3$	$a_{LH} = W_{13}$	$a_{LldQ} = W_{23}$	$\sqrt{2}a_{dG} = W_{41}$	$a^{(8)}_{QuQd} = W_{51}$
$a_{HD} = W_4$	$a_{dH} = W_{14}$	$a^{(1)}_{QuQd} = W_{24}$	$a_{QW} = W_{42}$	$a^{(1)}_{Ql} = W_{52}$
$a_{H\Box} = W_5$	$a_{uH} = W_{15}$	$a_{Ll} = W_{27}$	$a_{Hud} = W_{43}$	
$a_H = W_6$	$a_{HlA} = W_{16}$	$a_{LL} = W_{28}$	$\sqrt{2}a_{uG} = W_{44}$	
$a_{lWB} = W_7$	$a_{HlV} = W_{17}$	$a_{ll} = W_{32}$	$a^{(3)}_{QQ} = W_{45}$	
$a_{lBW} = W_8$	$a_{Hn} = W_{18}$	$a_{Ld} = W_{33}$	$a^{(3)}_{LQ} = W_{46}$	
$a_{dWB} = W_9$	$a_{HdA} = W_{19}$	$a^{(1)}_{QQ} = W_{37}$	$a^{(1)}_{LlQu} = W_{47}$	
$a_{dBW} = W_{10}$	$a_{HdV} = W_{20}$	$a^{(1)}_{ud} = W_{38}$	$a^{(3)}_{LlQu} = W_{48}$	

$a_{Lu} - a_{Ld} = W_{25}$	$-a_{ld} + a_{lu} = W_{31}$	$2a_{lu} = W_{34} + W_{31}$
$a_{Ld} + 2a^{(1)}_{LQ} = W_{26}$	$a_{ld} + a_{lu} = W_{34}$	$2a^{(1)}_{Qu} = W_{35} + W_{36}$
$a_{Ql} + a_{lu} = W_{29}$	$2a_{Ql} = W_{29} + W_{30} - W_{34}$	$2a^{(1)}_{Qd} = W_{35} - W_{36}$
$a_{Ql} + a_{ld} = W_{30}$	$2a_{ld} = W_{34} - W_{31}$	

```
#call Kahane(x)
#call Chisholm(x)
#call Trace(x)
```

where `x` indicates the spinorial line. It is worth noting that all is developed in dimension $n = 4 - \epsilon$.

The handling of loop integrals [36], [37] is addressed by calling the procedures

```
#call LoopInt
#call 'X'scalarizing
```

The first procedure reduces the tensorial integrals by introducing scalar quan-

tities. The second one solves the scalarization of the `X`-function and provides the output exclusively in terms of elementary functions such as A_0, B_0, C_0, etc.
Details about tensorial reduction and scalar functions are reported in the **Appendix C**.

The extrapolation of the UV and the finite parts is compiled from bubbles and tadpoles functions by requiring:

```
id B0(-s,[],m1?,m2?)= 1/epbM + B0f(-s,[],m1,m2);

id A0(M)= - M^2/epbM + M^2a0f(M).
```

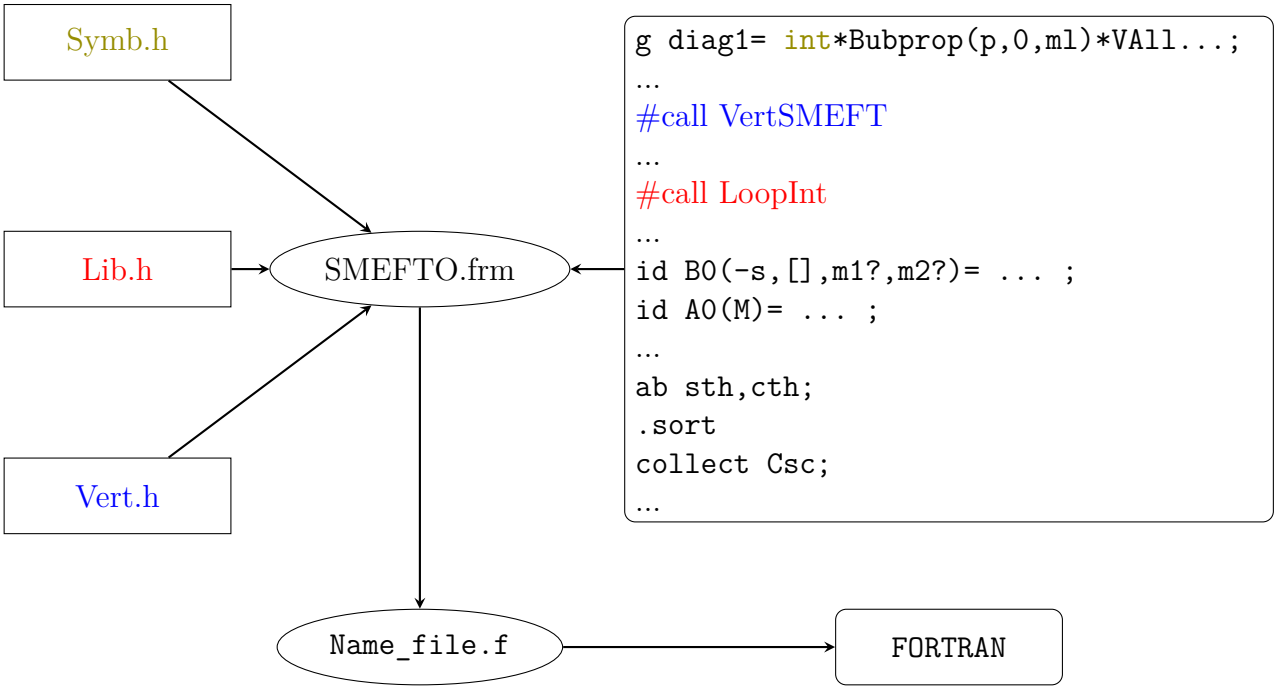
where `1/epbM` denotes the UV contribution.
The next step is to collect the terms involved, organizing and grouping similar algebraic expressions. This ensures that related terms are combined and simplified, guaranteeing a more readable output; quantities like sine (cosine) of weak mixing angle, the masses and the Wilson coefficients have to be collect in specific functions as:

```
ab sth,cth;
.sort
Collect Csc;

ab ml,mt,mb;
.sort
Collect Cm;
```

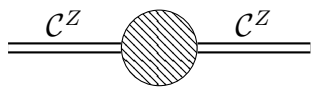
Once that the `FORM` manipulation has been completed, a series of procedures are defined in order to create files `Name_file.f` which can be easily incorporate into a `FORTRAN` program for the numerical analysis.

In the next page we have reported a simple flowchart which resumes our strategy.


Symb.h
Lib.h
Vert.h
SMEFTO.frm
g diag1= int*Bubprop(p,0,ml)*VAll...;
...
#call VertSMEFT
...
#call LoopInt
...
id B0(-s,[],m1?,m2?)= ... ;
id A0(M)= ... ;
...
ab sth,cth;
.sort
collect Csc;
...
Name_file.f
FORTRAN

3.2 Ward-Slavnov-Taylor identities in SMEFT

As a preliminary examination of the SMEFT consistency, our first task is to establish the validity of the well-known Ward-Slavnov-Taylor (WST) identities for the two-point Green functions.
WST identities hold deep significance in quantum field theory, serving as a critical tool for verifying the gauge invariance of the theory and ensuring the validity of the results obtained.
In accordance with the discussion presented in [38], defining a free field $\mathcal{C}$, it can be shown that any Green's function containing one or more external $\mathcal{C}$ sources is identically zero.
Once we have assembled the SM Lagrangian, we introduced gauge-fixing contributions as given in Eq.(2.13). Considering for example the neutral field $\mathcal{C}^Z$, our diagrammatic WST identity is given by:



$$= 0, \tag{3.1}$$

where the full blob indicate all the diagrams involved at one loop. The same discussion can be adopted for the charged field $\mathcal{C}^{\pm}$ and for the neutral $\mathcal{C}^A$.

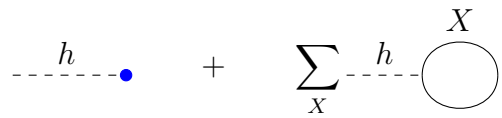
In order to extract these identities up to $d = 6$, we need to update the following ingredients:

- β_h-scheme,
- diagonalization of the neutral sector,
- self-energies.

3.2.1 β_h-scheme

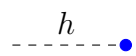
We know from the SM that the scalar potential induces tadpole terms. In the β_h-scheme, giving the Higgs VEV v, β_h is fixed order by order in perturbation theory by requiring $\langle 0| H |0\rangle = 0$.

In the NLO scenario, two different topologies contribute to the VEV: the LO tadpole and tadpole diagrams at one loop. The β_h-scheme implies the validity of the equation:



$$= 0, \tag{3.2}$$

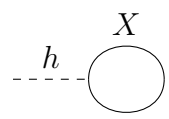
where X indicates any filed that contributes to the Higgs tadpole (gauge and fermionic fields, FP ghost and scalar bosons).
The LO satisfies the following condition:



$$= (2\pi)^4 i \left(-\beta_h \frac{2M}{g} \right). \tag{3.3}$$

and replacing it in the Eq.(3.2), we can write the one-loop equation for β_h as

$$\beta_h = \frac{1}{(2\pi)^4 i} \frac{g}{2M} \sum_X$$

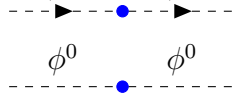


$$. \tag{3.4}$$

The net effect is the generation of new diagrams which must be included in the expression of the self-energies:

$$= (2\pi)^4 i(-\beta_h). \tag{3.5}$$

$$= (2\pi)^4 i(-\beta_h). \tag{3.6}$$



$$= (2\pi)^4 i(-\beta_h). \tag{3.7}$$

In the SMEFT we can store the full result of Eq.(3.4) in the following form:

$$\beta_h = \beta_h^{(4,fin)} + \beta_h^{(4,UV)} \Delta_{UV}(M_W^2) + \frac{g_6}{\sqrt{2}} \left[\beta_h^{(6,fin)} + \beta_h^{(6,UV)} \Delta_{UV}(M_W^2) \right]. \tag{3.8}$$

We remand to [26] for the complete results.

At this point, it becomes clear how the canonical normalization works.
Considering the Lagrangian that involves the β_h parameter

$$\mathcal{L}_{\beta_h}^{(4)} = -\beta_h \left[\frac{2M^2}{g^2} + \frac{2M}{g} H + \frac{1}{2} \left(H^2 + (\phi^0)^2 + 2\phi^+\phi^- \right) \right], \tag{3.9}$$

and performing the redefinition in Eq.(2.37) and Eq.(2.38), we obtain

$$\mathcal{L}^{(4)}_{\beta_h} \to \mathcal{L}_{\beta_h} = \mathcal{L}^{(4)}_{\beta_h} - \frac{g_6}{\sqrt{2}}\frac{\beta_h}{2}\left[\delta_H H^2 + \delta_{\phi^0}(\phi^0)^2 + 2\delta_\phi \phi^+\phi^-\right] \tag{3.10}$$

where the corrective terms δ_X are defined by:

$$\begin{aligned}
\delta_H &= a_{\Phi W} - \frac{1}{4}a_{\Phi D} + a_{\Phi\Box}, \\
\delta_{\phi^0} &= a_{\Phi W} - \frac{1}{4}a_{\Phi D} - a_{\Phi\Box}, \\
\delta_\phi &= a_{\Phi W} + \frac{1}{4}a_{\Phi D} - a_{\Phi\Box}.
\end{aligned} \tag{3.11}$$

Just to give an example, in the self energy $\phi^0 - \phi^0$ we have to add a diagram that contributes as

$$= (2\pi)^4 i\left[-\beta_h\left(1 + \frac{g_6}{\sqrt{2}}\,\delta_{\phi^0}\right)\right] \tag{3.12}$$

and the same for the self energies $H - H$ and $\phi - \phi$.

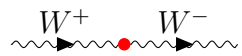
3.2.2 Diagonalization of the neutral sector

The correct treatment of the neutral sector requires a new definition of coupling constant g_0 in the expression of covariant derivative: by applying the Eq.(2.5), the constant Γ is fixed order by order to get a null amplitude for $Z - \gamma$ transition at $p^2 = 0$.

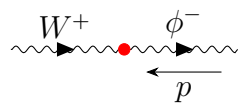
It is not our aim to discuss the full derivation of Γ [39]; instead, our focus is directed towards understanding its specific role within the context of the one-loop calculation.

Let us consider the WST identity for $\mathcal{C}^\pm$.The identity is spoiled when the Γ term is not included in the calculation and the correctness of the result

can be obtained only after the introduction of these new diagrams:



$$= (2\pi)^4 i \left[-2\delta_{\mu\nu} g^2 M^2 \Gamma \right],$$



$$= (2\pi)^4 i \left[i p_\mu M g^2 \Gamma \right]. \tag{3.13}$$

It is worth noting that Γ is fundamental in showing the ultraviolet finiteness of many results when we are dealing with the three-point Green functions.

The SMEFT extension reads as follows:

$$\Gamma = \Gamma^{(4)} \big(1 + 2 g_6 a_{\Phi W}\big), \tag{3.14}$$

where

$$\Gamma^{(4)} = -\frac{1}{8} \left[\frac{1}{\bar{\epsilon}} - 1 - a_0^{fin}(M) \right]. \tag{3.15}$$

3.2.3 Self energies

Scalar-scalar transition

The one loop self energy for a scalar-scalar transition can be written by the usual notation

$$S_{SS} = \frac{g^2}{16\pi^2} \Sigma_{SS} = \frac{g^2}{16\pi^2} \left[\Sigma_{SS}^{(4)} + \frac{g_6}{\sqrt{2}} \Sigma_{SS}^{(6)} \right], \tag{3.16}$$

where the UV pole and the finite part can be written as

$$\Sigma_{SS}^i = \Sigma_{SS;UV}^i \, \Delta_{UV}(M_W^2) + \Sigma_{SS;fin}^i, \quad i = 4, 6 \tag{3.17}$$

and SS=$\big\{ HH, \phi^0\phi^0, \phi\phi \big\}$.

Vector-vector transition

We have reported the vector vector transition self energy in Eq.(2.73).
We choose to adopt a different notation depending on the type of transition we are considering.

For the **photon self energy** we write:

$$S^{\mu\nu}_{\gamma\gamma} = \frac{g^2}{16\pi^2}\Sigma^{\mu\nu}_{\gamma\gamma} \tag{3.18}$$

with

$$\Sigma^{\mu\nu}_{\gamma\gamma} = \Pi_{\gamma\gamma}\big[p^2\delta^{\mu\nu} - p^\mu p^\nu\big]. \tag{3.19}$$

It is worth noting that the bare quantity Π_{AA} follows the usual decomposition

$$\Pi_{\gamma\gamma} = \Pi^{(4)}_{\gamma\gamma} + \frac{g_6}{\sqrt{2}}\Pi^{(6)}_{\gamma\gamma}. \tag{3.20}$$

The $Z-\gamma$ **transition** up to one loop is expressed by:

$$S^{\mu\nu}_{Z\gamma} = \frac{g^2}{16\pi^2}\Sigma^{\mu\nu}_{Z\gamma} + \frac{g_6}{\sqrt{2}}\big[p^2\delta^{\mu\nu} - p^\mu p^\nu\big]a_{AZ} \tag{3.21}$$

where

$$\Sigma^{\mu\nu}_{Z\gamma} = \Pi_{Z\gamma}\big[p^2\delta^{\mu\nu} - p^\mu p^\nu\big] + P_{Z\gamma}\, p^\mu p^\nu \tag{3.22}$$

Here we have included the $Z-\gamma$ transition term starting at $\mathcal{O}(g_6)$.

Denoting by VV the **Z-Z** and the **W-W transition**, we can write the self energy insertion up top one loop as:

$$S^{\mu\nu}_{VV} = \frac{g^2}{16\pi^2}\Sigma^{\mu\nu}_{VV}, \quad \Sigma^{\mu\nu}_{VV} = D_{VV}\delta^{\mu\nu} + P_{VV}p^\mu p^\nu \tag{3.23}$$

where each component follows the usual decomposition:

$$D_{VV} = D^{(4)}_{VV} + \frac{g_6}{\sqrt{2}}D^{(6)}_{VV}, \quad P_{VV} = P^{(4)}_{VV} + \frac{g_6}{\sqrt{2}}P^{(6)}_{VV}. \tag{3.24}$$

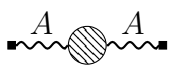
Vector-scalar transition

$$S^{\mu}_{VS} = \frac{g^2}{16\pi^2}\Sigma_{VS}\, ip^\mu = \frac{g^2}{16\pi^2}\left[\Sigma^{(4)}_{VS} + \frac{g_6}{\sqrt{2}}\Sigma^{(6)}_{VS}\right]ip^\mu. \tag{3.25}$$

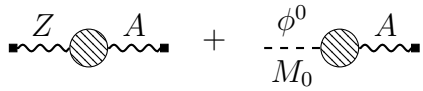
3.2.4 Assembling the WST identities

By assembling all the necessary ingredients, we can construct the WST identities for the two-point Green functions, checking the consistency of the SMEFT as gauge field theory.

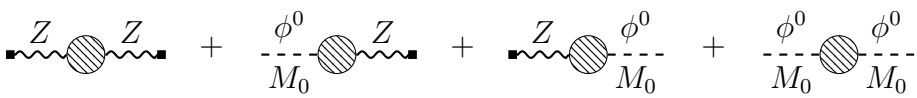
Our results are reported in the following diagrammatic equations:



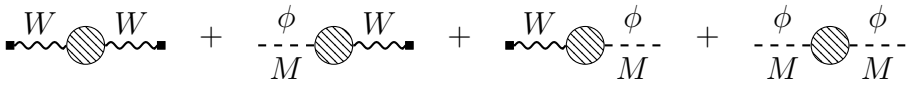
$= 0,$ (3.26)



$= 0,$ (3.27)



$= 0,$ (3.28)



$= 0,$ (3.29)

where the square dot indicates the multiplication by ip^μ while the full-blobs denote the sum of the needed Feynman diagrams at any order (we consider up to $\mathcal{O}(g^2 g_6)$).

We conclude by noting that extending WST identities to three-point Green functions is non-trivial and gives rise to anomalies that are UV-finite, mass-dependent, scheme-dependent, and local. [28].

3.3 Selecting the input parameter set (IPS)

3.3.1 W and Z masses

Let us summarize the counterterms language introduced in the previous chapter.
For a given field X with a bare mass m_X, the Dyson-resummed propagator

contains UV divergences that can be absorbed by specific counterterms.
From the inverse X-propagator, setting $p^2 = -s$ we can write:

$$\bar{\Delta}^{-1}(s) = -s + m_X^2 - \frac{g^2}{16\pi^2}\Sigma_X(s) \tag{3.30}$$

and the renormalization procedure implies the following chain:

$$m_X \to m_X^{\text{ren}} \to M_X. \tag{3.31}$$

where M_X is the physical mass.
From Eq.(2.63), the counterterms are fixed by requiring:

$$\bar{\Delta}_X^{-1}(M_X^2) = 0. \tag{3.32}$$

Subsequently, the WFR factor for the X field is extracted by the proper treatment of the sources.

We want to remark that our procedure does not involve counterterms for fields: for a given process, we introduce counterterms on the parameters and a WFR factor $Z_X^{-1/2}$ defined in Eq.(2.72) for each external leg. Adding the vertex correction, all the poles in $d = 4$ are canceled, while in $d = 6$ we have to require the mixing among Wilson coefficients.
In our results, we have followed the strategy adopted for LEP observables where the WFR factors are taken to be real, and the implementation of complex pole for unstable particles is not considered.

However, the cancellation of the UV poles does not imply the end of the renormalization procedure; rather, it is necessary to establish a connection between the renormalized quantities and the physical ones. In other words, we have to define the **IPS**: if a theory is defined by n independent parameters, any prediction can be made after choosing n experimental data points to be used as input.

Our strategy is based on the choice of M_W and M_Z with $c_\theta = M_W/M_Z$.
From the resummed W propagator

$$\bar{\Delta}_{WW}^{-1}(s) = -s + M^2 - S_{WW}^{\mu\nu}(s), \tag{3.33}$$

we derive

$$M \to M^{ren} \to M_W, \tag{3.34}$$

which guarantees to fix the counterterms $dZ_M^{(i)}$ and $dZ_{M_W}^{(i)}$, i= 4, 6.
From the inverse of Z propagator

$$\bar{\Delta}_{ZZ}^{-1}(s) = -s + \frac{M^2}{c_\theta} - S_{ZZ}^{\mu\nu}(s). \tag{3.35}$$

using Eq.(3.34) and setting

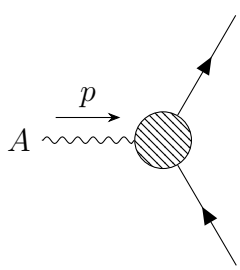
$$\bar{\Delta}_{ZZ}^{-1}(M_Z^2) = 0, \tag{3.36}$$

we obtain the full one loop correction for c_θ and the counterterm $dZ_{c_\theta}^{(i)}$.

The third data point is selected by connecting the renormalized SU(2) coupling constant g_R with the fine structure constant $\alpha(0)$ or the Fermi constant G_F.

3.3.2 Coupling constant: the scheme $g_R \to \alpha(0)$

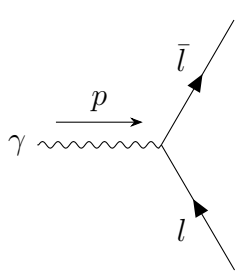
By a specific procedure in our code `SMEFTO.frm`, we have shown the validity of the diagrammatic equation



$$+ \quad \text{WFR factors} \Big|_{p^2=0} = 0, \tag{3.37}$$

also in $d = 6$.
Then, we have assembled the amplitude of Coulomb scattering by considering:

- The vertex $Al\bar{l}$



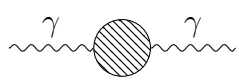
$$= -igs_\theta \left[1 + \frac{g_6}{\sqrt{2}}\,\Delta_\alpha\right]\gamma_\mu = \Gamma_{LO}\gamma_\mu, \tag{3.38}$$

where

$$\Delta_\alpha = \sqrt{2}\left[(s_\theta^2 + 2c_\theta^2)W_1 - c_\theta^2 W_2 + c_\theta^3 s_\theta W_3 - \frac{1}{4}c_\theta^2 s_\theta^2 W_4\right] \tag{3.39}$$

indicates the SMEFT contribution.

- The Dyson resummed propagator of the photon:



$$= \frac{1}{s}\left[S_{\gamma\gamma} - S_{\gamma Z}\frac{1}{s - M_Z^2}S_{Z\gamma}\right], \tag{3.40}$$

where the transition $\gamma - Z$ is taken into account.

The full amplitude reads as follows:

$$\mathcal{M}^c(g) = g^2 s_\theta^2\big(1 + \sqrt{2}g_6\ \Delta_\alpha\big)\frac{1}{s}\left[1 - \frac{g^2 s_\theta^2}{16\pi^2}\Pi_{\gamma\gamma}(s)\right]^{-1}\gamma_\mu \otimes \gamma_\mu. \tag{3.41}$$

where the function $\Pi_{\gamma\gamma}$ contains the $\gamma - \gamma$ and $Z - \gamma$ contributions.
Taking the residue of the pole at s= 0, the equation that connects g to α is expressed by

$$4\pi\alpha = g^2 s_\theta^2\big(1 + \sqrt{2}g_6\ \Delta_\alpha\big)\left[1 - \frac{g^2 s_\theta^2}{16\pi^2}\Pi_{AA}(0)\right]^{-1}, \tag{3.42}$$

from which we extract the renormalization equation for g

$$\frac{1}{g^2 s_\theta^2} = \frac{1}{4\pi\alpha(0)} + \frac{1}{16\pi^2}\Pi_{AA}(0) + \frac{1}{2\pi\alpha(0)}\frac{g_6}{\sqrt{2}}\Delta_\alpha. \tag{3.43}$$

The introduction of counterterms for g and the mix among the Wilson coefficients lead to an UV finite result in $d = 4$ and $d = 6$.
Finally we can write

$$g_R^2 = \frac{4\pi\alpha(0)}{s_\theta^2}\left[1 + \frac{g_6}{\sqrt{2}}\delta_\alpha^{(0,6)} + \frac{\alpha(0)}{\pi}\left(\delta_\alpha^{(1,4)} + \frac{g_6}{\sqrt{2}}\delta_\alpha^{(1,6)}\right)\right] \tag{3.44}$$

and fix the finite quantities δ_α.

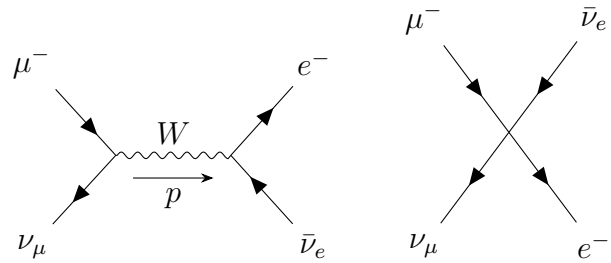


Figure 3: Diagrams involve in the LO computation of $\mu \to \nu_\mu \bar{\nu}_e e$.

3.3.3 Coupling constant: the scheme $g_R \to G_F$

Connecting g to G_F requires the muon decay:
Let us consider Fig.(3). The first diagram represents the classical LO SM-topology, whereas the second one involves the 4-fermion contact vertex, characterized by a spinorial structure given by

$$\begin{aligned} 4f &= \bar{u}_{\nu_\mu}\gamma^\mu(1+\gamma_5)v_{\bar{\nu}_e} \otimes \bar{u}_e\gamma^\mu(1+\gamma_5)u_\mu \qquad \text{(Fierz tr.)} \\ &= -\bar{u}_{\nu_\mu}\gamma^\mu(1+\gamma_5)u_\mu \otimes \bar{u}_e\gamma^\mu(1+\gamma_5)v_{\bar{\nu}_e}. \end{aligned} \tag{3.45}$$

Let us consider the W-propagator. The chain in Eq.(3.34) leads to:

$$\left[p^2 + M_W^2 - \frac{g^2}{16\pi^2}\bar{\Sigma}_{WW}(p^2)\right]^{-1}, \tag{3.46}$$

where $\bar{\Sigma}$ contains all the parameter counterterms.

Setting $p^2 = 0$, what we obtain is:

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}(1+\sqrt{2}g_6\Delta_G)\left[1 + \frac{g^2}{16\pi^2}\frac{\bar{\Sigma}_{WW}(0)}{M_W^2}\right] + \frac{g^2}{4M^2}\frac{g_6}{\sqrt{2}}W_{28}. \tag{3.47}$$

Δ_G derives from the left diagram in Fig.(3) and it is expressed by the combination

$$\Delta_G = 2s_\theta^2 W_1 + 2c_\theta^2 W_2 + 2s_\theta c_\theta W_3 + \sum_{i=16}^{18} W_i, \tag{3.48}$$

while the 4-fermion vertex gives the term with W_{28}.

The last step concerns the inclusion of the NLO correction δ_G which involves vertices, boxes etc.
In general we can write:

$$\delta_G = \delta_G^{(4)} + \frac{g_6}{\sqrt{2}}\delta_G^{(6)}, \tag{3.49}$$

where $\delta_G^{(4)}$ is known since long time [13] but we still miss a complete calculation of $\delta_G^{(6)}$. The complete equation results:

$$\frac{g^2}{8M_W^2}(1+\frac{g_6}{\sqrt{2}}\Delta_G)\left\{1+\frac{g^2}{16\pi^2}\left[\delta_G-\frac{\bar{\Sigma}_{WW}(0)}{M_W^2}\right]\right\}+\frac{g^2}{4M^2}\frac{g_6}{\sqrt{2}}W_{28}=\frac{G_F}{\sqrt{2}}. \tag{3.50}$$

The introduction of counterterms, renormalized quantities and mixing, allows us to write the equation

$$g_R^2 = 4\sqrt{2}G_F M_W^2\left\{1+\frac{g_6}{\sqrt{2}}\delta_g^{(0,6)}+\frac{4\sqrt{2}G_F M_W^2}{16\pi^2}\left(\delta_g^{(1,4)}+\frac{g_6}{\sqrt{2}}\delta_g^{(1,6)}\right)\right\}, \tag{3.51}$$

and fix the UV-finite δ_g (neglecting the terms proportional to M_e^2/M_W^2).

3.4 Decays in SMEFT

In this section, we analyze three distinct decays, laying the foundation for our investigation into potential deviations from the SM in the forthcoming chapter.
We will indicate by $\mathcal{M}_X^{(m,n)}$ the transition amplitude of a process X at the order m and mass dimension n, i.e. $\mathcal{M}_{Zll}^{(0,6)}$ indicate the LO in $d = 6$ for $Z \to l^+l^-$, while $\mathcal{M}_{H\gamma\gamma}^{(1,4)}$ is the NLO correction in $d = 4$ for $H \to \gamma\gamma$.

3.4.1 $Z \to l^+l^-$

The phenomenology of the Z-boson has been extensively investigated over the last 40 years and has consistently served as an excellent tool for verifying and confirming the predictability of the SM.

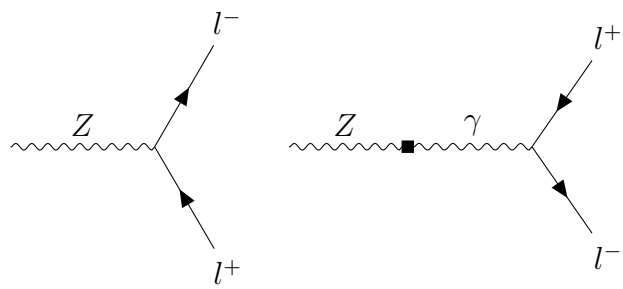


Figure 4: LO contributions in $Z \to l^+l^-$.

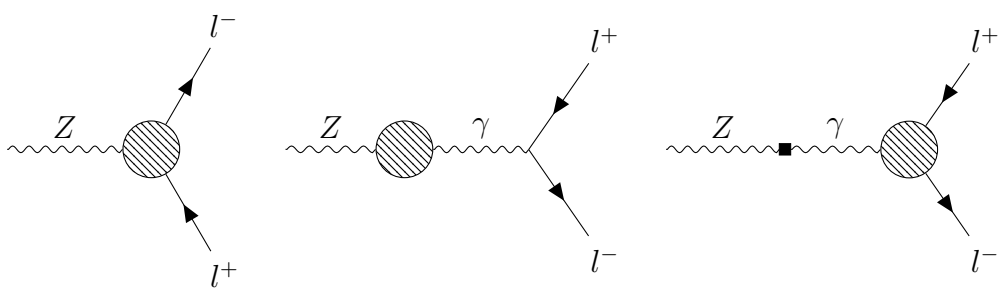


Figure 5: NLO calculation involves the vertex correction (left diagram) and the usual transition $Z - \gamma$ (central diagram). Extracting the transition at $\mathcal{O}(g_6)$, we have to take into account also the contribution of $\gamma \to l^+l^-$ up to $\mathcal{O}(g^3)$ (right diagram)

The information from Z decays are particularly interesting due to the accuracy of the LEP measurements and the low energy data set establishes stringent constraints on possible deviations from the SM.

Let us start from Fig.(4). At LO, for $p^2 = -M_Z^2$, the transition amplitude results:

$$\mathcal{M}_{Zll} = \bar{u}(p_1)\left[ig\left(\mathcal{M}^{(0,4)}_{Zll,\mu} + \frac{g_6}{\sqrt{2}}\mathcal{M}^{(0,6)}_{Zll,\mu}\right)\right]v(p_2)\mathcal{Z}_\mu(p), \tag{3.52}$$

where p is the 4-momentum associated to the Z polarization $\mathcal{Z}_\mu$.

The various sub-amplitude are defined as

$$\begin{aligned}\mathcal{M}_{\mu}^{(0,4)} &= -\frac{1}{4c_\theta}\gamma_\mu\big(v_l+\gamma_5\big),\\ \mathcal{M}_{\mu}^{(0,6)} &= \gamma_\mu\big(V_l+A_l\gamma_5\big) - i\frac{m_l}{M^2}[\gamma_\mu,\not{p}]\,W_8,\end{aligned} \tag{3.53}$$

with

$$\begin{aligned}v_l &= 1-4s_\theta^2,\\ V_l &= \frac{s_\theta^2}{4c_\theta^2}\big(7-4s_\theta^2\big)W_1 - \frac{c_\theta}{4}\big(1+4s_\theta^2\big)W_2 + \frac{s_\theta}{4}\big(3-4s_\theta^2\big)W_3\\ &\quad - \frac{1}{16c_\theta}\big(7-4s_\theta^2\big)W_4 - \frac{1}{2c_\theta}W_{17},\\ A_l &= -\frac{s_\theta^2}{4c_\theta}W_1 - \frac{c_\theta}{4}W_2 - \frac{s_\theta}{4}W_3 + \frac{1}{16c_\theta}W_4 - \frac{1}{2c_\theta}W_{16}.\end{aligned} \tag{3.54}$$

The complexity of the NLO calculation, see Fig.(5), demands a strategy to efficiently gather a substantial number of one-loop diagrams.
Various topologies are involved, including not only triangles but also bubbles, tadpoles, and transition diagrams.

We extract the WFR factors, associating to each external leg a factor defined in Eq.(2.72) and Eq(2.84). Omitting the spinors and the Z polarization $\mathcal{Z}_\mu$, the amplitude reads:

$$\mathcal{M}_{Zll,\mu} = ig\Big\{\, Z_{l^-}^{-\frac{1}{2}}\Big[\mathcal{M}_{Zll,\mu}^{(0,4)} + \frac{g_6}{\sqrt{2}}\mathcal{M}_{Zll,\mu}^{(0,6)}\Big] Z_{l^+}^{-\frac{1}{2}} Z_Z^{-\frac{1}{2}} + \mathcal{M}_{Zll,\mu}^{V} + \mathcal{M}_{Zll\mu}^{T}\Big\}, \tag{3.55}$$

where $\mathcal{M}_{Zll}^{V}$ indicates the vertex correction while $\mathcal{M}_{Zll}^{T}$ collects all the terms deriving from the $Z-\gamma$ transition.

We insert the counterterms for the parameters:

$$
\begin{aligned}
m_l &= m_{l,R}\left(1+\frac{g^2}{16\pi^2}dZ_{m_l}\ \Delta_{UV}\right),\\
M &= M_R\left(1+\frac{g^2}{16\pi^2}dZ_M\ \Delta_{UV}\right),\\
c_\theta &= c_{\theta,R}\left(1+\frac{g^2}{16\pi^2}dZ_{c_\theta}\ \Delta_{UV}\right),\\
s_\theta &= s_{\theta,R}\left(1-\frac{c_\theta^2}{s_\theta^2}\frac{g^2}{16\pi^2}dZ_{c_\theta}\ \Delta_{UV}\right),\\
g &= g_R\left(1+\frac{g^2}{16\pi^2}dZ_g\ \Delta_{UV}\right).
\end{aligned}
\tag{3.56}
$$

and the UV poles in $d = 4$ are removed.

We require the mixing by introducing:

$$
W_i = \left[\delta_{ij}+\frac{g^2}{16\pi^2}dZ_{ij}\Delta_{UV}\right]W_j^{(R)}. \tag{3.57}
$$

Observing the Wilson coefficients involved in the LO, the quantities $W_{1,\dots,4}$ are renormalized by using the matrix elements reported in [26], Sez.G2; the remaining Wilson coefficients W_8, W_{16}, W_{17} have to be mixed with other coefficients, defining the new entries in the mixing matrix $dZ_{a,j}$, with a= 8, 16, 17.
The mixing induces a logarithmic dependence in the amplitude, defined by the following relation:

$$
\mathcal{M}^{(1,6),UV}_{Zll,\mu} \to \mathcal{M}^{(1,6),R}_{Zll,\mu}\ln\frac{\mu_R^2}{M_W^2}. \tag{3.58}
$$

The next step provides the introduction of the finite counterterms:

$$
\begin{aligned}
m_{l,R} &= M_l\left(1+\frac{g^2}{16\pi^2}\delta_{M_l}\right),\\
M_R &= M_W\left(1+\frac{g^2}{16\pi^2}\delta_{M_W}\right),\\
c_{\theta,R} &= c_W\left(1+\frac{g^2}{16\pi^2}\delta_{c_\theta}\right),\\
s_{\theta,R} &= s_W\left(1-\frac{c_W^2}{s_W^2}\frac{g^2}{16\pi^2}\delta_{c_\theta}\right),\\
g_R &= g_F\left(1+\frac{g_F^2}{16\pi^2}\delta_g\right).
\end{aligned}
\tag{3.59}
$$

where we always assume the validity of the decomposition

$$
\delta_X = \delta_X^{(4)} + \frac{g_6}{\sqrt{2}}\delta_X^{(6)}. \tag{3.60}
$$

with $c_W = M_W/M_Z$ and $g_F^2 = 4\sqrt{2}G_F M_W^2$.
The complete result is reported in the **Appendix D, Sec.D.1**.

After the cancellation of UV poles, the full amplitude shows infrared/collinear poles due to the photon in the loops but the Kinoshita-Lee-Nauenberg theorem guarantees the finiteness of the result by adding the bremsstrahlung contribution.

IR/collinear part

To address the infrared (IR) sector, we have adopted the regularization scheme (ϵ,ϵ) by expanding around $n = 4+\epsilon$ and setting $m_l = 0$. Additionally, we have omitted the terms with a $Z-\gamma$ transitions that appears in the internal loops, due to their IR and collinear finiteness. Instead, we have considered the pure photon contribution and the Z_γ transition allowed by the external Z leg, as reported in Fig.(6).

Considering the massless case, no contribution comes from the lepton self

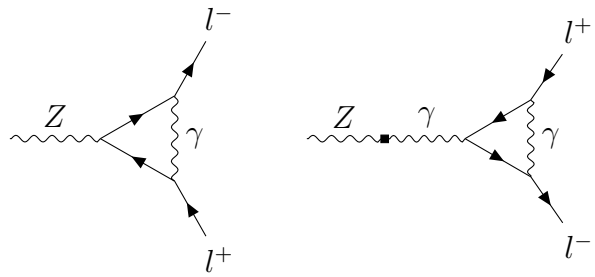


Figure 6: Virtual contribution in QED-SMEFT.

energy: it is possible to show that for $m_l = 0$ one obtains:

$$\Sigma(\not{p}) \sim \left(\frac{p^2}{\mu^2}\right)^{\epsilon/2} i\not{p} \to 0 \quad \text{if } p^2 = 0. \tag{3.61}$$

The virtual corrections can be cast in the form

$$\text{Virtual} = \frac{1}{n-1}\frac{1}{2M_Z}\int \mathrm{d}\Phi_{1\to 2} \sum_{spin} 2\mathrm{Re}\left[\mathcal{M}_{Zll}^{(0),\dagger}\, \mathcal{M}_{Zll}^{(1)}\right], \tag{3.62}$$

where $\mathcal{M}_{Zll}^{(0)}$ is the Born level amplitude, $\mathcal{M}_{Zll}^{(1)}$ collects the NLO-QED corrections.

Since

$$\int \mathrm{d}\Phi_{1\to 2} = \frac{1}{8\pi}\left(\frac{M_Z^2}{\mu^2}\right)^{\epsilon/2} \frac{(2\pi)^{-\epsilon}\pi^{\epsilon/2}\Gamma(1+\epsilon/2)}{(1+\epsilon)\Gamma(1+\epsilon)}, \tag{3.63}$$

it is possible to extract an overall factor $\mathcal{F}_V$ defined by

$$\mathcal{F}_V = -\frac{1}{\hat{\epsilon}^2} - \frac{2}{\hat{\epsilon}}\left(z - \frac{19}{12}\right) - 2z^2 - 2\bar{\gamma}z + 5\zeta(2) - \bar{\gamma}^2 + \frac{19}{3}z + \frac{19}{6}\bar{\gamma} - \frac{173}{18}, \tag{3.64}$$

with

$$\frac{1}{\hat{\epsilon}} = \frac{2}{\epsilon} + \bar{\gamma}, \qquad \bar{\gamma} = \gamma_E + \ln\pi, \qquad z = \ln\frac{M_Z^2}{(2\pi\mu)^2}. \tag{3.65}$$

Considering the real photon emission, the process under investigation is defined by $Z(p) \to l^-(p_1) + l^+(p_2) + \gamma(p_3)$.

The square amplitude is computed in terms of the new invariants x and y, defined by:

$$\begin{aligned} p_1 \cdot p_3 &= -\frac{M_Z^2}{2} x, \\ p \cdot p_3 &= -\frac{M_Z^2}{2} y. \end{aligned} \tag{3.66}$$

with $p^2 = -M_Z^2$ and $p_1^2 = p_2^2 = p_3^2 = 0$.
In this way we are able to express all the remaining scalar products in terms of the new invariants x and y.

The real part reads:

$$\text{Real} = \frac{1}{n-1} \frac{1}{2M_Z} \int \mathrm{d}\Phi_{1\to 3} \, |\mathcal{M}_{Zll}^{real}|^2, \tag{3.67}$$

Setting $z = (y-x)/(1-x)$, the structure of the phase space can be written as

$$\mathrm{d}\Phi_{1\to 3} = \frac{M_Z^2}{128\pi^3} \left(\frac{M_Z^2}{\mu^2}\right)^{\epsilon} \frac{(2\pi)^{-2\epsilon}\pi^{\epsilon}}{\Gamma(2+\epsilon)} x^{\epsilon/2}(1-x)^{1+\epsilon} \, [z(1-z)]^{\epsilon/2} \, \mathrm{d}x\mathrm{d}z, \tag{3.68}$$

with $x, z \in [0,1]$.
As for the virtual case, the expansion of all ingredients gives an overall factor that includes all the divergent terms:

$$\mathcal{F}_R = \frac{1}{\hat{\epsilon}^2} + \frac{2}{\hat{\epsilon}}\left(z - \frac{19}{12}\right) + 2z^2 + 2\bar{\gamma}z - 5\zeta(2) + \bar{\gamma}^2 - \frac{19}{3}z - \frac{19}{6}\bar{\gamma} + \frac{373}{36}. \tag{3.69}$$

The final result is the sum of Eq.(3.62) and Eq.(3.67), displaying a double factorization form as follows

$$\Gamma_{Zll}^{QED} = \frac{g^2}{16\pi^2} \Gamma_{Zll}^{(0,4)} \left[3s_\theta^2 \left(1 + \frac{g_6}{\sqrt{2}} \Delta\Gamma_{Zll}^{(1,6)}\right) + \frac{g_6}{\sqrt{2}} \delta_{Zll}^{(1,6)} \right]. \tag{3.70}$$

Here we report the various corrective terms:

$$\Gamma_{Zll}^{(0,4)} = \frac{g^2 M_Z^2}{192 c_\theta^2 \pi} (1 + v_l^2), \tag{3.71}$$

$$\Delta\Gamma_{Zll}^{(1,6)} = \frac{1}{s_\theta^2}\Bigg[\frac{1}{4}(10 - 5v_l - v_l^2)W_1 - 4s_\theta^2 c_\theta^2 W_2 + s_\theta c_\theta(2 + v_l)W_3 - \frac{1}{4}(4 + v_l)W_4\Bigg], \tag{3.72}$$

$$\delta_{Zll}^{(1,6)} = \frac{3}{(1+v_l^2)}\Bigg[-\frac{1}{2}(3+v_l)W_1 + 4s_\theta^2 c_\theta^2(1+v_l)W_2 - 2v_l s_\theta c_\theta W_3 + \frac{1}{2}(1+2v_l)W_4 + 4s_\theta^2 W_{16} + 4s_\theta^2 v_l W_{17}\Bigg]. \tag{3.73}$$

3.4.2 $H \to \bar{b}b$

The b-channel [40][41], being one of the most abundant Higgs decay modes (BR $\sim 58\%$), provides an excellent tool for studying the impact of Yukawa coupling in the SMEFT context.

Let us define the one loop amplitude as:

$$\mathcal{M}_{Hbb} = g\mathcal{M}_{Hbb}^{LO} + \frac{g^3}{16\pi^2}\mathcal{M}_{Hbb}^{1L} + \frac{g g_s^2}{16\pi^2}\mathcal{M}_{Hbb}^{1L,QCD} \tag{3.74}$$

where g_s gives the strong coupling in the QCD sector and

$$\begin{aligned}
\mathcal{M}_{Hbb}^{LO} &= \mathcal{M}_{Hbb}^{(0,4)} + \frac{g_6}{\sqrt{2}}\mathcal{M}_{Hbb}^{(0,6)}, \\
\mathcal{M}_{Hbb}^{1L} &= \mathcal{M}_{Hbb}^{(1,4)} + \frac{g_6}{\sqrt{2}}\mathcal{M}_{Hbb}^{(1,6)}, \\
\mathcal{M}_{Hbb}^{1L,QCD} &= \mathcal{M}_{Hbb}^{(1,4),QCD} + \frac{g_6}{\sqrt{2}}\mathcal{M}_{Hbb}^{(1,6),QCD}.
\end{aligned} \tag{3.75}$$

Just for completeness, we report here the LO contributions:

$$\mathcal{M}_{Hbb}^{(0,4)} = -\frac{m_b}{2M},$$

$$\mathcal{M}_{Hbb}^{(0,6)} = \mathcal{M}_{Hbb}^{(0,4)}\left(s_\theta^2 W_1 + c_\theta^2 W_2 + s_\theta c_\theta W_3 - \frac{1}{4}W_4 - W_5 + W_{14}\right). \tag{3.76}$$

The first step concerns the renormalization. We introduce the counterterms for the parameters

$$
\begin{aligned}
m_b &= m_{b,R}\left(1+\frac{g^2}{16\pi^2}dZ_{m_b}\,\Delta_{UV}+\frac{g_s^2}{16\pi^2}dZ_{m_b}^{QCD}\Delta_{UV}\right),\\
M &= M_R\left(1+\frac{g^2}{16\pi^2}dZ_M\,\Delta_{UV}\right),\\
c_\theta &= c_{\theta,R}\left(1+\frac{g^2}{16\pi^2}dZ_{c_\theta}\,\Delta_{UV}\right),\\
s_\theta &= s_{\theta,R}\left(1-\frac{c_\theta^2}{s_\theta^2}\frac{g^2}{16\pi^2}dZ_{c_\theta}\,\Delta_{UV}\right),\\
g &= g_R\left(1+\frac{g^2}{16\pi^2}dZ_g\,\Delta_{UV}\right).
\end{aligned}
\tag{3.77}
$$

and the WFR factors for the external legs.
We remark the fact that it is valid the decomposition

$$
dZ_X = dZ_X^{(4)} + \frac{g_6}{\sqrt{2}}dZ_X^{(6)} \tag{3.78}
$$

for each counterterms.
Once that the UV divergences are removed in $d=4$, we require the mixing in order to remove the remaining $d=6$ UV poles. The presence of a "strong" component, leads us to introduce the renormalized coefficients as

$$
W_i = \left[\delta_{ij}+\left(\frac{g^2}{16\pi^2}dZ_{ij}+\frac{g_s^2}{16\pi^2}dZ_{ij}^{QCD}\right)\Delta_{UV}\right]W_j^{(R)}. \tag{3.79}
$$

From [26], we use the mixing matrix elements for $i=1,...,5$, and the finiteness at $d=6$ is achieved through the mixing of W_{14} with other coefficients. At this point, we remain with a logarithmic part that rises after the mixing procedure:

$$
\mathcal{M}_{Hbb}^{(1,6),div} \to \left(\mathcal{M}_{Hbb}^{(R)}+\mathcal{M}_{Hbb}^{(R),QCD}\right)\ln\frac{\mu_R^2}{M_W^2} \tag{3.80}
$$

where the apex R indicates the renormalized part.
The introduction of the finite counterterms:

$$
\begin{aligned}
m_{b,R} &= M_b\left(1+\frac{g^2}{16\pi^2}\delta_{M_b}+\frac{g_s^2}{16\pi^2}\delta_{M_b}^{QCD}\right),\\
M_R &= M_W\left(1+\frac{g^2}{16\pi^2}\delta_{M_W}\right),\\
c_{\theta,R} &= c_W\left(1+\frac{g^2}{16\pi^2}\delta_{c_\theta}\right),\\
s_{\theta,R} &= s_W\left(1-\frac{c_W^2}{s_W^2}\frac{g^2}{16\pi^2}\delta_{c_\theta}\right),\\
g_R &= g_F\left(1+\frac{g^2}{16\pi^2}\delta_g\right).
\end{aligned}
\tag{3.81}
$$

conclude our procedure.
The complete result is reported in **Appendix D, Sec.D.2**.

IR/collinear part

As we have studied in the previous decay the correct strategy is perform the analitical continuation in $n = 4 + \epsilon$, choosing the more convenient regularization scheme (ϵ', ϵ').
We want to show the finitness of the SMEFT in the IR/collinear sector by considering for example the QED sector.
Taking into account only the one-loop diagrams with IR component, the integration over the full n-dimensional phase space $1 \to 2$ gives

$$
\text{Virtual} = \Gamma_{Hbb}^{(0,4)}\Phi_{res}\frac{g^2}{16\pi^2}\mathcal{F}_V\left[\Delta\Gamma_{Hbb}^{(1,4)}+\frac{g_6}{\sqrt{2}}\Delta\Gamma_{Hbb}^{(1,6)}\right], \tag{3.82}
$$

where the quantity Φ_{res} is expressed by

$$
\Phi_{res} = \left(\frac{1}{2}\right)^{1+\epsilon}\frac{\pi^{1/2+\epsilon/2}}{\Gamma(3/2+\epsilon/2)}\left(\frac{M_H}{\mu}\right)^{\epsilon} \to 1, \qquad \epsilon \to 0. \tag{3.83}
$$

The overall factor

$$
\mathcal{F}_V = -\frac{1}{\hat{\epsilon}^2}+\frac{1}{\hat{\epsilon}}\left(\bar{\gamma}-z\right)-\frac{1}{2}\bar{\gamma}^2-\frac{1}{2}z^2+\frac{7}{2}\zeta(2)-1, \tag{3.84}
$$

exhibits both a single pole and a double pole, with the latter indicating the combined IR×Collinear behavior.
Here our notation:

$$\frac{1}{\hat{\epsilon}} = \frac{2}{\epsilon} + \bar{\gamma}, \qquad \bar{\gamma} = \gamma_E + \ln \pi, \qquad z = \ln \frac{M_H^2}{(2\pi\mu)^2}. \tag{3.85}$$

The bremsstrahlung part is performed by integrating the quantity $|\mathcal{M}_{Hbb}^{real}|^2$ over the full n-dimensional phase space $1 \to 3$. Just considering the IR/collinear diagrams, we specify the kinematic in terms of the two invariants x and y:

$$\begin{aligned} (p_1 + p_3)^2 &= -xM_H^2, \\ (p + p_3)^2 &= -(y-1)M_H^2. \end{aligned} \tag{3.86}$$

for $p^2 = -M_H^2$, with $x \in [0,1]$ and $y \in [x,1]$.
We extract from the squared amplitude the spectrum function

$$f(x,y) = \frac{4 - 4y + (2+\epsilon)y^2}{x(y-x)}, \tag{3.87}$$

where its integration can be expressed in terms of master integrals $I(a,b,c)$:

$$\begin{aligned} I(a,b,c) &= \int_0^1 \mathrm{d}x \int_x^1 \mathrm{d}y \; x^{a+\epsilon/2} y^b (y-x)^{x+\epsilon/2} (1-y)^{\epsilon/2} \\ &= \mathrm{B}(a+b+c+2+\epsilon)\mathrm{B}(a+1+\epsilon/2, c+1+\epsilon/2). \end{aligned} \tag{3.88}$$

In analogy with Eq.(3.67), after the integration we obtain:

$$\text{Real} = \Gamma_{Hbb}^{(0,4)} \Phi_{res} \frac{g^2}{16\pi^2} \mathcal{F}_R \left[\Delta\Gamma_{Hbb}^{(1,4)} + \frac{g_6}{\sqrt{2}} \Delta\Gamma_{Hbb}^{(1,6)} \right] \tag{3.89}$$

and an overall factor is extracted

$$\mathcal{F}_R = \frac{1}{\hat{\epsilon}^2} + \frac{1}{\hat{\epsilon}}\left(z - \bar{\gamma} - \frac{3}{2}\right) + \frac{1}{2}z^2 - \frac{3}{2}z - \frac{7}{2}\zeta(2) + \frac{1}{2}\bar{\gamma}^2 + \frac{21}{4}. \tag{3.90}$$

The final sum of virtual and real contributions gives

$$\mathcal{F} = \mathcal{F}_V + \mathcal{F}_R = \frac{17}{4} - \frac{3}{2}z - \frac{3}{2}\frac{1}{\hat{\epsilon}}, \tag{3.91}$$

where a residual dependence from the pole $1/\hat{\epsilon}$ is still present.
Using the correspondence [15],[42]

$$\frac{1}{\hat{\epsilon}} = \frac{4}{3} - \ln \frac{M_b^2}{(2\pi\mu)^2}, \tag{3.92}$$

we obtain

$$\mathcal{F} = \frac{9}{4} + \frac{3}{2} \ln \frac{M_b^2}{M_H^2}. \tag{3.93}$$

The residual logarithmic term is then removed by expressing the pole mass of b-quark in terms of running mass $M_b^2(M_H)$ and the same procedure can be applied for the QCD case.
The total QED correction can be cast in the form:

$$\Gamma_{Hbb} = \Gamma_{Hbb}^{(0,4)} \left[1 + \frac{g_6}{\sqrt{2}} \Delta\Gamma_{Hbb}^{(0,6)} + \frac{g^2}{16\pi^2} \Delta\Gamma_{Hbb}^{(1,4)} \left(1 + \frac{g_6}{\sqrt{2}} \Delta\Gamma_{Hbb}^{(1,6)} \right) \mathcal{F} \right]. \tag{3.94}$$

where

$$\Gamma_{Hbb}^{(0,4)} = \frac{G_F M_b^2(M_H)}{4\sqrt{2}\pi} M_H N_c, \quad N_c = \text{color factor} \tag{3.95}$$

$$\Gamma_{Hbb}^{(0,6)} = 2\left(s_\theta^2 W_1 + c_\theta^2 W_2 + s_\theta c_\theta W_3 - \frac{1}{4} W_4 + W_5 - W_{14} \right), \tag{3.96}$$

$$\Delta\Gamma_{Hbb}^{(1,4)} = \frac{4}{9} s_\theta^2, \tag{3.97}$$

$$\Delta\Gamma_{Hbb}^{(1,6)} = 4W_1 + \frac{2c_\theta}{s_\theta} W_3 - \frac{1}{2s_\theta^2} W_4 + 2W_5 - 2W_{14}. \tag{3.98}$$

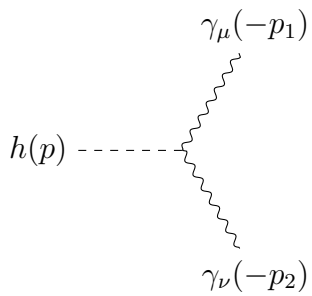
3.4.3 $H \to \gamma\gamma$

One of the main way to search the Higgs boson at LHC was to investigate its decay in two photon [43].
The BR of this decay is relatively rare compared to other decay modes (BR$\sim$ 0.2% for a SM Higgs of mass 125 GeV). Its significance lies in the clarity of its final state and the precision with which the Higgs mass signature can be measured [44].

Considering the SM scenario, the decay is one-loop induced, receiving the main contribution from W boson and fermions, in particular from the top

quark.

The SMEFT approach allows to define a LO term with no equivalent in SM, given by



$$= i\ g\ \frac{g_6}{\sqrt{2}}\ \frac{1}{M_H^2}\Big[p_{1,\nu}p_{2,\mu} - p_1 \cdot p_2 \delta_{\mu\nu}\Big]\epsilon_\mu(p_1)\epsilon_\nu(p_2)\mathcal{M}^{(0,6)}_{H\gamma\gamma}, \tag{3.99}$$

where the quantity $\mathcal{M}^{(0,6)}_{hAA}$ is equal to

$$\mathcal{M}^{(0,6)}_{H\gamma\gamma} = 2\frac{M_H^2}{M}W_1. \tag{3.100}$$

The investigation up to one loop correction, gives the well known SM contribution and the new part that describes the SM deviation in the SMEFT language.
To get an idea about the complexity of Feynman diagrams involved, see [27].

Considering the SM part, the full result is UV-finite, while the SMEFT amplitude shows divergences.
That implies writing the full amplitude as

$$\mathcal{M}_{H\gamma\gamma} = ig\frac{g_6}{\sqrt{2}}\mathcal{M}^{(0,6)}_{H\gamma\gamma} + i\frac{g^3}{16\pi^2}\mathcal{M}^{(1)}_{H\gamma\gamma}, \tag{3.101}$$

with

$$\mathcal{M}^{(1)}_{H\gamma\gamma} = \mathcal{M}^{(1,4)}_{H\gamma\gamma} + \frac{g_6}{\sqrt{2}}\Big[\mathcal{M}^{(1,6),f}_{H\gamma\gamma} + \mathcal{M}^{(1,6),UV}_{H\gamma\gamma}\Delta_{UV}(M_W^2)\Big]. \tag{3.102}$$

We introduce the counterterms for the parameters

$$\begin{aligned} c_\theta &= c_{\theta_R}\left(1+\frac{g^2}{16\pi^2}dZ_{c_\theta}\ \Delta_{UV}\right), \\ M &= M_R\left(1+\frac{g^2}{16\pi^2}dZ_M\ \Delta_{UV}\right), \\ g &= g_R\left(1+\frac{g_R^2}{16\pi^2}dZ_g\ \Delta_{UV}\right) \end{aligned} \tag{3.103}$$

and the WFR factors for the external legs [5].

The residual UV-pole in $d = 6$ require the mixing among the Wilson coefficients:

$$W_i = \sum_j Z_{ij} W_i^{ren} \tag{3.104}$$

where the matrix Z_{ij} is defined by Eq.(2.55).
Fixing the enters of the matrix in order to obtain a finite result, we can write:

$$dZ_{1,1} = -\frac{M_l^2}{4M_W^2} + \frac{1}{12c_\theta^2}\left(3 - 134c_\theta^2 + 212c_\theta^4 - 72c_\theta^6\right) + \frac{M_H^2}{4M_W^2}\left(3 - 4c_\theta^2 + 4c_\theta^4\right),$$

$$dZ_{1,2} = c_\theta^2 s_\theta^2\left(-6 + \frac{M_H^2}{M_W^2}\right),$$

$$dZ_{1,3} = c_\theta s_\theta\left[-\frac{M_H^2}{2M_W^2}(-1+2c_\theta^2) - 5 + 6c_\theta^2\right],$$

$$dZ_{1,42} = -9s_\theta^2. \tag{3.105}$$

The renormalized Wilson coefficients give a logarithmic scale dependence defined by the relation

$$\mathcal{M}_{H\gamma\gamma}^{(1,6),div} \to \mathcal{M}_{H\gamma\gamma}^{(1,6),R} \ln\frac{\mu_R^2}{M_W^2} \tag{3.106}$$

[5] For the photon, we have to consider the resummed propagator in Eq.(3.40).

and the complete one loop amplitude for the process results:

$$\mathcal{M}_{H\gamma\gamma} = ig\frac{g_6}{\sqrt{2}}\mathcal{M}^{(0,6)}_{H\gamma\gamma} + i\frac{g^3}{16\pi^2}\left\{\mathcal{M}^{(1,4)}_{H\gamma\gamma} + \frac{g_6}{\sqrt{2}}\left[\mathcal{M}^{(1,6),f}_{H\gamma\gamma} + \mathcal{M}^{(1,6),R}_{H\gamma\gamma}\ln\frac{\mu_R^2}{M_W^2}\right]\right\}, \tag{3.107}$$

The final step concerns the insertion of the finite counterterms:

$$\begin{aligned} c_{\theta_R} &= c_W\left(1 + \frac{g_R^2}{16\pi^2}\delta_{c_\theta}\right), \\ M_R &= M_W\left(1 + \frac{g_R^2}{16\pi^2}\delta_{M_W}\right), \\ g_R &= g_F\left(1 + \frac{g_F^2}{16\pi^2}\delta_g\right), \end{aligned} \tag{3.108}$$

where we defined $c_W = M_W/M_Z$. The renormalized coupling constant g_R is connected to G_F by using the Eq.(3.51) and the relation $g_F^2 = 4\sqrt{2}G_F M_W^2$. The partial width decay amplitude is obtained by integrating over the full phase-space:

$$\Gamma_{H\gamma\gamma} = \frac{1}{2M_H}\int \mathrm{d}\Phi_2\, |\mathcal{M}_{H\gamma\gamma}|^2. \tag{3.109}$$

The final result in the **Appendix D, Sec.D.3**.

Chapter 4

SMEFT deviations

4.1 Introduction and operative strategy

In the previous chapter we have explained how our `FORM` code `SMEFTO.frm` works: for a given process, the code performs the renormalization procedure and introduces the mixing for the coefficients in order to guarantee the correct finiteness of the results.
For example, by calling

```
#call HdecAA
```

the code compiles the finite matrix elements for the process $H(p) \to \gamma(p_1)\gamma(p_2)$ up to one-loop in the SMEFT; in a subsequent stage, the code employs specific abbreviation procedures which guarantee a more efficient numerical implementation in `FORTRAN`.

We have assembled three different procedures to compile the following processes:

- $Z \to l^+ l^-$,
- $H \to \bar{b}b$,
- $H \to \gamma\gamma$.

For each of them, we have calculated the transition amplitude $\mathcal{M}_{\text{decay}}$, isolating the IR/collinear part (if present), splitting the final results in a SM one

plus its $d = 6$ part and integrating over the phase space. This has defined our POs.
However, after calculating a PO [6] , we face specific challenges: What should be our next step? How should we analyze the SMEFT results? What values should we assign to the Wilson coefficients?

In this regard, we dedicate this chapter to analyze our previous results, working in an equiprobable space of Wilson coefficients. The randomly generated Wilson coefficients (within Warsaw basis) are treated as pseudo-data, allowing us to compute the corresponding probability density function (pdf) for each PO and study how large are the deviations from the SM [47].
Our computation is based on a strictly bottom-up approach: the Wilson coefficients are produced using the random number generator `RANLUX` [48] in order to delineate an equiprobable space of Wilson coefficients wherein any coefficient is as likely as any other.
It is not our goal to describe the use of the SMEFT in fits [49, 50, 51]; instead our aim is to compute deviations from SM, understand how large they can be and the possible correlation among different POs.

Let us take a generic PO $\mathcal{O}$. If $\mathcal{M}_{\mathcal{O}}$ is the corresponding matrix element, then

$$\mathcal{O} = \int \mathrm{dLIPS}_n \sum_{\mathrm{spin}} |\mathcal{M}_{\mathcal{O}}|^2, \tag{4.1}$$

where

$$\mathcal{M}_{\mathcal{O}} = \mathcal{M}_{\mathcal{O}}^{(4)} + \frac{g_6}{\sqrt{2}} \mathcal{M}_{\mathcal{O}}^{(6)}. \tag{4.2}$$

Working in the linear approximation, we obtain the following expression for the squared amplitude :

$$|\mathcal{M}_{\mathcal{O}}|^2 = |\mathcal{M}_{\mathcal{O}}^{(4)}|^2 + \sqrt{2}\, g_6 \mathrm{Re}\big[\mathcal{M}_{\mathcal{O}}^{(4),\dagger} \mathcal{M}_{\mathcal{O}}^{(6)}\big]. \tag{4.3}$$

[6] Quoting [45], "*what the experiments do is to collapse (and/or) transform some "primordial quantities" (such as the number of observed events in some pre-defined set-up) into some "secondary quantities" which we feel that are closer to the theoretical description of the phenomena".*
The quantities that can be directly measured in the experiments are cross-sections, asymmetries, etc., and are called Realistic Observables (ROs). To determine, from ROs, quantities as Higgs mass, width or couplings, a deconvolution procedure is needed: it is necessary to unfold some of the higher order corrections and interference contributions. These secondary quantities are called Pseudo-Observables (POs), [15, 46].

The numerical results have been obtained by using our `FORTRAN` code `SMEFTO.f`, which has enabled us to define deviations

$$\Delta\mathcal{O} = \frac{\mathcal{O}^{(6)}}{\mathcal{O}^{(4)}} - 1, \tag{4.4}$$

plot the relative pdf and display the relationships between the POs.

4.2 Computation in the making

4.2.1 $\Gamma(Z \to \mu^-\mu^+)$ and $sin\theta^{\mu}_{eff}$

Let us consider the decay $Z \to l^+l^-$. The finite amplitude $\mathcal{M}_{Zll}$ can be re-written in the form:

$$\mathcal{M}_{Zll} = \bar{u}(p_1)\mathcal{M}_\mu v(p_2)\mathcal{Z}_\mu(-p_1 - p_2), \tag{4.5}$$

where $\mathcal{Z}_\mu$ denotes the polarization of Z boson.
We split the quantity $\mathcal{M}_\mu$ is a vectorial and axial component:

$$\mathcal{M}_\mu = \gamma_\mu\Big(\mathcal{G}^l_V + \gamma_5\,\mathcal{G}^l_A\Big), \tag{4.6}$$

in accordance to the LEP strategy [52], where the terms $\mathcal{G}_{V,A}$ collect all the correction factors evaluated at $p^2 = -M_Z^2$. It follows the decomposition:

$$\mathcal{G}_{V,A} = \mathcal{G}^{(4)}_{V,A} + \frac{g_6}{\sqrt{2}}\,\mathcal{G}^{(6)}_{V,A}. \tag{4.7}$$

The integration over the full phase space gives the corresponding width $\Gamma(Z \to l^-l^+)$ (following the LEP1 conventions):

$$\Gamma(Z \to l^-l^+) = 4\Gamma_0\Big[|\mathcal{G}^l_V|^2 R^l_V + |\mathcal{G}^l_A|^2 R^l_A\Big], \tag{4.8}$$

where the LO is defined as

$$\Gamma_0 = \frac{G_F M_Z^3}{24\sqrt{2}\pi}, \tag{4.9}$$

and the radiators factors $R^l_{V,A}$ describes the final state QED corrections, taking into account the lepton mass.

In the numerical analysis, we will consider the case $l = \mu$.
Now we define $g^{\mu}_{V,A}$ as the real part of $\mathcal{G}^{\mu}_{V,A}$ and construct the next PO associated to the *effective sinus* of the weak mixing angle:

$$sin^2\theta^{\mu}_{eff} = \frac{1}{4}\left(1 - \frac{g^{\mu}_V}{g^{\mu}_A}\right). \tag{4.10}$$

4.2.2 $\Gamma(H \to \bar{b}b)$ and $\Gamma(H \to \gamma\gamma)$

The process $H \to \gamma\gamma$ does not require additional comments.

Considering the decay $h \to \bar{b}b$, we have to integrate the squared amplitude over the full phase space of the real photon (gluon) in order to remove the IR/ collinear poles.
In this regard, we will consider only the deconvoluted result in our simulations.

4.2.3 Anomalous magnetic moment a_μ

Another PO at our disposal is the anomalous magnetic moment of the muon, the so-called $g - 2$.
For an elementary particle with spin $\vec{S}$, charge q and mass m, the magnetic momentum is given by

$$\vec{\mu} = g\frac{q}{2m}\vec{S}, \tag{4.11}$$

where g is the gyromagnetic ratio.

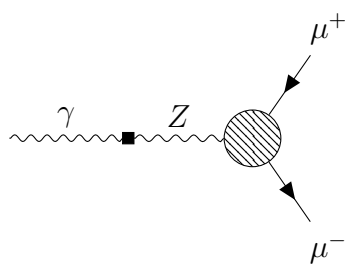
Historically Dirac predicted $g = 2$ for the electron in 1928 [53] but a QFT description implies additional contributions from the radiative corrections, changing the value of g.
The first order correction from QED was calculated by J. Schwinger in 1948 [54]:

$$a^{QED}_{\mu} = \frac{\alpha}{2\pi}. \tag{4.12}$$

Starting from $\gamma \to \mu^+\mu^-$, our strategy can be resume as follows:

- computation of the Schwinger term (QED),
- inclusion of one-loop SM contributions (Z and W contributions),
- addition of $d = 6$ operators,
- renormalization procedure,
- separation of IR/collinear QED correction,
- set the limit $p^2 \to 0$,
- take into account the terms proportional to $\sigma_{\mu\nu}$, with $\sigma_{\mu\nu} = \frac{i}{4}[\gamma_\mu, \gamma_\nu]$.

With reference to Eq.(2.42), the $Z - \gamma$ transition is now multiplied by the photon polarization $\varepsilon^\lambda_\mu(p)$, thus we can write:

$\varepsilon^\lambda_\mu(p) \times$ 

$= 0 \quad \text{with} \;\; p^2 \to 0,$ (4.13)

where the full circle indicates one loop contributions.
For a model-independent analysis of the magnetic and electric dipole moments of the muon and electron see Ref.[55]. Our result is presented in **Appendix D, Sec.D.4**.

4.3 Numerical results

4.3.1 SMEFT and a_μ

We want to study the deviations Δa_μ defined by

$$\Delta a_\mu = \frac{a_\mu^{(6)}}{a_\mu^{(4)}} - 1, \tag{4.14}$$

$a_\mu^{\rm th}$	0.00116591810(43)	Ref.[57]
$a_\mu^{\rm exp}$	0.00116592061(41)	Ref.[58]
$a_\mu^{\rm EW}$	194.8 $\times 10^{-11}$	Ref.[59]
	153.6 $\times 10^{-11}$(h.o)	Ref.[60]

Table 2: a_μ state of art.

where $a_\mu^{(4)}$ is given by the one-loop EW contributions

$$a_\mu^{(4)} \equiv a_\mu^{EW} = \frac{G_F M_\mu^2}{24\sqrt{2}\pi^2}\left[5 + \left(1 - 4s_\theta^2\right)^2\right]. \tag{4.15}$$

The current state of the a_μ is summarized in Table 2 [7]:
We observe that $a_\mu^{\rm exp} - a_\mu^{\rm th} = 251 \times 10^{-11}$, while for the EW corrections we have $a_\mu^{\rm EW} = 194.8 \times 10^{-11}$, thus the novel experimental value requires deviations of $\mathcal{O}(100)$ percent compared to the SM, a goal that appears to be very challenging to achieve within the SMEFT framework.

Selecting the scale Λ and generating the Wilson coefficients in a specific range $[W_{i,min}; W_{i,max}]$, we distinguish three different scenarios:

- S1 $\Rightarrow \Lambda = 1$ TeV, $W_i \in [-0.1; +0.1]$,
- S2 $\Rightarrow \Lambda = 2$ TeV, $W_i \in [-0.1; +0.1]$,
- S3 $\Rightarrow \Lambda = 1$ TeV, $W_i \in [-0.5; +0.5]$.

[7] Recent results at the Fermilab (August 2023) gave a new experimental average value:

$$a_\mu^{\rm exp,new} = 0.00116592059(22) \times 10^{-11}, \tag{4.16}$$

as reported in Ref.[56].

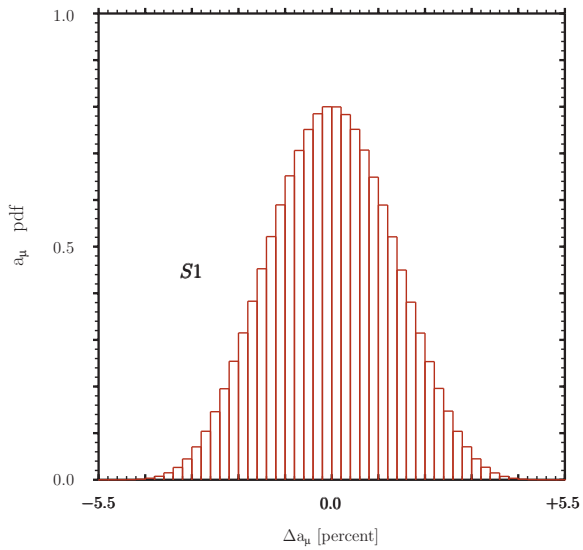


Figure 7: The pdf for Δa_μ, scenario S1.

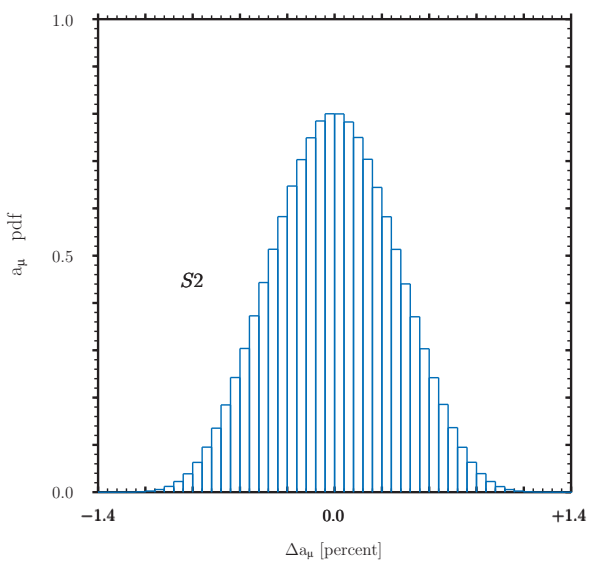


Figure 8: The pdf for Δa_μ, scenario S2.

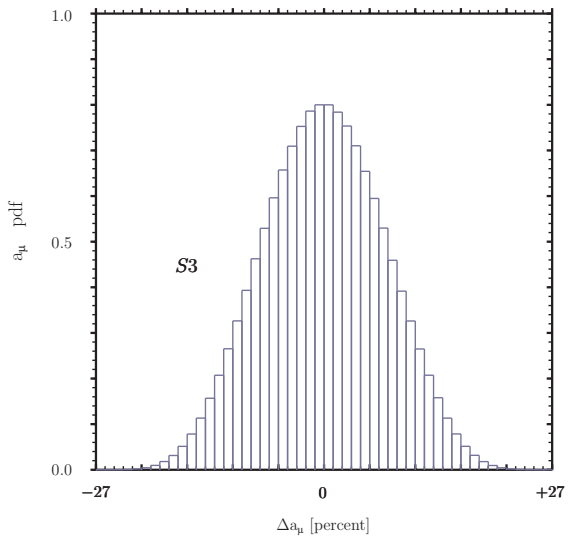


Figure 9: The pdf for Δa_μ at $\Lambda = 1$ TeV and $W_i \in [-0.5\ ;\ +0.5]$.

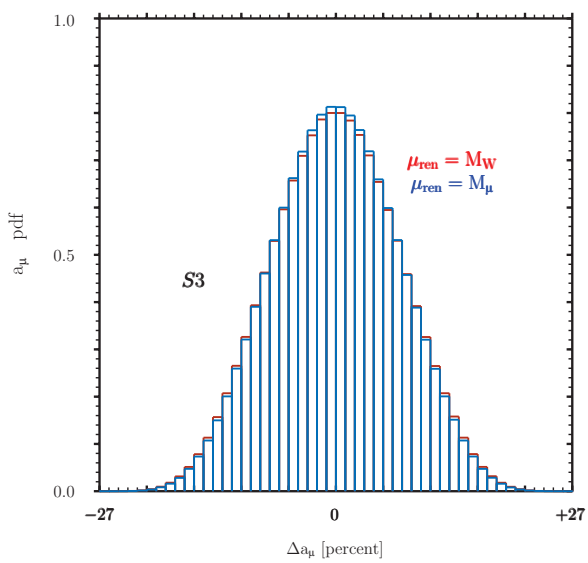


Figure 10: Same scenario of Fig.(9), different renormalization scales.

The numerical results for S1 are reported in Fig.(7) while for S2 are reported in Fig.(8).

The difficult to reach a level of deviations at $\mathcal{O}(100\%)$ becomes evident when the scenario S3 is investigated. We can reach a 50% deviations but only in the corners of the space of Wilson coefficients, as reported in Fig.(9).

In Fig.(10), we have selected two different renormalization scales to accentuate potential deviations, but what we observe is a slight discrepancy in the peaks and a negligible separation in the tails of the distributions.

Comparing Fig.(7) and Fig.(9), we have set the same scale Λ but a smaller range of coefficients in S1: what we observe are deviations in S3 at the level of $\pm 25\%$, while in S1 we reach deviations Δa_μ at the level of $\pm 5\%$.

We have also investigated a different scenario with the scale $\Lambda = 1$ TeV and the Wilson coefficients $W_i \in [-1; +1]$. Discarding the points in the space of Wilson coefficients where $|\Delta sin^2\theta^\mu_{eff}| > 10^{-3}$, we have observed that the largest fraction of Δa_μ deviations is compatible with zero.

Our results are collected in the following table:

Λ(TeV)	W_i	Δa_μ	σ
1	$[-0.1; +0.1]$	$\pm 5\%$	1.428
	$[-0.5; +0.5]$	$\pm 25\%$	7.142
2	$[-0.1; +0.1]$	$\pm 1\%$	0.357

Table 3: Standard deviation σ and deviations Δa_μ for different energy scales and different spaces of Wilson coefficients.

We want to remark the fact that the Wilson coefficients in our simulation are randomly generated, defining a space of coefficients where each of them is treated as a pseudo-data.
In order to investigate the SMEFT deviations, we can explore the case where some constraints are put on the space of Wilson coefficients.

Let us consider the tree level of $\gamma \to \mu^+\mu^-$. There is a combination of Wilson coefficients (in the Warsaw basis), $a_{lWB} = s_\theta a_{lW} - c_\theta a_{lB} \equiv W_7$, which appears in the calculation of a_μ but only at one-loop in the other POs. The strategy could be assigning large values to W_7, accompanied by the explicit reduction of all other coefficients to zero. It is worth noting that a_{lWB} is the coefficient of a LG operator and a loop suppression factor $1/(16\pi^2)$ is required.

To give an example, let us suppose that we use the SMEFT at tree level and define the following coefficients (in the Warsaw basis):

$$\begin{aligned} a_{\Phi lV} &= a^{(3)}_{\Phi l} - a^{(1)}_{\Phi l} - a_{\Phi l}, \\ a_{\Phi lA} &= a^{(3)}_{\Phi l} - a^{(1)}_{\Phi l} + a_{\Phi l}. \end{aligned} \tag{4.17}$$

We can derive both of them in terms of other Wilson coefficients by asking zero deviation in the vector and axial couplings for $Z\mu^+\mu^-$interaction. After that, we fix $a_{lWB} = s_\theta a_{lW} - c_\theta a_{lB}$ in order to reproduce the a_μ deviation with the rest of coefficients left free.
By constraining the free coefficients within the interval $[-0.1; +0.1]$, we require values of a_{lWB} such that

$$\frac{a_{lWB}}{16\pi^2} > 0.01, \tag{4.18}$$

to reach 50% deviations for a_μ.
Alternatively, it is possible to adopt a different strategy by decomposing a generic observable as follows:

$$\mathcal{O} = \mathcal{O}^{(4)} + \frac{g_6}{\sqrt{2}}\left[\mathcal{O}^{(0,6)} + \frac{G_F M_W^2}{\pi^2}\,\mathcal{O}^{(1,6)}\right]. \tag{4.19}$$

The term $\mathcal{O}^{(0,6)}$ indicates the tree level contribution in $d = 6$ while $\mathcal{O}^{(1,6)}$ takes into account all the one-loop terms.
Setting to zero the $\mathcal{O}^{(0,6)}$ terms in the vector and axial coupling of $Z\mu^+\mu^-$, we find two linear conditions among the Wilson coefficients.

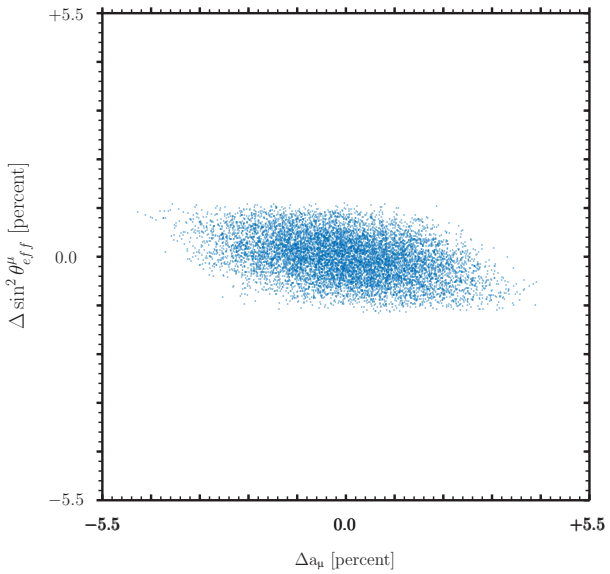


Figure 11: Scattered plot Δa_μ vs $\Delta sin^2\theta^{\mu}_{eff}$, S1.

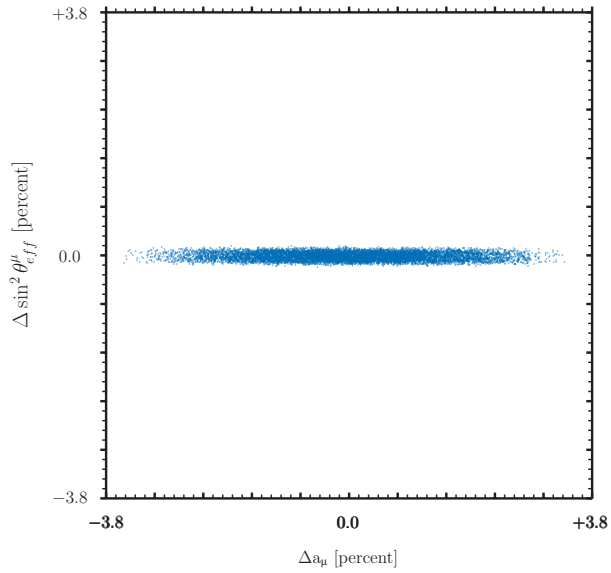


Figure 12: Scattered plot Δa_μ vs $\Delta sin^2\theta^{\mu}_{eff}$, with constraints defined by Eq.(4.19), S1.

The resulting scattered plots are reported in Fig.(11) and Fig.(12), where the deviations of $sin^2\theta^{\mu}_{eff}$ are defined by:

$$\Delta sin^2\theta^{\mu}_{eff} = \frac{sin^2\theta^{\mu}_{eff}|_{dim=6}}{sin^2\theta^{\mu}_{eff}|_{dim=4}} - 1. \tag{4.20}$$

In Fig.(11) we have a scattered plot for the tree-level SMEFT, while Fig.(12) displays the same relation by taking into account the constraint given by Eq.(4.19). Comparing the two plots we can conclude that the tree-level is dominant but the one-loop contribution is not negligible [27, 61].

What we observe is a remarkably low level of correlation between Δa_μ and $\Delta sin^2\theta^{\mu}_{eff}$ in the tree-level set-up, with a low deviation for $sin^2\theta^{\mu}_{eff}$ at the level of $\pm 1\%$ while we reach the level of $\pm 5\%$ for a_μ.
Taking into account the constraint in Eq.(4.19), we are able to reduce the deviation for $sin^2\theta^{\mu}_{eff}$ at the level of $\pm 0.2\%$ while it is still possible to accommodate large deviation for a_μ at level of $\pm 4\%$.

4.3.2 SMEFT and $Z \to \mu^+\mu^-$

In the previous analysis we have displayed the scattered plot for the deviations $\Delta sin^2\theta^{\mu}_{eff}$. In particular, we have observed how the inclusion of constraints leads to a different level of deviations.

From the LEP data [62] we are able to extract the value

$$sin^2\theta^{l}_{eff} = 0.23153 \pm 0.00016, \tag{4.21}$$

with an error of 0.07%.
From this result, it is clear that it is extremely difficult to evade this bound. It is necessary to observe that the LEP1 data were used to predict the W mass $M_W = 80.363 \pm 0.032$ GeV (we use 80.379 GeV) and the Higgs boson mass was not an input parameter, it was obtained $M_H < 285$ GeV at 95 % C.L.

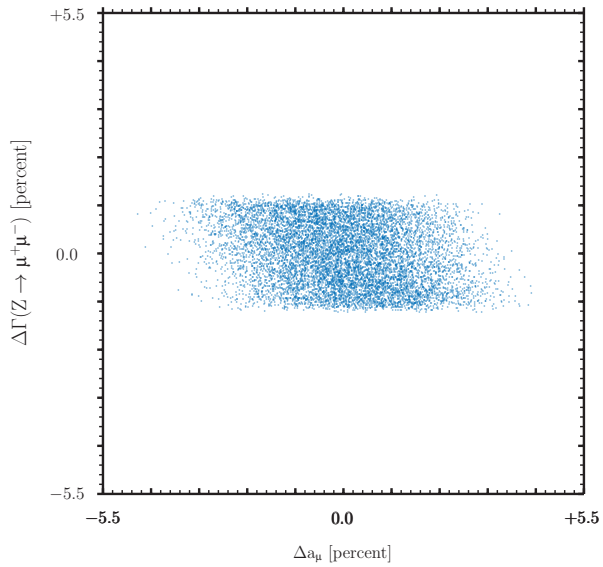


Figure 13: Scattered plot $\Delta\Gamma(Z \to \mu^+\mu^-)$ vs Δa_μ, scenario S1.

Let us consider the partial width amplitude $\Gamma(Z \to \mu^+\mu^-)$ and construct deviations:

$$\Delta\Gamma(Z \to \mu^+\mu^-) = \frac{\Gamma(Z \to \mu^+\mu^-)|_{dim=6}}{\Gamma(Z \to \mu^+\mu^-)|_{dim=4}} - 1. \tag{4.22}$$

Combining the deviations $\Delta\Gamma(Z \to \mu^+\mu^-)$ with Δa_μ we obtain the scattered plot in Fig.(13).

We observe that the largest fraction of $\Delta\Gamma(Z \to \mu^+\mu^-)$ is relevant around the zero percent, localized in the range $[-1\%; +1\%]$, implying deviations of a_μ at the level of $\pm(1-2)\%$. Furthermore, we note that deviations at the level of $\pm(3-4)\%$ for a_μ are extremely rare for any value of $\Delta\Gamma(Z \to l^+l^-)$.

In the previous section we have introduced the definition of PO as a quantity relates to measured cross-section and asymmetries by some deconvolution or unfolding procedure (i.e undressing of QED corrections).
When we consider LEP within the SMEFT framework, it becomes evident that the QED corrections have to be handled carefully: the SMEFT is a new theory and the introduction of operators in $d = 6$ changes not only the EW part, but also the QED(QCD) one.

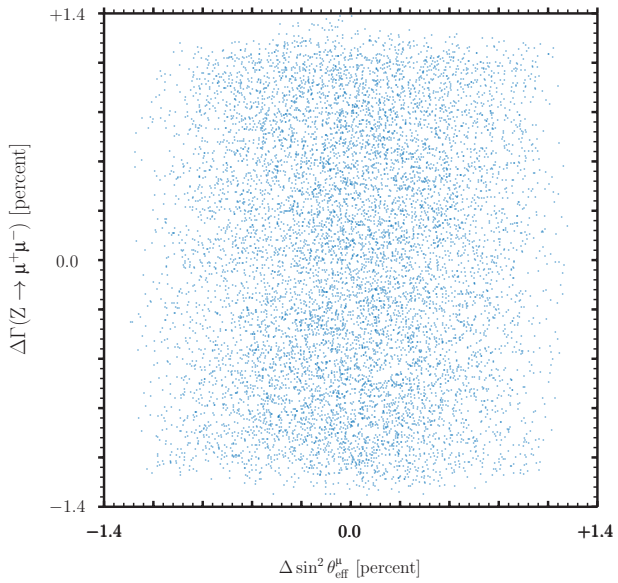


Figure 14: Scattered plot $\Delta\Gamma(Z \to \mu^+\mu^-)$ vs $\Delta sin^2\theta^{\mu}_{eff}$, scenario S1.

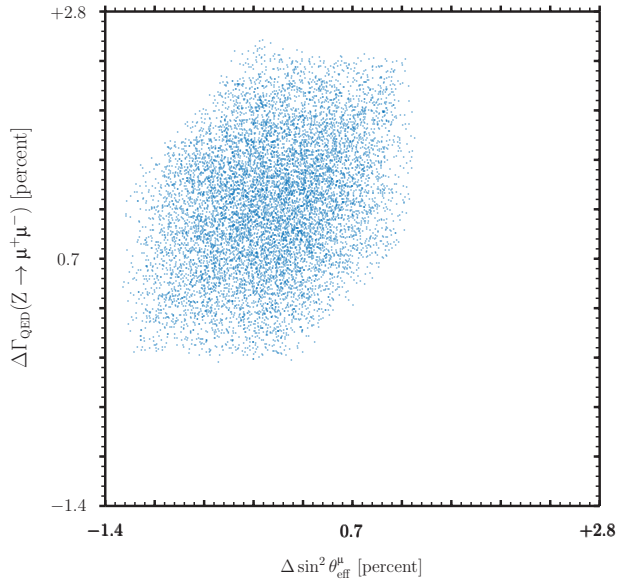


Figure 15: Scattered plot $\Delta\Gamma_{QED}(Z \to \mu^+\mu^-)$ vs $\Delta sin^2\theta^{\mu}_{eff}$, scenario S1.

Once that the full amplitude in Eq.(4.5) is assembled for $l = \mu$ and all the UV poles are removed, we have to consider the analytical continuation in $n = 4 + \epsilon$ in order to extract the IR/collinear part. By adding up the bremsstrahlung parts, the final result is a IR/collinear free quantity, both in the SM and in $d = 6$ (Kinoshita-Lee-Nauenberg theorem).

In Fig.(14) we have reported the scattered plot of deviations $\Delta\Gamma(Z \to \mu^+\mu^-)$ in relation to deviations $\Delta sin^2\theta^{\mu}_{eff}$. What we observe is a conspicuous absence of any significant correlation between these two distinct quantities.

The Fig.(15) displays the relationship between the QED component of deviations $\Delta\Gamma(Z \to \mu^+\mu^-)$ and $\Delta \sin^2\theta^{\mu}_{\rm eff}$. In comparison to the previous plot, this scattered plot exhibits a reduced dispersion, with a largest fraction of "data" points into a localized cluster with a slight positive trend.
The important fact here is that the large number of deviations of the PO $\Gamma_{QED}(Z \to \mu^+\mu^-)$ are not around the zero percent but they are confined in the region between the 0.7% and 2.5%.
This result is important, not only for the finiteness of IR/collinear sector in the SMEFT, but also because it shows that higher dimensional operators enter everywhere: signal, background and radiation.
The latter one is particular relevant when one wants to include (SM-deconvoluted) EW precision observable constraints in a fit to Higgs data.

We want to remark that since LEP POs are (mostly) SM-deconvoluted, the effect of $d = 6$ operators on the deconvolution procedure should be checked carefully.

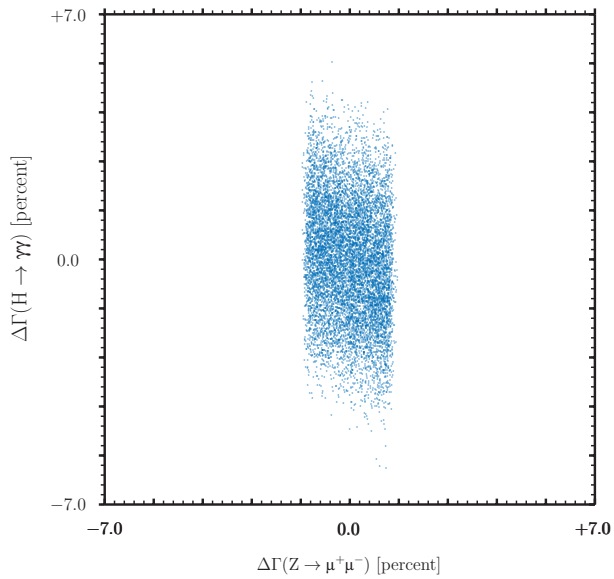


Figure 16: Scattered plot $\Delta\Gamma(H \to \gamma\gamma)$ vs $\Delta\Gamma(Z \to \mu^+\mu^-)$, scenario S1.

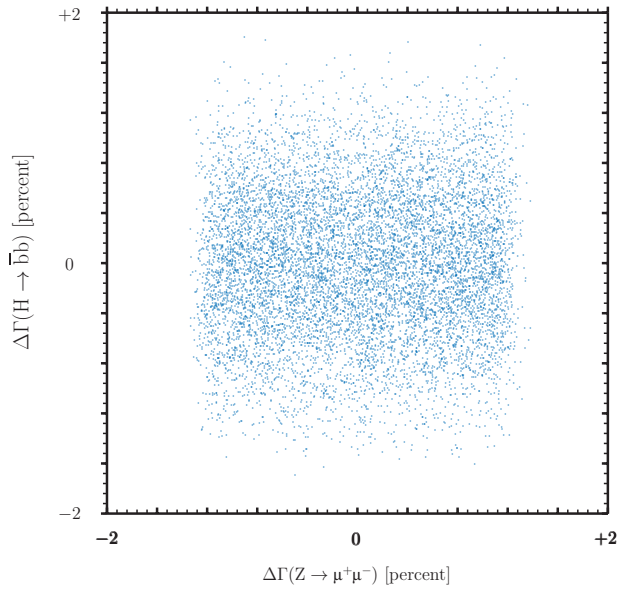


Figure 17: Scattered plot $\Delta\Gamma(H \to \bar{b}b$ vs $\Delta\Gamma(Z \to \mu^+\mu^-)$, scenario S1.

4.3.3 SMEFT and $H \to \gamma\gamma$, $H \to \bar{b}b$

Starting from the squared amplitudes $|\mathcal{M}_{H\gamma\gamma}|^2$ and $|\mathcal{M}_{Hbb}|^2$, we have integrated over the LIPS, obtaining the POs $\Gamma(H \to \gamma\gamma)$ and $\Gamma(H \to \bar{b}b)$.
In the numerical implementation we have used the deconvoluted result for $H \to \bar{b}b$, both QED and QCD.

At this point, we have constructed deviations

$$\Delta\Gamma(h \to X) = \frac{\Gamma(h \to X)|_{d=6}}{\Gamma(h \to X)|_{d=4}} - 1, \qquad X = \gamma\gamma, \bar{b}b. \tag{4.23}$$

The scattered plot between the deviations $\Delta\Gamma(H \to \gamma\gamma)$ and $\Delta\Gamma(Z \to \mu^+\mu^-)$ is reported in Fig.(16).
The simulation shows deviations of $\Gamma(Z \to \mu^+\mu^-)$ well bounded in the range $[-1\%; +1\%]$ and deviations of $\Gamma(H \to \gamma\gamma)$ at the level of $\pm 3\%$.

A largest degree of dispersion is found in the scattered plot between the deviations of $\Gamma(H \to \bar{b}b)$ and $\Gamma(Z \to \mu^+\mu^-)$, as reported in Fig.(17).

Chapter 5

EW precision data: the W mass

5.1 LEP renormalization equations

We direct our focus towards the challenge of predicting the W boson mass within the SMEFT framework, employing LEP1 IPS data.

In Sec.3.3 we have selected the IPS containing M_W, M_Z where the weak mixing angle defined by $\cos\theta_W = M_W/M_Z$.
Predicting the W mass means changing the IPS [8]. Therefore we start from the following renormalization equation:

$$\begin{aligned}
&(a) \quad \frac{1}{g^2 s_\theta^2} = \frac{1}{4\pi\alpha} + \frac{1}{16\pi^2}\Pi_{\gamma\gamma}(0), \\
&(b) \quad 8\frac{M^2}{g^2} = \frac{\sqrt{2}}{G_F} + \frac{1}{2\pi^2}\left[\Sigma_{WW}(0) + M^2 s_\theta^2 \delta_G\right] = X, \\
&(c) \quad \frac{M^2}{c_\theta^2} = M_Z^2 + \frac{g^2}{16\pi^2 c_\theta^2}\mathrm{Re}\Sigma_{ZZ}(M_Z^2) = Y.
\end{aligned} \tag{5.1}$$

The last equation is specific for the Z-resonance IPS, where the Z mass is defined as the zero of the real part of the inverse of the Z-propagator.

[8] We will use the following definitions for the two point functions:

$$S_{\gamma\gamma} = \frac{g^2 s_\theta^2}{16\pi^2}\Pi_{\gamma\gamma}(p^2)p^2, \qquad S_{Z\gamma} = \frac{g^2 s_\theta}{16\pi^2 c_\theta}\Sigma_{Z\gamma}(p^2),$$
$$S_{ZZ} = \frac{g^2}{16\pi^2 c_\theta^2}\Sigma_{ZZ}(p^2), \qquad S_{WW} = \frac{g^2}{16\pi^2}\Sigma_{WW}(p^2)$$

By solving the set of renormalization equations we obtain bare parameters as a function of experimental data. It is always convenient to resum large logarithmic corrections so that the lowest-order resummed solution for the weak mixing angle will be

$$s_\theta^2 = \bar{s}^2 = \frac{1}{2}\left[1 - \sqrt{1 - 4\frac{\pi\alpha(M_Z^2)}{\sqrt{2}G_F M_Z^2}}\right], \tag{5.2}$$

where s_θ denotes the Lagrangian parameter and $\alpha(M_Z^2)$ the running value of α constant. However, in our discussion we will use $\alpha = \alpha(0)$.
Inserting the solutions into the inverse of W propagator returns M_W with a LO solution given by

$$M_W = \bar{c} M_Z, \tag{5.3}$$

which receives corrections in perturbation theory, including the ones due to $d = 6$ operators.

It is worth noting that the inclusion of $d = 6$ operators touches all the ingredients of the calculation. For example if we can consider

$$\alpha(M_Z) = \alpha(0)\left[1 - \Delta\alpha_l - \Delta\alpha_t - \Delta\alpha_{\text{had}}^{(5)} + \mathcal{O}(\alpha\alpha_s)\right]^{-1}. \tag{5.4}$$

the inclusion of the SMEFT effects leads to

$$\begin{aligned} \Delta\alpha_l &= \Delta\alpha_l^{(4)} + \frac{g_6}{\sqrt{2}}\Delta\alpha_l^{(6)}, \quad \text{l= lepton} \\ \Delta\alpha_t &= \Delta\alpha_t^{(4)} + \frac{g_6}{\sqrt{2}}\Delta\alpha_t^{(6)}, \quad \text{t= top} \end{aligned} \tag{5.5}$$

The hadronic contribution $\alpha_{\text{had}}^{(5)}$ is extracted from data (the apex (5) indicates the number of light quarks) and the default value is $\Delta\alpha_{\text{had}}^{(5)}$= 0.0280398.

5.2 Solution of renormalization equations

Let us start by considering Eq.(5.1). Following the strategy in [15], we derive M^2 from Eq.(5.1b) and substitute the solution in Eq.(5.1c), which becomes

$$\frac{c_\theta^2}{g^2} M_Z^2 = \frac{\sqrt{2}}{8G_F} + \frac{1}{16\pi^2}\left[\Sigma_{WW}(0) - \text{Re}\Sigma_{ZZ}(M_Z^2) + M^2 s_\theta^2 \delta_G\right]. \tag{5.6}$$

We convert α into G_F using the following identity

$$\pi\alpha = \sqrt{2}G_F M_Z^2 \bar{s}^2 \bar{c}^2, \tag{5.7}$$

and what we obtain is an equation for s_θ^2:

$$\begin{aligned} s_\theta^2 c_\theta^2 = \bar{s}_\theta^2 \bar{c}_\theta^2 \Bigg\{ & 1 + \frac{G_F}{2\sqrt{2}\pi^2} \Bigg[\Sigma_{WW}(0) - \text{Re } \Sigma_{ZZ}(M_Z^2) \\ & - M_Z^2 s_\theta^2 c_\theta^2 \Pi_{AA}(0) + M^2 s_\theta^2 \delta_G \Bigg]_{s_\theta = \bar{s}_\theta} \Bigg\} \end{aligned} \tag{5.8}$$

The solution can be cast in the form:

$$s_\theta^2 = \bar{s}^2 + \frac{G_F M_Z^2}{2\sqrt{2}\pi^2} \tilde{\Delta} s^2, \tag{5.9}$$

where $\tilde{\Delta} s^2$ contains a finite part and an ultraviolet divergence.
Therefore we can re-write

$$s_\theta^2 = \bar{s}^2 + \frac{1}{4\pi} \Big(s_{\text{div}} + \Delta s^2 \Big) \tag{5.10}$$

where the divergent part is given by

$$s_{\text{div}} = \text{Re } \Pi_{ZA}(M_Z^2) \mid_{s_\theta = \bar{s}}, \tag{5.11}$$

and the finite one results to be

$$\Delta s^2 = \frac{1}{\bar{c}^2 - \bar{s}^2} \left[\frac{1}{M_Z^2} \Sigma_F + \bar{s}^2 \bar{c}^2 \text{Re } \Pi_F(M_Z^2) \right]. \tag{5.12}$$

We have adopted the notation X_F to indicate that a generic function X is devoid of UV poles.

The new auxiliary functions Σ_F and Π_F are defined by

$$\begin{aligned} \Sigma_F &= \Sigma_{WW}^G(0) - \text{Re } \Sigma_{33}(M_Z^2) + \text{Re } \Sigma_{3Q}(M_Z^2), \\ \Pi_F &= \Pi_{AA}(s) - \Pi_{AA}(0) \end{aligned} \tag{5.13}$$

where

$$\begin{aligned} \Sigma_{ZZ}(p^2) &= \Sigma_{33}(p^2) - 2 s_\theta^2 \Sigma_{3Q}(p^2) + s_\theta^4 \Pi_{AA}(p^2) p^2, \\ \Sigma_{ZA}(p^2) &= \Sigma_{3Q}(p^2) - s_\theta^2 \Pi_{AA}(p^2) p^2, \\ \Sigma_{WW}^G(0) &= \Sigma_{WW}(0) + M^2 s_\theta^2 \delta_G. \end{aligned} \tag{5.14}$$

We only have to fix a counterterm to cancel the UV pole in $s_{\rm div}$.
By inserting the solution for s_θ^2 into Eqs.(5.1a) and (5.1b), we derive solutions for g^2 and M^2 respectively, e.g

$$8\frac{M^2}{g^2} = X, \qquad \frac{M^2}{c_\theta^2} = Y, \tag{5.15}$$

from which we obtain

$$8c_\theta^2 Y|_{c_\theta=\bar{c}} = g^2 X. \tag{5.16}$$

5.2.1 SM and M_W

Working in $d = 4$, it is easy to show that all the terms in Eq.(5.12) are finite quantities. However, the extension to $d = 6$ does not assure the finiteness of Δs^2 and a change in notation is needed.

The function Σ_F in Eq.(5.13) is only UV-finite for $d = 4$, therefore we define

$$\Sigma = \Sigma^G_{WW}(0) - \text{Re }\Sigma_{33}(M_Z^2) + \text{Re }\Sigma_{3Q}(M_Z^2), \tag{5.17}$$

and assume the decomposition

$$\Sigma = \Sigma_F^{(4)} + \frac{g_6}{\sqrt{2}}\left[\Sigma^{(6)}_{\rm div} + \Sigma^{(6)}_{\rm F}\right]. \tag{5.18}$$

Inserting the previous quantities in Eq.(5.10) we find [9]:

$$s_\theta^2 = \bar{s}^2 + \frac{\alpha}{4\pi}\left\{\text{Re }\Pi_{ZA}(M_Z^2) + \frac{1}{\bar{c}^2 - \bar{s}^2}\left[\frac{1}{M_Z^2}\Sigma + \bar{s}^2\bar{c}^2 \text{Re }\Pi_F(M_Z^2)\right]\right\}_{s_\theta=\bar{s}}. \tag{5.20}$$

By explicitly separating the finite and divergent components from Σ and Re Π_{ZA} as

$$\begin{aligned} \text{Re }\Pi_{ZA}(M_Z^2) &= \text{Re }\Pi^F_{ZA}(M_Z^2) + \frac{\Delta_1}{\bar{\epsilon}}, \\ \Sigma &= \Sigma_F + \frac{\Delta_2}{\bar{\epsilon}}. \end{aligned} \tag{5.21}$$

[9] The quantity s_θ is a bare parameter, therefore

$$s_\theta^2 = [s_\theta^2]_{\rm LEP1} + \delta s_\theta \frac{1}{\bar{\epsilon}} = s_{\theta 1}^2 + \delta s_\theta \frac{1}{\bar{\epsilon}}. \tag{5.19}$$

the final expression of the Eq.(5.20) is

$$\begin{aligned} s_\theta^2 &= \bar{s}^2 + \frac{\alpha}{4\pi}\Bigg\{\text{Re }\Pi_{ZA}^F(M_Z^2) + \frac{\Delta_1}{\bar{\epsilon}} \\ &+ \frac{1}{\bar{c}^2 - \bar{s}^2}\left[\frac{1}{M_Z^2}\left(\Sigma_F + \frac{\Delta_2}{\bar{\epsilon}}\right) + \bar{s}^2\bar{c}^2\text{Re }\Pi_F(M_Z^2)\right]\Bigg\} \\ &= s_{\theta 1}^2 + \delta s_\theta \frac{1}{\bar{\epsilon}}, \end{aligned} \tag{5.22}$$

In order to extract the W mass, we have to consider the Eq.(5.1c) and write the c_θ in terms of the finite part plus the divergent one:

$$M^2 = M_Z^2\left[\bar{c}_{\theta 1}^2 + \delta c_\theta \frac{1}{\bar{\epsilon}}\right] + \frac{g^2}{16\pi^2}\Sigma_{ZZ}(M_Z^2), \tag{5.23}$$

where

$$\delta c_\theta = -\frac{1}{2}\frac{\delta s_\theta}{c_{\theta 1}}. \tag{5.24}$$

From the inverse of W propagator reported in Eq.(3.33) we substitute the expression in Eq.(5.23), obtaining

$$\Delta_{WW}^{-1}(p^2) = p^2 + M_Z^2\left[\bar{c}_{\theta 1}^2 + \frac{\delta c_\theta}{\bar{\epsilon}}\right] + \frac{g^2}{16\pi^2}\left[\text{Re }\Sigma_{ZZ}(M_Z^2) - \Sigma_{WW}(p^2)\right]. \tag{5.25}$$

where it is understood

$$\begin{aligned} \Sigma_{ZZ}(M_Z^2) &= \Sigma_{ZZ}(M_Z^2)^F + \frac{\Delta_3}{\bar{\epsilon}}, \\ \Sigma_{WW}(p^2) &= \Sigma_{WW}(p^2)^F + \frac{\Delta_4}{\bar{\epsilon}}. \end{aligned} \tag{5.26}$$

The W mass is determined by setting the zero of the function $\Delta_{WW}^{-1}(p^2)$ and imposing the on-shell condition $p^2 = -M_W^2$. The final expression is as follows

$$-M_W^2 + M_Z^2\bar{c}_{\theta 1}^2 + \frac{4\pi\alpha}{16\pi^2\bar{s}_\theta^2}\left[\text{Re }\Sigma_{ZZ}^F(M_Z^2) - \text{Re }\Sigma_{WW}^F(M_W^2)\right]_{s_\theta = \bar{s}} = 0. \tag{5.27}$$

We have confirmed the UV finiteness of Eq. (5.27) at $d = 4$ by implementing it in a `FORM` procedure, which allows us to make a prediction for M_W

5.2.2 Equation for s_θ^2: how does it change?

Let's consider the $Z-\gamma$ transition: when $g_6 \neq 0$, this transition introduces a term at $\mathcal{O}(g_6)$, leading to the presence of the Wilson coefficient W_3. Defining

$$S_{ZA}(p^2) = \frac{g^2}{16\pi^2}\Sigma_{ZA}(p^2) \tag{5.28}$$

and setting $p^2 = -M_Z^2$, the SMEFT extension gives

$$\Sigma_{ZA}(M_Z^2) = ... - \frac{16\pi^2}{g^2}\frac{g_6}{\sqrt{2}}\frac{c_\theta}{s_\theta}M_Z^2 W_3. \tag{5.29}$$

As a consequence, by replacing g for α, we obtain

$$\begin{aligned}
\Pi_{ZA}(M_Z^2) &= \Pi_{ZA}^\alpha(M_Z^2) + \frac{4\pi}{\alpha}\frac{g_6}{\sqrt{2}}s_\theta c_\theta W_3,\\
\Sigma_{3Q}(M_Z^2) &= \Sigma_{3Q}^\alpha(M_Z^2) - \frac{4\pi}{\alpha}\frac{g_6}{\sqrt{2}}\frac{c_\theta}{s_\theta}M_Z^2 W_3,\\
\Sigma_{33}(M_Z^2) &= \Sigma_{33}^\alpha(M_Z^2) - \frac{8\pi}{\alpha}\frac{g_6}{\sqrt{2}}s_\theta c_\theta M_Z^2 W_3,
\end{aligned} \tag{5.30}$$

thus the equation for s_θ has to be reconsidered.
Starting from Eq.(5.1) and using the set of equations in Eq.(5.14), we can write

$$\begin{aligned}
s_\theta^2 c_\theta^2 = \bar{s}_\theta^2\bar{c}_\theta^2 + \frac{\alpha}{4\pi}\Bigg\{\frac{1}{M_Z^2}\Bigg[&\Sigma_{WW}^G(0) - \mathrm{Re}\ \Sigma_{33}(M_Z^2) + 2s_\theta^2\mathrm{Re}\ \Sigma_{3Q}(M_Z^2)\\
&+ s_\theta^4\Pi_{AA}(M_Z^2)M_Z^2\Bigg] - \bar{s}_\theta^2\bar{c}_\theta^2\Pi_{AA}(0)\Bigg\},
\end{aligned} \tag{5.31}$$

which leads to

$$\begin{aligned}
s_\theta^2 c_\theta^2 &= \bar{s}_\theta^2\bar{c}_\theta^2\\
&+ \frac{\alpha}{4\pi}\Bigg\{\frac{1}{M_Z^2}\Bigg[\Sigma_{WW}^G(0) - \mathrm{Re}\ \Sigma_{33}^\alpha(M_Z^2) + \frac{8\pi}{\alpha}\frac{g_6}{\sqrt{2}}s_\theta^2 c_\theta^2 M_Z^2 W_3\\
&+ 2s_\theta^2\left(\Sigma_{3Q}^\alpha(M_Z^2) - \frac{4\pi}{\alpha}\frac{g_6}{\sqrt{2}}\frac{c_\theta}{s_\theta}M_Z^2 W_3\right) + s_\theta^4\Pi_{AA}(M_Z^2)M_Z^2 - \bar{s}_\theta^2\bar{c}_\theta^2\Pi_{AA}(0)\Bigg]\Bigg\}.
\end{aligned} \tag{5.32}$$

All the terms with W_3 cancel out and rearranging the equation appropriately, we have

$$s_\theta^2 c_\theta^2 = \bar{s}_\theta^2 \bar{c}_\theta^2 + \frac{\alpha}{4\pi}\left\{\frac{1}{M_Z^2}\Sigma + \frac{\bar{s}_\theta^2 - \bar{c}_\theta^2}{M_Z^2}\left[\Sigma_{ZA}^{\alpha}(M_Z^2) - \bar{s}_\theta^2 \Pi_{AA}(M_Z^2) M_Z^2\right] + \bar{s}_\theta^4 \Pi_{AA}(M_Z^2) - \bar{s}_\theta^2 \bar{c}_\theta^2 \Pi_{AA}(0)\right\}. \tag{5.33}$$

After a few straightforward algebraic steps, it becomes evident that it remains unaltered even with the incorporation of $d = 6$ terms.

5.3 Renormalization equations in SMEFT

The insertion of the SMEFT operators changes the renormalization equations and we must update them.
We have briefly addressed this topic during our discussion of the IPS definition in Sec.3.3.

Form the Coulomb scattering, we arrive at the equation

$$\frac{1}{g^2 s_\theta^2} = \frac{1}{4\pi\alpha} + \frac{1}{16\pi^2}\Pi_{AA}(0) + \frac{1}{2\pi\alpha}\frac{g_6}{\sqrt{2}}\Delta_\alpha, \tag{5.34}$$

where Δ_α indicates the combination of Wilson coefficients involved in the LO vertex $\gamma \bar{l} l$, as shown in Eq.(3.39) .

The muon decay gives the second renormalization equation

$$8\frac{M^2}{g^2} = \frac{\sqrt{2}}{G_F} + \frac{1}{2\pi^2}\Sigma_{WW}^{G}(0) + 2\frac{g_6}{G_F}(\Delta_G + W_{28}), \tag{5.35}$$

where the term Δ_G is defined in Eq.(3.48), while the coefficient W_{28} derives from the vertex which involves 4 fermion lines.

The equation Eq.(5.1c) involves the Z mass and it remains unchanged. We re-write it as

$$\frac{c_\theta^2}{g^2} M_Z^2 = \frac{M^2}{g^2} - \frac{1}{16\pi^2}\text{Re } \Sigma_{ZZ}(M_Z^2) \tag{5.36}$$

and inserting the expression for M^2/c_θ^2, we find

$$\frac{c_\theta^2}{g^2} = \frac{1}{4\sqrt{2}G_F M_Z^2} + \frac{1}{16\pi^2 M_Z^2}\left[\Sigma_{WW}^G(0) - \text{Re }\Sigma_{ZZ}(M_Z^2)\right] + \frac{1}{4}\frac{g_6}{G_F M_Z^2}(\Delta_G + W_{28}). \tag{5.37}$$

The whole equation for s_θ^2 is now derived by using Eq.(5.34):

$$\begin{aligned}\left[\frac{1}{4\pi\alpha}(1+\sqrt{2}g_6\Delta_\alpha) + \frac{1}{16\pi^2}\Pi_{AA}(0)\right]s_\theta^2 c_\theta^2 &= \frac{1}{4\sqrt{2}G_F M_Z^2} \\ &+ \frac{1}{16\pi^2 M_Z^2}\left[\Sigma_{WW}^G(0) - \text{Re }\Sigma_{ZZ}(M_Z^2)\right] + \frac{1}{4}\frac{g_6}{G_F M_Z^2}(\Delta_G + W_{28}).\end{aligned} \tag{5.38}$$

By using the Eq.(5.7) and isolating the s_θ^2, the previous equation becomes

$$\begin{aligned}s_\theta^2 c_\theta^2 = &\left[1 - \sqrt{2}g_6\Delta_\alpha - \frac{\alpha}{4\pi}\Pi_{AA}(0)\right]\Bigg\{\bar{s}_\theta^2\bar{c}_\theta^2 \\ &+ \frac{\alpha}{4\pi M_Z^2}\left[\Sigma_{WW}^G(0) - \text{Re }\Sigma_{ZZ}(M_Z^2)\right] + \sqrt{2}g_6 s_\theta^2 c_\theta^2(\Delta_G + W_{28})\Bigg\}.\end{aligned} \tag{5.39}$$

In the final stage of the calculation, we introduce

$$s_\theta^2 = \bar{s}_\theta^2 + \frac{g_6}{\sqrt{2}}\Delta s_{(0,6)}^2 + \frac{\alpha}{4\pi}\left[\Delta s^2 + \frac{g_6}{\sqrt{2}}\Delta s_{(1,6)}^2\right] \tag{5.40}$$

and derive the corrective terms $\Delta s_{(0,6)}^2$, Δs^2 and $\Delta s_{(1,6)}^2$.

There is an additional complication: the LO terms involve the combination of Wilson coefficients Δ_α and Δ_G, which can be rewrite as

$$\begin{aligned}\Delta_G &= \sqrt{2}\left[s_\theta^2 W_1 + c_\theta^2 W_2 + s_\theta c_\theta W_3 + \frac{1}{2}\big(W_{16} + W_{17} + W_{18}\big)\right], \\ \Delta_\alpha &= \sqrt{2}\left[\big(s_\theta^2 + 2c_\theta^2\big)W_1 - c_\theta^2 W_2 + c_\theta^3 s_\theta W_3 - \frac{1}{4}c_\theta^2 s_\theta^2 W_4\right].\end{aligned} \tag{5.41}$$

In this case, we adopt the following notation

$$\begin{aligned}s_\theta &= \bar{s}_\theta + \frac{\alpha}{4\pi}\frac{\Delta s^2}{2\bar{s}_\theta}, \\ c_\theta &= \bar{c}_\theta - \frac{\alpha}{4\pi}\frac{\Delta s^2}{2\bar{c}_\theta}\end{aligned} \tag{5.42}$$

and define

$$\Delta_G = \bar{\Delta}_G + \frac{\alpha}{\pi}\Delta_G^{(1)}\Delta s^2, \quad \Delta_\alpha = \bar{\Delta}_\alpha + \frac{\alpha}{\pi}\Delta_\alpha^{(1)}\Delta s^2, \tag{5.43}$$

where the new combinations of Wilson coefficients result to be

$$\begin{aligned}
\bar{\Delta}_G &= \frac{1}{\sqrt{2}}\left(2\bar{s}_\theta^2 W_1 + 2\bar{c}_\theta^2 W_2 + 2\bar{c}_\theta\bar{s}_\theta W_3 + \sum_{i=16}^{18} W_i\right),\\
\bar{\Delta}_\alpha &= \sqrt{2}\left[\left(\bar{s}_\theta^2 + 2\bar{c}_\theta^2\right)W_1 - \bar{c}_\theta^2 W_2 + \bar{c}_\theta^3\bar{s}_\theta W_3 - \frac{1}{4}\bar{c}_\theta^2\bar{s}_\theta^2 W_4\right],\\
\Delta_G^{(1)} &= \frac{1}{4\sqrt{2}}\left(2W_1 - 2W_2 + \frac{\bar{c}_\theta^2 - \bar{s}_\theta^2}{\bar{c}_\theta\bar{s}_\theta}W_3\right),\\
\Delta_\alpha^{(1)} &= \frac{1}{8\sqrt{2}}\left[-4W_1 + 4W_2 - 2\frac{\bar{c}_\theta}{\bar{s}_\theta}\left(3s_\theta^2 + 2\bar{c}_\theta^2\right)W_3 + W_4\right].
\end{aligned} \tag{5.44}$$

Finally, we are able to give the full corrective term of s_θ^2, i.e we find:

$$\Delta s^2_{(0,6)} = \frac{2\bar{c}_\theta^2\bar{s}_\theta^2}{\bar{c}_\theta^2 - \bar{s}_\theta^2}\left(\bar{\Delta}_G - \bar{\Delta}_\alpha\right), \tag{5.45}$$

$$\Delta s^2 = \frac{1}{\bar{c}_\theta^2 - \bar{s}_\theta^2}\left[\frac{1}{M_Z^2}\Sigma_F + \bar{s}_\theta^2\bar{c}_\theta^2\Pi_F\right] - \frac{1}{M_Z^2}\Sigma_{ZA}(M_Z^2)\Big|_{s_\theta=\bar{s}_\theta}. \tag{5.46}$$

The corrective term $\Delta s^2_{(1,6)}$ is not reported here.

5.3.1 SMEFT and M_W

Let us restart from

$$p^2 + M^2 + \frac{g^2}{16\pi^2}\Sigma_{WW}(p^2), \quad M^2 = c_\theta^2 M_Z^2 + \frac{g^2}{16\pi^2}\text{Re }\Sigma_{ZZ}(M_Z^2). \tag{5.47}$$

Our result will be given by the following equation

$$-M_W^2 + c_\theta^2 M_Z^2 + \frac{g^2}{16\pi^2}\left[\text{Re }\Sigma_{ZZ}(M_Z^2) - \text{Re }\Sigma_{WW}(M_W^2)\right] = 0, \tag{5.48}$$

The coupling constant has to be expressed as $g^2 = 4\pi\alpha/s_\theta^2$, with s_θ^2 and c_θ^2 expanded according to Eq.(5.40). The function $\Sigma_{ZZ}(M_Z^2)$ follows the decomposition given by Eq.(5.14).

It should be clear that the prediction for M_W is contingent on the selection of the IPS. By opting for the set $\{\alpha, G_F, M_Z\}$ and abstaining from introducing counterterms on the parameters, we have established a direct connection between the bare parameters and the IPS.
Nevertheless, the mixing of Wilson coefficients is required to guarantee the finiteness also for $d = 6$. After that, a prediction for M_W in the SMEFT will follow.
The full result of M_W is reported in **Appendix D, Sec.D.5**.

Nevertheless, this result remains incomplete. When we examine Eq. (5.39), it becomes apparent that we should have incorporated the one-loop corrections of the muon, which includes vertices, boxes, and other components in $d = 6$. This correction is denoted by $\delta_G^{(6)}$ and should be included in the function $\Sigma_{WW}^G(0)$. At the moment the term $\delta_G^{(6)}$ is not at our disposal.
Therefore, all the terms of $\mathcal{O}(g^2 g_6)$ coming from Δ_α and Δ_G will be discarded, or, better phrased, included in the unknown $\delta_G^{(6)}$. This is not a problem as far as mixing is concerned; once that $\delta_G^{(6)}$ will be available, we will "simply" update the mixing (if needed), and include the finite part into the result.

To summarize our result, we obtain a finite prediction in the form:

$$M_W^2 = \mathcal{O}(1) + \mathcal{O}(g_6) + \mathcal{O}(\alpha) + \mathcal{O}(\alpha g_6). \tag{5.49}$$

By setting $\alpha = 0$ and $g_6 = 0$, we have

$$M_W = \bar{c}_\theta M_Z, \tag{5.50}$$

which indicates the $\mathcal{O}(1)$ contribution at $d = 4$.
The terms at $\mathcal{O}(g_6)$ contain the Wilson coefficients involved in the mixing procedure to cancel the residual divergence at $\mathcal{O}(\alpha g_6)$. The mixing is not complete since the term $\delta_G^{(6)}$ is unknown yet. At this regard, we discarded all the terms proportional to Δ_α and Δ_G at $\mathcal{O}(\alpha g_6)$.
The parts at $\mathcal{O}(\alpha)$ are UV-finite and give the well known SM result.

With a finite (partial) prediction of M_W we can assess deviations by defining the quantity

$$\Delta M_W = \frac{M_W^{(6)}}{M_W^{(4)}} - 1. \tag{5.51}$$

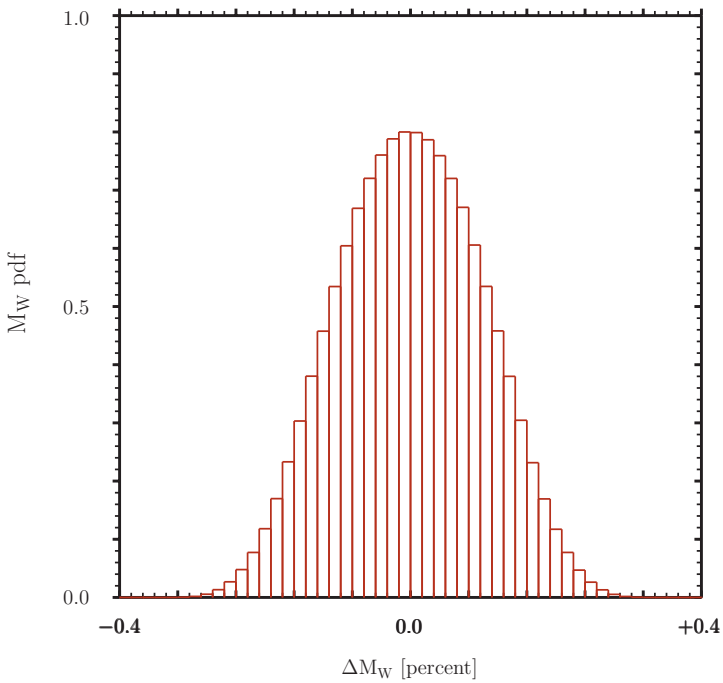


Figure 18: The pdf for ΔM_W, scenario S1

The result from CDF II detector is $M_W = 80.4335 \pm 0.0094\ GeV$ [64] to be compared with the previous world-average, $M_W = 80.379 \pm 0.012\ GeV$ [65]. Therefore we have to require a deviation of $+0.068\%$.

Our simulation for ΔM_W is displayed in Fig.(18), where the pdf is computed by setting $\Lambda = 1\ TeV$ and $W_i \in [-0.1\ ;\ +0.1]$.

Chapter 6

Conclusions

The use of an EFT to investigate new frontiers in high energy physics is well justified by its range of application. In a *bottom-up* approach, the EFTs guarantee a deep investigation of unknown phenomena without reliance on high-energy scale. A minimal set of operators in dimension higher than four is incorporated in the low-energy Lagrangian, with each operator suppressed by a cut off scale Λ. To characterized the effects of new physics, a Wilson coefficient is associated to each individual operator.

Following this approach, we can define the SMEFT: after the introduction of operators in $d = 6$, we have to perform a redefinition of fields in order to eliminate the redundancy among the operators. At this point, we give the correct normalization of the kinetic terms by applying the canonical normalization and extract the Feynman rules.

The remarkable precision of experimental data motivates us to expand our theoretical framework into the NLO region. As for the SM, the NLO corrections are necessary to show the consistency of the new theory as a well defined and predictive field theory.
Performing an NLO calculation for a given process entails building a generic observable with the complete one-loop correction part, including all operators at a fixed order in mass dimension. To achieve this goal, we have checked the WST identities and computed a set of POs, developing a renormalization procedure in order to reabsorb the UV divergences and ensure a finite result. Furthermore, the closure in the IR/collinear sector has been verified.

Without looking at the experimental data, we have treated the randomly generated Wilson coefficients of the SMEFT (in the Warsaw basis) as pseudo-data, and for each PO, we have computed the corresponding pdf. That has meant working in an equiprobable space of coefficients, where any Wilson coefficient is as likely as any other.

Our goal has been to understand how large are the deviations from the SM once the SMEFT scale Λ and the range of the Wilson coefficients are selected. The set of POs analyzed in this dissertation is given by: $\Gamma(Z \to \bar{l}l)$, $\Gamma(H \to \bar{b}b)$, $\Gamma(H \to \gamma\gamma)$, $g-2$, $\sin\theta^{\mu}_{\text{eff}}$ and M_W. The corresponding results can be compared with experimental data to understand how easy or difficult will be to explain (possible) SM deviations in terms of the SMEFT language.

We have shown that for different scenarios the deviations from the SM are generally $< 10\%$, with the largest fraction of them compatibles with zero. By studying the correlation between various POs, we have not observed specific trends. However, putting constraints on the Wilson coefficients and analyzing the scattered plots, it has been possible to explain how the magnitude of deviations changes: in particular, constraints in the space of Wilson coefficients have been introduced to explain large deviations for one PO and very small deviations for a second PO.
Furthermore, we have remarked the non-negligible impact of NLO corrections in the SMEFT, discussing how the deviations for $sin\theta^{\mu}_{\text{eff}}$ and a_μ are modified by including the one-loop results into a LO scenario.

We want to stress that our discussion about SMEFT deviations should be regarded as a preliminary analysis due to the missing terms $\delta_G^{(6)}$. The logical progression of this work may involve a more in-depth investigation into muon physics in order to obtain the complete expression of δ_G. Currently, a complete one-loop correction term within linear $d = 6$ extension is **unavailable** and its calculation will be necessary to achieve comprehensive results.
The mixing matrix of Wilson coefficients is an essential elements in order to obtain a finite result at one-loop once that the three-point Green functions are introduced in the calculation and this update to δ_G could lead to a modification of the entries in the matrix, prompting a critical review of all the theoretical results at our disposal.
To undertake this calculation, a well defined strategy is essential to create an efficient `FORM` code capable of handling intricate Feynman diagrams and

extensive strings of Dirac matrices. Special consideration is needed for the QED sector, where the incorporation of operators in $d = 6$ markedly complicates the integration over the entire phase space.
Once that a complete corrective term of the G_F constant is obtained, all the results of this thesis will need to be updated. However, work in progress...

The future challenge in the field of fundamental particle physics entails advancing the already impressive levels of experimental precision achieved in recent years. This pursuit demands a well-defined and robust theoretical framework as its foundation in order to explain possible new physics signatures and the EFT approach, such as the SMEFT, presents itself as an ideal candidate.

Appendix A

Field Redefinitions

Following the approach in [8], we give the proof that, under field redefinitions, the S-matrix elements do not change.
We start from the functional integral

$$Z[J] = \int \mathcal{D}\phi \ \exp\left\{ i \int \mathrm{d}^4x \ \mathcal{L}[\phi] + \mathcal{J}\phi \right\}, \tag{A.1}$$

that allows to write the Green's functions as

$$\begin{aligned} \langle 0| \, \mathrm{T} \prod_{i=1}^{r} \phi(x_i) \, |0\rangle &= \frac{\int \mathcal{D}\phi \ \prod_{i=1}^{r} \phi(x_i) \ e^{iS[\phi]}}{\int \mathcal{D}\phi \ e^{iS[\phi]}} \\ &= \frac{1}{Z[\mathcal{J}]} \ \prod_{i=1}^{r} \frac{\delta}{i\delta\mathcal{J}(x_i)} \ Z[\mathcal{J}]\Big|_{\mathcal{J}=0}. \end{aligned} \tag{A.2}$$

Under local transformation of fileds

$$\phi(x) = F[\phi'(x)], \tag{A.3}$$

the Lagrangian is redefined as

$$\mathcal{L}[\phi(x)] = \mathcal{L}[F[\phi'(x)]] = \mathcal{L}'[\phi'(x)], \tag{A.4}$$

where the functional defined by F involves powers of ϕ and its derivatives.
In this way we are able to define a new functional $Z'[J]$

$$\begin{aligned} Z'[\mathcal{J}] &= \int \mathcal{D}\phi' \ \exp\left\{ i \int \mathrm{d}^4x \ \mathcal{L}'[\phi'] + \mathcal{J}\phi' \right\} \\ &= \int \mathcal{D}\phi \ \exp\left\{ i \int \mathrm{d}^4x \ \mathcal{L}'[\phi] + \mathcal{J}\phi \right\}, \end{aligned} \tag{A.5}$$

which gives the correlation functions of ϕ' with Langrangian $\mathcal{L}'[\phi']$. At the same time, it gives the correlation function of ϕ with Lagrangian $\mathcal{L}'[\phi]$, since the variable ϕ is a dummy integration variable and the replacement $\phi' \to \phi$ is well justified.
Taking into account the transformation given in Eq.(A.3), the functional integral $Z[\mathcal{J}]$ results:

$$Z[\mathcal{J}] = \int \mathcal{D}\phi' \left|\frac{\delta F}{\delta \phi'}\right| \exp\left\{ i \int \mathrm{d}^4x \; \mathcal{L}'[\phi'] + \mathcal{J} F[\phi'] \right\}. \tag{A.6}$$

Working in dimensional regularization, the Jacobian can be considered as unity, thus setting $\phi = \phi'$ we have

$$Z[\mathcal{J}] = \int \mathcal{D}\phi \exp\left\{ i \int \mathrm{d}^4x \; \mathcal{L}'[\phi] + \mathcal{J} F[\phi] \right\}. \tag{A.7}$$

We observe that Z and Z' are two different correlation functions and this implies different Green's functions. Z leads us to S-matrix using $\mathcal{L}'[\phi]$ and $F[\phi]$ as interpolating field Z' leads us to S-matrix using $\mathcal{L}'[\phi]$ and ϕ as interpolating field.
However, the S-matrix elements do not change; the LSZ relates the functional integral with the S-matrix elements:

$$\prod_{a=1}^{m} Z_a^{1/2} \prod_{b=1}^{n} Z_b^{1/2} {}_{\text{out}} < q_1 ... q_m | p_1 ... p_n >_{\text{in}}, \tag{A.8}$$

where $Z_{a,b}$ are the WFR factors. The important point to stress is that the particle states are given by the physical spectrum of the theory and the S-matrix elements depend on the particle states, not on fields.
The LSZ reduction does not take into account what field is used, so, after a redefinition of fields the S-matrix is unchanged.

Appendix B

The Warsaw basis in d= 6

$1: X^3$	
Q_G	$f^{ABC} G_\mu^{A\nu} G_\nu^{B\rho} G_\rho^{C\mu}$
$Q_{\widetilde{G}}$	$f^{ABC} \widetilde{G}_\mu^{A\nu} G_\nu^{B\rho} G_\rho^{C\mu}$
Q_W	$\epsilon^{IJK} W_\mu^{I\nu} W_\nu^{J\rho} W_\rho^{K\mu}$
$Q_{\widetilde{W}}$	$\epsilon^{IJK} \widetilde{W}_\mu^{I\nu} W_\nu^{J\rho} W_\rho^{K\mu}$

$2: H^6$	
Q_H	$(H^\dagger H)^3$

$3: H^4 D^2$	
$Q_{H\Box}$	$(H^\dagger H)\Box(H^\dagger H)$
Q_{HD}	$\left(H^\dagger D_\mu H\right)^* \left(H^\dagger D_\mu H\right)$

$4: X^2 H^2$	
Q_{HG}	$H^\dagger H\, G_{\mu\nu}^A G^{A\mu\nu}$
$Q_{H\widetilde{G}}$	$H^\dagger H\, \widetilde{G}_{\mu\nu}^A G^{A\mu\nu}$
Q_{HW}	$H^\dagger H\, W_{\mu\nu}^I W^{I\mu\nu}$
$Q_{H\widetilde{W}}$	$H^\dagger H\, \widetilde{W}_{\mu\nu}^I W^{I\mu\nu}$
Q_{HB}	$H^\dagger H\, B_{\mu\nu} B^{\mu\nu}$
$Q_{H\widetilde{B}}$	$H^\dagger H\, \widetilde{B}_{\mu\nu} B^{\mu\nu}$
Q_{HWB}	$H^\dagger \tau^I H\, W_{\mu\nu}^I B^{\mu\nu}$
$Q_{H\widetilde{W}B}$	$H^\dagger \tau^I H\, \widetilde{W}_{\mu\nu}^I B^{\mu\nu}$

$5: \psi^2 H^3 + \text{h.c.}$	
Q_{eH}	$(H^\dagger H)(\bar{l}_p e_r H)$
Q_{uH}	$(H^\dagger H)(\bar{q}_p u_r \widetilde{H})$
Q_{dH}	$(H^\dagger H)(\bar{q}_p d_r H)$

$6: \psi^2 X H + \text{h.c.}$	
Q_{eW}	$(\bar{l}_p \sigma^{\mu\nu} e_r) \tau^I H W_{\mu\nu}^I$
Q_{eB}	$(\bar{l}_p \sigma^{\mu\nu} e_r) H B_{\mu\nu}$
Q_{uG}	$(\bar{q}_p \sigma^{\mu\nu} T^A u_r) \widetilde{H}\, G_{\mu\nu}^A$
Q_{uW}	$(\bar{q}_p \sigma^{\mu\nu} u_r) \tau^I \widetilde{H}\, W_{\mu\nu}^I$
Q_{uB}	$(\bar{q}_p \sigma^{\mu\nu} u_r) \widetilde{H}\, B_{\mu\nu}$
Q_{dG}	$(\bar{q}_p \sigma^{\mu\nu} T^A d_r) H\, G_{\mu\nu}^A$
Q_{dW}	$(\bar{q}_p \sigma^{\mu\nu} d_r) \tau^I H\, W_{\mu\nu}^I$
Q_{dB}	$(\bar{q}_p \sigma^{\mu\nu} d_r) H\, B_{\mu\nu}$

$7:\psi^2H^2D$	
$Q_{Hl}^{(1)}$	$(H^\dagger i\overleftrightarrow{D}_\mu H)(\bar{l}_p\gamma^\mu l_r)$
$Q_{Hl}^{(3)}$	$(H^\dagger i\overleftrightarrow{D}^I_\mu H)(\bar{l}_p\tau^I\gamma^\mu l_r)$
Q_{He}	$(H^\dagger i\overleftrightarrow{D}_\mu H)(\bar{e}_p\gamma^\mu e_r)$
$Q_{Hq}^{(1)}$	$(H^\dagger i\overleftrightarrow{D}_\mu H)(\bar{q}_p\gamma^\mu q_r)$
$Q_{Hq}^{(3)}$	$(H^\dagger i\overleftrightarrow{D}^I_\mu H)(\bar{q}_p\tau^I\gamma^\mu q_r)$
Q_{Hu}	$(H^\dagger i\overleftrightarrow{D}_\mu H)(\bar{u}_p\gamma^\mu u_r)$
Q_{Hd}	$(H^\dagger i\overleftrightarrow{D}_\mu H)(\bar{d}_p\gamma^\mu d_r)$
Q_{Hud} + h.c.	$i(\widetilde{H}^\dagger D_\mu H)(\bar{u}_p\gamma^\mu d_r)$

$8:(\bar{L}L)(\bar{L}L)$	
Q_{ll}	$(\bar{l}_p\gamma_\mu l_r)(\bar{l}_s\gamma^\mu l_t)$
$Q_{qq}^{(1)}$	$(\bar{q}_p\gamma_\mu q_r)(\bar{q}_s\gamma^\mu q_t)$
$Q_{qq}^{(3)}$	$(\bar{q}_p\gamma_\mu\tau^I q_r)(\bar{q}_s\gamma^\mu\tau^I q_t)$
$Q_{lq}^{(1)}$	$(\bar{l}_p\gamma_\mu l_r)(\bar{q}_s\gamma^\mu q_t)$
$Q_{lq}^{(3)}$	$(\bar{l}_p\gamma_\mu\tau^I l_r)(\bar{q}_s\gamma^\mu\tau^I q_t)$

$8:(\bar{R}R)(\bar{R}R)$	
Q_{ee}	$(\bar{e}_p\gamma_\mu e_r)(\bar{e}_s\gamma^\mu e_t)$
Q_{uu}	$(\bar{u}_p\gamma_\mu u_r)(\bar{u}_s\gamma^\mu u_t)$
Q_{dd}	$(\bar{d}_p\gamma_\mu d_r)(\bar{d}_s\gamma^\mu d_t)$
Q_{eu}	$(\bar{e}_p\gamma_\mu e_r)(\bar{u}_s\gamma^\mu u_t)$
Q_{ed}	$(\bar{e}_p\gamma_\mu e_r)(\bar{d}_s\gamma^\mu d_t)$
$Q_{ud}^{(1)}$	$(\bar{u}_p\gamma_\mu u_r)(\bar{d}_s\gamma^\mu d_t)$
$Q_{ud}^{(8)}$	$(\bar{u}_p\gamma_\mu T^A u_r)(\bar{d}_s\gamma^\mu T^A d_t)$

$8:(\bar{L}L)(\bar{R}R)$	
Q_{le}	$(\bar{l}_p\gamma_\mu l_r)(\bar{e}_s\gamma^\mu e_t)$
Q_{lu}	$(\bar{l}_p\gamma_\mu l_r)(\bar{u}_s\gamma^\mu u_t)$
Q_{ld}	$(\bar{l}_p\gamma_\mu l_r)(\bar{d}_s\gamma^\mu d_t)$
Q_{qe}	$(\bar{q}_p\gamma_\mu q_r)(\bar{e}_s\gamma^\mu e_t)$
$Q_{qu}^{(1)}$	$(\bar{q}_p\gamma_\mu q_r)(\bar{u}_s\gamma^\mu u_t)$
$Q_{qu}^{(8)}$	$(\bar{q}_p\gamma_\mu T^A q_r)(\bar{u}_s\gamma^\mu T^A u_t)$
$Q_{qd}^{(1)}$	$(\bar{q}_p\gamma_\mu q_r)(\bar{d}_s\gamma^\mu d_t)$
$Q_{qd}^{(8)}$	$(\bar{q}_p\gamma_\mu T^A q_r)(\bar{d}_s\gamma^\mu T^A d_t)$

$8:(\bar{L}R)(\bar{R}L)$ + h.c.	
Q_{ledq}	$(\bar{l}_p^j e_r)(\bar{d}_s q_{tj})$

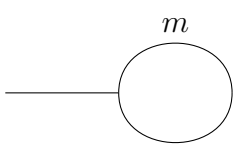
$8:(\bar{L}R)(\bar{L}R)$ + h.c.	
$Q_{quqd}^{(1)}$	$(\bar{q}_p^j u_r)\epsilon_{jk}(\bar{q}_s^k d_t)$
$Q_{quqd}^{(8)}$	$(\bar{q}_p^j T^A u_r)\epsilon_{jk}(\bar{q}_s^k T^A d_t)$
$Q_{lequ}^{(1)}$	$(\bar{l}_p^j e_r)\epsilon_{jk}(\bar{q}_s^k u_t)$
$Q_{lequ}^{(3)}$	$(\bar{l}_p^j\sigma_{\mu\nu}e_r)\epsilon_{jk}(\bar{q}_s^k\sigma^{\mu\nu}u_t)$

Appendix C

One loop functions

C.1 Tadpole-function

Let us start from the simplest topology:



$$\equiv \quad A_0(m^2) = \frac{\mu^{4-n}}{i\pi^2} \int d^n q \, \frac{1}{q^2 + m^2 - i\eta} \, . \tag{C.1}$$

Expanding around $n = 4 - \epsilon$, and introducing the following regulator

$$\Delta_{UV}(x) = \frac{2}{\epsilon} - \gamma_E - \ln \pi - \ln \frac{x}{\mu^2} = \frac{1}{\bar{\epsilon}} - \ln \frac{x}{\mu^2}, \tag{C.2}$$

we obtain

$$A_0(m) = -m^2 \left[\Delta_{UV}(M_W^2) + a_0^{fin}(m) \right] + \mathcal{O}(\epsilon) \tag{C.3}$$

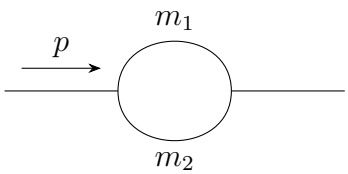
where

$$a_0^{fin}(m) = 1 - \ln \frac{m^2}{M_W^2}. \tag{C.4}$$

We have set the EW scale $x = M_W^2$ in Eq.(C.2).

C.2 Bubbles, reduction and derivates

The typical bubble diagram is given by



$$\equiv \quad B_0(p^2; m_1, m_2) = \frac{\mu^{4-n}}{i\pi^2} \int d^n q \, \frac{1}{d_0 d_1} \, , \tag{C.5}$$

where $d_0 = q^2 + m_1^2 - i\eta$ and $d_1 = (q+p)^2 + m_2^2 - i\eta$.

What we obtain is the B_0 function defined by:

$$B_0(p^2; m_1, m_2) = \Delta_{UV}(M_W^2) + B_0^{fin}(p^2; m_1, m_2), \tag{C.6}$$

with the finite contribution given by

$$B_{0f}(p^2; m_1, m_2) = 2 - \ln \frac{m_1 m_2}{M_W^2} + \frac{m_1^2 - m_2^2}{2p^2} \ln \frac{m_1^2}{m_2^2} - R. \tag{C.7}$$

Here, we have introduced the R function

$$R = -\frac{\lambda^{1/2}(-p^2, m_1^2, m_2^2)}{p^2} \ln \frac{p^2 - i\eta + m_1^2 + m_2^2 - \lambda^{1/2}(-p^2, m_1^2, m_2^2)}{2m_1 m_2}, \tag{C.8}$$

where $\lambda^{1/2}(-p^2, m_1^2, m_2^2)$ is the well-known Källen function.

For different values of masses in input we have:

$$\begin{aligned}
m_1 = m_2 = m \Rightarrow \quad & B_0^{fin}(p^2; m, m) = 2 - \ln \frac{m^2}{M_W^2} - \beta \ln \frac{\beta + 1}{\beta - 1}, \\
m_1 = 0, \; m_2 = m \Rightarrow \quad & B_0^{fin}(p^2; 0, m) = 2 - \ln \frac{m^2}{M_W^2} - \left(1 + \frac{m^2}{p^2}\right) \ln \left(1 + \frac{p^2 - i\eta}{m^2}\right), \\
m_1 = m_2 = 0 \Rightarrow \quad & B_0^{fin}(p^2; 0, m) = 2 - \ln \frac{p^2 - i\eta}{M_W^2}.
\end{aligned} \tag{C.9}$$

with $\beta^2 = 1 + 4m^2/(p^2 - i\eta)$.

C.2.1 QED case

When we take into account QED corrections, we have the presence of infrared divergences in the derivative of the scalar function B_0 (see Sec.5.1.3 of Ref.[15]).
Starting from the integral representation

$$B_0(p^2;m,0) = \pi^{n/2-2}\Gamma\left(2-\frac{n}{2}\right)\int_0^1 dx\left[\frac{(1-x)(p^2x+m^2)}{\mu^2}\right]^{n/2-2}, \quad \text{(C.10)}$$

with $n = 4+\epsilon'$, we obtain the derivative function

$$B_{0p}(p^2;m,0) = \frac{\partial}{\partial p^2}B_0(p^2;m,0)\bigg|_{p^2=-m^2} = -\frac{1}{2m^2}\left[\Delta_{IR}(M_W^2) - 2 + \ln\frac{m^2}{M_W^2}\right], \quad \text{(C.11)}$$

where we have introduced the regulator

$$\Delta_{IR}(x) = \frac{2}{\epsilon'} + \gamma_E + \ln\pi + \ln\frac{x}{\mu^2} = \frac{1}{\hat{\epsilon}} + \ln\frac{x}{\mu^2}. \quad \text{(C.12)}$$

C.2.2 Tensorial reduction

Let us consider the function defined by

$$B_\mu(p^2;m_1^2,m_2) = \frac{\mu^{4-n}}{i\pi^2}\int d^nq\,\frac{q_\mu}{d_0d_1} = B_1(p^2;m_1^2,m_2)p_\mu. \quad \text{(C.13)}$$

Contracting with p_μ, we obtain the scalar function B_1:

$$B_1(p^2;m_1,m_2) = \frac{1}{2p^2}\big[A_0(m_1) - A_0(m_2) + (-p^2+m_1^2-m_2^2)B_0(p^2;m_1,m_2)\big]. \quad \text{(C.14)}$$

The rank-two integral is reduced as

$$\begin{aligned} B_{\mu\nu}(p^2;m_1,m_2) &= \frac{\mu^{4-n}}{i\pi^2}\int d^nq\,\frac{q_\mu q_\nu}{d_0d_1} \\ &= B_{21}(p^2;m_1,m_2)p_\mu p_\nu + B_{22}(p^2;m_1,m_2)\delta_{\mu\nu} \end{aligned} \quad \text{(C.15)}$$

and the contraction with $p_\mu p_\nu$ and $\delta_{\mu\nu}$ allows to obtain B_{21} and B_{22} as functions of elementary scalar quantities A_0 and B_0. The same procedure is applicable for the higher rank integrals.

C.2.3 $\mathbf{p^2 = 0}$ case

In this particular case, the reduction algorithm fails.
The strategy is to consider the integral representation of $B_0(p^2; m_1^2, m_2^2)$ given by

$$B_0(p^2; m_1^2, m_2^2) = \Delta_{UV}(M_W^2) + \int_0^1 dx \ln \frac{-p^2x^2 + (p^2 + m_2^2 - m_1^2)x + m_1^2}{M_W^2} \tag{C.16}$$

and take the limit $p^2 \to 0$. The same has to be done for B_{0p} function.
In this way we obtain the correct finite part for B_0 in that limit, simply expanding around $p^2 = 0$

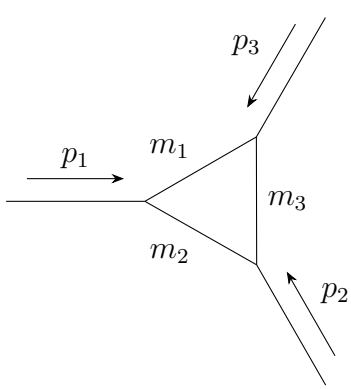
$$B_0^{fin}(p^2; m_1, m_2) = B_0^{fin}(0; m_1, m_2) + p^2 B_{0p}^{fin}(0; m_1, m_2) + \frac{1}{2} p^4 B_{0s}^{fin}(0; m_1, m_2) + \mathcal{O}(p^6). \tag{C.17}$$

Here we present some results for equal values of internal masses:

$$B_0^{fin}(0; m, m) = 1 - a_0^{fin}(m), \quad B_{0p}^{fin}(0, m, m) = -\frac{1}{6m^2}, \quad B_{0s}^{fin}(0; m, m) = \frac{1}{30m^4}. \tag{C.18}$$

C.3 Triangle function

The triangle function is given by the following diagram



$$\equiv C_0(p_1^2, p_2^2, Q^2; m_1, m_2, m_3) = \frac{\mu^{4-n}}{i\pi^2} \int d^n q \, \frac{1}{d_0 d_1 d_2} \tag{C.19}$$

with

$$d_0 = q^2 + m_1^2 - i\eta, \quad d_1 = (q + p_1)^2 + m_2^2 - i\eta, \quad d_2 = (q + Q)^2 + m_3^2 - i\eta,$$

and $Q = p_1 + p_2$.
The general case is discussed in [15], where the solution is given in terms of 12-di-logarithms plus a collection of simple logarithms.

As the B_{0p} function, we can extract the IR behaviour from the C_0.
Let us consider the massless case:

$$C_0(0,0,Q^2;0,0,0) = \frac{\mu^{4-n}}{i\pi^2}\int d^n\,\frac{1}{q^2(q+p_1)^2(q+Q)^2}. \tag{C.20}$$

Expanding around $n = 4 + 2\hat{\epsilon}$ we obtain:

$$Q^2 C_0(0,0,Q^2;0,0,0) = \frac{1}{\hat{\epsilon}^2} + \frac{1}{\hat{\epsilon}}\big(\ln z_Q - \bar{\gamma}\big) + \frac{1}{2}\ln^2 z_Q + \frac{1}{2}\bar{\gamma}^2 - \frac{\pi^2}{12}, \tag{C.21}$$

with $z_Q = (Q^2 - i\eta)/\mu^2$, $\bar{\gamma} = \gamma_E + \ln\pi$ and $1/\hat{\epsilon} = \Delta_{IR}(M_W^2) - \ln M_W^2/\mu^2$.

Here we have use the scheme (ϵ', ϵ'), where the double and the single poles in reflect the infrared and collinear ($m = 0$) singularities. Adopting a different strategy and working in (ϵ', m) scheme, the collinear divergence is controlled by terms like $\ln m^2$.
If $m_1 = m_3$ and $m_2 = 0$ we have:

$$Q^2 C_0(-m^2,-m^2,Q^2;m,0,m) = \left(\frac{1}{\hat{\epsilon}} + \ln\frac{m^2}{\mu^2}\right)\ln\frac{Q^2 - i\eta}{m^2} + \frac{1}{2}\ln^2\frac{Q^2 - i\eta}{m^2} - \frac{\pi^2}{6}. \tag{C.22}$$

Appendix D

Notations, renormalized amplitudes & results

D.1 Zll

D.1.1 Notations

Starting from the renormalized amplitude

$$i\mathcal{M}_\mu = g\left[\gamma_\mu\left(\mathcal{G}_V^l + \gamma_5\,\mathcal{G}_A^l\right) + \frac{i\sqrt{2}G_F}{2M_l\pi^2}\sigma_{\mu,(p_1+p_2)}\tilde{\mathcal{G}}\right], \tag{D.1}$$

we can express the form factors by the following decomposition

$$\mathcal{G}_{V,A} = \mathcal{G}_{V,A}^{(0,4)} + \frac{g_6}{\sqrt{2}}\mathcal{G}_{V,A}^{(0,6)} + \frac{\sqrt{2}G_F}{\pi^2}\left[\mathcal{G}_{V,A}^{(1,4)} + \frac{g_6}{\sqrt{2}}\mathcal{G}_{V,A}^{(1,6)}\right], \tag{D.2}$$

$$\tilde{\mathcal{G}} = \tilde{\mathcal{G}}^{(4)} + \frac{g_6}{\sqrt{2}}\tilde{\mathcal{G}}^{(6)} \tag{D.3}$$

where the order in perturbation theory and the mass dimension are indicated by the upper indices.

To ”simplify” our result, we report the deconvoluted amplitude.

Loop scalar function

We have used the shorthand notation

$$CX(p^2; m_1, m_2) = \mathrm{Re}X(p^2; m_1, m_2) + i\ \mathrm{Im}X(p^2; m_1, m_2), \tag{D.4}$$

where $X = \left\{B_{0f},\ B_{0p}\right\}$ and

$$B_{0p}(p^2; m_1, m_2) = \frac{d}{dp^2} B_0(p^2; m_1, m_2). \tag{D.5}$$

We have also adopted the following convention for the scalar functions involved:

$$\begin{aligned} a_{0f}(M_t) &= a_{0f}(1) & a_{0f}(M_b) &= a_{0f}(2) \\ a_{0f}(M_W) &= a_{0f}(3) & a_{0f}(M_Z) &= a_{0f}(4) \\ a_{0f}(M_H) &= a_{0f}(5) \end{aligned}$$

$$\begin{aligned} B_{0f}(-M_Z^2; M_l, M_l) &= B_{0f}(1) & CB_{0f}(-M_Z^2; M_l, M_l) &= CB_{0f}(1) \\ B_{0f}(-M_Z^2; M_t, M_t) &= B_{0f}(2) & CB_{0f}(-M_Z^2; M_t, M_t) &= CB_{0f}(2) \\ B_{0f}(-M_Z^2; M_b, M_b) &= B_{0f}(3) & CB_{0f}(-M_Z^2; M_b, M_b) &= CB_{0f}(3) \\ B_{0f}(-M_Z^2; 0, 0) &= B_{0f}(4) & CB_{0f}(-M_Z^2; 0, 0) &= CB_{0f}(4) \end{aligned}$$

$$\begin{aligned} \sum_{gen} B_{0f}(-M_Z^2; M_{l,g}, M_{l,g}) &= B_{0f}(5) & \sum_{gen} CB_{0f}(-M_Z^2; M_{l,g}, M_{l,g}) &= CB_{0f}(5) \\ \sum_{gen} B_{0f}(-M_Z^2; M_{t,g}, M_{t,g}) &= B_{0f}(6) & \sum_{gen} CB_{0f}(-M_Z^2; M_{t,g}, M_{t,g}) &= CB_{0f}(6) \\ \sum_{gen} B_{0f}(-M_Z^2; M_{b,g}, M_{b,g}) &= B_{0f}(7) & \sum_{gen} CB_{0f}(-M_Z^2; M_{b,g}, M_{b,g}) &= CB_{0f}(7) \end{aligned}$$

$$\begin{aligned} B_{0f}(-M_Z^2; M_W, M_W) &= B_{0f}(8) & CB_{0f}(-M_Z^2; M_W, M_W) &= CB_{0f}(8) \\ B_{0f}(-M_Z^2; M_H, M_Z) &= B_{0f}(9) & CB_{0f}(-M_Z^2; M_H, M_Z) &= CB_{0f}(9) \\ B_{0f}(-M_W^2; 0, M_l) &= B_{0f}(10) & CB_{0f}(-M_W^2; 0, M_l) &= CB_{0f}(10) \\ B_{0f}(-M_W^2; M_t, M_b) &= B_{0f}(11) & CB_{0f}(-M_W^2; M_t, M_b) &= CB_{0f}(11) \end{aligned}$$

$$\sum_{gen} B_{0f}(-M_W^2;0,M_{l,g}) = B_{0f}(12) \qquad \sum_{gen} CB_{0f}(-M_W^2;0,M_{l,g}) = CB_{0f}(12)$$

$$\sum_{gen} B_{0f}(-M_W^2;M_{t,g},M_{b,g}) = B_{0f}(13) \qquad \sum_{gen} CB_{0f}(-M_W^2;M_{t,g},M_{b,g}) = CB_{0f}(13)$$

$$B_{0f}(-M_W^2;M_W,M_Z) = B_{0f}(14) \qquad CB_{0f}(-M_W^2;M_W,M_Z) = CB_{0f}(14)$$

$$B_{0f}(-M_W^2;M_W,M_H) = B_{0f}(15) \qquad CB_{0f}(-M_W^2;M_W,M_H) = CB_{0f}(15)$$

$$B_{0f}(-M_W^2;0,M_W) = B_{0f}(16) \qquad CB_{0f}(-M_W^2;0,M_W) = CB_{0f}(16)$$

$$CB_{0f}(-M_l^2;M_Z,M_l) = CB_{0f}(17) \qquad CB_{0f}(-M_l^2;M_H,M_l) = CB_{0f}(18)$$

$$CB_{0f}(-M_l^2;0,M_W) = CB_{0f}(19) \qquad B_{0p}(-M_l^2;M_Z,M_l) = B_{0p}(1)$$

$$B_{0p}(-M_l^2;0,M_W) = B_{0p}(2) \qquad B_{0p}(-M_Z^2;M_l,M_l) = B_{0p}(3)$$

$$B_{0p}(-M_Z^2;M_t,M_t) = B_{0p}(4) \qquad B_{0p}(-M_Z^2;M_b,M_b) = B_{0p}(5)$$

$$\sum_{gen} B_{0p}(-M_Z^2;M_{l,g},M_{l,g}) = B_{0p}(6) \qquad \sum_{gen} B_{0p}(-M_Z^2;M_{t,g},M_{t,g}) = B_{0p}(7)$$

$$\sum_{gen} B_{0p}(-M_Z^2;M_{b,g},M_{b,g}) = B_{0p}(8) \qquad B_{0p}(-M_Z^2;0,0) = B_{0p}(9)$$

$$B_{0p}(-M_Z^2;M_W,M_W) = B_{0p}(10) \qquad B_{0p}(-M_Z^2;M_H,M_Z) = B_{0p}(11)$$

$$C_0(-M_Z^2,-M_l^2,-M_l^2;M_l,M_Z,M_l) = C_0(1) \qquad C_0(-M_Z^2,-M_l^2,-M_l^2;0,M_W,0) = C_0(2)$$

$$C_0(-M_Z^2,-M_l^2,-M_l^2;M_W,0,M_W) = C_0(3)$$

where $\sum_{gen}$ indicates the sum over the three fermionic generations.

$\mathbf{A}_i$ functions

$A_0 = 29 + (-96 + 64c_\theta^2)c_\theta^2$

$A_1 = 73 + (-744 + (864 - 256c_\theta^2)c_\theta^2)c_\theta^2$

$A_2 = 205 + (-840 + (864 - 256c_\theta^2)c_\theta^2)c_\theta^2$

$A_3 = 73 + (46 + (-512 + 384c_\theta^2)c_\theta^2)c_\theta^2$

$A_4 = 247 + (318 + (-2576 + (2496 - 512c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_5 = 7 - 64c_\theta^2$

$A_6 = 21 + (-160 + (-288 + 256c_\theta^2)c_\theta^2)c_\theta^2$

$A_7 = 7 + (-80 + 64c_\theta^2)c_\theta^2$

$A_8 = 21 + (-532 + (704 - 256c_\theta^2)c_\theta^2)c_\theta^2$

$A_9 = 3 + 4c_\theta^2$

$A_{10} = 5 - 12c_\theta^2$

$A_{11} = 3 + (16 - 16c_\theta^2)c_\theta^2$

$A_{12} = 25 + (-312 + (288 - 64c_\theta^2)c_\theta^2)c_\theta^2$

$A_{13} = 109 + (-360 + (288 - 64c_\theta^2)c_\theta^2)c_\theta^2$

$A_{14} = 7 + (222 + (-896 + (768 - 128c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{15} = 25 + (-50 + (-80 + 96c_\theta^2)c_\theta^2)c_\theta^2$

$A_{16} = 17 + 16c_\theta^2$

$A_{17} = 51 + (16 + (96 - 64c_\theta^2)c_\theta^2)c_\theta^2$

$A_{18} = 17 + (8 - 16c_\theta^2)c_\theta^2$

$A_{19} = 51 + (124 + (-176 + 64c_\theta^2)c_\theta^2)c_\theta^2$

$A_{20} = 1 - 8c_\theta^2$

$A_{21} = 1 + (48 + (-4 - 144c_\theta^2)c_\theta^2)c_\theta^2$

$A_{22} = 9 + (104 + (-572 + (1344 - 192c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{23} = 5 + (-32 + (-828 + (1344 - 192c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{24} = 1 + (-18 + (-228 + (56 + 288c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{25} = 1 + (434 + (908 + (1240 + (-3264 + 384c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{26} = 5 + (-8 - 12c_\theta^2)c_\theta^2$

$A_{27} = 8 + (11 - 6c_\theta^2)c_\theta^2$

$A_{28} = 13 + (-112 + (-132 + 48c_\theta^2)c_\theta^2)c_\theta^2$

$A_{29} = 5 - 28c_\theta^2$

$A_{30} = 3 - c_\theta^2$

$A_{31} = 3 + (17 - 8c_\theta^2)c_\theta^2$

$A_{32} = 9 + (17 - 8c_\theta^2)c_\theta^2$

$A_{33} = 1 + (-14 + 16c_\theta^2)c_\theta^2$

$A_{34} = 3 + 2c_\theta^2$

$A_{35} = 3 + 8c_\theta^2$

$A_{36} = 9 + 4c_\theta^2$

$A_{37} = 29 + (-68 + (64 - 32c_\theta^2)c_\theta^2)c_\theta^2$

$A_{38} = 25 + (-60 + (64 - 32c_\theta^2)c_\theta^2)c_\theta^2$

$A_{39} = 5 + (-22 + (32 - 16c_\theta^2)c_\theta^2)c_\theta^2$

$A_{40} = 5 + (62 + (-160 + (32 + 64c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{41} = 3 - 2c_\theta^2$

$A_{42} = 97 + (-208 + (176 - 128c_\theta^2)c_\theta^2)c_\theta^2$

$A_{43} = 85 + (-160 + (176 - 128c_\theta^2)c_\theta^2)c_\theta^2$

$A_{44} = 17 + (-74 + (112 - 64c_\theta^2)c_\theta^2)c_\theta^2$

$A_{45} = 17 + (202 + (-736 + (288 + 256c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{46} = 1 + 4c_\theta^2$

$A_{47} = 13 + (-28 + (-16 - 32c_\theta^2)c_\theta^2)c_\theta^2$

$A_{48} = 25 + (-4 + (-16 - 32c_\theta^2)c_\theta^2)c_\theta^2$

$A_{49} = 5 + (-26 + (-208 + (192 + 64c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{50} = 5 + (-14 + (16 - 16c_\theta^2)c_\theta^2)c_\theta^2$

$A_{51} = 19 + (-16 - 36c_\theta^2)c_\theta^2$

$A_{52} = 61 + (104 + (-204 - 192c_\theta^2)c_\theta^2)c_\theta^2$

$A_{53} = 65 + (184 + (-156 - 192c_\theta^2)c_\theta^2)c_\theta^2$

$A_{54} = 19 + (-54 + (-4 + 72c_\theta^2)c_\theta^2)c_\theta^2$

$A_{55} = 61 + (-22 + (-492 + (168 + 384c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{56} = 1 + (20 + 12c_\theta^2)c_\theta^2$

$A_{57} = 3 + (68 + (196 + 96c_\theta^2)c_\theta^2)c_\theta^2$

$A_{58} = 3 + (64 + (116 + 48c_\theta^2)c_\theta^2)c_\theta^2$

$A_{59} = 2 - c_\theta^2$

$A_{60} = 6 + c_\theta^2$

$A_{61} = 9 + (2 - 4c_\theta^2)c_\theta^2$

$A_{62} = 7 + 2c_\theta^2$

$A_{63} = 5 - 4c_\theta^2$

$A_{64} = 13 - 10c_\theta^2$

$A_{65} = 19 + (-8 - 8c_\theta^2)c_\theta^2$

$A_{66} = 1 - 2c_\theta^2$

$A_{67} = 5 + 16c_\theta^2$

$A_{68} = 1 + 16c_\theta^2$

$A_{69} = 5 + (10 - 32c_\theta^2)c_\theta^2$

$A_{70} = 1 - 5c_\theta^2$

$A_{71} = 3 + (5 - 36c_\theta^2)c_\theta^2$

$A_{72} = 25 - 36c_\theta^2$

$A_{73} = 1 + (-11 + 12c_\theta^2)c_\theta^2$

$A_{74} = 3 - 5c_\theta^2$

$A_{75} = 2 + 3c_\theta^2$

$A_{76} = 6 + (9 + 4c_\theta^2)c_\theta^2$

$A_{77} = 4 + (-6 + (1 + 4c_\theta^2)c_\theta^2)c_\theta^2$

$A_{78} = 4 + (2 - 3c_\theta^2)c_\theta^2$

$A_{79} = 2 - 3c_\theta^2$

$A_{80} = 2 + (5 + 4c_\theta^2)c_\theta^2$

$A_{81} = 2 + c_\theta^2$

$A_{82} = 2 + (9 + 4c_\theta^2)c_\theta^2$

$A_{83} = 11 + (48 - 80c_\theta^2)c_\theta^2$

$A_{84} = 61 + (-144 + 80c_\theta^2)c_\theta^2$

$A_{85} = 25 + (-96 + 80c_\theta^2)c_\theta^2$

$A_{86} = 7 + (-12 + 8c_\theta^2)c_\theta^2$

$A_{87} = 1 + (2 + c_\theta^2)c_\theta^2$

$A_{88} = 5 + (14 + (13 + 4c_\theta^2)c_\theta^2)c_\theta^2$

$A_{89} = 1 + (6 + (9 + 4c_\theta^2)c_\theta^2)c_\theta^2$

$A_{90} = 27 + (-44 + (-48 + 64c_\theta^2)c_\theta^2)c_\theta^2$

$A_{91} = 37 + (-148 + (176 - 64c_\theta^2)c_\theta^2)c_\theta^2$

$A_{92} = 5 + (-52 + (112 - 64c_\theta^2)c_\theta^2)c_\theta^2$

$A_{93} = 9 - 20c_\theta^2$

$A_{94} = 17 - 20c_\theta^2$

$A_{95} = 13 - 20c_\theta^2$

$A_{96} = 7 + 4c_\theta^2$

$A_{97} = 67 + (-140 + (16 + 64c_\theta^2)c_\theta^2)c_\theta^2$

$A_{98} = 2 + (-19 + 8c_\theta^2)c_\theta^2$

$A_{99} = 1 + (-14 + 4c_\theta^2)c_\theta^2$

$A_{100} = 19 + (384 - 544c_\theta^2)c_\theta^2$

$A_{101} = 159 + (-172 + (-288 + 384c_\theta^2)c_\theta^2)c_\theta^2$

$A_{102} = 11 + (18 + 24c_\theta^2)c_\theta^2$

$A_{103} = 31 + (-2 + (-80 - 32c_\theta^2)c_\theta^2)c_\theta^2$

$A_{104} = 1 + (18 + (-76 + 72c_\theta^2)c_\theta^2)c_\theta^2$

$A_{105} = 3 + (58 + (132 + (8 + 96c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{106} = 15 + (32 + 32c_\theta^2)c_\theta^2$

$A_{107} = 55 + (272 - 368c_\theta^2)c_\theta^2$

$A_{108} = 15 + (22 + 8c_\theta^2)c_\theta^2$

$A_{109} = 11 + 4c_\theta^2$

$A_{110} = 13 + 6c_\theta^2$

$A_{111} = 5 + (-19 + 8c_\theta^2)c_\theta^2$

$A_{112} = 2 + (-7 + 2c_\theta^2)c_\theta^2$

$A_{113} = 4 + (413 + (-960 + 544c_\theta^2)c_\theta^2)c_\theta^2$

$A_{114} = 12 + (297 + (-980 + (1056 - 384c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{115} = 21 + (14 - 24c_\theta^2)c_\theta^2$

$A_{116} = 57 + (-110 + (16 + 32c_\theta^2)c_\theta^2)c_\theta^2$

$A_{117} = 1 + (66 + (-172 + 72c_\theta^2)c_\theta^2)c_\theta^2$

$A_{118} = 5 + (134 + (-260 + (184 - 96c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{119} = 15 + (-32 + 32c_\theta^2)c_\theta^2$

$A_{120} = 305 + (-688 + 368c_\theta^2)c_\theta^2$

$A_{121} = 5 + (-8 + 4c_\theta^2)c_\theta^2$

$A_{122} = 11 + 6c_\theta^2$

$A_{123} = 1 + (2 - 6c_\theta^2)c_\theta^2$

$A_{124} = 4 + (7 + (-31 + 8c_\theta^2)c_\theta^2)c_\theta^2$

$A_{125} = 1 + (-1 - 6c_\theta^2)c_\theta^2$

$A_{126} = 1 + (7 + (-40 + 8c_\theta^2)c_\theta^2)c_\theta^2$

$A_{127} = 81 + (-404 + (672 - 384c_\theta^2)c_\theta^2)c_\theta^2$

$A_{128} = 185 + (-656 + 544c_\theta^2)c_\theta^2$

$A_{129} = 5 + (-2 - 24c_\theta^2)c_\theta^2$

$A_{130} = 13 + (-54 + (48 + 32c_\theta^2)c_\theta^2)c_\theta^2$

$A_{131} = 1 + (38 + (-196 + (88 - 96c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{132} = 1 + (42 + (-124 + 72c_\theta^2)c_\theta^2)c_\theta^2$

$A_{133} = 15 + 32c_\theta^2$

$A_{134} = 125 + (-480 + 368c_\theta^2)c_\theta^2$

$A_{135} = 3 + (14 + 8c_\theta^2)c_\theta^2$

$A_{136} = 1 + (-12 + 8c_\theta^2)c_\theta^2$

$A_{137} = 5 + 2c_\theta^2$

$A_{138} = 17 - 32c_\theta^2$

$A_{139} = 51 + (-44 + (-144 + 128c_\theta^2)c_\theta^2)c_\theta^2$

$A_{140} = 5 - 8c_\theta^2$

$A_{141} = 15 + (28 + (-48 + 32c_\theta^2)c_\theta^2)c_\theta^2$

$A_{142} = 163 - 288c_\theta^2$

$A_{143} = 11 + (50 + 24c_\theta^2)c_\theta^2$

$A_{144} = 31 + (14 + (-80 - 32c_\theta^2)c_\theta^2)c_\theta^2$

$A_{145} = 1 + (18 + (20 + 72c_\theta^2)c_\theta^2)c_\theta^2$

$A_{146} = 3 + (58 + (36 + (8 + 96c_\theta^2)c_\theta^2)c_\theta^2)c_\theta^2$

$A_{147} = 135 - 208c_\theta^2$

$A_{148} = 1 + c_\theta^2$

$A_{149} = 3 + (-4 + (2 - 4c_\theta^2)c_\theta^2)c_\theta^2$

$A_{150} = 13 - 4c_\theta^2$

$A_{151} = 29 - 20c_\theta^2$

$A_{152} = 3 + (-28 + 16c_\theta^2)c_\theta^2$

$A_{153} = 3 + (14 - 8c_\theta^2)c_\theta^2$

$A_{154} = 41 + (-96 + 64c_\theta^2)c_\theta^2$

$A_{155} = 2 + (-3 - 2c_\theta^2)c_\theta^2$

$A_{156} = 1 + 2c_\theta^2$

$A_{157} = 1 + (18 - 16c_\theta^2)c_\theta^2$

$A_{158} = 5 - 2c_\theta^2$

$A_{159} = 5 + 6c_\theta^2$

$A_{160} = 13 - 16c_\theta^2$

$A_{161} = 2 + (-5 + 2c_\theta^2)c_\theta^2$

$A_{162} = 16 + (-47 + 30c_\theta^2)c_\theta^2$

$A_{163} = 3 + (-6 + 4c_\theta^2)c_\theta^2$

$A_{164} = 9 + (-10 + (-12 + 16c_\theta^2)c_\theta^2)c_\theta^2$

$A_{165} = 11 + (-38 + (44 - 16c_\theta^2)c_\theta^2)c_\theta^2$

$A_{166} = 1 + (-14 + (28 - 16c_\theta^2)c_\theta^2)c_\theta^2$

$A_{167} = 9 + (-30 + (34 - 16c_\theta^2)c_\theta^2)c_\theta^2$

$A_{168} = 9 + (-30 + (38 - 20c_\theta^2)c_\theta^2)c_\theta^2$

$A_{169} = 5 + (-12 + 8c_\theta^2)c_\theta^2$

$A_{170} = 17 + (-40 + 32c_\theta^2)c_\theta^2$

$A_{171} = 5 + (-4 + 8c_\theta^2)c_\theta^2$

$A_{172} = 1 + (-72 + 96c_\theta^2)c_\theta^2$

$A_{173} = 73 + (-168 + 96c_\theta^2)c_\theta^2$

$A_{174} = 37 + (-120 + 96c_\theta^2)c_\theta^2$

$A_{175} = 7 + (-18 + 12c_\theta^2)c_\theta^2$

$A_{176} = 1 + (-24 + 32c_\theta^2)c_\theta^2$

$A_{177} = 25 + (-56 + 32c_\theta^2)c_\theta^2$

$A_{178} = 13 + (-40 + 32c_\theta^2)c_\theta^2$

$A_{179} = 7 + (28 + (52 + 48c_\theta^2)c_\theta^2)c_\theta^2$

$A_{180} = 11 + (8 - 32c_\theta^2)c_\theta^2$

$A_{181} = 5 + (-432 + 576c_\theta^2)c_\theta^2$

$A_{182} = 5 + (-360 + 416c_\theta^2)c_\theta^2$

$A_{183} = 19 + 6c_\theta^2$

$A_{184} = 4 + (437 + (-1008 + 576c_\theta^2)c_\theta^2)c_\theta^2$

$A_{185} = 365 + (-776 + 416c_\theta^2)c_\theta^2$

$A_{186} = 17 + 6c_\theta^2$

$A_{187} = 209 + (-704 + 576c_\theta^2)c_\theta^2$

$A_{188} = 185 + (-568 + 416c_\theta^2)c_\theta^2$

$A_{189} = 35 + (-82 + 52c_\theta^2)c_\theta^2$

$A_{190} = 109 + (-254 + 156c_\theta^2)c_\theta^2$

$A_{191} = 89 + (-216 + 144c_\theta^2)c_\theta^2$

$A_{192} = 5 + (-10 - 8c_\theta^2)c_\theta^2$

$A_{193} = 1 + (18 + (-4 + 24c_\theta^2)c_\theta^2)c_\theta^2$

$A_{194} = 75 + (-164 + 104c_\theta^2)c_\theta^2$

$\mathbf{B}_i$ functions

$$B_0 = B_1^a$$
$$B_1 = CB_{0f}(9)B_0^a - 6CB_{0f}(18)M_H^2 M_Z^2$$

$B_2 = B_2^a$ $\quad B_3 = B_3^a$ $\quad B_4 = B_4^a$ $\quad B_5 = B_5^a$

$B_6 = B_6^a$ $\quad B_7 = B_7^a$ $\quad B_8 = CB_{0f}(4)$ $\quad B_9 = B_8^a$

$B_{10} = B_{10}^a$ $\quad B_{11} = B_{11}^a$ $\quad B_{12} = B_{12}^a$ $\quad B_{13} = B_{16}^a$

$B_{14} = B_{13}^a$ $\quad B_{15} = B_{14}^a$ $\quad B_{16} = C_0(3)$ $\quad B_{17} = C_0(1)$

$B_{18} = C_0(2)$ $\quad B_{19} = B_{15}^a$ $\quad B_{20} = B_{17}^a$ $\quad B_{21} = CB0f(17)$

$B_{22} = CB0f(19)$ $\quad B_{23} = CB0f(8)$ $\quad B_{24} = B_{156}^a$ $\quad B_{25} = CB0f(1)$

$$B_{26} = 36C_0(1)M_Z^6 - CB_{0f}(9)B_0^a + 6CB_{0f}(18)M_H^2 M_Z^2$$
$$B_{27} = 12CB_{0f}(9)B_0^a - 72CB_{0f}(18)M_H^2 M_Z^2 - B_{161}^a M_Z^2$$

$B_{28} = B^a_9$ $B_{29} = B^a_{18}$ $B_{30} = B^a_{19}$ $B_{31} = B^a_{20}$
$B_{32} = B^a_{21}$ $B_{33} = B^a_{22}$ $B_{34} = B^a_{23}$ $B_{35} = B^a_{24}$
$B_{36} = B^a_{25}$ $B_{37} = B^a_{26}$ $B_{38} = B^a_{27}$ $B_{39} = B^a_{28}$
$B_{40} = B^a_{157}$ $B_{41} = B^a_{158}$ $B_{42} = B^a_{159}$ $B_{43} = B^a_{160}$
$B_{44} = B^a_{162}$ $B_{45} = B^a_{163}$ $B_{46} = B^a_{29}$ $B_{47} = B^a_{30}$

$$B_{48} = 432C_0(1)M_Z^6 - 12CB_{0f}(9)B_0^a + 72CB_{0f}(18)M_H^2M_Z^2 + B^a_{33}$$
$$B_{49} = CB_{0f}(9)B_0^a - B^a_{165}$$
$$B_{50} = 24CB_{0f}(18)M_H^2 + B^a_{166}$$

$B_{51} = B^a_{31}$ $B_{52} = B^a_{32}$ $B_{53} = B^a_{34}$ $B_{54} = B^a_{35}$
$B_{55} = B^a_{36}$ $B_{56} = B^a_{37}$ $B_{57} = B^a_{38}$ $B_{58} = B^a_{39}$
$B_{59} = B^a_{40}$ $B_{60} = B^a_{41}$ $B_{61} = B^a_{42}$ $B_{62} = B^a_{164}$
$B_{63} = B^a_{167}$ $B_{64} = B^a_{168}$ $B_{65} = B^a_{169}$ $B_{66} = B^a_{43}$
$B_{67} = B^a_{44}$

$$B_{68} = 144C_0(1)M_Z^6 - 4CB_{0f}(9)B_0^a + 24CB_{0f}(18)M_H^2M_Z^2 + B^a_{45}M_Z^2$$
$$B_{69} = 4CB_{0f}(9)B_0^a - 24CB_{0f}(18)M_H^2M_Z^2 - B^a_{170}M_Z^2$$

$B_{70} = B^a_{46}$ $B_{71} = B^a_{47}$ $B_{72} = B^a_{48}$ $B_{73} = B^a_{49}$
$B_{74} = B^a_{50}$ $B_{75} = B^a_{51}$ $B_{76} = B^a_{52}$ $B_{77} = B^a_{53}$
$B_{78} = B^a_{54}$ $B_{79} = B^a_{171}$ $B_{80} = B^a_{172}$ $B_{81} = 36C_0(1) + B^a_2$

$$B_{82} = CB_{0f}(9)B_0^a - 6CB_{0f}(18)M_H^2M_Z^2 + 3B^a_{55}M_Z^2$$

$B_{83} = B^a_{56}$ $B_{84} = B^a_{57}$ $B_{85} = B^a_{173}$ $B_{86} = B^a_{58}$
$B_{87} = B^a_{174}$ $B_{88} = B^a_{59}$ $B_{89} = B^a_{175}$ $B_{90} = B^a_{176}$

$$B_{91} = 432C_0(1)M_Z^6 - 12CB_{0f}(9)B_0^a + 72CB_{0f}(18)M_H^2M_Z^2 + B^a_{61}$$
$$B_{92} = 12CB_{0f}(9)B_0^a - 72CB_{0f}(18)M_H^2M_Z^2 - B^a_{177}$$

$B_{93} = B^a_{60}$ $B_{94} = B^a_{62}$ $B_{95} = B^a_{63}$ $B_{96} = B^a_{64}$
$B_{97} = B^a_{65}$ $B_{98} = B^a_{66}$ $B_{99} = B^a_{67}$ $B_{100} = B^a_{68}$
$B_{101} = B^a_{69}$ $B_{102} = B^a_{178}$ $B_{103} = B^a_{179}$ $B_{104} = B^a_{180}$
$B_{105} = B^a_{181}$ $B_{106} = B^a_{70}$ $B_{107} = B^a_{71}$ $B_{108} = B^a_{72}$

$B_{109} = B^a_{73}$ $B_{110} = B^a_{74}$ $B_{111} = B^a_{75}$ $B_{112} = B^a_{182}$
$B_{113} = -CB_{0f}(19) + 1_c$ $B_{114} = CB_{0f}(17) + B^a_{54}$ $B_{115} = B^a_{76}$ $B_{116} = B^a_{77}$
$B_{117} = B^a_{78}$ $B_{118} = B^a_{79}$ $B_{119} = B^a_{80}$ $B_{120} = B^a_{81}$
$B_{121} = B^a_{82}$ $B_{122} = B^a_{183}$ $B_{123} = B^a_{83}$ $B_{124} = B^a_{184}$
$B_{125} = B^a_{84}$ $B_{126} = B^a_{85}$ $B_{127} = 3CB_{0f}(19) - 8B^a_8$

$$B_{128} = 8C_0(1)M_Z^2 - 5CB_{0f}(1) \qquad B_{129} = 4C_0(1)M_Z^2 + 3CB_{0f}(17)$$

$B_{130} = B^a_{86}$ $B_{131} = B^a_{87}$ $B_{132} = B^a_{88}$ $B_{133} = B^a_{89}$
$B_{134} = B^a_{90}$ $B_{135} = B^a_{91}$ $B_{136} = B^a_{92}$ $B_{137} = B^a_{93}$
$B_{138} = B^a_{125}$ $B_{139} = B^a_{185}$ $B_{140} = B^a_{186}$ $B_{141} = B^a_{187}$
$B_{142} = B^a_{188}$

$$B_{143} = 72C_0(1)M_Z^2 - 45CB_{0f}(1) + 51CB_{0f}(17) + 2B^a_{99}$$

$B_{144} = B^a_{94}$ $B_{145} = B^a_{95}$ $B_{146} = B^a_{96}$ $B_{147} = B^a_{97}$
$B_{148} = B^a_{98}$ $B_{149} = B^a_{100}$ $B_{150} = B^a_{101}$ $B_{151} = B^a_{189}$
$B_{152} = B^a_{190}$ $B_{153} = B^a_{191}$ $B_{154} = B^a_{102}$ $B_{155} = B^a_{103}$
$B_{156} = B^a_{104}$ $B_{157} = B^a_{105}$ $B_{158} = B^a_{106}$ $B_{159} = B^a_{107}$
$B_{160} = B^a_{108}$ $B_{161} = B^a_{151}$ $B_{162} = B^a_{192}$ $B_{163} = B^a_{193}$
$B_{164} = B^a_{109}$ $B_{165} = B^a_{110}$ $B_{166} = B^a_{194}$ $B_{167} = B^a_{111}$
$B_{168} = B^a_{112}$ $B_{169} = B^a_{113}$ $B_{170} = B^a_{114}$ $B_{171} = B^a_{115}$
$B_{172} = B^a_{195}$ $B_{173} = B^a_{196}$ $B_{174} = B^a_{116}$ $B_{175} = B^a_{197}$
$B_{176} = B^a_{117}$ $B_{177} = B^a_{118}$ $B_{178} = B^a_{119}$ $B_{179} = B^a_{120}$
$B_{180} = B^a_{121}$ $B_{181} = B^a_{198}$ $B_{182} = B^a_{199}$

$$B_{183} = 2CB_{0f}(3)M_b^2 + CB_{0f}(7)M_Z^2 + B^a_{122}$$
$$B_{184} = 2CB_{0f}(2)M_t^2 + CB_{0f}(6)M_Z^2 + B^a_{123}$$
$$B_{185} = 4CB_{0f}(2)M_t^2 - 2CB_{0f}(3)M_b^2 + 2CB_{0f}(6)M_Z^2 - CB_{0f}(7)M_Z^2 + B^a_{124}$$
$$B_{186} = CB_{0f}(1) + CB_{0f}(5) - B^a_{74}$$
$$B_{187} = CB_{0f}(1) + 2CB_{0f}(5)$$

$B_{189} = B^a_{126}$ $B_{190} = B^a_{127}$ $B_{191} = B^a_{128}$ $B_{192} = B^a_{129}$
$B_{193} = B^a_{130}$ $B_{194} = B^a_{131}$ $B_{195} = B^a_{200}$ $B_{196} = B^a_{201}$

$$B_{197} = 4CB_{0f}(3)M_b^2 + 2CB_{0f}(7)M_Z^2 + B^a_{132}$$
$$B_{198} = 4CB_{0f}(2)M_t^2 + 2CB_{0f}(6)M_Z^2 + B^a_{133}$$
$$B_{199} = 6CB_{0f}(1) + 12CB_{0f}(5) - 5B^a_{134}$$
$$B_{200} = 2CB_{0f}(2)M_t^2 + 2CB_{0f}(3)M_b^2 + CB_{0f}(6)M_Z^2 + CB_{0f}(7)M_Z^2 + 2B^a_{135}$$

$B_{201} = B^a_{136}$ $B_{202} = B^a_{137}$ $B_{203} = B^a_{138}$ $B_{204} = B^a_{202}$ $B_{205} = B^a_{203}$

$$B_{206} = 4CB_{0f}(2)M_t^2 + 2CB_{0f}(3)M_b^2 + 2CB_{0f}(6)M_Z^2 + CB_{0f}(7)M_Z^2 + B^a_{139}$$
$$B_{207} = 12CB_{0f}(9)B^a_0 - 72CB_{0f}(18)M_H^2 M_Z^2 - B^a_{204}$$

$B_{208} = B^a_{140}$ $B_{209} = B^a_{141}$ $B_{210} = B^a_{142}$ $B_{211} = B^a_{205}$
$B_{212} = B^a_{206}$ $B_{213} = B^a_{207}$ $B_{214} = B^a_{208}$ $B_{215} = B^a_{143}$
$B_{216} = B^a_{144}$ $B_{217} = B^a_{145}$ $B_{218} = B^a_{146}$ $B_{219} = B^a_{147}$
$B_{220} = B^a_{148}$ $B_{221} = B^a_{149}$ $B_{222} = B^a_{150}$ $B_{223} = B^a_{152}$
$B_{224} = CB_{0f}(17) - B^a_{153}$ $B_{225} = B^a_{154}$ $B_{226} = B^a_{155}$

$\mathbf{B}^a_i$

$$B^a_0 = B^b_0$$
$$B^a_1 = B_{0f}(2)$$
$$B^a_2 = 3B_{0p}(1) - B_{0p}(3) - B_{0p}(6)$$
$$B^a_3 = B_{0f}(3)$$
$$B^a_4 = 6B_{0f}(4)M_Z^4 + B_{0f}(9)B^b_0 + 6B_{0f}(11)B^b_2 M_Z^2$$
$$B^a_5 = 6B_{0f}(11)B^b_3 M_t^2 + B_{0f}(15)M_H^4 + 6a_{0f}(1)B^b_4 M_t^2 + 6a_{0f}(2)M_t^2 M_b^2$$
$$B^a_6 = B_{0f}(13)$$
$$B^a_7 = B_{0f}(8)$$
$$B^a_8 = B_{0f}(1) + B_{0f}(5)$$
$$B^a_9 = B_{0f}(15)$$
$$B^a_{10} = B_{0f}(6)$$
$$B^a_{11} = B_{0f}(7)$$
$$B^a_{12} = B_{0f}(14)$$
$$B^a_{13} = B_{0f}(16)$$
$$B^a_{14} = B_{0f}(10) + B_{0f}(12)$$
$$B^a_{15} = B_{0p}(2)$$
$$B^a_{16} = 1 + a_{0f}(1)$$
$$B^a_{17} = 1 + a_{0f}(2)$$
$$B^a_{18} = 32M_b^2 - 144B_{0f}(16)M_Z^2 - 9B_{0p}(2)M_Z^4 + 32a_{0f}(2)M_b^2$$
$$B^a_{19} = 24B_{0f}(4)M_Z^4 + 4B_{0f}(9)B^b_0 + 24B_{0f}(11)B^b_2 M_Z^2 + 8B^c_0 M_H^2 M_Z^2 - B^c_5 M_Z^4$$
$$B^a_{20} = 24B_{0f}(11)B^b_3 M_t^2 + 4B_{0f}(15)M_H^4 + 24a_{0f}(1)B^b_4 M_t^2 + 24a_{0f}(2)M_t^2 M_b^2 + 4B^c_1 M_H^2 M_Z^2 - B^c_6 M_Z^4$$
$$B^a_{21} = B^c_0$$
$$B^a_{22} = 12B^c_1 M_H^2 - B^c_9 M_Z^2$$
$$B^a_{23} = 8B^c_2 M_H^2 - B^c_8 M_Z^2$$

$$B^a_{24} = 12B^c_3 M^2_H - B^c_{10} M^2_Z$$
$$B^a_{25} = B^c_4$$
$$B^a_{26} = B^c_7$$
$$B^a_{27} = B^c_{11}$$
$$B^a_{28} = B^c_{12}$$
$$B^a_{29} = 2B_{0f}(9) - B^c_0$$
$$B^a_{30} = B_{0f}(9)$$
$$B^a_{31} = 2M^2_b - 9B_{0f}(16)M^2_Z + 2a_{0f}(2)M^2_b$$
$$B^a_{32} = 6B_{0f}(4)M^4_Z + B_{0f}(9)M^4_H + 6B_{0f}(11)B^b_2 M^2_Z$$
$$B^a_{33} = 72B_{0f}(4)M^4_Z + 12B_{0f}(9)M^4_H + 72B_{0f}(11)B^b_2 M^2_Z - 96B_{0f}(15)M^2_H M^2_Z - B^c_{17} M^4_Z$$
$$B^a_{34} = 5B_{0f}(10) + 5B_{0f}(12) + 12B_{0f}(13) - 12B_{0f}(15)$$
$$B^a_{35} = 6B_{0f}(11)B^b_3 M^2_t + B_{0f}(15)M^4_H + 6a_{0f}(1)B^b_4 M^2_t + 6a_{0f}(2)M^2_t M^2_b - B^c_0 M^4_H - B^c_{14} M^4_Z$$
$$B^a_{36} = 18B^c_0 M^2_H + B^c_{15} M^2_Z$$
$$B^a_{37} = 12B^c_3 M^2_H - B^c_{20} M^2_Z$$
$$B^a_{38} = 24B^c_{13} M^2_H + B^c_{19} M^2_Z$$
$$B^a_{39} = B^c_{16}$$
$$B^a_{40} = B^c_{18}$$
$$B^a_{41} = B^c_{21}$$
$$B^a_{42} = B^c_{22}$$
$$B^a_{43} = B_{0p}(1)$$
$$B^a_{44} = B_{0p}(3) + B_{0p}(6)$$
$$B^a_{45} = 8B^c_0 M^2_H - B^c_{24} M^2_Z$$
$$B^a_{46} = B^c_0 M^4_H - B^c_{23} M^4_Z$$
$$B^a_{47} = 12B^c_3 M^2_H - B^c_{30} M^2_Z$$
$$B^a_{48} = 24B^c_{25} M^2_H - B^c_{31} M^2_Z$$
$$B^a_{49} = B^c_{26}$$
$$B^a_{50} = B^c_{27}$$
$$B^a_{51} = B^c_{28}$$
$$B^a_{52} = B^c_{29}$$
$$B^a_{53} = B^c_{32}$$
$$B^a_{54} = B^c_{33}$$
$$B^a_{55} = 2B_{0f}(9)M^2_H - B_{0f}(11)B^b_2$$
$$B^a_{56} = B_{0f}(10) + B_{0f}(12) + 2B_{0f}(13)$$
$$B^a_{57} = B_{0f}(11)B^b_3 + a_{0f}(1)B^b_4 + a_{0f}(2)M^2_b$$
$$B^a_{58} = a_{0f}(1)$$
$$B^a_{59} = a_{0f}(2)$$
$$B^a_{60} = 72B_{0f}(4)M^2_Z - 108B_{0f}(15)M^2_Z + 24B^c_{36} M^2_b - 24B^c_{39} M^2_t + B^c_{41} M^2_Z - 24B^c_{47} M^2_H$$
$$B^a_{61} = 24B_{0f}(9)B^b_1 + 36B_{0f}(11)B^b_2 M^2_Z - 48B_{0f}(15)M^2_H M^2_Z + 12B^c_{37} M^2_H M^2_Z - 8B^c_{38} M^2_b M^2_Z$$
$$- 8B^c_{40} M^2_t M^2_Z + 3B^c_{42} M^4_Z$$
$$B^a_{62} = 9B_{0f}(10) + 9B_{0f}(12) + 18B_{0f}(13) - 36B_{0f}(15) + B^c_{43}$$

$$B^a_{63} = 3B_{0f}(11)B^b_3M^2_t - B_{0f}(15)M^4_H + 3a_{0f}(1)B^b_4M^2_t + 3a_{0f}(2)M^2_tM^2_b - B^c_{35}M^4_Z$$
$$B^a_{64} = 48B_{0f}(15)M^2_H - 12B^c_{37}M^2_H + 8B^c_{38}M^2_b + 8B^c_{40}M^2_t + B^c_{46}M^2_Z$$
$$B^a_{65} = B_{0f}(15)M^4_H + B^c_{35}M^4_Z$$
$$B^a_{66} = 144B_{0f}(16)M^2_Z - 9B_{0p}(2)M^4_Z - 32B^b_5$$
$$B^a_{67} = B^c_{34}$$
$$B^a_{68} = B^c_{44}$$
$$B^a_{69} = B^c_{45}$$
$$B^a_{70} = 3B_{0f}(9)M^2_Z + B_{0f}(15)M^2_H$$
$$B^a_{71} = 2B_{0f}(9)M^4_H - 2B^c_{33}M^2_HM^2_Z - B^c_{49}M^4_Z$$
$$B^a_{72} = 2B_{0f}(9)B^b_1 + 4B_{0f}(15)M^2_HM^2_Z - 2B^c_{33}M^2_HM^2_Z - B^c_{49}M^4_Z$$
$$B^a_{73} = 18B_{0f}(15) - 5B^c_{48}$$
$$B^a_{74} = B^c_{48}$$
$$B^a_{75} = B^c_{50}$$
$$B^a_{76} = B^c_{35}$$
$$B^a_{77} = B^c_{51}$$
$$B^a_{78} = B^c_{52}$$
$$B^a_{79} = B_{0f}(11)$$
$$B^a_{80} = B_{0f}(11) + a_{0f}(1)$$
$$B^a_{81} = B^c_{53}$$
$$B^a_{82} = B_{0f}(11) + B^c_{53}$$
$$B^a_{83} = B_{0f}(11)B^b_4 - a_{0f}(1)M^2_t + a_{0f}(2)M^2_b$$
$$B^a_{84} = 6B_{0f}(9)M^2_Z + 3B_{0f}(11)B^b_2 + 2B_{0f}(15)M^2_H$$
$$B^a_{85} = B_{0f}(13) + B_{0f}(15)$$
$$B^a_{86} = 12B_{0f}(4) - B^c_{54}$$
$$B^a_{87} = 18B_{0f}(9)M^4_H - 18B^c_{33}M^2_HM^2_Z + B^c_{57}M^4_Z$$
$$B^a_{88} = 2B_{0f}(9)B^b_1 + 6B_{0f}(11)B^b_2M^2_Z + 4B_{0f}(15)M^2_HM^2_Z + 12B_{0p}(1)M^6_Z - 4B_{0p}(3)M^6_Z - 4B_{0p}(6)M^6_Z - 2B^c_{33}M^2_HM^2_Z - B^c_{60}M^4_Z$$
$$B^a_{89} = 3B_{0f}(10) + 3B_{0f}(12) + 9B_{0f}(13) + 9B_{0f}(15) - B^c_{59}$$
$$B^a_{90} = 6B_{0f}(11)B^b_3M^2_t + B_{0f}(15)M^4_H + 6a_{0f}(1)B^b_4M^2_t + 6a_{0f}(2)M^2_tM^2_b - B^c_{35}M^4_Z$$
$$B^a_{91} = 2B_{0f}(16) + B^c_{55}$$
$$B^a_{92} = B^c_{56}$$
$$B^a_{93} = B^c_{58}$$
$$B^a_{94} = 36B_{0f}(4) + 3B_{0f}(10) + 3B_{0f}(12) - B^c_{65}$$
$$B^a_{95} = 6B_{0f}(9)M^4_H - 6B^c_{33}M^2_HM^2_Z + B^c_{66}M^4_Z$$
$$B^a_{96} = 6B_{0f}(9)B^b_1 + 18B_{0f}(11)B^b_2M^2_Z + 12B_{0f}(15)M^2_HM^2_Z - 6B^c_{33}M^2_HM^2_Z - B^c_{67}M^4_Z$$
$$B^a_{97} = 3B_{0f}(13) + 3B_{0f}(15) - B^c_{63}$$
$$B^a_{98} = 4B_{0f}(16) + B^c_{61}$$
$$B^a_{99} = 3B_{0p}(1)M^2_Z - B_{0p}(3)M^2_Z - B_{0p}(6)M^2_Z + 3B^c_{33}$$
$$B^a_{100} = B^c_{62}$$
$$B^a_{101} = B^c_{64}$$
$$B^a_{102} = 24B_{0f}(4) + B^c_{72}$$

$B^a_{103} = 18B_{0f}(9)M_H^4 - 18B^c_{33}M_H^2M_Z^2 + B^c_{74}M_Z^4$
$B^a_{104} = 6B_{0f}(9)B^b_1 + 18B_{0f}(11)B^b_2M_Z^2 + 12B_{0f}(15)M_H^2M_Z^2 - 6B^c_{33}M_H^2M_Z^2 - B^c_{73}M_Z^4$
$B^a_{105} = 6B_{0f}(13) + 6B_{0f}(15) - B^c_{70}$
$B^a_{106} = 2B_{0f}(16) + B^c_{68}$
$B^a_{107} = B^c_{69}$
$B^a_{108} = B^c_{71}$
$B^a_{109} = 8B_{0f}(3)M_b^2 - B_{0f}(11)B^b_2 + 2a_{0f}(1)M_t^2 - 2a_{0f}(2)M_b^2 - B^c_{36}M_b^2 + B^c_{75}M_t^2$
$B^a_{110} = B_{0f}(13) - B^c_{48}$
$B^a_{111} = B_{0f}(11)B^b_2 - 2a_{0f}(1)M_t^2 - B^c_{75}M_t^2$
$B^a_{112} = 8M_b^2 - 3B_{0f}(13)M_Z^2 + 3B^c_{48}M_Z^2$
$B^a_{113} = 3B_{0f}(11)B^b_2 - 6a_{0f}(1)M_t^2 - 3B^c_{75}M_t^2 - B^c_{77}M_b^2$
$B^a_{114} = B^c_{76}$
$B^a_{115} = B^c_{77}$
$B^a_{116} = 8B_{0f}(2)M_t^2 - B_{0f}(11)B^b_2 - 2a_{0f}(1)M_t^2 + 2a_{0f}(2)M_b^2 - B^c_{36}M_t^2 + B^c_{75}M_b^2$
$B^a_{117} = B_{0f}(11)B^b_2 - 2a_{0f}(2)M_b^2 - B^c_{75}M_b^2$
$B^a_{118} = 16M_t^2 - 3B_{0f}(13)M_Z^2 + 3B^c_{48}M_Z^2$
$B^a_{119} = 3B_{0f}(11)B^b_2 - 6a_{0f}(2)M_b^2 - 3B^c_{75}M_b^2 - B^c_{79}M_t^2$
$B^a_{120} = B^c_{78}$
$B^a_{121} = B^c_{79}$
$B^a_{122} = 2M_b^2 + 2a_{0f}(2)M_b^2 - B^c_{48}M_Z^2$
$B^a_{123} = 2M_t^2 + 2a_{0f}(1)M_t^2 - B^c_{48}M_Z^2$
$B^a_{124} = 4a_{0f}(1)M_t^2 - 2a_{0f}(2)M_b^2 - 2B^b_6 - B^c_{48}M_Z^2$
$B^a_{125} = B_{0f}(10) + B_{0f}(12) + 3B_{0f}(13) + 3B_{0f}(15)$
$B^a_{126} = 36B_{0f}(4) - B^c_{83}$
$B^a_{127} = 18B_{0f}(9)M_H^4 - 18B^c_{33}M_H^2M_Z^2 + B^c_{85}M_Z^4$
$B^a_{128} = 6B_{0f}(9)B^b_1 + 18B_{0f}(11)B^b_2M_Z^2 + 12B_{0f}(15)M_H^2M_Z^2 - 6B^c_{33}M_H^2M_Z^2 - B^c_{84}M_Z^4$
$B^a_{129} = 3B_{0f}(10) + 3B_{0f}(12) + 9B_{0f}(13) + 9B_{0f}(15) - B^c_{81}$
$B^a_{130} = B^c_{80}$
$B^a_{131} = B^c_{82}$
$B^a_{132} = 4M_b^2 + 4a_{0f}(2)M_b^2 - B^c_{48}M_Z^2$
$B^a_{133} = 4M_t^2 + 4a_{0f}(1)M_t^2 - 3B^c_{48}M_Z^2$
$B^a_{134} = B^c_{86}$
$B^a_{135} = a_{0f}(1)M_t^2 + a_{0f}(2)M_b^2 + B^b_2 - B^c_{48}M_Z^2$
$B^a_{136} = 6B_{0f}(8) + 5B^c_{87}$
$B^a_{137} = B^c_{88}$
$B^a_{138} = B^c_{89}$
$B^a_{139} = 4a_{0f}(1)M_t^2 + 2a_{0f}(2)M_b^2 + 2B^b_7 - 3B^c_{48}M_Z^2$
$B^a_{140} = 36B_{0f}(4) - B^c_{92}$
$B^a_{141} = 6B_{0f}(9)M_H^4 - 6B^c_{33}M_H^2M_Z^2 + B^c_{91}M_Z^4$
$B^a_{142} = 2B_{0f}(9)B^b_1 + 6B_{0f}(11)B^b_2M_Z^2 + 4B_{0f}(15,1)M_H^2M_Z^2 - 2B^c_{33}M_H^2M_Z^2 - B^c_{90}M_Z^4$

$B^a_{143} = B^c_{55}$
$B^a_{144} = B^c_{93}$
$B^a_{145} = B^c_{94}$
$B^a_{146} = B^c_{95}$
$B^a_{147} = B^c_{96}$
$B^a_{148} = B^c_{97}$
$B^a_{149} = B^c_{98}$
$B^a_{150} = B^c_{99}$
$B^a_{151} = B^c_{68}$
$B^a_{152} = B^c_{100}$
$B^a_{153} = 1 - B^c_{33}$
$B^a_{154} = B^c_{101}$
$B^a_{155} = B^c_{102}$
$B^a_{156} = B_{0p}(10)$
$B^a_{157} = 32M_b^2 - 144B_{0f}(16)M_Z^2 + 3B_{0p}(2)M_Z^4 + 32a_{0f}(2)M_b^2$
$B^a_{158} = 6B_{0f}(11)B^b_3M_t^2 + B_{0f}(15)M_H^4 + 6a_{0f}(1)B^b_4M_t^2 + 6a_{0f}(2)M_t^2M_b^2 + B^b_9B^c_{33}M_Z^2$
$B^a_{159} = 2B_{0p}(4)M_t^2 + B_{0p}(7)M_Z^2$
$B^a_{160} = 2B_{0p}(5)M_b^2 + B_{0p}(8)M_Z^2$
$B^a_{161} = 12B^c_{103}M_H^2 - B^c_{104}M_Z^2$
$B^a_{162} = B^c_{105}$
$B^a_{163} = B^c_{106}$
$B^a_{164} = 6M_t^2 - 2B_{0p}(5)M_b^2M_Z^2 - B_{0p}(8)M_Z^4 + 6a_{0f}(1)M_t^2$
$B^a_{165} = 6B_{0f}(4)M_Z^4 + B_{0f}(9)M_H^4 + 6B_{0f}(11)B^b_2M_Z^2 - 8B_{0f}(15)M_H^2M_Z^2$
$B^a_{166} = 48B_{0f}(9)M_Z^2 + 4B^c_{103}M_H^2 - B^c_{108}M_Z^2$
$B^a_{167} = 72B_{0f}(11)B^b_3M_t^2 + 12B_{0f}(15)M_H^4 + 144B_{0p}(11)B^b_8M_Z^4 + 72a_{0f}(1)B^b_4M_t^2 + 72a_{0f}(2)M_t^2M_b^2$
$\quad - 12B^c_0M_H^4 + 12B^c_{107}M_H^2M_Z^2 + B^c_{109}M_Z^4$
$B^a_{168} = B^c_{110}$
$B^a_{169} = B^c_{111}$
$B^a_{170} = 4B^c_{103}M_H^2 - B^c_{112}M_Z^2$
$B^a_{171} = B^c_{113}$
$B^a_{172} = B_{0f}(2) - B_{0p}(4)M_Z^2$
$B^a_{173} = B_{0f}(3) - B_{0p}(5)M_Z^2$
$B^a_{174} = B_{0p}(7)M_Z^4 - 2a_{0f}(1)M_t^2$
$B^a_{175} = B_{0p}(8)M_Z^4 - 2a_{0f}(2)M_b^2$
$B^a_{176} = B^b_5$
$B^a_{177} = 24B_{0f}(9)B^b_1 - 48B_{0f}(15)M_H^2M_Z^2 + 72B_{0p}(9)M_Z^2M_Z^4 - 12B_{0p}(11)B^b_0M_Z^2 + 12B^c_{37}M_H^2M_Z^2$
$\quad + 24B^c_{39}M_t^2M_Z^2 + 24B^c_{114}M_b^2M_Z^2 - B^c_{117}M_Z^4$
$B^a_{178} = B_{0f}(10) + B_{0f}(12) - 12B_{0f}(15)$
$B^a_{179} = 16B_{0f}(16) - B_{0p}(2)M_Z^2$
$B^a_{180} = B^c_{115}$
$B^a_{181} = B^c_{116}$

$$B^a_{182} = 6B_{0f}(9)B^b_1 + 12B_{0f}(15)M^2_H M^2_Z - 3B_{0p}(11)B^b_0 M^2_Z + 3B^c_{118}M^2_H M^2_Z - B^c_{119}M^4_Z$$
$$B^a_{183} = B_{0p}(5)$$
$$B^a_{184} = B_{0p}(4)$$
$$B^a_{185} = 6B_{0f}(9)B^b_1 + 18B_{0f}(11)B^b_2 M^2_Z + 12B_{0f}(15)M^2_H M^2_Z - 18B_{0p}(9)M^2_Z M^4_Z - 3B_{0p}(11)B^b_0 M^2_Z + 3B^c_{118}M^2_H M^2_Z + B^c_{120}M^4_Z$$
$$B^a_{186} = B_{0p}(7)$$
$$B^a_{187} = B_{0p}(8)$$
$$B^a_{188} = B^c_{121}$$
$$B^a_{189} = 6B_{0f}(9)B^b_1 + 18B_{0f}(11)B^b_2 M^2_Z + 12B_{0f}(15)M^2_H M^2_Z - 18B_{0p}(9)M^2_Z M^4_Z - 3B_{0p}(11)B^b_0 M^2_Z + 3B^c_{118}M^2_H M^2_Z + B^c_{123}M^4_Z$$
$$B^a_{190} = B^c_{122}$$
$$B^a_{191} = B^c_{124}$$
$$B^a_{192} = 6B_{0f}(9)B^b_1 + 18B_{0f}(11)B^b_2 M^2_Z + 12B_{0f}(15)M^2_H M^2_Z + 18B_{0p}(9)M^2_Z M^4_Z - 3B_{0p}(11)B^b_0 M^2_Z + 3B^c_{118}M^2_H M^2_Z - B^c_{126}M^4_Z$$
$$B^a_{193} = B^c_{125}$$
$$B^a_{194} = 8B_{0f}(3)M^2_b - B_{0f}(11)B^b_2 - 8B_{0p}(5)M^2_b M^2_Z + 2B_{0p}(8)M^4_Z + 2a_{0f}(1)M^2_t - 2a_{0f}(2)M^2_b - B^c_{36}M^2_b + B^c_{75}M^2_t$$
$$B^a_{195} = 2B_{0f}(3)M^2_b - 2B_{0p}(5)M^2_b M^2_Z - B_{0p}(8)M^4_Z$$
$$B^a_{196} = B_{0f}(11)B^b_2 - 2a_{0f}(1)M^2_t - B^c_{75}M^2_t + B^c_{127}M^2_b$$
$$B^a_{197} = 8B_{0f}(2)M^2_t - B_{0f}(11)B^b_2 - 8B_{0p}(4)M^2_t M^2_Z + 2B_{0p}(7)M^4_Z - 2a_{0f}(1)M^2_t + 2a_{0f}(2)M^2_b - B^c_{36}M^2_t + B^c_{75}M^2_b$$
$$B^a_{198} = 2B_{0f}(2)M^2_t - 2B_{0p}(4)M^2_t M^2_Z - B_{0p}(7)M^4_Z$$
$$B^a_{199} = B_{0f}(11)B^b_2 - 2a_{0f}(2)M^2_b - B^c_{75}M^2_b + B^c_{127}M^2_t$$
$$B^a_{200} = 2B_{0f}(9)B^b_1 + 6B_{0f}(11)B^b_2 M^2_Z + 4B_{0f}(15,1)M^2_H M^2_Z - 6B_{0p}(9)M^2_Z M^4_Z - B_{0p}(11)B^b_0 M^2_Z + B^c_{118}M^2_H M^2_Z - B^c_{128}M^4_Z$$
$$B^a_{201} = B^c_{129}$$
$$B^a_{202} = B^c_{87}$$
$$B^a_{203} = B^c_{130}$$
$$B^a_{204} = 12B_{0f}(9)B^b_1 + 36B_{0f}(11)B^b_2 M^2_Z + 24B_{0f}(15)M^2_H M^2_Z - 36B_{0p}(9)M^2_Z M^4_Z - 6B_{0p}(11)B^b_0 M^2_Z + B^c_{132}M^4_Z - 6B^c_{133}M^2_H M^2_Z$$
$$B^a_{205} = B_{0f}(10) + B_{0f}(12) + 2B_{0f}(13) + 2B_{0f}(15)$$
$$B^a_{206} = 6B_{0f}(11)B^b_3 M^2_t + B_{0f}(15)M^4_H + 6a_{0f}(1)B^b_4 M^2_t + 6a_{0f}(2)M^2_t M^2_b - B^c_0 M^4_H - B^c_{35}M^4_Z$$
$$B^a_{207} = B^c_{131}$$
$$B^a_{208} = B^c_{134}$$

$\mathbf{B}_i^b$

$$B_0^b = M_H^4 + 12M_Z^4 - 4M_H^2 M_Z^2 \qquad B_1^b = M_H^4 + 6M_Z^4 - 3M_H^2 M_Z^2 \qquad B_2^b = M_b^2 + M_t^2$$
$$B_3^b = 2M_b^2 - M_t^2 \qquad B_4^b = M_b^2 - M_t^2 \qquad B_5^b = M_b^2 + 4M_t^2$$
$$B_6^b = M_b^2 - 2M_t^2 \qquad B_7^b = M_b^2 + 2M_t^2 \qquad B_8^b = -2M_Z^2 + M_H^2$$
$$B_9^b = -17M_Z^2 + M_H^2$$

$\mathbf{B}_i^c$

$$L_z = \ln \frac{M_Z^2}{M_W^2} \qquad L_h = \ln \frac{M_H^2}{M_W^2} \qquad L_R = \ln \frac{\mu_R^2}{M_W^2} \tag{D.6}$$

$$B_0^c = 1 - L_h \qquad B_1^c = 2 + L_z - 2L_h \qquad B_2^c = 3 - 2L_z - 3L_h$$
$$B_3^c = 6 - L_z - 4L_h \qquad B_4^c = 76 + 324L_z + 27L_R \qquad B_5^c = 125 + 4L_z + 54L_R$$
$$B_6^c = 125 + 72L_z + 54L_R \qquad B_7^c = 389 - 432L_z - 276L_R \qquad B_8^c = 623 - 820L_z + 162L_R$$
$$B_9^c = 623 - 344L_z + 162L_R \qquad B_{10}^c = 1241 - 1644L_z - 363L_R \qquad B_{11}^c = 3827 - 3096L_z - 408L_R$$
$$B_{12}^c = 6379 - 5184L_z - 2208L_R \qquad B_{13}^c = 11 - 2L_z - 11L_h \qquad B_{14}^c = 35 + 13L_z$$
$$B_{15}^c = 59 - 11L_z \qquad B_{16}^c = 67 + 324L_z + 27L_R \qquad B_{17}^c = 535 + 12L_z + 30L_R$$
$$B_{18}^c = 823 - 864L_z - 552L_R \qquad B_{19}^c = 1111 - 612L_z + 30L_R \qquad B_{20}^c = 1792 - 1572L_z - 519L_R$$
$$B_{21}^c = 6097 - 4464L_z - 816L_R \qquad B_{22}^c = 6541 - 5184L_z - 2208L_R \qquad B_{23}^c = 19 - 36L_z$$
$$B_{24}^c = 29 - 236L_z + 14L_R \qquad B_{25}^c = 31 - 3L_z - 29L_h \qquad B_{26}^c = 41 - 204L_z - 55L_R$$
$$B_{27}^c = 61 - 112L_z \qquad B_{28}^c = 715 - 864L_z - 552L_R \qquad B_{29}^c = 4379 - 2640L_z - 2848L_R$$
$$B_{30}^c = 5614 - 3156L_z - 3885L_R \qquad B_{31}^c = 5989 - 3708L_z - 2754L_R \qquad B_{32}^c = 6325 - 5184L_z - 2208L_R$$
$$B_{33}^c = L_z \qquad B_{34}^c = 1 - 24L_z \qquad B_{35}^c = 1 - L_z$$
$$B_{36}^c = 2 - 9L_R \qquad B_{37}^c = 4 - L_z + 6L_R - 4L_h \qquad B_{38}^c = 10 - 27L_R$$
$$B_{39}^c = 10 + 9L_R \qquad B_{40}^c = 34 - 27L_R \qquad B_{41}^c = 115 - 36L_z - 741L_R$$
$$B_{42}^c = 127 - 184L_z - 54L_R \qquad B_{43}^c = 178 - 288L_z - 129L_R \qquad B_{44}^c = 371 - 384L_z$$
$$B_{45}^c = 797 - 912L_z - 156L_R \qquad B_{46}^c = 1147 - 1144L_z + 162L_R \qquad B_{47}^c = 3L_R - L_h$$
$$B_{48}^c = 1 - 3L_R \qquad B_{49}^c = 5 - 3L_z - 9L_R \qquad B_{50}^c = 25 + 6L_z - 39L_R$$
$$B_{51}^c = 1 - 2L_z \qquad B_{52}^c = 7 - 4L_z \qquad B_{53}^c = L_R$$
$$B_{54}^c = 1 - 80L_z - 5L_R \qquad B_{55}^c = 1 - 3L_z \qquad B_{56}^c = 55 - 48L_z$$
$$B_{57}^c = 61 - 543L_z + 6L_R \qquad B_{58}^c = 109 - 18L_z - 36L_R \qquad B_{59}^c = 112 - 9L_z$$
$$B_{60}^c = 122 + 23L_z \qquad B_{61}^c = 5 - 6L_z \qquad B_{62}^c = 5 - 4L_z$$
$$B_{63}^c = 58 - 3L_z - 5L_R \qquad B_{64}^c = 136 - 207L_z - 54L_R \qquad B_{65}^c = 337 - 456L_z$$
$$B_{66}^c = 466 - 541L_z \qquad B_{67}^c = 613 - 111L_z + 6L_R \qquad B_{68}^c = 2 - 3L_z$$
$$B_{69}^c = 7 - 6L_z \qquad B_{70}^c = 53 - 6L_z - 10L_R \qquad B_{71}^c = 77 - 9L_z - 18L_R$$
$$B_{72}^c = 107 - 160L_z - 10L_R \qquad B_{73}^c = 370 - 111L_z + 6L_R \qquad B_{74}^c = 466 - 543L_z + 6L_R$$
$$B_{75}^c = 2 - 3L_R \qquad B_{76}^c = 2 + L_R \qquad B_{77}^c = 10 - 9L_R$$

$B^c_{78} = 10 + 3L_R$ $B^c_{79} = 26 - 9L_R$ $B^c_{80} = 17 - 12L_z$
$B^c_{81} = 62 - 9L_z + 60L_R$ $B^c_{82} = 107 - 9L_z$ $B^c_{83} = 143 - 240L_z + 60L_R$
$B^c_{84} = 266 - 111L_z + 120L_R$ $B^c_{85} = 362 - 543L_z + 120L_R$ $B^c_{86} = 5 - 6L_R$
$B^c_{87} = 1 - 6L_R$ $B^c_{88} = 5 + 24L_R$ $B^c_{89} = 25 + 24L_z + 30L_R$
$B^c_{90} = 122 - 37L_z$ $B^c_{91} = 154 - 181L_z$ $B^c_{92} = 193 - 240L_z$
$B^c_{93} = 4 + 9L_z$ $B^c_{94} = 31 - 30L_z$ $B^c_{95} = 1 + L_z$
$B^c_{96} = 7 - 2L_z$ $B^c_{97} = 7 - 3L_z$ $B^c_{98} = 19 - 9L_z$
$B^c_{99} = 47 - 30L_z$ $B^c_{100} = 13 - 12L_z$ $B^c_{101} = 8 - 5L_z$
$B^c_{102} = 11 - 6L_z$ $B^c_{103} = 4 - 3L_h$ $B^c_{104} = 71 - 1284L_z + 174L_R$
$B^c_{105} = 313 - 756L_z - 273L_R$ $B^c_{106} = 607 - 864L_z - 552L_R$ $B^c_{107} = 2 - 3L_h$
$B^c_{108} = 89 - 428L_z + 54L_R$ $B^c_{109} = 115 - 144L_z + 150L_R$ $B^c_{110} = 335 - 432L_z - 276L_R$
$B^c_{111} = 689 - 1512L_z - 546L_R$ $B^c_{112} = 29 - 236L_z + 54L_R$ $B^c_{113} = 41 - 204L_z - 45L_R$
$B^c_{114} = 2 + 9L_R$ $B^c_{115} = 17 + 48L_z$ $B^c_{116} = 187 - 504L_z - 129L_R$
$B^c_{117} = 331 - 600L_z - 354L_R$ $B^c_{118} = 2 - 2L_z - L_h$ $B^c_{119} = 5 - 9L_z + 3L_R$
$B^c_{120} = 82 + 75L_z - 30L_R$ $B^c_{121} = 188 + 90L_z - 15L_R$ $B^c_{122} = 11 - 12L_z$
$B^c_{123} = 83 + 75L_z - 96L_R$ $B^c_{124} = 86 + 30L_z - 15L_R$ $B^c_{125} = 29 + 30L_z - 15L_R$
$B^c_{126} = 52 - 75L_z + 96L_R$ $B^c_{127} = 2 + 3L_R$ $B^c_{128} = 6 - 25L_z - 30L_R$
$B^c_{129} = 88 + 90L_z + 105L_R$ $B^c_{130} = 1 - 6L_z + 30L_R$ $B^c_{131} = 7 - 24L_z$
$B^c_{132} = 13 + 198L_z - 462L_R$ $B^c_{133} = 20 + 6L_R - 23L_h$ $B^c_{134} = 157 + 90L_z - 66L_R$

$\mathbf{C}_i$ functions

$C_0 = A_{106} - A_{107}$ $C_1 = A_{41}M_H^2 - 3A_{59}M_Z^2c_\theta^2$ $C_2 = 3A_{60}M_Z^2c_\theta^2 - A_{61}M_H^2$
$C_3 = A_{119} + A_{120}$ $C_4 = 33M_Z^2c_\theta^2 - 2A_{62}M_H^2$ $C_5 = 3A_{31}M_Z^2 - A_{32}M_H^2$
$C_6 = A_{133} + A_{134}$ $C_7 = 3A_{41}M_Z^2c_\theta^2 - A_{63}M_H^2$ $C_8 = 3A_{64}M_Z^2c_\theta^2 - A_{65}M_H^2$
$C_9 = 17M_Z^2 - M_H^2c_\theta^2$ $C_{10} = A_{133} - A_{147}$ $C_{11} = A_{34}M_H^4 + 6A_{35}M_Z^4$
$C_{12} = 64A_{20}M_t^2 + 9A_{109}M_Z^2s_\theta^4$ $C_{13} = M_H^4 + 12c_\theta^4M_Z^4$ $C_{14} = A_9M_H^4 + 12A_9c_\theta^4M_Z^4$

We have introduced

$$v_f = 1 - 2\frac{Q_f}{I_f^3}s_\theta^2,$$

where $Q_l = -1$, $Q_b = -\frac{1}{3}$ and $Q_t = \frac{2}{3}$; I_f^3 is the third component of isospin.

D.1.2 $\mathcal{G}_V$ terms

$$\mathcal{G}_V^{(0,4)} = \frac{v_l}{4c_\theta}.$$

$$\begin{aligned}\mathcal{G}_V^{(0,6)} = &\frac{s_\theta^2}{4c_\theta^2}\big(7-4s_\theta^2\big)W_1 - \frac{c_\theta}{4}\big(1+4s_\theta^2\big)W_2 + \frac{s_\theta}{4}\big(3-4s_\theta^2\big)W_3 \\ &- \frac{1}{16c_\theta}\big(7-4s_\theta^2\big)W_4 - \frac{1}{2c_\theta}W_{17}.\end{aligned}$$

$$\begin{aligned}\mathcal{G}_V^{(1,4)} = &\Bigg\{\frac{M_t^2}{144}\Big[A_7B_{80} - 24B_{13}s_\theta^2(1+v_t)\Big] \\ &- \frac{M_b^2}{144}\Big[A_{18}B_{85} + 12B_{20}s_\theta^2(1+v_b)\Big] \\ &+ \frac{M_Z^2}{288c_\theta^2}\Big[9A_{56}B_{12}v_b + 6(2A_{26}B_7c_\theta^2 + 6A_{34}B_8c_\theta^2 - 12A_{155}B_{22}c_\theta^2 \\ &- 12A_{156}B_{23}c_\theta^4 + 6A_{175}B_{129} - 3A_{189}B_{25} - 2B_{138}c_\theta^2)c_\theta^2 \\ &- 2(18A_{81}s_\theta^4 - 72B_{14}c_\theta^2 - 108B_{78}s_\theta^2 + B_{142})c_\theta^2s_\theta^2\Big] \\ &- \frac{M_H^4}{192c_\theta^2M_Z^2}B_{32}(1+v_l) - \frac{1}{288c_\theta^2M_Z^2}(3B_{134} - B_{139}c_\theta^2) \\ &+ \frac{M_Z^4}{96}\Big[(3A_{56}B_{24}v_b + B_2(1+v_l^2) - 3B_{140}(1+v_t^2) - 3B_{141}(1+v_b^2)) \\ &- 12(4A_{81}B_{16}c_\theta^4 + 2A_{87}B_{18} - B_{19}c_\theta^2)c_\theta^2\Big]\Bigg\}\,\mathcal{G}_V^{(0,4)} \\ &- \frac{1}{192c_\theta}(A_9B_{125} - M_H^2A_{36}B_{47}) - \frac{M_H^4}{768c_\theta^3M_Z^2}\Big[2A_{41}B_{32} - B_{32}v_l(1+v_l)\Big] \\ &+ \frac{M_Z^2}{1152c_\theta^3}\Bigg\{4\Big[9c_\theta^2s_\theta^4 - 36A_9B_{14}c_\theta^2 - 9A_{148}v_lc_\theta^2s_\theta^2 + 9B_{128}v_l(1+v_l) \\ &- 18B_{135}v_lc_\theta^2 + 18B_{193}c_\theta^2s_\theta^2 - 2B_{194}c_\theta^2\Big]c_\theta^2s_\theta^2\end{aligned}$$

$$
\begin{aligned}
&+ 4\Big[3A_9B_{138}c_\theta^2 - 3A_{26}B_7v_lc_\theta^2 - 3A_{28}B_7c_\theta^2 + 9A_{34}B_8c_\theta^2 \\
&- 9A_{34}B_8v_lc_\theta^2 + 18A_{155}B_{22}v_lc_\theta^2 - 18A_{156}B_{23}c_\theta^4 + 18A_{156}B_{23}v_lc_\theta^4 \\
&- 18A_{161}B_{22}c_\theta^2 - 6B_9c_\theta^2(1+v_l) - 18B_{10}c_\theta^2(1+v_t) - 18B_{11}c_\theta^2(1+v_b) \\
&- 18B_{21}v_l^3 + B_{133}v_lc_\theta^2 - B_{208}c_\theta^2\Big]c_\theta^2 - 9(A_{56}B_{12}v_lv_b + A_{58}B_{12}v_b + B_{115})\Big\} \\
&- \frac{M_t^2}{576c_\theta}\Big[(A_7B_0v_l + A_8B_0) + 8(2A_{152}B_{13} - 3B_{13}v_l(1+v_t))s_\theta^2\Big] \\
&+ \frac{1}{384c_\theta^3M_Z^2}\Big[(A_9B_5 + B_{134}v_l) - (B_{209} + B_{210}v_l)c_\theta^2\Big] \\
&+ \frac{M_b^2}{576c_\theta}\Big[(A_{18}B_3v_l + A_{19}B_3) + 4(2A_{153}B_{20} + 3B_{20}v_l(1+v_b))s_\theta^2\Big] \\
&- \frac{c_\theta M_Z^4}{16}\Big[2(A_{81}B_{16} - A_{81}B_{16}v_l)c_\theta^4 + (A_{87}B_{18} - A_{87}B_{18}v_l)\Big].
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{G}_V^{(1,6)} = &\Big\{\frac{M_t^2}{144}\Big[A_7B_{80} - 24B_{13}s_\theta^2(1+v_t)\Big] \\
&- \frac{M_b^2}{144}\Big[A_{18}B_{85} + 12B_{20}s_\theta^2(1+v_b)\Big] \\
&+ \frac{M_Z^2}{288c_\theta^2}\Big[9A_{56}B_{12}v_b + 3(4A_{26}B_7c_\theta^2 + 6A_{34}B_8c_\theta^2 + 3A_{191}B_{21} - 3A_{192}B_{22}c_\theta^2 \\
&- A_{193}B_{23} - 3A_{194}B_{25} - 6B_{211}c_\theta^2)c_\theta^2 \\
&- (9A_{34}s_\theta^4 - 144B_{14}c_\theta^2 + 9B_{213}s_\theta^2 + 2B_{214})c_\theta^2s_\theta^2\Big] \\
&- \frac{1}{576c_\theta^2M_Z^2}(B_{207}c_\theta^2 + 6B_{212}) \\
&+ \frac{M_Z^4}{96}\Big[3A_{56}B_{24}v_b + 4A_{163}B_2 + 36B_{17}(1+v_l^2) - 3B_{140}(1+v_t^2) \\
&- 3B_{141}(1+v_b^2) - 6(4A_{81}B_{16}c_\theta^4 + 2A_{87}B_{18} - 3B_{19}c_\theta^2)c_\theta^2\Big]\Big\}\,\mathcal{G}_V^{(0,6)} \\
&+ \sum_i W_i\mathcal{F}_{V,i}
\end{aligned}
$$

We report the functions $\mathcal{F}_{V,i}$.

$$
\begin{aligned}
\mathcal{F}_{V,1} = \Bigg\{ & -\frac{1}{24}A_{11}B_3c_\theta^2M_b^2 \\
& + \frac{1}{24}\Big[A_0B_0 + 32A_{66}B_{13}s_\theta^2\Big]c_\theta^2M_t^2 \\
& - \frac{1}{288}\Big[18A_{51}B_{12}v_b - 2(3A_{21}B_7 + 3A_{70}B_{15} + 6A_{169}B_9 + 2A_{170}B_{10} \\
& + 2A_{171}B_{11} - 36B_6c_\theta^2)c_\theta^2 \\
& - (18A_{34}B_8c_\theta^2 - 9A_{102}B_{22}c_\theta^2 + 3A_{104}B_{23} - 9A_{181}B_{21} + 9A_{182}B_{25} + 9A_{183}s_\theta^6 \\
& + 96B_{42}c_\theta^2v_t + 48B_{43}c_\theta^2v_b - 2B_{44}s_\theta^2 + B_{45}s_\theta^4)s_\theta^2\Big]M_Z^2 \\
& + \frac{1}{576M_Z^2}(12B_4c_\theta^2 + B_{27}s_\theta^2 - 12B_{41}) \\
& + \frac{1}{48}\Big[6(2A_{75}B_{16}c_\theta^4 - A_{87}B_{18} - 2A_{159}B_{24}v_b)c_\theta^2 \\
& - 12A_{172}B_{17} + A_{176}B_2\Big]s_\theta^2M_Z^4 + \frac{1}{48}(4B_{28}C_1 - B_{40}c_\theta^4s_\theta^2)\Bigg\}\,\mathcal{G}_V^{(0,4)} \\
& + \frac{M_H^4}{48c_\theta M_Z^2}B_{32}s_\theta^4 - \frac{1}{288}\Big[(16A_{99}B_{20}s_\theta^2 - (3A_{11}B_3v_l - A_{12}B_3)\Big]c_\theta M_b^2 \\
& - \frac{1}{2304c_\theta}\Big[3B_{34}c_\theta^2 + 3(16B_{28}C_1v_l - 16B_{28}C_2 - B_{33}) - 4(3B_{29}v_lc_\theta^2 + B_{35})c_\theta^2s_\theta^2\Big] \\
& - \frac{1}{288}\Big[3A_0B_0v_l + A_1B_0 + 16(2A_{98}B_{13} - 3B_{13}v_l[(1+v_l))s_\theta^2\Big]c_\theta M_t^2 \\
& - \frac{1}{768c_\theta M_Z^2}\Big[4(A_9B_1 - B_{26}v_l)s_\theta^2 - (4A_{20}B_4 - B_{30}v_l)c_\theta^2 + 4A_{20}B_5 - B_{31}v_l\Big] \\
& + \frac{M_Z^4}{192c_\theta}\Big[A_9B_2v_l^2 - 12A_{83}B_{17}v_l - A_{90}B_2 \\
& - 3(4A_{75}B_{16}v_lc_\theta^4 + 4A_{76}B_{16}c_\theta^4 - 2A_{87}B_{18}v_l + 2A_{88}B_{18} + A_{93}B_{19}c_\theta^2)c_\theta^2\Big]s_\theta^2
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{M_Z^2}{1152 c_\theta}\Bigg[2(36A_{20}B_6c_\theta^2 + 3A_{21}B_7v_l - 3A_{22}B_7 - 6A_{37}B_9 - 2A_{42}B_{10} \\
&- 2A_{47}B_{11} + 3A_{70}B_{15}v_l + 3A_{71}B_{15} - 36B_6v_lc_\theta^2 + 3B_9v_l(1+v_l^2) \\
&+ 9B_{10}v_l(1+v_t^2) + 9B_{11}v_l(1+v_b^2))c_\theta^2 \\
&+ (18A_{34}B_8v_lc_\theta^2 + 288A_{67}B_{14}c_\theta^4 + 9A_{100}B_{21}v_l - 9A_{101}B_{21} - 9A_{102}B_{22}v_lc_\theta^2 \\
&+ 9A_{103}B_{22}c_\theta^2 + 3A_{104}B_{23}v_l + 3A_{105}B_{23} - 18A_{108}B_8c_\theta^2 + 18A_{109}c_\theta^2s_\theta^6 \\
&- 9A_{110}v_lc_\theta^2s_\theta^4 + 9B_{25}C_0v_l - 2B_{36}v_lc_\theta^2 - 2B_{37}v_lc_\theta^2s_\theta^2 - 2B_{38}c_\theta^2s_\theta^2 + B_{39}c_\theta^2s_\theta^4)s_\theta^2 \\
&- 18(A_{51}B_{12}v_+A_{52}B_{12})v_b\Bigg].
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{V,2} = &\Bigg\{\frac{1}{24}A_{11}B_3c_\theta^2M_b^2 - \frac{1}{12}A_{30}B_{47}M_H^2 \\
&- \frac{1}{24}(A_0B_0 - 8B_{13}v_ls_\theta^2)c_\theta^2M_t^2 \\
&+ \frac{1}{48}\Bigg[12A_{159}B_{24}s_\theta^2v_b - 3(4A_{78}B_{16}c_\theta^2 + 2A_{87}B_{18} + B_{19}c_\theta^2)c_\theta^2 \\
&- (12A_{173}B_{17} + A_{177}B_2)\Bigg]c_\theta^2M_Z^4 \\
&+ \frac{1}{576}(12B_{49}c_\theta^2 + B_{63})M_Z^{-2} \\
&+ \frac{1}{192}\Bigg[B_{50} + 32(2B_{51}c_\theta^2 + B_{62}v_b)c_\theta^2\Bigg]s_\theta^2 \\
&- \frac{1}{288}\Bigg[(6A_{21}B_7 - 18A_{34}B_8c_\theta^2 - 9A_{115}B_{22}c_\theta^2 - 3A_{117}B_{23} \\
&+ 12A_{169}B_9 + 4A_{170}B_{10} + 4A_{171}B_{11} - 9A_{185}B_{25} - 6B_{53}c_\theta^2)c_\theta^2 \\
&- 9(2A_{51}B_{12}v_b - A_{184}B_{21}) \\
&+ (9A_{186}s_\theta^6 + 96B_{42}c_\theta^2v_t + 2B_{64}s_\theta^4 - B_{65}s_\theta^2)s_\theta^2\Bigg]M_Z^2\Bigg\}\mathcal{G}_V^{(0,4)} \\
&- \frac{1}{192}A_{121}B_{32}c_\theta^{-1}M_Z^{-2}M_H^4 \\
&+ \frac{1}{96}A_{30}B_{46}v_lc_\theta^{-1}M_H^2
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{1}{288}\Big[32A_{112}B_{20}s_\theta^2 - (3A_{11}B_3v_l - A_{13}B_3)\Big]c_\theta M_b^2 \\
&+ \frac{1}{288}\Big[3A_0B_0v_l + A_2B_0 + 4(8A_{111}B_{13} - 9B_{13}v_l(1+vt) - 6B_{13}v_l^2)s_\theta^2\Big]c_\theta M_t^2 \\
&+ \frac{1}{1152}\Big[(6A_{21}B_7v_l - 6A_{23}B_7 - 12A_{38}B_9 - 4A_{43}B_{10} - 4A_{48}B_{11} \\
&+ 6A_{72}B_{15}c_\theta^2 - 9A_{115}B_{22}v_lc_\theta^2 + 9A_{116}B_{22}c_\theta^2 - 3A_{117}B_{23}v_l + 3A_{118}B_{23} \\
&- 216B_6c_\theta^2v_t + 6B_9v_l(1+vl^2) + 18B_{10}v_l(1+v_t^2) + 18B_{11}v_l(1+v_b^2) \\
&- 9B_{25}C_3v_l - 6B_{53}v_lc_\theta^2)c_\theta^2 - 9(2A_{51}B_{12}v_lv_b + 2A_{53}B_{12}v_b - A_{113}B_{21}v_l + A_{114}B_{21}) \\
&+ (288A_{68}B_{14}c_\theta^4 + 18A_{96}c_\theta^2s_\theta^6 - 9A_{122}v_lc_\theta^2s_\theta^4 - 72B_{47}v_l - 2B_{58}v_lc_\theta^2 \\
&- B_{59}v_lc_\theta^2s_\theta^2 - B_{60}c_\theta^2s_\theta^2 + B_{61}c_\theta^2s_\theta^4)s_\theta^2\Big]c_\theta^{-1}M_Z^2 \\
&+ \frac{1}{2304c_\theta M_Z}\Big[(12A_{63}B_1 + B_{48}v_l + 36B_{52}v_t)c_\theta^2 - 12(3B_5v_t + B_{54}v_l)\Big] \\
&- \frac{1}{192}\Big[A_{63}B_2v_l^2 - 12A_{84}B_{17}v_l - A_{91}B_2 \\
&+ 3(4A_{77}B_{16} - 4A_{78}B_{16}v_l + A_{94}B_{19} + 3B_{19}v_l)c_\theta^4\Big]c_\theta M_Z^4 \\
&+ \frac{1}{2304c_\theta}\Big[(48B_{28}C_4 + B_{57})c_\theta^2 - 12(4B_{47}C_5 + B_{55}) - 4(48B_{51}v_lc_\theta^2 + B_{56})c_\theta^2s_\theta^2\Big].
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{V,3} = &\Big\{\frac{1}{144}\Big[A_3B_0 + 64A_{123}B_{13}s_\theta^2\Big]c_\theta s_\theta^{-1}M_t^2 \\
&+ \frac{1}{144}\Big[A_{15}B_3 + 16A_{125}B_{20}s_\theta^2\Big]c_\theta s_\theta^{-1}M_b^2 \\
&+ \frac{1}{288}\Big[9A_{54}B_{12}v_b - (3A_{24}B_7 + 6A_{39}B_9 + 2A_{44}B_{10} + 2A_{50}B_{11} \\
&- 36A_{66}B_6c_\theta^2 - 6A_{74}B_{15}c_\theta^2)c_\theta^2 + (18A_{34}B_8c_\theta^2 - 432A_{66}B_{14}c_\theta^2 + 9A_{129}B_{22}c_\theta^2 \\
&+ 3A_{132}B_{23} + 27A_{137}s_\theta^6 - 9A_{187}B_{21} + 9A_{188}B_{25} + 72B_{42}v_t[1+vb] \\
&+ 36B_{43}v_b(1+v_b) + B_{75}s_\theta^4 - 6B_{79}s_\theta^2)c_\theta^2s_\theta^2\Big]c_\theta^{-1}s_\theta^{-1}M_Z^2
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{1}{192}\Big[B_4 c_\theta^2(1+v_l) - B_5(1+v_l) + (B_{69}c_\theta^2 - 2B_{70})s_\theta^2\Big] c_\theta^{-1} s_\theta^{-1} M_Z^{-2} \\
&- \frac{1}{48}\Big[3(4A_{79}B_{16}c_\theta^2 + 2A_{87}B_{18} + B_{19}c_\theta^2)c_\theta^2 + 12A_{174}B_{17} + 3A_{178}B_{66} \\
&+ 2A_{179}B_{24} + A_{180}B_{67}\Big] c_\theta s_\theta M_Z^4 - \frac{1}{96}(4B_{28}C_7c_\theta^2 - B_{78}C_9)c_\theta^{-1}s_\theta^{-1}\Big\} \mathcal{G}_V^{(0,4)} \\
&- \frac{1}{384}A_{136}B_{32}c_\theta^{-2}s_\theta M_Z^{-2}M_H^4 \\
&- \frac{1}{576}\Big[A_3B_0v_l + A_4B_0 + 64(A_{123}B_{13}v_l + A_{124}B_{13})s_\theta^2\Big] s_\theta^{-1}M_t^2 \\
&- \frac{1}{576}\Big[A_{14}B_3 + A_{15}B_3v_l + 16(A_{125}B_{20}v_l + A_{126}B_{20})s_\theta^2\Big] s_\theta^{-1}M_b^2 \\
&+ \frac{1}{1152}\Big[(3A_{24}B_7v_l - 3A_{25}B_7 + 36A_{33}B_6c_\theta^2 + 6A_{39}B_9v_l - 6A_{40}B_9 \\
&+ 2A_{44}B_{10}v_l - 2A_{45}B_{10} - 2A_{49}B_{11} + 2A_{50}B_{11}v_l - 36A_{66}B_6v_lc_\theta^2 \\
&+ 18A_{73}B_{15}c_\theta^2 - 6A_{74}B_{15}v_lc_\theta^2)c_\theta^2 - (18A_{34}B_8v_lc_\theta^4 - 432A_{66}B_{14}v_lc_\theta^4 \\
&- 144A_{69}B_{14}c_\theta^4 - 18A_{109}c_\theta^2s_\theta^8 + 9A_{127}B_{21}c_\theta^2 - 9A_{128}B_{21}v_lc_\theta^2 + 9A_{129}B_{22}v_lc_\theta^4 \\
&- 9A_{130}B_{22}c_\theta^4 - 3A_{131}B_{23}c_\theta^2 + 3A_{132}B_{23}v_lc_\theta^2 - 18A_{135}B_8c_\theta^4 + 27A_{137}v_lc_\theta^2s_\theta^6 \\
&+ 9B_{25}C_6v_lc_\theta^2 - 6B_{73}v_lc_\theta^2s_\theta^2 + 3B_{74} + B_{75}v_lc_\theta^2s_\theta^4 + 3B_{76}c_\theta^2s_\theta^4 - B_{77}c_\theta^2s_\theta^6)s_\theta^2 \\
&- 9(A_{54}B_{12}v_l + A_{55}B_{12})v_b\Big] c_\theta^{-2}s_\theta^{-1}M_Z^2 \\
&- \frac{1}{768}\Big[(2A_{33}B_4 + B_4v_l[1+v_l])c_\theta^2 - (2A_{33}B_5 + B_5v_l(1+v_l)) \\
&+ (6B_1c_\theta^2(1+v_t) - B_{68}v_lc_\theta^2 - 2B_{70}v_l)s_\theta^2\Big] c_\theta^{-2}s_\theta^{-1}M_Z^{-2} \\
&+ \frac{1}{384}\Big[6(4A_{79}B_{16}v_lc_\theta^2 - 4A_{80}B_{16}c_\theta^2 + 2A_{87}B_{18}v_l - 2A_{89}B_{18} \\
&- A_{95}B_{19}c_\theta^2 - 3B_{19}v_lc_\theta^2)c_\theta^2 + (24A_{85}B_{17}v_l + 6A_{92}B_{66} - 2A_{96}B_{67}v_l^2 \\
&+ 2A_{97}B_{67} + 9B_{66}v_l^2[1+v_t])\Big] s_\theta M_Z^4 \\
&+ \frac{1}{2304}\Big[24(B_{28}C_7v_l + B_{28}C_8)c_\theta^2 - (4B_{71}s_\theta^2 - B_{72})c_\theta^2s_\theta^2 \\
&- 6(3B_{78}C_9 + B_{78}C_9v_l)\Big] c_\theta^{-2}s_\theta^{-1}.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{V,4} = \Bigg\{ & -\frac{1}{288}A_5B_{80}M_t^2 + 1/288A_{16}B_{85}M_b^2 - 1/192A_{148}B_{32}c_\theta^{-2}M_Z^{-2}M_H^4 \\
& + \frac{1}{1152}\Bigg[18A_{56}B_{12}v_b + 3(8A_{26}B_7c_\theta^2 - 6A_{34}B_8c_\theta^2 - 3A_{142}B_{21} \\
& + 3A_{143}B_{22}c_\theta^2 - A_{145}B_{23} + 3A_{147}B_{25} - 2B_{102}c_\theta^2 - 6B_{103}c_\theta^4)c_\theta^2 \\
& - (9A_{183}s_\theta^4 - 9B_{104}s_\theta^2 - 2B_{105})c_\theta^2s_\theta^2\Bigg]c_\theta^{-2}M_Z^2 \\
& - \frac{1}{288}\Bigg[16A_{148}B_{90}s_\theta^2 - (A_{138}B_{87} + A_{140}B_{89})\Bigg] \\
& - \frac{1}{2304}\Bigg[B_{92}c_\theta^2 - 12B_{97}\Bigg]c_\theta^{-2}M_Z^{-2} \\
& + \frac{1}{192}\Bigg[(A_9B_{81}v_l + 3A_{56}B_{24}v_b) - 6(6A_{81}B_{16}c_\theta^4 - A_{87}B_{18})c_\theta^2\Bigg]M_Z^4\Bigg\}\mathcal{G}_V^{(0,4)} \\
& + \frac{1}{256}A_9B_{84}c_\theta^{-3}M_t^2M_Z^{-2} \\
& - \frac{1}{1152}\Bigg[16c_\theta^2s_\theta^2(1+v_l) + A_{16}B_3v_l + A_{17}B_3 - 2A_{140}B_{88}v_l - 2A_{141}B_{88}\Bigg]c_\theta^{-1}M_b^2 \\
& + \frac{1}{1152}\Bigg[A_5B_0v_l + A_6B_0 + 2A_{138}B_{86}v_l + 2A_{139}B_{86}\Bigg]c_\theta^{-1}M_t^2 \\
& + \frac{1}{768}\Bigg[3A_{148}B_{32} + A_{148}B_{32}v_l\Bigg]c_\theta^{-3}M_Z^{-2}M_H^4 \\
& + \frac{1}{9216}\Bigg[3B_{96} + 4(B_{93} + B_{98}v_lc_\theta^2)c_\theta^2 + 4C_{12}c_\theta^2s_\theta^2\Bigg]c_\theta^{-1} \\
& - \frac{1}{768}\Bigg[3(A_9B_{19}c_\theta^2 - 12A_{81}B_{16}v_lc_\theta^4 - 4A_{82}B_{16}c_\theta^4 + 2A_{87}B_{18}v_l \\
& - 2A_{88}B_{18})c_\theta^2 + (A_9B_{81}v_l^2 - A_{90}B_2)\Bigg]c_\theta^{-1}M_Z^4 \\
& + \frac{1}{9216}\Bigg[(12A_9B_{82} - 24B_{47}C_{11} - B_{91}v_l)c_\theta^2 + 12(B_{95}v_l - 3B_{97})\Bigg]c_\theta^{-3}M_Z^{-2} \\
& + \frac{1}{4608}\Bigg[(18A_9B_{83}c_\theta^2 - 288A_{20}B_{14}c_\theta^4 - 24A_{26}B_7v_lc_\theta^2 + 144A_{27}B_7c_\theta^2v_b \\
& + 18A_{34}B_8v_lc_\theta^2 - 48A_{46}B_{10}c_\theta^2 - 9A_{101}B_{21} - 18A_{108}B_8c_\theta^2 + 9A_{142}B_{21}v_l
\end{aligned}
$$

$$- 9A_{143}B_{22}v_l c_\theta^2 + 9A_{144}B_{22}c_\theta^2 + 3A_{145}B_{23}v_l + 3A_{146}B_{23} - 72B_9 c_\theta^2[1+v_b]$$
$$- 24B_{11}c_\theta^2[1+v_l] + 9B_{25}C_{10}v_l + 2B_{94}v_l c_\theta^2)c_\theta^2 - 18(A_{56}B_{12}v_l + A_{57}B_{12})v_b$$
$$- 3(3A_{110}v_l s_\theta^2 + 6B_{99}v_l - 3B_{100}s_\theta^2 + 2B_{101})c_\theta^4 s_\theta^2\Big] c_\theta^{-3}M_Z^2.$$

$$\begin{aligned}\mathcal{F}_{V,5} = \Big\{ &- \frac{5}{72}B_{110}s_\theta^2 M_Z^2 - \frac{1}{96}B_{32}[1+v_l]c_\theta^{-2}M_Z^{-2}M_H^4 \\ &+ \frac{1}{144}\Big[B_{112}c_\theta^2 - 3B_{28}C_{13}\Big]c_\theta^{-2}M_Z^{-2}\Big\}\ \mathcal{G}_V^{(0,4)} \\ &- \frac{1}{48}A_9B_{106}c_\theta^{-1} + \frac{1}{96}A_{36}B_{47}c_\theta^{-1}M_H^2 \\ &- \frac{1}{384}\Big[(2A_{41}B_{32} - 2B_{28}v_l - B_{32}v_l(1+v_l)\Big]c_\theta^{-3}M_Z^{-2}M_H^4 \\ &+ \frac{1}{288}\Big[20B_{110}s_\theta^2 + B_{109}v_l - B_{111}\Big]c_\theta M_Z^2 \\ &+ \frac{1}{192}\Big[B_{28}C_{14} - (3B_{107} + B_{108}v_l)c_\theta^2\Big]c_\theta^{-3}M_Z^{-2}.\end{aligned}$$

$$\begin{aligned}\mathcal{F}_{V,7} = &\frac{1}{16}A_{66}B_{113}c_\theta^2 s_\theta M_Z^2\ \mathcal{G}_V^{(0,4)} \\ &- \frac{1}{128}\Big[(B_{113} - B_{113}v_l)(1+vl)\Big]c_\theta s_\theta M_Z^2.\end{aligned}$$

$$\begin{aligned}\mathcal{F}_{V,8} = &-\frac{1}{32}\Big[2A_{66}B_{22}c_\theta^4 - A_{149} + B_{114}v_l\Big]c_\theta^{-1}M_Z^2\ \mathcal{G}_V^{(0,4)} \\ &- \frac{1}{256}\Big[4(2s_\theta^2 - B_{117})c_\theta^2 s_\theta^2 + 4(A_{66}B_{22}c_\theta^2 - A_{66}B_{22}v_l c_\theta^2 - B_{116})c_\theta^2 \\ &+ (2A_{149}v_l - B_{21}[1+v_l^2] + 2B_{114}v_l^2 + 10B_{115})\Big]c_\theta^{-2}M_Z^2.\end{aligned}$$

$$\mathcal{F}_{V,9} = -\frac{3}{16}\Big[B_{119}c_\theta^{-2}s_\theta M_t^2 M_b^2 M_Z^{-2} + B_{121}s_\theta M_b^2\Big]\ \mathcal{G}_V^{(0,4)}$$
$$+ 1/64\Big[3A_9B_{118} + A_{150}B_{120} - 6B_3(1+v_t) + 3B_{121}v_l\Big]c_\theta^{-1}s_\theta M_b^2$$
$$+ 3/64\Big[(A_9B_{119} + B_{119}v_l\Big]c_\theta^{-3}s_\theta M_t^2 M_b^2 M_Z^{-2}.$$

$$\mathcal{F}_{V,10} = -\frac{3}{32}\Big[2B_{119}c_\theta^{-1}M_t^2 M_b^2 M_Z^{-2} + 2B_{121}c_\theta M_b^2 + B_{122}(1+v_t)c_\theta^{-1}M_b^2 M_Z^2\Big]\ \mathcal{G}_V^{(0,4)}$$
$$+ 3/128\Big[2A_9B_{118} - 8B_3(1+v_b) - B_{120}(1+v_t) + 2B_{121}v_l\Big]M_b^2$$
$$+ 3/64\Big[(A_9B_{119} + B_{119}v_l\Big]c_\theta^{-2}M_t^2 M_b^2 M_Z^{-2}.$$

$$\mathcal{F}_{V,11} = \frac{3}{16}\Big[B_{121}s_\theta M_t^2 + B_{123}c_\theta^{-2}s_\theta M_t^2 M_Z^{-2}\Big]\mathcal{G}_V^{(0,4)}$$
$$- \frac{1}{64}\Big[3A_9B_{118} + A_{151}B_{120} - 12B_0v_t + 3B_{121}v_l\Big]c_\theta^{-1}s_\theta M_t^2$$
$$- \frac{3}{64}\Big[(A_9B_{123} + B_{123}v_l\Big]c_\theta^{-3}s_\theta M_t^2 M_Z^{-2}.$$

$$\mathcal{F}_{V,12} = \frac{3}{16}\Big[B_{121}c_\theta M_t^2 + B_{123}c_\theta^{-1}M_t^2 M_Z^{-2} + 6CF1(B_{124}v_t)c_\theta^{-1}M_t^2 M_Z^2\Big]\mathcal{G}_V^{(0,4)}$$
$$- \frac{1}{64}\Big[3A_9B_{118} + A_{94}B_{120} - 12B_0(1+v_t) + 3B_{121}v_l\Big]M_t^2$$
$$- \frac{3}{64}3/64\Big[A_9B_{123} + B_{123}v_l\Big]c_\theta^{-2}M_t^2 M_Z^{-2}.$$

$$
\begin{aligned}
\mathcal{F}_{V,16} = \Bigg\{ & -\frac{1}{144}\Big[A_7 B_{80} - 24 B_{13} s_\theta^2 [1+v_t]\Big] M_t^2 \\
& + \frac{1}{144}\Big[A_{18} B_{85} + 12 B_{20} s_\theta^2 [1+v_b]\Big] M_b^2 \\
& - \frac{1}{288}\Big[9 A_{56} B_{12} v_b + 6(2 A_{26} B_7 c_\theta^2 + 6 A_{34} B_8 c_\theta^2 - 12 A_{155} B_{22} c_\theta^2 \\
& - 12 A_{156} B_{23} c_\theta^4 + 6 A_{175} B_{129} - 3 A_{189} B_{25} - 2 B_{138} c_\theta^2) c_\theta^2 \\
& - 2(18 A_{81} s_\theta^4 - 72 B_{14} c_\theta^2 - 108 B_{78} s_\theta^2 + B_{142}) c_\theta^2 s_\theta^2\Big] c_\theta^{-2} M_Z^2 \\
& + \frac{1}{192} B_{32}(1+v_l) c_\theta^{-2} M_Z^{-2} M_H^4 + \frac{1}{288}\Big[3 B_{134} - B_{139} c_\theta^2\Big] c_\theta^{-2} M_Z^{-2} \\
& - \frac{1}{96}\Big[3 A_{56} B_{24} v_b + B_2(1+v_l^2) - 3 B_{140}(1+v_t^2) - 3 B_{141}(1+v_b^2) \\
& - 12(4 A_{81} B_{16} c_\theta^4 + 2 A_{87} B_{18} - B_{19} c_\theta^2) c_\theta^2\Big] M_Z^4 \Bigg\} \mathcal{G}_V^{(0,4)} \\
& + \frac{1}{192} A_9 B_{125} c_\theta^{-1} - \frac{1}{192} A_{36} B_{47} c_\theta^{-1} M_H^2 \\
& + \frac{1}{768}\Big[2 A_{41} B_{32} - B_{32} v_l (1+v_l)\Big] c_\theta^{-3} M_Z^{-2} M_H^4 \\
& - \frac{1}{32}\Big[2 A_{87} B_{18} v_l + (4 A_{81} B_{16} c_\theta^2 - B_{19}) v_l c_\theta^2\Big] c_\theta M_Z^4 \\
& + \frac{1}{576}\Big[(A_7 B_0 v_l + A_8 B_0) + 8(2 A_{152} B_{13} - 3 B_{13} v_l (1+v_t)) s_\theta^2\Big] c_\theta^{-1} M_t^2 \\
& - \frac{1}{1152}\Big[3(A_9 B_5 + B_{134} v_l) - (B_{131} + 3 B_{132} v_l) c_\theta^2\Big] c_\theta^{-3} M_Z^{-2} \\
& + \frac{1}{1152}\Big[2(72 A_9 B_{14} + 18 A_{148} v_l s_\theta^2 + 36 B_{135} v_l - 9 B_{136} s_\theta^2 + 2 B_{137}) c_\theta^4 s_\theta^2 \\
& - 2(18 A_9 B_{126} c_\theta^2 - 6 A_{26} B_7 v_l c_\theta^2 - 6 A_{28} B_7 c_\theta^2 - 18 A_{34} B_8 v_l c_\theta^2 - 3 A_{41} B_{127} c_\theta^2 \\
& - 9 A_{86} B_{128} v_l - 9 A_{154} B_{21} v_l + 36 A_{155} B_{22} v_l c_\theta^2 + 36 A_{156} B_{23} v_l c_\theta^4 - 3 A_{157} B_{23}
\end{aligned}
$$

$$
\begin{aligned}
&- 36B_{10}c_\theta^2(1+v_t) - 36B_{11}c_\theta^2(1+v_b) - 6B_{130}c_\theta^2 + 2B_{133}v_l c_\theta^2)c_\theta^2 \\
&+ 9(A_{56}B_{12}v_l v_b + A_{58}B_{12}v_b + B_{115})\Big] c_\theta^{-3} M_Z^2 \\
&- \frac{1}{576}\Big[(A_{18}B_3 v_l + A_{19}B_3) + 4(2A_{153}B_{20} + 3B_{20}v_l[1+v_b])s_\theta^2\Big] c_\theta^{-1} M_b^2.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{V,17} = \Big\{ &- \frac{1}{144}\Big[A_7 B_{80} - 24B_{13}s_\theta^2(1+v_t)\Big] M_t^2 \\
&+ \frac{1}{144}\Big[A_{18}B_{85} + 12B_{20}s_\theta^2(1+v_b)\Big] M_b^2 \\
&- \frac{1}{96}\Big[3A_{56}B_{12}v_b + 2(3s_\theta^4 + 24B_{14}c_\theta^2 - 3B_{152}s_\theta^2 - B_{153})c_\theta^2 s_\theta^2 \\
&+ 2(2A_{26}B_7 c_\theta^2 - 3A_{41}B_{22}c_\theta^2 - A_{157}B_{23} + 6A_{175}B_{129} - 3A_{189}B_{25} \\
&- 6B_{126}c_\theta^2 - B_{143}v_l)c_\theta^2\Big] c_\theta^{-2} M_Z^2 \\
&- \frac{1}{96}\Big[3A_{56}B_{24}v_b + B_2[1+v_l^2] - 3B_{140}(1+v_t^2) - 3B_{141}(1+v_b^2)\Big] M_Z^4 \\
&+ \frac{1}{192}B_{32}(1+v_l)c_\theta^{-2}M_Z^{-2}M_H^4 + \frac{1}{288}(3B_{134} - B_{151}c_\theta^2)c_\theta^{-2}M_Z^{-2}\Big\}\, \mathcal{G}_V^{(0,4)} \\
&+ \frac{1}{192}A_9 B_{125}c_\theta^{-1} - 1/192 A_{36}B_{47}c_\theta^{-1}M_H^2 \\
&+ \frac{1}{768}\Big[2A_{41}B_{32} - B_{32}v_l[1+v_l]\Big] c_\theta^{-3}M_Z^{-2}M_H^4 \\
&- \frac{1}{1152}\Big[2(-18c_\theta^2 s_\theta^4 + 9v_l c_\theta^2 s_\theta^2 - 72A_9B_{14}c_\theta^2 + 18B_{128}v_l[1+v_l] \\
&- 18B_{148}v_l c_\theta^2 + 108B_{149}c_\theta^2 s_\theta^2 + 4B_{150}c_\theta^2)c_\theta^2 s_\theta^2 \\
&+ 2(18A_9B_{126}c_\theta^2 - 6A_{26}B_7 v_l c_\theta^2 - 6A_{28}B_7 c_\theta^2 + 18A_{34}B_8 c_\theta^2 + 9A_{41}B_{22}v_l c_\theta^2 \\
&- 36A_{155}B_{22}c_\theta^2 - 36A_{156}B_{23}c_\theta^4 + 3A_{157}B_{23}v_l + 12B_9 c_\theta^2(1+v_l) \\
&- 36B_{10}c_\theta^2[1+v_t] - 36B_{11}c_\theta^2[1+v_b] + 54B_{21}(1+v_l^2) - 36B_{21}v_l^3 + 3B_{143}v_l^2 \\
&- 2B_{144}c_\theta^2 + 6B_{147}v_l c_\theta^2)c_\theta^2 - 9(A_{56}B_{12}v_l v_b + A_{58}B_{12}v_b + B_{115})\Big] c_\theta^{-3}M_Z^2
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{576}\Big[(A_7B_0v_l+A_8B_0)+8(2A_{152}B_{13}-3B_{13}v_l[1+v_t])s_\theta^2\Big]c_\theta^{-1}M_t^2\\
&-\frac{1}{1152}\Big[3(A_9B_5+B_{134}v_l)-(3B_{145}+B_{146}v_l)c_\theta^2\Big]c_\theta^{-3}M_Z^{-2}\\
&-\frac{1}{576}\Big[A_{18}B_3v_l+A_{19}B_3+4(2A_{153}B_{20}+3B_{20}v_l(1+v_b))s_\theta^2\Big]c_\theta^{-1}M_b^2\\
&+\frac{1}{96}\Big[3(4A_{81}B_{16}c_\theta^4+2A_{87}B_{18}-B_{19}v_lc_\theta^2)c_\theta^2-4(B_2(1+v_l)+3B_{19}c_\theta^4)s_\theta^2\Big]c_\theta^{-1}M_Z^4.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{V,18}=\Big\{&-\frac{1}{144}\Big[A_7B_{80}-24B_{13}s_\theta^2(1+v_t)\Big]M_t^2\\
&+\frac{1}{144}\Big[A_{18}B_{85}+12B_{20}s_\theta^2(1+v_b)\Big]M_b^2\\
&-\frac{1}{96}\Big[3A_{56}B_{12}v_b+2(2A_{26}B_7c_\theta^2-6A_{34}B_8c_\theta^2-A_{157}B_{23}+3A_{159}B_{22}c_\theta^2\\
&+6A_{175}B_{129}-3A_{189}B_{25}-6B_{126}c_\theta^2)c_\theta^2+2(24B_{14}c_\theta^2-12B_{161}s_\theta^2-B_{163})c_\theta^2s_\theta^2\Big]c_\theta^{-2}M_Z^2\\
&-\frac{1}{96}\Big[24A_{87}B_{18}c_\theta^2+3A_{56}B_{24}v_b+B_2(1+v_l^2)-3B_{140}(1+v_t^2)-3B_{141}(1+v_b^2)\Big]M_Z^4\\
&+\frac{1}{192}B_{32}(1+v_l)c_\theta^{-2}M_Z^{-2}M_H^4+\frac{1}{288}(3B_{134}-B_{162}c_\theta^2)c_\theta^{-2}M_Z^{-2}\Big\}\mathcal{G}_V^{(0,4)}\\
&+\frac{1}{192}A_9B_{125}c_\theta^{-1}-\frac{1}{192}A_{36}B_{47}c_\theta^{-1}M_H^2\\
&+\frac{1}{768}\Big[2A_{41}B_{32}-B_{32}v_l(1+v_l)\Big]c_\theta^{-3}M_Z^{-2}M_H^4\\
&-\frac{1}{16}\Big[A_{87}B_{18}-A_{87}B_{18}v_l\Big]c_\theta M_Z^4\\
&+\frac{1}{576}\Big[A_7B_0v_l+A_8B_0+8(2A_{152}B_{13}-3B_{13}v_l[1+v_t])s_\theta^2\Big]c_\theta^{-1}M_t^2\\
&-\frac{1}{1152}\Big[3(A_9B_5+B_{134}v_l)-(B_{155}+B_{156}v_l)c_\theta^2\Big]c_\theta^{-3}M_Z^{-2}
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{1152}\Big[4(36A_9B_{14}c_\theta^2-9B_{128}v_l(1+v_l)+18B_{158}v_lc_\theta^2-36B_{159}c_\theta^2s_\theta^2\\
&+2B_{160}c_\theta^2)c_\theta^2s_\theta^2-6(6A_9B_{126}c_\theta^2-2A_{26}B_7v_lc_\theta^2-2A_{28}B_7c_\theta^2-6A_{34}B_8c_\theta^2\\
&+6A_{34}B_8v_lc_\theta^2-A_{157}B_{23}+A_{157}B_{23}v_l+3A_{158}B_{22}c_\theta^2-3A_{159}B_{22}v_lc_\theta^2\\
&-4B_9c_\theta^2(1+v_l)-12B_{10}c_\theta^2(1+v_t)-12B_{11}c_\theta^2(1+v_b)-12B_{21}v_l^3+B_{154}c_\theta^2\\
&+B_{157}v_lc_\theta^2)c_\theta^2+9(A_{56}B_{12}v_lv_b+A_{58}B_{12}v_b+B_{115})\Big]c_\theta^{-3}M_Z^2\\
&-\frac{1}{576}\Big[A_{18}B_3v_l+A_{19}B_3+4(2A_{153}B_{20}+3B_{20}v_l(1+v_b))s_\theta^2\Big]c_\theta^{-1}M_b^2.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{V,19}=&\Big[-\frac{1}{16}B_{84}c_\theta^{-2}M_t^2M_Z^{-2}-\frac{1}{8}B_{165}c_\theta^2M_Z^2-\frac{1}{16}B_{166}\Big]\,\mathcal{G}_V^{(0,4)}\\
&+\frac{1}{32}(A_9B_6+A_{63}B_{110}-8B_{11}+B_{165}v_l)c_\theta M_Z^2\\
&+\frac{1}{64}(A_9B_{84}+B_{84}v_l)c_\theta^{-3}M_t^2M_Z^{-2}\\
&+\frac{1}{64}(A_9B_{164}+B_{164}v_l)c_\theta^{-1}.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{V,20}=&\Big\{-\frac{1}{16}B_{84}c_\theta^{-2}M_t^2M_Z^{-2}-\frac{1}{8}B_{165}c_\theta^2M_Z^2-\frac{1}{24}(8s_\theta^2-3B_{88}v_t)M_b^2\\
&+\frac{1}{16}\Big[B_{172}(1+v_t)+B_{173}\Big]\Big\}\mathcal{G}_V^{(0,4)}\\
&+\frac{1}{64}\Big[2A_9B_6-8B_{11}(1+v_b)+B_{110}(1+v_t)\Big]c_\theta M_Z^2\\
&+\frac{1}{64}(A_9B_{84}+B_{84}v_l)c_\theta^{-3}M_t^2M_Z^{-2}\\
&-\frac{1}{192}\Big[2B_{168}v_lc_\theta^2+(3A_9B_{167}+B_{169}v_l)\Big]c_\theta^{-1}\\
&-\frac{1}{64}\Big[12B_{170}c_\theta^2-(2A_{10}B_{88}+4A_{20}B_3-2B_3v_l(1+v_t)\\
&-2B_{88}v_lv_t+B_{171})\Big]c_\theta^{-1}M_b^2.
\end{aligned}
$$

$$\begin{aligned}\mathcal{F}_{V,21} = &\left[-\frac{1}{16}B_{84}c_\theta^{-2}M_t^2M_Z^{-2} - \frac{1}{8}B_{165}c_\theta^2M_Z^2 - \frac{1}{16}B_{175}\right]\mathcal{G}_V^{(0,4)}\\ &+\frac{1}{32}(A_9B_6 + A_{63}B_{110} - 8B_{10} + B_{165}v_l)c_\theta M_Z^2\\ &+\frac{1}{64}(A_9B_{84} + B_{84}v_l)c_\theta^{-3}M_t^2M_Z^{-2}\\ &+\frac{1}{64}(A_9B_{174} + B_{174}v_l)c_\theta^{-1}.\end{aligned}$$

$$\begin{aligned}\mathcal{F}_{V,22} = &\left[-\frac{1}{16}B_{84}c_\theta^{-2}M_t^2M_Z^{-2} - \frac{1}{8}B_{165}c_\theta^2M_Z^2\right.\\ &\left.-\frac{1}{24}(16s_\theta^2 + A_{160}B_{86})M_t^2 + \frac{1}{16}(2B_{181}v_t + B_{182})\right]\mathcal{G}_V^{(0,4)}\\ &+\frac{1}{96}\left[3A_9B_6 - A_{94}B_{110} - 12B_{10}(1+v_t)\right]c_\theta M_Z^2\\ &+\frac{1}{64}(A_9B_{84} + B_{84}v_l)c_\theta^{-3}M_t^2M_Z^{-2}\\ &-\frac{1}{192}(2B_{177}v_lc_\theta^2 + 3A_9B_{176} + B_{178}v_l)c_\theta^{-1}\\ &-\frac{1}{192}\left[12B_{179}c_\theta^2 - (12A_{10}B_0 + 6A_{95}B_{86} + 2A_{160}B_{86}v_l - 12B_0v_lv_t + 3B_{180})\right]c_\theta^{-1}M_t^2.\end{aligned}$$

$$\mathcal{F}_{V,23} = -\frac{1}{24}B_{183}s_\theta.$$

$$\mathcal{F}_{V,25} = \frac{1}{3}B_{184}c_\theta s_\theta\,\mathcal{G}_V^{(0,4)} + \frac{1}{12}(B_{184} - B_{184}v_l)s_\theta.$$

$$\mathcal{F}_{V,26} = \frac{1}{12}B_{185}c_\theta s_\theta\mathcal{G}_V^{(0,4)} + \frac{1}{48}(B_{185} - B_{185}v_l)s_\theta.$$

$$\mathcal{F}_{V,27} = \frac{1}{12} B_{186} M_Z^2 \mathcal{G}_V^{(0,4)} - \frac{1}{24} B_{186} v_l c_\theta^- 1 M_Z^2.$$

$$\begin{aligned}
\mathcal{F}_{V,28} = \Bigg\{ & -\frac{1}{72}\Big[A_7 B_{80} - 24 B_{13} s_\theta^2 (1+v_t)\Big] M_t^2 \\
& + \frac{1}{72}\Big[A_{18} B_{85} + 12 B_{20} s_\theta^2 (1+v_b)\Big] M_b^2 \\
& - \frac{1}{144}\Big[9 A_{56} B_{12} v_b + 6(2 A_{26} B_7 c_\theta^2 + 6 A_{34} B_8 c_\theta^2 - 12 A_{155} B_{22} c_\theta^2 \\
& - 12 A_{156} B_{23} c_\theta^4 + 6 A_{175} B_{129} - A_{190} B_{25} - 2 B_{138} c_\theta^2 + 4 B_{188}(1+v_l)) c_\theta^2 \\
& - 2(18 A_{81} s_\theta^4 - 72 B_{14} c_\theta^2 - 108 B_{78} s_\theta^2 + B_{196}) c_\theta^2 s_\theta^2\Big] c_\theta^{-2} M_Z^2 \\
& + \frac{1}{96} B_{32}(1+v_l) c_\theta^{-2} M_Z^{-2} M_H^4 + \frac{1}{48}(B_{134} - B_{195} c_\theta^2) c_\theta^{-2} M_Z^{-2} \\
& - \frac{1}{48}\Big[3 A_{56} B_{24} v_b + B_2(1+v_l^2) - 3 B_{140}(1+v_t^2) - 3 B_{141}(1+v_b^2) \\
& - 12(4 A_{81} B_{16} c_\theta^4 + 2 A_{87} B_{18} - B_{19} c_\theta^2) c_\theta^2\Big] M_Z^4 \Bigg\} \mathcal{G}_V^{(0,4)} \\
& + \frac{1}{96} A_9 B_{125} c_\theta^{-1} - \frac{1}{96} A_{36} B_{47} c_\theta^{-1} M_H^2 \\
& + \frac{1}{384}\Big[2 A_{41} B_{32} - B_{32} v_l (1+v_l)\Big] c_\theta^{-3} M_Z^{-2} M_H^4 \\
& - \frac{1}{8}\Big[4 B_{17} v_l s_\theta^2 (1+v_l) \\
& - (2 A_{81} B_{16} c_\theta^4 - 2 A_{81} B_{16} v_l c_\theta^4 + A_{87} B_{18} - A_{87} B_{18} v_l) c_\theta^2\Big] c_\theta^{-1} M_Z^4
\end{aligned}$$

$$
\begin{aligned}
&- \frac{1}{576}\Big[4(9s_\theta^4 - 36A_9B_{14} - 9A_{148}v_l s_\theta^2 - 18B_{135}v_l + 18B_{193}s_\theta^2 - 2B_{194})c_\theta^4 s_\theta^2 \\
&+ 4(3A_9B_{138}c_\theta^2 - 3A_{26}B_7 v_l c_\theta^2 - 3A_{28}B_7 c_\theta^2 + 9A_{34}B_8 c_\theta^2 \\
&- 9A_{34}B_8 v_l c_\theta^2 + 18A_{155}B_{22}v_l c_\theta^2 - 18A_{156}B_{23}c_\theta^4 + 18A_{156}B_{23}v_l c_\theta^4 \\
&- 18A_{161}B_{22}c_\theta^2 + 6A_{162}B_{25}v_l - 6B_9 c_\theta^2(1+v_l) - 18B_{10}c_\theta^2(1+v_t) \\
&- 18B_{11}c_\theta^2(1+v_b) - 18B_{21}v_l^3 + 3B_{187}(1+v_l) - 6B_{188}v_l(1+v_l) - B_{189}c_\theta^2 \\
&+ B_{192}v_l c_\theta^2)c_\theta^2 - 9(A_{56}B_{12}v_l v_b + A_{58}B_{12}v_b + B_{115})\Big]c_\theta^{-3}M_Z^2 \\
&+ \frac{1}{288}\Big[(A_7B_0v_l + A_8B_0) + 8(2A_{152}B_{13} - 3B_{13}v_l(1+v_t))s_\theta^2\Big]c_\theta^{-1}M_t^2 \\
&- \frac{1}{576}\Big[3(A_9B_5 + B_{134}v_l) - (B_{190} + B_{191}v_l)c_\theta^2\Big]c_\theta^{-3}M_Z^{-2} \\
&- \frac{1}{288}\Big[(A_{18}B_3v_l + A_{19}B_3) + 4(2A_{153}B_{20} + 3B_{20}v_l(1+v_b))s_\theta^2\Big]c_\theta^{-1}M_b^2.
\end{aligned}
$$

$$
\mathcal{F}_{V,29} = -\frac{1}{12}B_{183}c_\theta s_\theta\ \mathcal{G}_V^{(0,4)} + \frac{1}{48}B_{110}s_\theta M_Z^2 + \frac{1}{48}B_{183}v_l s_\theta.
$$

$$
\mathcal{F}_{V,30} = -\frac{1}{12}B_{183}c_\theta s_\theta \mathcal{G}_V^{(0,4)} + \frac{1}{48}(B_{183}v_l - B_{197})s_\theta.
$$

$$
\mathcal{F}_{V,31} = \frac{1}{12}B_{185}c_\theta s_\theta\ \mathcal{G}_V^{(0,4)} - \frac{1}{48}(B_{185}v_l - B_{198})s_\theta.
$$

$$
\mathcal{F}_{V,32} = -\frac{1}{18}B_{199}s_\theta^2 M_Z^2 \mathcal{G}_V^{(0,4)} + \frac{1}{72}(B_{199} + B_{199}v_l)c_\theta^- 1 s_\theta^2 M_Z^2.
$$

$$
\mathcal{F}_{V,33} = \frac{1}{12}B_{185}c_\theta s_\theta \mathcal{G}_V^{(0,4)} + \frac{1}{48}(B_{185} - B_{185}v_l)s_\theta.
$$

$$
\mathcal{F}_{V,34} = \frac{1}{6}B_{200}c_\theta s_\theta \mathcal{G}_V^{(0,4)} + \frac{1}{24}(B_{184} - B_{200}v_l)s_\theta.
$$

$$
\begin{aligned}
\mathcal{F}_{V,42} = \Bigg\{ & \frac{3}{8}\Bigg[(3B_{24}v_b - 2(B_{16})c_\theta^4\Bigg]c_\theta^2 M_Z^4 \\
& + \frac{1}{16}\Bigg[5B_{204}s_\theta^2 + 6(A_{66}B_{23} - B_7 + 2B_{22}c_\theta^2)c_\theta^2 \\
& + 9B_{12}(1+v_t) + B_{205}\Bigg]M_Z^2\Bigg\}\ \mathcal{G}_V^{(0,4)} \\
& - \frac{3}{16}(B_{16} - B_{16}v_l)c_\theta^5 M_Z^4 \\
& - \frac{1}{64}\Bigg[3(6A_9B_{12}v_b + 3B_{12}v_l(1+v_t) + 6B_{115} + 2B_{115}v_l) \\
& + (6A_{29}B_7 - 6B_{22}c_\theta^2 + 12B_{22}v_l c_\theta^2 + 3B_{23}(1+v_l) \\
& - 3B_{23}v_l(1+v_l) - B_{201}v_l - B_{203})c_\theta^2 + 6B_{202}c_\theta^2 s_\theta^2\Bigg]c_\theta^{-1}M_Z^2.
\end{aligned}
$$

$$
\mathcal{F}_{V,43} = \frac{3}{8}B_{121}M_tM_b\mathcal{G}_V^{(0,4)} - \frac{3}{32}(A_9B_{121} + B_{121}v_l)c_\theta^{-1}M_tM_b.
$$

$$
\mathcal{F}_{V,46} = -\frac{1}{6}B_{206}c_\theta s_\theta \mathcal{G}_V^{(0,4)} - \frac{1}{24}(B_{206} - B_{206}v_l)s_\theta.
$$

$$
\mathcal{F}_{V,47} = -\frac{1}{12}B_{184}s_\theta.
$$

$$
\mathcal{F}_{V,52} = -\frac{1}{3}B_{184}c_\theta s_\theta \mathcal{G}_V^{(0,4)} + \frac{1}{12}(B_{184} + B_{184}v_l)s_\theta.
$$

D.1.3 $\mathcal{G}_A$ terms

$$
\mathcal{G}_A^{(0,4)} = \frac{1}{4c_\theta}.
$$

$$
\mathcal{G}_A^{(0,6)} = -\frac{s_\theta^2}{4c_\theta}W_1 - \frac{c_\theta}{4}W_2 - \frac{s_\theta}{4}W_3 + \frac{1}{16c_\theta}W_4 - \frac{1}{2c_\theta}W_{16}.
$$

$$
\begin{aligned}
\mathcal{G}_A^{(1,4)} = \Bigg\{ & \frac{M_t^2}{144}\Big[A_7 B_{80} - 24 B_{13} s_\theta^2 (1+v_t)\Big] \\
& - \frac{M_b^2}{144}\Big[A_{18} B_{85} + 12 B_{20} s_\theta^2 (1+v_b)\Big] \\
& + \frac{M_Z^2}{288 c_\theta^2}\Big[9 A_{56} B_{12} v_b + 6(2 A_{26} B_7 c_\theta^2 + 6 A_{34} B_8 c_\theta^2 - 12 A_{155} B_{22} c_\theta^2 \\
& - 12 A_{156} B_{23} c_\theta^4 + 6 A_{175} B_{129} - 3 A_{189} B_{25} - 2 B_{138} c_\theta^2) c_\theta^2 \\
& - 2(18 A_{81} s_\theta^4 - 72 B_{14} c_\theta^2 - 108 B_{78} s_\theta^2 + B_{142}) c_\theta^2 s_\theta^2\Big] - \frac{M_H^4}{192 c_\theta^2 M_Z^2} B_{32}(1+v_l) \\
& - \frac{1}{288 c_\theta^2 M_Z^2}(3 B_{134} - B_{139} c_\theta^2) \\
& + \frac{M_Z^4}{96}\Big[3 A_{56} B_{24} v_b + B_2(1+v_l^2) - 3 B_{140}(1+v_t^2) - 3 B_{141}(1+v_b^2) \\
& - 12(4 A_{81} B_{16} c_\theta^4 + 2 A_{87} B_{18} - B_{19} c_\theta^2) c_\theta^2\Big]\Bigg\} \mathcal{G}_A^{(0,4)}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{G}_A^{(1,6)} = \Bigg\{ & \frac{1}{144}\Big[A_7 B_{80} - 24 B_{13} s_\theta^2 (1+v_t)\Big] M_t^2 \\
& - \frac{1}{144}\Big[A_{18} B_{85} + 12 B_{20} s_\theta^2 (1+v_b)\Big] M_b^2 \\
& + \frac{1}{288}\Big[9 A_{56} B_{12} v_b + 3(4 A_{26} B_7 c_\theta^2 + 6 A_{34} B_8 c_\theta^2 + 3 A_{191} B_{21} \\
& - 3 A_{192} B_{22} c_\theta^2 - A_{193} B_{23} - 3 A_{194} B_{25} - 6 B_{211} c_\theta^2) c_\theta^2 \\
& - (9 A_{34} s_\theta^4 - 144 B_{14} c_\theta^2 + 9 B_{213} s_\theta^2 + 2 B_{214}) c_\theta^2 s_\theta^2\Big] c_\theta^{-2} M_Z^2 \\
& - \frac{1}{576}(B_{207} c_\theta^2 + 6 B_{212}) c_\theta^{-2} M_Z^{-2} \\
& + \frac{1}{96}\Big[3 A_{56} B_{24} v_b + 4 A_{163} B_2 + 36 B_{17}(1+v_l^2) \\
& - 3 B_{140}(1+v_t^2) - 3 B_{141}(1+v_b^2) \\
& - 6(4 A_{81} B_{16} c_\theta^4 + 2 A_{87} B_{18} - 3 B_{19} c_\theta^2) c_\theta^2\Big] M_Z^4 + \sum_i W_i \, \mathcal{F}_{A,i}\Bigg\} \mathcal{G}_A^{(0,6)},
\end{aligned}
$$

where the functions $\mathcal{F}_{A,i}$ are defined as

$$
\begin{aligned}
\mathcal{F}_{A,1} = & -\frac{1}{24}A_{11}B_3c_\theta^2M_b^2 \\
& + \frac{1}{24}(A_0B_0 + 32A_{66}B_{13}s_\theta^2)c_\theta^2M_t^2 \\
& - \frac{1}{288}\Big[18A_{51}B_{12}v_b - 2(3A_{21}B_7 + 3A_{70}B_{15} + 6A_{169}B_9 + 2A_{170}B_{10} \\
& + 2A_{171}B_{11} - 36B_6c_\theta^2)c_\theta^2 - (18A_{34}B_8c_\theta^2 - 9A_{102}B_{22}c_\theta^2 + 3A_{104}B_{23} - 9A_{181}B_{21} \\
& + 9A_{182}B_{25} + 9A_{183}s_\theta^6 + 96B_{42}c_\theta^2v_t + 48B_{43}c_\theta^2v_b - 2B_{44}s_\theta^2 + B_{45}s_\theta^4)s_\theta^2\Big]M_Z^2 \\
& + \frac{1}{576}\Big[12B_4c_\theta^2 + B_{27}s_\theta^2 - 12B_{41}\Big]M_Z^{-2} \\
& + \frac{1}{48}\Big[6(2A_{75}B_{16}c_\theta^4 - A_{87}B_{18} - 2A_{159}B_{24}v_b)c_\theta^2 - (12A_{172}B_{17} + A_{176}B_2)\Big]s_\theta^2M_Z^4 \\
& + \frac{1}{48}(4B_{28}C_1 - B_{40}c_\theta^4s_\theta^2).
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{A,2} = & \frac{1}{24}A_{11}B_3c_\theta^2M_b^2 - \frac{1}{12}A_{30}B_{47}M_H^2 - \frac{1}{24}(A_0B_0 - 8B_{13}v_ls_\theta^2)c_\theta^2M_t^2 \\
& + \frac{1}{48}\Big[12A_{159}B_{24}s_\theta^2v_b - 3(4A_{78}B_{16}c_\theta^2 + 2A_{87}B_{18} + B_{19}c_\theta^2)c_\theta^2 \\
& - 12A_{173}B_{17} + A_{177}B_2\Big]c_\theta^2M_Z^4 + \frac{1}{576}(12B_{49}c_\theta^2 + B_{63})M_Z^{-2} \\
& + \frac{1}{192}\Big[B_{50} + 32(2B_{51}c_\theta^2 + B_{62}v_b)c_\theta^2\Big]s_\theta^2 \\
& - \frac{1}{288}\Big[(6A_{21}B_7 - 18A_{34}B_8c_\theta^2 - 9A_{115}B_{22}c_\theta^2 - 3A_{117}B_{23} + 12A_{169}B_9 \\
& + 4A_{170}B_{10} + 4A_{171}B_{11} - 9A_{185}B_{25} - 6B_{53}c_\theta^2)c_\theta^2 - 9(2A_{51}B_{12}v_b - A_{184}B_{21}) \\
& + (9A_{186}s_\theta^6 + 96B_{42}c_\theta^2v_t + 2B_{64}s_\theta^4 - B_{65}s_\theta^2)s_\theta^2\Big]M_Z^2.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{A,3} = & \frac{1}{144}(A_3B_0 + 64A_{123}B_{13}s_\theta^2)c_\theta s_\theta^{-1}M_t^2 \\
& + \frac{1}{144}(A_{15}B_3 + 16A_{125}B_{20}s_\theta^2)c_\theta s_\theta^{-1}M_b^2 \\
& + \frac{1}{288}\Big[9A_{54}B_{12}v_b - (3A_{24}B_7 + 6A_{39}B_9 + 2A_{44}B_{10} + 2A_{50}B_{11} \\
& - 36A_{66}B_6c_\theta^2 - 6A_{74}B_{15}c_\theta^2)c_\theta^2 + (18A_{34}B_8c_\theta^2 - 432A_{66}B_{14}c_\theta^2 + 9A_{129}B_{22}c_\theta^2 \\
& + 3A_{132}B_{23} + 27A_{137}s_\theta^6 - 9A_{187}B_{21} + 9A_{188}B_{25} + 72B_{42}v_t(1+v_b) \\
& + 36B_{43}v_b(1+v_b) + B_{75}s_\theta^4 - 6B_{79}s_\theta^2)c_\theta^2 s_\theta^2\Big]c_\theta^{-1}s_\theta^{-1}M_Z^2 \\
& + \frac{1}{192}\Big[B_4c_\theta^2(1+v_l) - B_5(1+v_l) + (B_{69}c_\theta^2 - 2B_{70})s_\theta^2\Big]c_\theta^{-1}s_\theta^{-1}M_Z^{-2} \\
& - \frac{1}{48}\Big[(3(4A_{79}B_{16}c_\theta^2 + 2A_{87}B_{18} + B_{19}c_\theta^2)c_\theta^2 \\
& + (12A_{174}B_{17} + 3A_{178}B_{66} + 2A_{179}B_{24} + A_{180}B_{67})\Big]c_\theta s_\theta M_Z^4 \\
& - \frac{1}{96}(4B_{28}C_7c_\theta^2 - B_{78}C_9)c_\theta^{-1}s_\theta^{-1}.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{A,4} = & -\frac{1}{288}A_5B_{80}M_t^2 + 1/288A_{16}B_{85}M_b^2 - 1/192A_{148}B_{32}c_\theta^{-2}M_Z^{-2}M_H^4 \\
& + \frac{1}{1152}\Big[18A_{56}B_{12}v_b + 3(8A_{26}B_7c_\theta^2 - 6A_{34}B_8c_\theta^2 - 3A_{142}B_{21} \\
& + 3A_{143}B_{22}c_\theta^2 - A_{145}B_{23} + 3A_{147}B_{25} - 2B_{102}c_\theta^2 - 6B_{103}c_\theta^4)c_\theta^2 \\
& - (9A_{183}s_\theta^4 - 9B_{104}s_\theta^2 - 2B_{105})c_\theta^2 s_\theta^2\Big]c_\theta^{-2}M_Z^2 \\
& - \frac{1}{288}\Big[16A_{148}B_{90}s_\theta^2 - (A_{138}B_{87} + A_{140}B_{89})\Big] \\
& - \frac{1}{2304}(B_{92}c_\theta^2 - 12B_{97})c_\theta^{-2}M_Z^{-2} \\
& + \frac{1}{192}\Big[(A_9B_{81}v_l + 3A_{56}B_{24}v_b) - 6(6A_{81}B_{16}c_\theta^4 - A_{87}B_{18})c_\theta^2\Big]M_Z^4.
\end{aligned}
$$

$$\mathcal{F}_{A,5} = -\frac{5}{72}B_{110}s_\theta^2 M_Z^2 - \frac{1}{96}B_{32}(1+v_l)c_\theta^{-2}M_Z^{-2}M_H^4 + \frac{1}{144}(B_{112}c_\theta^2 - 3B_{28}C_{13}c_\theta^{-2}M_Z^{-2}.$$

$$\mathcal{F}_{A,7} = \frac{1}{16}A_{66}B_{113}c_\theta^2 s_\theta M_Z^2.$$

$$\mathcal{F}_{A,8} = -\frac{1}{32}\Big[(2A_{66}B_{22}c_\theta^4 - (A_{149}+B_{114}v_l)\Big]c_\theta^-1M_Z^2$$

$$\mathcal{F}_{A,9} = -\frac{3}{16}B_{119}c_\theta^{-2}s_\theta M_t^2 M_b^2 M_Z^{-2} - \frac{3}{16}B_{121}s_\theta M_b^2.$$

$$\mathcal{F}_{A,10} = -\frac{3}{16}B_{119}c_\theta^{-1}M_t^2 M_b^2 M_Z^{-2} - \frac{3}{16}B_{121}c_\theta M_b^2 - \frac{3}{32}B_{122}(1+v_t)c_\theta^{-1}M_b^2 M_Z^2.$$

$$\mathcal{F}_{A,11} = +\frac{3}{16}B_{121}s_\theta M_t^2 + \frac{3}{16}B_{123}c_\theta^{-2}s_\theta M_t^2 M_Z^{-2}.$$

$$\mathcal{F}_{A,12} = +\frac{3}{16}B_{121}c_\theta M_t^2 + \frac{3}{16}B_{123}c_\theta^{-1}M_t^2 M_Z^{-2} + \frac{3}{16}B_{124}v_t c_\theta^{-1}M_t^2 M_Z^2.$$

$$\begin{aligned}
\mathcal{F}_{A,16} = &-\frac{1}{144}\Big[A_7B_{80} - 24B_{13}s_\theta^2(1+v_t)\Big]M_t^2 \\
&+\frac{1}{144}\Big[A_{18}B_{85} + 12B_{20}s_\theta^2(1+v_b)\Big]M_b^2 \\
&-\frac{1}{288}\Big[9A_{56}B_{12}v_b + 6(2A_{26}B_7c_\theta^2 + 6A_{34}B_8c_\theta^2 - 12A_{155}B_{22}c_\theta^2 \\
&-12A_{156}B_{23}c_\theta^4 + 6A_{175}B_{129} - 3A_{189}B_{25} - 2B_{138}c_\theta^2)c_\theta^2 \\
&-2(18A_{81}s_\theta^4 - 72B_{14}c_\theta^2 - 108B_{78}s_\theta^2 + B_{142})c_\theta^2 s_\theta^2\Big]c_\theta^{-2}M_Z^2 \\
&+\frac{1}{192}B_{32}(1+v_l)c_\theta^{-2}M_Z^{-2}M_H^4 + 1/288(3B_{134} - B_{139}c_\theta^2)c_\theta^{-2}M_Z^{-2} \\
&-\frac{1}{96}\Big[3A_{56}B_{24}v_b + B_2(1+v_l^2) - 3B_{140}(1+v_t^2) - 3B_{141}(1+v_b^2) \\
&-12(4A_{81}B_{16}c_\theta^4 + 2A_{87}B_{18} - B_{19}c_\theta^2)c_\theta^2\Big]M_Z^4.
\end{aligned}$$

$$
\begin{aligned}
\mathcal{F}_{A,17} = & -\frac{1}{144}\Big[A_7 B_{80} - 24 B_{13} s_\theta^2 (1+v_t)\Big] M_t^2 \\
& + \frac{1}{144}\Big[A_{18} B_{85} + 12 B_{20} s_\theta^2 (1+v_b)\Big] M_b^2 \\
& - \frac{1}{96}\Big[3 A_{56} B_{12} v_b + 2(3 s_\theta^4 + 24 B_{14} c_\theta^2 - 3 B_{152} s_\theta^2 - B_{153}) c_\theta^2 s_\theta^2 \\
& + 2(2 A_{26} B_7 c_\theta^2 - 3 A_{41} B_{22} c_\theta^2 - A_{157} B_{23} + 6 A_{175} B_{129} - 3 A_{189} B_{25} \\
& - 6 B_{126} c_\theta^2 - B_{143} v_l) c_\theta^2\Big] c_\theta^{-2} M_Z^2 \\
& - \frac{1}{96}\Big[3 A_{56} B_{24} v_b + B_2 (1+v_l^2) - 3 B_{140} (1+v_t^2) - 3 B_{141} (1+v_b^2)\Big] M_Z^4 \\
& + \frac{1}{192} B_{32} (1+v_l) c_\theta^{-2} M_Z^{-2} M_H^4 + 1/288 \Big[3 B_{134} - B_{151} c_\theta^2\Big] c_\theta^{-2} M_Z^{-2}.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{F}_{A,18} = & -\frac{1}{144}\Big[A_7 B_{80} - 24 B_{13} s_\theta^2 (1+v_t)\Big] M_t^2 \\
& + \frac{1}{144}\Big[A_{18} B_{85} + 12 B_{20} s_\theta^2 (1+v_b)\Big] M_b^2 \\
& - \frac{1}{96}\Big[3 A_{56} B_{12} v_b + 2(2 A_{26} B_7 c_\theta^2 - 6 A_{34} B_8 c_\theta^2 - A_{157} B_{23} \\
& + 3 A_{159} B_{22} c_\theta^2 + 6 A_{175} B_{129} - 3 A_{189} B_{25} - 6 B_{126} c_\theta^2) c_\theta^2 \\
& + 2(24 B_{14} c_\theta^2 - 12 B_{161} s_\theta^2 - B_{163}) c_\theta^2 s_\theta^2\Big] c_\theta^{-2} M_Z^2 \\
& - \frac{1}{96}\Big[24 A_{87} B_{18} c_\theta^2 + 3 A_{56} B_{24} v_b + B_2 (1+v_l^2) - 3 B_{140} (1+v_t^2) - 3 B_{141} (1+v_b^2)\Big] M_Z^4 \\
& + \frac{1}{192} B_{32} (1+v_l) c_\theta^{-2} M_Z^{-2} M_H^4 + \frac{1}{288} (3 B_{134} - B_{162} c_\theta^2) c_\theta^{-2} M_Z^{-2}.
\end{aligned}
$$

$$\mathcal{F}_{A,19} = -\frac{1}{16}B_{84}c_\theta^{-2}M_t^2M_Z^{-2} - \frac{1}{8}B_{165}c_\theta^2M_Z^2 - \frac{1}{16}B_{166}.$$

$$\begin{aligned}\mathcal{F}_{A,20} = &-\frac{1}{16}B_{84}c_\theta^{-2}M_t^2M_Z^{-2} - \frac{1}{8}B_{165}c_\theta^2M_Z^2 \\ &- \frac{1}{24}(8s_\theta^2 - 3B_{88}v_t)M_b^2 + \frac{1}{16}\Big[B_{172}(1+v_t) + B_{173}\Big].\end{aligned}$$

$$\mathcal{F}_{A,21} = -\frac{1}{16}B_{84}c_\theta^{-2}M_t^2M_Z^{-2} - \frac{1}{8}B_{165}c_\theta^2M_Z^2 - \frac{1}{16}B_{175}.$$

$$\begin{aligned}\mathcal{F}_{A,22} = &-\frac{1}{16}B_{84}c_\theta^{-2}M_t^2M_Z^{-2} - \frac{1}{8}B_{165}c_\theta^2M_Z^2 \\ &- \frac{1}{24}(16s_\theta^2 + A_{160}B_{86})M_t^2 + \frac{1}{16}(2B_{181}v_t + B_{182}).\end{aligned}$$

$$\mathcal{F}_{A,25} = \frac{1}{3}B_{184}c_\theta s_\theta.$$

$$\mathcal{F}_{A,26} = \frac{1}{12}B_{185}c_\theta s_\theta.$$

$$\mathcal{F}_{A,27} = \frac{1}{12}B_{186}M_Z^2.$$

$$\begin{aligned}\mathcal{F}_{A,28} = &-1/72\Big[A_7B_{80} - 24B_{13}s_\theta^2(1+v_t)\Big]M_t^2 \\ &+ 1/72\Big[A_{18}B_{85} + 12B_{20}s_\theta^2(1+v_b)\Big]M_b^2 \\ &- 1/144\Big[9A_{56}B_{12}v_b + 6(2A_{26}B_7c_\theta^2 + 6A_{34}B_8c_\theta^2 - 12A_{155}B_{22}c_\theta^2\end{aligned}$$

$$- 12A_{156}B_{23}c_\theta^4 + 6A_{175}B_{129} - A_{190}B_{25} - 2B_{138}c_\theta^2 + 4B_{188}(1+v_l))c_\theta^2$$
$$- 2(18A_{81}s_\theta^4 - 72B_{14}c_\theta^2 - 108B_{78}s_\theta^2 + B_{196})c_\theta^2 s_\theta^2\Big]c_\theta^{-2}M_Z^2$$
$$+ 1/96B_{32}(1+v_l)c_\theta^{-2}M_Z^{-2}M_H^4 + 1/48(B_{134} - B_{195}c_\theta^2)c_\theta^{-2}M_Z^{-2}$$
$$- 1/48\Big[3A_{56}B_{24}v_b + B_2(1+v_l^2) - 3B_{140}(1+v_t^2) - 3B_{141}(1+v_b^2)$$
$$- 12(4A_{81}B_{16}c_\theta^4 + 2A_{87}B_{18} - B_{19}c_\theta^2)c_\theta^2\Big]M_Z^4.$$

$$\mathcal{F}_{A,29} = -\frac{1}{12}B_{183}c_\theta s_\theta.$$

$$\mathcal{F}_{A,30} = -\frac{1}{12}B_{183}c_\theta s_\theta.$$

$$\mathcal{F}_{A,31} = \frac{1}{12}B_{185}c_\theta s_\theta.$$

$$\mathcal{F}_{A,32} = -\frac{1}{18}B_{199}s_\theta^2 M_Z^2.$$

$$\mathcal{F}_{A,33} = \frac{1}{12}B_{185}c_\theta s_\theta.$$

$$\mathcal{F}_{A,34} = \frac{1}{6}B_{200}c_\theta s_\theta.$$

$$\mathcal{F}_{A,42} = \frac{3}{8}(3B_{24}v_b - 2B_{16}c_\theta^4)c_\theta^2 M_Z^4$$
$$+ \frac{1}{16}\Big[5B_{204}s_\theta^2 + 6(A_{66}B_{23} - B_7 + 2B_{22}c_\theta^2)c_\theta^2 + 9B_{12}(1+v_t) + B_{205}\Big]M_Z^2.$$

$$\mathcal{F}_{A,43} = \frac{3}{8}B_{121}M_t M_b$$

$$\mathcal{F}_{A,46} = -\frac{1}{6}B_{206}c_\theta s_\theta$$

$$\mathcal{F}_{A,52} = -\frac{1}{3}B_{184}c_\theta s_\theta$$

D.1.4 $\sigma_{\mu,(p_1+p_2)}$ terms

$$\tilde{\mathcal{G}}^{(4)} = -\frac{i}{8}\Big[2A_{66}B_{22}c_\theta^4 - (A_{163}B_{114}v_l + A_{168})\Big]c_\theta^{-1}M_Z^2$$

$$\tilde{\mathcal{G}}^{(6)} = i\sum_i W_i\,\mathcal{F}_{\sigma,i}$$

where the functions $\mathcal{F}_{\sigma,i}$ are defined as

$$\begin{aligned}\mathcal{F}_{\sigma,1} = -\frac{1}{8}\Big[&2(A_{156}B_{22}c_\theta^2 - B_{216})c_\theta^2 + 2A_{163}B_{114}v_l + 3A_{164}B_{21} - 9B_{215}\\ &- 2(6B_{117}s_\theta^2 - B_{217})c_\theta^2 s_\theta^2\Big]c_\theta^{-1}s_\theta^2 M_Z^2\end{aligned}$$

$$\begin{aligned}\mathcal{F}_{\sigma,2} = \frac{1}{8}\Big[&2A_{41}B_{22}c_\theta^4 - (2A_{163}B_{114}v_l - 3A_{165}B_{21} + 3B_{115})\\ &- 6(2B_{117}s_\theta^4 + B_{218} - B_{219}s_\theta^2)s_\theta^2\Big]c_\theta M_Z^2\end{aligned}$$

$$\begin{aligned}\mathcal{F}_{\sigma,3} = \frac{1}{8}\Big[&2A_{66}B_{22}c_\theta^4 - (2A_{163}B_{114}v_l - 3A_{166}B_{21} - B_{220})\\ &- 2(6B_{117}s_\theta^4 + B_{221} - B_{222}s_\theta^2)s_\theta^2\Big]s_\theta M_Z^2\end{aligned}$$

$$\begin{aligned}\mathcal{F}_{\sigma,4} = \frac{1}{32}\Big[&2(A_{156}B_{22}c_\theta^2 + 3B_{161})c_\theta^2 + 3(A_{164}B_{21} - 9B_{115})\\ &+ 2(5B_{159} - 2B_{223}s_\theta^2)c_\theta^2 s_\theta^2\Big]c_\theta^{-1}M_Z^2\end{aligned}$$

$$\mathcal{F}_{\sigma,16} = \frac{1}{4}A_{41}B_{224}v_l c_\theta M_Z^2$$

$$\mathcal{F}_{\sigma,17} = \frac{1}{16}\Big[4B_{226}c_\theta^2 + 8(4s_\theta^2 + B_{159})c_\theta^2 s_\theta^2 - (2A_{163}B_{114}v_l - 3B_{21}(1+v_l^2) + 6B_{225})\Big]c_\theta^{-1}M_Z^2$$

$$\mathcal{F}_{\sigma,18} = -\frac{1}{8}(A_{163}B_{114}v_l + A_{167} + 2B_{22}c_\theta^4)c_\theta^{-1}M_Z^2$$

$$\mathcal{F}_{\sigma,28} = \frac{1}{4}\Big[2A_{66}B_{22}c_\theta^4 - (A_{163}B_{114}v_l + A_{168})\Big]c_\theta^{-1}M_Z^2$$

$$\mathcal{F}_{\sigma,42} = -\frac{3}{8}B_{113}c_\theta^3 M_Z^2$$

D.2 Hbb

D.2.1 Notations

The renormalized amplitude can be written in the form

$$\mathcal{M}_{Hbb} = gM_b\left[\mathcal{M}_{LO} + \frac{\sqrt{2}G_F}{\pi^2}\mathcal{M}_{EW} + \frac{g_s^2}{\pi^2}\mathcal{M}_{QCD}\right],$$

where the following decomposition is adopted

$$\begin{aligned}\mathcal{M}_{LO} &= \mathcal{M}_{LO}^{(0,4)} + \frac{g_6}{\sqrt{2}}\mathcal{M}_{LO}^{(0,6)}\\ \mathcal{M}_{EW} &= \mathcal{M}_{EW}^{(1,4)} + \frac{g_6}{\sqrt{2}}\mathcal{M}_{EW}^{(1,6)}\\ \mathcal{M}_{QCD} &= \mathcal{M}_{QCD}^{(1,4)} + \frac{g_6}{\sqrt{2}}\mathcal{M}_{QCD}^{(1,6)}.\end{aligned}$$

The full result reported is QED/QCD deconvoluted.

Loop scalar functions

We have used the shorthand notation

$$CX(p^2; m_1, m_2) = \mathrm{Re}X(p^2; m_1, m_2) + i\ \mathrm{Im}X(p^2; m_1, m_2), \tag{D.7}$$

where $X = \left\{B_{0f}, B_{1f}, B_{21f}, B_{22f}, B_{0p}\right\}$ and

$$B_{0p}(p^2; m_1, m_2) = \frac{d}{dp^2}B_0(p^2; m_1, m_2). \tag{D.8}$$

The sum over the fermionic generations is indicated by $\sum_{gen}$.

$$a_{0f}(M_t) = a_{0f}(1) \qquad a_{0f}(M_b) = a_{0f}(2)$$
$$a_{0f}(M_W) = a_{0f}(3) \qquad a_{0f}(M_Z) = a_{0f}(4)$$
$$a_{0f}(M_H) = a_{0f}(5)$$

$$B_{0f}(-M_Z^2; M_l, M_l) = B_{0f}(1) \qquad B_{0f}(-M_Z^2; M_t, M_t) = B_{0f}(2)$$
$$B_{0f}(-M_Z^2; M_b, M_b) = B_{0f}(3) \qquad B_{0f}(-M_Z^2; 0, 0) = B_{0f}(4)$$
$$\sum_{gen} B_{0f}(-M_Z^2; M_{l,g}, M_{l,g}) = B_{0f}(5) \qquad \sum_{gen} B_{0f}(-M_Z^2; M_{t,g}, M_{t,g}) = B_{0f}(6)$$
$$\sum_{gen} B_{0f}(-M_Z^2; M_{b,g}, M_{b,g}) = B_{0f}(7) \qquad B_{0f}(-M_Z^2; M_W, M_W) = B_{0f}(8)$$
$$B_{0f}(-M_Z^2; M_H, M_Z) = B_{0f}(9) \qquad B_{0f}(-M_W^2; 0, M_l) = B_{0f}(10)$$
$$B_{0f}(-M_W^2; M_t, M_b) = B_{0f}(11) \qquad \sum_{gen} B_{0f}(-M_W^2; 0, M_{l,g}) = B_{0f}(11)$$
$$\sum_{gen} B_{0f}(-M_W^2; M_{t,g}, M_{b,g}) = B_{0f}(12) \qquad B_{0f}(-M_W^2; M_W, M_Z) = B_{0f}(13)$$
$$B_{0f}(-M_W^2; M_W, M_H) = B_{0f}(14) \qquad B_{0f}(-M_W^2; 0, M_W) = B_{0f}(15)$$
$$B_{0f}(-M_H^2; M_t, M_t) = B_{0f}(16) \qquad B_{0f}(-M_H^2; M_b, M_b) = B_{0f}(17)$$
$$B_{0f}(-M_H^2; M_W, M_W) = B_{0f}(18) \qquad B_{0f}(-M_H^2; M_Z, M_Z) = B_{0f}(19)$$
$$B_{0f}(-M_H^2; M_H, M_H) = B_{0f}(20) \qquad B_{0f}(-M_W^2; 0, 0) = B_{0f}(21)$$
$$B_{0f}(-M_b^2; M_W, M_t) = B_{0f}(22) \qquad B_{0f}(-M_b^2; M_Z, M_b) = B_{0f}(23)$$
$$B_{0f}(-M_b^2; M_H, M_b) = B_{0f}(24)$$

$$CB_{0f}(-M_H^2; M_t, M_t) = CB_{0f}(16) \qquad CB_{0f}(-M_H^2; M_b, M_b) = CB_{0f}(17)$$
$$CB_{0f}(-M_H^2; M_W, M_W) = CB_{0f}(18) \qquad CB_{0f}(-M_H^2; M_Z, M_Z) = CB_{0f}(19)$$
$$CB_{0f}(-M_H^2; M_H, M_H) = CB_{0f}(20) \qquad CB_{0f}(-M_H^2; 0, M_Z) = CB_{0f}(25)$$

$$CB_{0f}(-M_b^2; M_W, M_t) = CB_{0f}(22) \qquad CB_{0f}(-M_b^2; M_Z, M_b) = CB_{0f}(23)$$
$$CB_{0f}(-M_b^2; M_H, M_b) = CB_{0f}(24) \qquad CB_{0f}(-M_b^2; 0, M_b) = CB_{0f}(26)$$

$$CB_{1f}(-M_b^2; M_W, M_t) = CB_{1f}(22) \qquad CB_{1f}(-M_b^2; M_Z, M_b) = CB_{1f}(23)$$
$$CB_{1f}(-M_b^2; M_H, M_b) = CB_{1f}(24) \qquad CB_{1f}(-M_b^2; 0, M_b) = CB_{1f}(26)$$

$$CB_{21f}(-M_b^2; M_W, M_t) = CB_{21f}(22) \qquad CB_{21f}(-M_b^2; M_Z, M_b) = CB_{21f}(23)$$
$$CB_{21f}(-M_b^2; M_H, M_b) = CB_{21f}(24) \qquad CB_{21f}(-M_b^2; 0, M_b) = CB_{21f}(26)$$

$$B_{0p}(-M_H^2; M_t, M_t) = B_{0p}(14) \qquad B_{0p}(-M_H^2; M_b, M_b) = B_{0p}(15)$$
$$B_{0p}(-M_H^2; M_W, M_W) = B_{0p}(16) \qquad B_{0p}(-M_H^2; M_Z, M_Z) = B_{0p}(17)$$
$$B_{0p}(-M_H^2; M_H, M_H) = B_{0p}(28) \qquad B_{0p}(-M_b^2; M_W, M_t) = B_{0p}(29)$$
$$B_{0p}(-M_b^2; M_Z, M_b) = B_{0p}(30) \qquad B_{0p}(-M_b^2; M_H, M_b) = B_{0p}(31)$$
$$B_{0p}(0; M_W, M_t) = B_{0p}(32) \qquad B_{0p}(0; M_Z, M_b) = B_{0p}(33)$$

$$C_0(-M_H^2,-M_b^2,-M_b^2;M_b,M_Z,M_b)=C_0(7) \qquad C_0(-M_H^2,-M_b^2,-M_b^2;M_b,M_H,M_b)=C_0(8)$$
$$C_0(-M_H^2,-M_b^2,-M_b^2;M_t,M_W,M_t)=C_0(9) \qquad C_0(-M_H^2,-M_b^2,-M_b^2;M_W,M_t,M_W)=C_0(10)$$
$$C_0(-M_H^2,-M_b^2,-M_b^2;M_Z,M_b,M_Z)=C_0(11) \qquad C_0(-M_H^2,-M_b^2,-M_b^2;M_Z,M_b,0)=C_0(12)$$
$$C_0(-M_H^2,-M_b^2,-M_b^2;M_H,M_b,M_H)=C_0(13) \qquad C_0(-M_H^2,-M_b^2,-M_b^2;0,M_b,0)=C_0(14)$$

$\mathbf{D}_i$ functions

$$D_0=9+(39+(-112+64s_\theta^2)s_\theta^2)s_\theta^2 \qquad D_1=9+(30+(-160+128s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_2=1+(-1+2s_\theta^2)s_\theta^2 \qquad D_3=1+(-3+(4-4s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_4=1+(-4+8s_\theta^2)s_\theta^2 \qquad D_5=1+(-6+(16-16s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_6=9+(-24+32s_\theta^2)s_\theta^2 \qquad D_7=9+(-42+(80-64s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_8=9+(-12+8s_\theta^2)s_\theta^2 \qquad D_9=9+(-30+(32-16s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_{10}=99+(-264+(212-48s_\theta^2)s_\theta^2)s_\theta^2 \qquad D_{11}=99+(-462+(740+(-472+96s_\theta^2)s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_{12}=1+(-2+s_\theta^2)s_\theta^2 \qquad D_{13}=1-2s_\theta^2$$
$$D_{14}=1+(-3+2s_\theta^2)s_\theta^2 \qquad D_{15}=2-s_\theta^2$$
$$D_{16}=198+(-891+(1448+(-996+240s_\theta^2)s_\theta^2)s_\theta^2)s_\theta^2 \qquad D_{17}=33+(-176+(236-80s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_{18}=99+(-495+(988+(-852+240s_\theta^2)s_\theta^2)s_\theta^2)s_\theta^2 \qquad D_{19}=3-4s_\theta^2$$
$$D_{20}=1+2s_\theta^2 \qquad D_{21}=1+4s_\theta^2$$
$$D_{22}=2+3s_\theta^2 \qquad D_{23}=1+3s_\theta^2$$
$$D_{24}=6-5s_\theta^2 \qquad D_{25}=3-5s_\theta^2$$
$$D_{26}=7-s_\theta^2 \qquad D_{27}=33+(-44+24s_\theta^2)s_\theta^2$$
$$D_{28}=3+(-4+8s_\theta^2)s_\theta^2 \qquad D_{29}=21+(-28+24s_\theta^2)s_\theta^2$$
$$D_{30}=51+(-20+8s_\theta^2)s_\theta^2 \qquad D_{31}=9+(12-8s_\theta^2)s_\theta^2$$
$$D_{32}=6+(-11+6s_\theta^2)s_\theta^2 \qquad D_{33}=3+(-7+6s_\theta^2)s_\theta^2$$
$$D_{34}=6+(-5+2s_\theta^2)s_\theta^2 \qquad D_{35}=3-2s_\theta^2$$
$$D_{36}=15+(-44+24s_\theta^2)s_\theta^2 \qquad D_{37}=39+(-88+48s_\theta^2)s_\theta^2$$
$$D_{38}=27+(-33+(-40+48s_\theta^2)s_\theta^2)s_\theta^2 \qquad D_{39}=27+(-21+(-20+24s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_{40}=3+(-28+24s_\theta^2)s_\theta^2 \qquad D_{41}=15+(-56+48s_\theta^2)s_\theta^2$$
$$D_{42}=9+(-24+16s_\theta^2)s_\theta^2 \qquad D_{43}=3+(-7+4s_\theta^2)s_\theta^2$$
$$D_{44}=3+(-10+8s_\theta^2)s_\theta^2 \qquad D_{45}=5-s_\theta^2$$
$$D_{46}=3+(4-8s_\theta^2)s_\theta^2 \qquad D_{47}=3+(20-8s_\theta^2)s_\theta^2$$
$$D_{48}=34+(-335+(652+(-448+96s_\theta^2)s_\theta^2)s_\theta^2)s_\theta^2 \qquad D_{49}=102+(-480+(823+(-496+48s_\theta^2)s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_{50}=28+(61-92s_\theta^2)s_\theta^2 \qquad D_{51}=403+(-1032+(776-144s_\theta^2)s_\theta^2)s_\theta^2$$
$$D_{52}=83+(-300+(352-96s_\theta^2)s_\theta^2)s_\theta^2 \qquad D_{53}=4+(-93+(144-16s_\theta^2)s_\theta^2)s_\theta^2$$

$D_{54} = 3 - s_\theta^2$

$D_{55} = 9 + (-21 + (-20 + 24s_\theta^2)s_\theta^2)s_\theta^2$

$D_{56} = 28 + (-49 + (-61 + 92s_\theta^2)s_\theta^2)s_\theta^2$

$D_{57} = 101 + (-503 + (1024 + (-776 + 144s_\theta^2)s_\theta^2)s_\theta^2)s_\theta^2$

$D_{58} = 17 + (-209 + (476 + (-400 + 96s_\theta^2)s_\theta^2)s_\theta^2)s_\theta^2$

$D_{59} = 17 + (-86 + (193 + (-160 + 16s_\theta^2)s_\theta^2)s_\theta^2)s_\theta^2$

$D_{60} = 3 + (-12 + 8s_\theta^2)s_\theta^2$

$D_{61} = 23 + (15 - 92s_\theta^2)s_\theta^2$

$D_{62} = 123 + (-350 + (352 - 72s_\theta^2)s_\theta^2)s_\theta^2$

$D_{63} = 19 - 20s_\theta^2$

$D_{64} = 6 + (109 + (-128 + 16s_\theta^2)s_\theta^2)s_\theta^2$

$D_{65} = 15 + (-20 + 8s_\theta^2)s_\theta^2$

$D_{66} = 32 - 29s_\theta^2$

$D_{67} = 85 + (-80 - 8s_\theta^2)s_\theta^2$

$D_{68} = 57 - 56s_\theta^2$

$D_{69} = 72 + (-171 + (112 - 16s_\theta^2)s_\theta^2)s_\theta^2$

$D_{70} = 1 + s_\theta^2$

$D_{71} = 159 + (-164 + 8s_\theta^2)s_\theta^2$

$D_{72} = 9 + (-16 + 6s_\theta^2)s_\theta^2$

$D_{73} = 45 + (-82 + 33s_\theta^2)s_\theta^2$

$D_{74} = 61 - 60s_\theta^2$

$D_{75} = 14 - 11s_\theta^2$

$D_{76} = 155 + (-160 + 8s_\theta^2)s_\theta^2$

$D_{77} = 149 + (-154 + 8s_\theta^2)s_\theta^2$

$D_{78} = 19 - 7s_\theta^2$

$D_{79} = 9 - 5s_\theta^2$

$\mathbf{E}_i$ functions

$E_0 = E_{12}^a$

$E_1 = E_{13}^a$

$E_2 = E_{14}^a$

$E_3 = E_{15}^a$

$E_4 = 8C_0(9)M_t^6 + 4C_0(10)E_3^a M_t^2 M_H^2 - 4CB_{0f}(16)E_{10}^a M_t^2 - 2CB_{0f}(18)E_4^a M_H^2 - 2CB_{0f}(19)M_H^4$
$\quad - 18CB_{0f}(20)M_H^4 + 8CB_{0f}(22)E_{11}^a M_t^2 - E_{17}^a M_H^4$

$E_5 = E_{16}^a$

$E_6 = E_{18}^a$

$E_7 = E_{19}^a$

$E_8 = 24C_0(9)M_t^6 M_W^2 + 12C_0(10)E_3^a M_t^2 M_W^2 M_H^2 - 12CB_{0f}(16)E_{10}^a M_t^2 M_W^2 - 6CB_{0f}(18)E_4^a M_W^2 M_H^2$
$\quad - 6CB_{0f}(19)M_W^2 M_H^4 - 54CB_{0f}(20)M_W^2 M_H^4 + 24CB_{0f}(22)E_{11}^a M_t^2 M_W^2 - E_{20}^a M_H^2$

$E_9 = 24C_0(9)M_t^2 M_W^2 M_t^4 + 12C_0(10)E_3^a M_t^2 M_W^2 M_H^2 - 12CB_{0f}(16)E_{10}^a M_t^2 M_W^2 - 6CB_{0f}(18)E_4^a M_W^2 M_H^2$
$\quad - 6CB_{0f}(19)M_W^2 M_H^4 - 54CB_{0f}(20)M_W^2 M_H^4 + 24CB_{0f}(22)E_{11}^a M_t^2 M_W^2 - E_{21}^a M_H^2$

$E_{10} = E_{22}^a$

$E_{11} = E_{23}^a$

$E_{12} = 24C_0(9)M_t^2 M_W^2 M_t^4 + 12C_0(10)E_3^a M_t^2 M_W^2 M_H^2 - 12CB_{0f}(16)E_{10}^a M_t^2 M_W^2 - 6CB_{0f}(18)E_4^a M_W^2 M_H^2$
$\quad - 6CB_{0f}(19)M_W^2 M_H^4 - 54CB_{0f}(20)M_W^2 M_H^4 + 24CB_{0f}(22)E_{11}^a M_t^2 M_W^2 - E_{24}^a M_H^2$

$E_{13} = E_{25}^a$

$E_{14} = 6C_0(9)M_t^4 + 2C_0(10)E_4^a M_t^2 - 6CB_{0f}(16)M_t^2 - 4CB_{0f}(18)M_t^2 - 2CB_{0f}(22)M_t^2 + E_{26}^a M_t^2 + 24CB_{22f}(22)$

$E_{15} = CB_{22f}(23)$

$E_{16} = E_{27}^a$

$E_{17} = 3C_0(9)E_0^a M_t^2 M_W^2 + C_0(10)E_4^a M_W^2 M_H^2 + 3CB_{0f}(16)E_0^a M_W^2 - 2CB_{0f}(18)M_W^2 M_H^2$
$\quad - CB_{0f}(22)E_{28}^a M_W^2 - E_{29}^a M_H^2 - 6CB_{1f}(22)M_W^2 M_H^2$

$E_{18} = 2C_0(9)M_t^4 - CB_{0f}(16)E_{10}^a - 2CB_{0f}(22)M_t^2 + 3E_{30}^a M_H^2 + CB_{1f}(22)M_H^2$

$$E_{19} = 48C_0(9)M_t^2M_W^2M_t^4 + 24C_0(10)E_3^aM_t^2M_W^2M_H^2 - 24CB_{0f}(16)E_{10}^aM_t^2M_W^2 - 24CB_{0f}(18)M_t^2M_W^2M_H^2 + 24CB_{0f}(22)E_{10}^aM_t^2M_W^2 - E_{31}^aM_H^2$$
$$E_{20} = E_{32}^a$$
$$E_{21} = 8C_0(9)M_t^2M_W^2M_t^4 + 4C_0(10)E_3^aM_t^2M_W^2M_H^2 - 4CB_{0f}(16)E_{10}^aM_t^2M_W^2 - 2CB_{0f}(18)E_4^aM_W^2M_H^2 - 4CB_{0f}(19)M_W^2M_H^4 + 4CB_{0f}(22)E_{10}^aM_t^2M_W^2 + E_{34}^aM_t^2M_H^2 + 4CB_{1f}(22)M_t^2M_W^2M_H^2 + 8CB_{22f}(22)M_W^2M_H^2 + 16CB_{22f}(23)M_W^2M_H^2$$
$$E_{22} = E_4^a$$
$$E_{23} = E_{33}^a$$
$$E_{24} = 8C_0(9)M_t^2M_W^2M_t^4 + 4C_0(10)E_3^aM_t^2M_W^2M_H^2 - 4CB_{0f}(16)E_{10}^aM_t^2M_W^2 - 2CB_{0f}(18)E_4^aM_W^2M_H^2 + 4CB_{0f}(22)E_{10}^aM_t^2M_W^2 + E_{34}^aM_t^2M_H^2 + 4CB_{1f}(22)M_t^2M_W^2M_H^2 + 8CB_{22f}(22)M_W^2M_H^2$$
$$E_{25} = E_{35}^a$$
$$E_{26} = -2CB_{0f}(17) + 1_c$$
$$E_{27} = 2C_0(9)M_t^4 + 2C_0(10)M_t^2M_H^2 - 2CB_{0f}(16)M_t^2 - 2CB_{0f}(18)M_H^2 - 2CB_{0f}(22)M_t^2 - E_{37}^aM_t^2$$
$$E_{28} = E_3^a$$
$$E_{29} = E_{36}^a$$
$$E_{30} = CB_{0f}(16)E_0^a - 30E_{38}^aM_t^2$$
$$E_{31} = E_{39}^a$$
$$E_{32} = 12CB_{0f}(18)M_W^2M_H^2 + 12CB_{0f}(19)M_W^2M_H^2 + 108CB_{0f}(20)M_W^2M_H^2 - 24CB_{0f}(22)M_t^2M_W^2 - E_{40}^a$$
$$E_{33} = E_{41}^a$$
$$E_{34} = E_{44}^a$$
$$E_{35} = E_{45}^a$$
$$E_{36} = E_{46}^a$$
$$E_{37} = E_{47}^a$$
$$E_{38} = E_{48}^a$$
$$E_{39} = E_{49}^a$$
$$E_{40} = E_{50}^a$$
$$E_{41} = E_{51}^a$$
$$E_{42} = E_{52}^a$$
$$E_{43} = C_0(7)$$
$$E_{44} = 4CB_{0f}(19) - 4CB_{0f}(23) + E_{53}^aM_H^2$$
$$E_{45} = CB_{0f}(23)$$
$$E_{46} = E_{54}^a$$
$$E_{47} = 18C_0(9)E_1^aM_t^2 - 18C_0(10)E_5^a + 16C_0(14)M_H^4 - 72CB_{0f}(16)M_t^2 - 54CB_{0f}(18)E_{42}^a + 36CB_{0f}(22)E_{43}^a + 9E_{55}^aM_H^2 - 72CB_{1f}(22)M_H^2$$
$$E_{48} = C_0(11)$$
$$E_{49} = CB_{0f}(19) - E_{56}^a$$
$$E_{50} = E_{57}^a$$
$$E_{51} = CB_{1f}(23)$$
$$E_{52} = E_{56}^a$$
$$E_{53} = CB_{0f}(19) - CB_{0f}(23)$$
$$E_{54} = E_{53}^a$$
$$E_{55} = CB_{0f}(19)$$
$$E_{56} = 6C_0(9)E_1^aM_t^2 - 6C_0(10)E_5^a - 24CB_{0f}(16)M_t^2 - 18CB_{0f}(18)E_{42}^a + 12CB_{0f}(22)E_{43}^a + E_{58}^aM_H^2 - 24CB_{1f}(22)M_H^2$$
$$E_{57} = C_0(12)$$
$$E_{58} = C_0(12)M_H^2 - CB_{1f}(23)$$
$$E_{59} = 2C_0(12)M_H^2 - 2CB_{0f}(25) + CB_{0f}(26)$$

$$E_{60} = 4CB_{0f}(19) + E^a_{53}M_H^2$$
$$E_{61} = 2C_0(9)E^a_1M_t^2 - 2C_0(10)E^a_5 - 8CB_{0f}(16)M_t^2 - 6CB_{0f}(18)E^a_{42} + 4CB_{0f}(22)E^a_{43} + E^a_{55}M_H^2 - 8CB_{1f}(22)M_H^2$$
$$E_{62} = 6C_0(9)E^a_1M_t^2 - 6C_0(10)E^a_6 - 24CB_{0f}(16)M_t^2 - 6CB_{0f}(18)E^a_{42} + 12CB_{0f}(22)E^a_{59} - E^a_{60}M_H^2$$
$$E_{63} = 3CB_{0f}(19) - E^a_{61}$$
$$E_{64} = CB_{0f}(19) - 3E^a_{56}$$
$$E_{65} = E^a_{62}$$
$$E_{66} = 6C_0(9)M_t^2 + 4C_0(10)E^a_7 - 8CB_{0f}(18) - 4CB_{0f}(22) + E^a_{63}$$
$$E_{67} = 2CB_{0f}(19) + CB_{0f}(23)$$
$$E_{68} = 6C_0(9)M_t^2 + 4C_0(10)E^a_7 - 8CB_{0f}(18) - 4CB_{0f}(22) + 3E^a_{64}$$
$$E_{69} = 3C_0(9)E^a_0 + 4C_0(10)E^a_8 - 12CB_{0f}(18) + 12CB_{0f}(22) - 3E^a_{65}M_H^2$$
$$E_{70} = C_0(7)M_H^2 + 2CB_{0f}(23)$$
$$E_{71} = E^a_{66}$$
$$E_{72} = C_0(9)E^a_1 - 4CB_{0f}(16) + 4CB_{0f}(22)$$
$$E_{73} = 2C_0(9)E^a_1M_t^2 - 2C_0(10)E^a_6 - 8CB_{0f}(16)M_t^2 - 2CB_{0f}(18)E^a_{42} + 4CB_{0f}(22)E^a_{59} - E^a_{67}M_H^2$$
$$E_{74} = C_0(7)M_H^2 + 2CB_{0f}(19) + 4CB_{0f}(23) - E^a_{70} + 2CB_{1f}(23)$$
$$E_{75} = 4C_0(9)E^a_1M_t^2 - 4C_0(10)E^a_6 - 16CB_{0f}(16)M_t^2 - 4CB_{0f}(18)E^a_{42} - 8CB_{0f}(22)E^a_{68} - E^a_{69}M_H^2 - 8CB_{1f}(22)M_H^2$$
$$E_{76} = C_0(7)M_H^2 + 4CB_{0f}(23) - E^a_{71} - 2CB_{1f}(23)$$
$$E_{77} = 4C_0(11)M_H^2 - 4CB_{0f}(19) + 4CB_{0f}(23) - E^a_{53}M_H^2$$
$$E_{78} = 4C_0(9)E^a_1M_t^2 - 4C_0(10)E^a_6 - 16CB_{0f}(16)M_t^2 - 4CB_{0f}(18)E^a_{42} - 8CB_{0f}(22)E^a_{68} + E^a_{72}M_H^2 - 8CB_{1f}(22)M_H^2$$
$$E_{79} = 6C_0(9)E^a_1M_t^2 - 6C_0(10)E^a_6 - 24CB_{0f}(16)M_t^2 - 6CB_{0f}(18)E^a_{42} + 12CB_{0f}(22)E^a_{59} - E^a_{66}M_H^2$$
$$E_{80} = E^a_{73}$$
$$E_{81} = 2C_0(9)E^a_2 - 4C_0(10)E^a_9 + 4CB_{0f}(18) - E^a_{74}$$

$\mathbf{E}^a_i$ functions

$E^a_0 = E^b_4$	$E^a_1 = E^b_5$	$E^a_2 = E^b_6$	$E^a_3 = E^b_7$
$E^a_4 = E^b_8$	$E^a_5 = E^b_9$	$E^a_6 = E^b_{10}$	$E^a_7 = E^b_0$
$E^a_8 = E^b_{11}$	$E^a_9 = E^b_{12}$	$E^a_{10} = E^b_{13}$	$E^a_{11} = E^b_{14}$
$E^a_{12} = B_{0f}(2)$	$E^a_{13} = B_{0f}(9)$	$E^a_{14} = B_{0f}(14)$	$E^a_{15} = 1 + a_{0f}(1)$

$$E^a_{16} = 4B_{0f}(9)M_W^2M_H^2 - 6B_{0f}(11)E^b_0M_t^2 + B_{0f}(14)M_H^4 - 6a_{0f}(1)M_t^4 - a_{0f}(3)M_W^2M_H^2 + a_{0f}(4)M_W^2M_H^2$$
$$E^a_{17} = 4B_{0f}(24) + 17a_{0f}(5)$$
$$E^a_{18} = a_{0f}(4)$$
$$E^a_{19} = a_{0f}(5)$$

$$
\begin{aligned}
E^a_{20} &= 8B_{0f}(9)M_W^2M_H^2 - 12B_{0f}(11)E^b_0M_t^2 + 2B_{0f}(14)E^b_2M_H^2 + 12B_{0f}(24)M_W^2M_H^2 \\
&\quad - 12a_{0f}(1)M_t^4 - 2a_{0f}(3)M_W^2M_H^2 + 2a_{0f}(4)M_W^2M_H^2 + 51a_{0f}(5)M_W^2M_H^2 \\
E^a_{21} &= 4M_W^2M_H^2 - 2B_{0f}(14)E^b_3M_H^2 - 36B_{0f}(16)M_t^2M_W^2 - 12B_{0f}(18)M_W^2M_H^2 + 6B_{0f}(19)M_W^2M_H^2 \\
&\quad - 18B_{0f}(20)M_W^2M_H^2 - 36B_{0p}(14)E^b_4M_t^2M_W^2 - 6B_{0p}(16)M_W^2M_H^4 + 3B_{0p}(17)M_W^2M_H^4 \\
&\quad - 81B_{0p}(28)M_W^2M_H^4 + 2a_{0f}(3)M_W^2M_H^2 - a_{0f}(5)E^b_{15}M_H^2 \\
E^a_{22} &= 4B_{0f}(23) + 3a_{0f}(4) \\
E^a_{23} &= E^b_{16} \\
E^a_{24} &= 4M_W^2M_H^2 - 2B_{0f}(14)E^b_3M_H^2 - 36B_{0f}(16)M_t^2M_W^2 - 12B_{0f}(18)M_W^2M_H^2 - 6B_{0f}(19)M_W^2M_H^2 \\
&\quad - 18B_{0f}(20)M_W^2M_H^2 - 36B_{0p}(14)E^b_4M_t^2M_W^2 - 6B_{0p}(16)M_W^2M_H^4 - 3B_{0p}(17)M_W^2M_H^4 \\
&\quad - 81B_{0p}(28)M_W^2M_H^4 + 2a_{0f}(3)M_W^2M_H^2 - a_{0f}(5)E^b_{15}M_H^2 \\
E^a_{25} &= E^b_{17} \\
E^a_{26} &= 11 + 3B_{0f}(22) - 3a_{0f}(1) \\
E^a_{27} &= 11 - 9B_{0f}(23) - 3a_{0f}(4) \\
E^a_{28} &= E^b_{18} \\
E^a_{29} &= 2M_W^2 + 3B_{0f}(11)E^b_1 + 3B_{0p}(29)M_t^2M_W^2 + 3B_{0p}(32)M_t^2M_W^2 - 3a_{0f}(1)M_t^2 \\
E^a_{30} &= B_{0f}(16) + B_{0p}(14)E^b_4 \\
E^a_{31} &= 12B_{0f}(11)E^b_0M_t^2 - 2B_{0f}(14)E^b_3M_H^2 - 36B_{0f}(16)M_t^2M_W^2 - 12B_{0f}(22)M_t^2M_W^2 - 12B_{0f}(24)M_W^2M_H^2 \\
&\quad - 36B_{0p}(14)E^b_4M_t^2M_W^2 - 6B_{0p}(16)M_W^2M_H^4 - 3B_{0p}(17)M_W^2M_H^4 - 27B_{0p}(28)M_W^2M_H^4 - 12B_{0p}(29)M_W^2M_t^4 \\
&\quad - 12B_{0p}(32)M_W^2M_t^4 - 12a_{0f}(1)E^b_1M_t^2 + 2a_{0f}(3)M_W^2M_H^2 - 2a_{0f}(5)E^b_{19}M_H^2 + 4E^b_{20}M_W^2 \\
E^a_{32} &= B_{0f}(23) + a_{0f}(4) \\
E^a_{33} &= 1 + 3B_{0f}(23) + a_{0f}(4) \\
E^a_{34} &= 5M_W^2 - 2B_{0f}(11)E^b_0 + B_{0f}(22)M_W^2 + 2B_{0p}(29)M_t^2M_W^2 + 2B_{0p}(32)M_t^2M_W^2 + a_{0f}(1)E^b_{21} \\
E^a_{35} &= E^b_{22} \\
E^a_{36} &= B_{0f}(12) \\
E^a_{37} &= B_{0f}(22) + 3a_{0f}(1) \\
E^a_{38} &= a_{0f}(1) \\
E^a_{39} &= E^b_{23} \\
E^a_{40} &= 12B_{0f}(11)E^b_0M_t^2 - 2B_{0f}(14)E^b_3M_H^2 - 36B_{0f}(16)M_t^2M_W^2 - 12B_{0f}(22)M_t^2M_W^2 - 36B_{0f}(24)M_W^2M_H^2 \\
&\quad - 36B_{0p}(14)E^b_4M_t^2M_W^2 - 6B_{0p}(16)M_W^2M_H^4 - 3B_{0p}(17)M_W^2M_H^4 - 27B_{0p}(28)M_W^2M_H^4 - 12B_{0p}(29)M_W^2M_t^4 \\
&\quad - 12B_{0p}(32)M_W^2M_t^4 - 12a_{0f}(1)E^b_1M_t^2 + 2a_{0f}(3)M_W^2M_H^2 - 2a_{0f}(5)E^b_{24}M_H^2 + 4E^b_{20}M_W^2 \\
E^a_{41} &= 2B_{0f}(23) + 3a_{0f}(4)
\end{aligned}
$$

$$
\begin{aligned}
&E^a_{42} = E^b_{28} && E^a_{43} = E^b_{29} && E^a_{44} = B_{0f}(4) && E^a_{45} = B_{0f}(5) \\
&E^a_{46} = B_{0f}(6) && E^a_{47} = B_{0f}(7) && E^a_{48} = B_{0f}(8) && E^a_{49} = B_{0f}(11) + 3B_{0f}(12) + B_{0f}(21) \\
&E^a_{50} = B_{0f}(13) && E^a_{51} = B_{0f}(15) && E^a_{52} = B_{0f}(23) && E^a_{53} = B_{0p}(30) + B_{0p}(33)
\end{aligned}
$$

$$
\begin{aligned}
E^a_{54} &= a_{0f}(3)\\
E^a_{55} &= 14B_{0f}(18) - B_{0f}(22) + 2B_{0p}(16)E^b_{25} + B_{0p}(29)E^b_{27} + B_{0p}(32)E^b_{27}\\
E^a_{56} &= B_{0f}(19) + B_{0p}(17)M_H^2\\
E^a_{57} &= B_{0p}(17)\\
E^a_{58} &= 2B_{0f}(11) + 6B_{0f}(12) - 18B_{0f}(14) + 42B_{0f}(18) + 2B_{0f}(21) - 3B_{0f}(22) + 6B_{0p}(16)E^b_{25}\\
&\quad + 3B_{0p}(29)E^b_{27} + 3B_{0p}(32)E^b_{27}\\
E^a_{59} &= E^b_{30}\\
E^a_{60} &= 2B_{0f}(11) + 6B_{0f}(12) + 18B_{0f}(14) - 18B_{0f}(18) + 2B_{0f}(21) + 3B_{0f}(22) - 18B_{0p}(16)E^b_{26}\\
&\quad - 3B_{0p}(29)E^b_{27} - 3B_{0p}(32)E^b_{27}\\
E^a_{61} &= B_{0f}(19) - B_{0p}(17)M_H^2\\
E^a_{62} &= B_{0p}(28)\\
E^a_{63} &= 11 + 3B_{0f}(22) - 3a_{0f}(3)\\
E^a_{64} &= B_{0f}(22) - a_{0f}(3)\\
E^a_{65} &= B_{0p}(29) + B_{0p}(32)\\
E^a_{66} &= 2B_{0f}(11) + 6B_{0f}(12) + 6B_{0f}(14) - 6B_{0f}(18) + 2B_{0f}(21) + 3B_{0f}(22) - 6B_{0p}(16)E^b_{26}\\
&\quad - 3B_{0p}(29)E^b_{27} - 3B_{0p}(32)E^b_{27}\\
E^a_{67} &= 2B_{0f}(12) + 2B_{0f}(14) - 2B_{0f}(18) + B_{0f}(22) - 2B_{0p}(16)E^b_{26} - B_{0p}(29)E^b_{27} - B_{0p}(32)E^b_{27}\\
E^a_{68} &= E^b_{27}\\
E^a_{69} &= 4B_{0f}(12) + B_{0f}(22) - 2B_{0p}(29)E^b_{27} - 2B_{0p}(32)E^b_{27} - a_{0f}(3)\\
E^a_{70} &= 3B_{0f}(23) + a_{0f}(4)\\
E^a_{71} &= B_{0f}(23) - a_{0f}(4)\\
E^a_{72} &= 7 - 4B_{0f}(12) - B_{0f}(22) + 2B_{0p}(29)E^b_{27} + 2B_{0p}(32)E^b_{27} + a_{0f}(3)\\
E^a_{73} &= 5 + 3a_{0f}(3)\\
E^a_{74} &= 5B_{0f}(22) - a_{0f}(3)
\end{aligned}
$$

$\mathbf{E}^b_i$ functions

$$
\begin{aligned}
E^b_0 &= M_W^2 + M_t^2 & E^b_1 &= M_W^2 - M_t^2 & E^b_2 &= M_H^2 + 8M_W^2\\
E^b_3 &= M_H^2 - 4M_W^2 & E^b_4 &= M_H^2 - 4M_t^2 & E^b_5 &= M_H^2 + 4M_W^2 - 2M_t^2\\
E^b_6 &= 2M_H^2 - 7M_t^2 & E^b_7 &= M_H^2 + M_t^2 & E^b_8 &= M_H^2 + 2M_t^2
\end{aligned}
$$

$$
\begin{aligned}
E^b_9 &= 6M_t^4 - 12M_W^4 + 6M_W^2M_H^2 - M_t^2M_H^2 + 6M_t^2M_W^2\\
E^b_{10} &= 2M_t^4 - 4M_W^4 + 2M_W^2M_H^2 - M_t^2M_H^2 + 2M_t^2M_W^2
\end{aligned}
$$

$$E^b_{11} = 2M_H^2 - 3M_W^2 + 3M_t^2 \qquad E^b_{12} = 2M_W^2 + M_t^2 \qquad E^b_{13} = M_H^2 - 2M_t^2$$
$$E^b_{14} = M_H^2 - M_t^2 \qquad E^b_{15} = 2M_H^2 - 21M_W^2 \qquad E^b_{16} = 3M_H^2 - 4M_t^2$$
$$E^b_{17} = 5M_H^2 - 12M_t^2 \qquad E^b_{18} = M_H^2 - 12M_t^2 \qquad E^b_{19} = M_H^2 + 3M_W^2$$
$$E^b_{20} = M_H^2 - 6M_t^2 \qquad E^b_{21} = 3M_W^2 - 2M_t^2 \qquad E^b_{22} = M_H^2 + 6M_t^2$$
$$E^b_{23} = M_H^2 + 26M_t^2 \qquad E^b_{24} = M_H^2 + 24M_W^2 \qquad E^b_{25} = 7M_H^2 - 15M_W^2$$
$$E^b_{26} = M_H^2 - 3M_W^2 \qquad E^b_{27} = 2M_W^2 - M_t^2 \qquad E^b_{28} = M_H^2 - 4M_W^2 - 2M_t^2$$
$$E^b_{29} = M_H^2 - 6M_W^2 - M_t^2 \qquad E^b_{30} = M_H^2 - 2M_W^2 + M_t^2$$

$\mathbf{F}_i$ functions

$$L_R = \ln \frac{\mu_R^2}{M_W^2} \tag{D.9}$$

$$F_0 = -17L_R M_H^2 c_\theta^2 + M_W^2 L_R \qquad F_1 = L_R$$
$$F_2 = 6M_t^2 L_R c_\theta^4 - D_{19} M_W^2 L_R \qquad F_3 = -19L_R M_H^2 c_\theta^2 + M_W^2 L_R$$
$$F_4 = 2D_{36} M_W^2 - D_{37} M_H^2 c_\theta^2 \qquad F_5 = 3D_{50} L_R + D_{51}$$
$$F_6 = 4D_{35} s_\theta^2 c_\theta^4 M_H^4 + D_{38} M_W^2 M_H^2 c_\theta^2 - 2D_{39} M_W^4 \qquad F_7 = 3D_{56} L_R + D_{57}$$
$$F_8 = 2D_{40} M_W^2 - D_{41} M_H^2 c_\theta^2 \qquad F_9 = 16M_W^2 s_\theta^2 + D_{60} M_H^2$$
$$F_{10} = 3D_{61} L_R + 2D_{62} \qquad F_{11} = 3D_{66} L_R - D_{67}$$
$$F_{12} = 2D_8 M_W^2 - D_{42} M_H^2 c_\theta^2 \qquad F_{13} = 9D_{70} L_R + D_{71}$$
$$F_{14} = 6D_{72} L_R - D_{73} \qquad F_{15} = 3D_{75} L_R - D_{76}$$
$$F_{16} = 3L_R c_\theta^2 + D_{77} \qquad F_{17} = 12D_{15} L_R - D_{78}$$
$$F_{18} = 12L_R s_\theta^2 - D_{79} \qquad F_{19} = 15L_R c_\theta^2 - D_{76}$$

D.2.2 $\mathcal{M}_{LO}$ term

$$\mathcal{M}_{LO}^{(0,4)} = -\frac{1}{2M_W}$$

$$\mathcal{M}_{LO}^{(0,6)} = \mathcal{M}_{LO}^{(0,4)} \left[s_\theta^2 W_1 + c_\theta^2 W_2 + s_\theta c_\theta W_3 - \frac{1}{4}(W_4 - 4W_5 + 4W_{14}) \right]$$

D.2.3 $\mathcal{M}_{EW}$ terms

$$\begin{aligned}\mathcal{M}_{EW}^{(1,4)} = &-\frac{1}{192}(E_{19}c_\theta^2 + 12E_{20}M_W^4M_H^2)c_\theta^{-2}M_W^{-2}M_H^{-2}\mathcal{M}_{LO}^{(0,4)}\\ &-\frac{1}{576}\Big\{4E_{48}F_{12}M_W^2 - 8(D_{35}E_{43}M_H^2 - 18E_{41}c_\theta^2)c_\theta^4s_\theta^2M_H^2\\ &+\Big[2D_8E_{44}M_W^2 + 3D_{10}E_{40}M_H^2 + D_{69}E_6M_H^2 - 54E_{50}M_W^2M_H^2\\ &+(4D_8E_{45}M_H^2 + 2D_{31}E_{42}M_H^2 + 3D_{74}E_{46}M_H^2 - 18E_{49}M_H^2 + 6E_{79}c_\theta^2\\ &-2F_{19}M_H^2)c_\theta^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_WM_H^{-2}\end{aligned}$$

The SMEFT part of the EW amplitude is decomposed as follows

$$\mathcal{M}_{EW}^{(1,6)} = \mathcal{M}_{LO}^{(0,4)}\sum_i W_i\mathcal{H}_{a,i} + \sum_i W_i\mathcal{H}_{b,i} + \mathcal{M}_{LO}^{(0,6)}\mathcal{H}_c \tag{D.10}$$

where we have introduced

$\mathcal{H}_{a,i}$ functions

$$\begin{aligned}\mathcal{H}_{a,1} = &-\frac{1}{288}\Big\{2(3E_5c_\theta^2 + 2D_0E_0M_t^2M_W^2 - 3D_{12}E_1M_H^4 - 12D_{20}E_2M_W^2M_H^2)c_\theta^2M_H^2\\ &+\Big[(64D_{43}E_3M_t^2M_W^2M_H^2 + 9E_4M_W^2 + 6E_7c_\theta^2M_H^6)c_\theta^2\\ &+9(-E_6M_W^2 + F_0)M_W^2M_H^2\Big]s_\theta^2\Big\}c_\theta^{-2}M_W^{-2}M_H^{-2}\end{aligned}$$

$$\begin{aligned}\mathcal{H}_{a,2} = &-\frac{1}{288}\Big[3E_8c_\theta^2 - 2(32D_{43}E_3M_t^2M_W^2 + 3E_7c_\theta^2M_H^4)s_\theta^2M_H^2\\ &-(4D_0E_0M_t^2M_W^2 - 6D_{12}E_1M_H^4 + 9E_6M_W^4 - 9F_0M_W^2)M_H^2\Big]M_W^{-2}M_H^{-2}\end{aligned}$$

$$\mathcal{H}_{a,3} = -\frac{1}{288}\Bigg\{(2D_1E_0M_t^2M_W^2 + 3D_{13}E_5 - 3D_{14}E_1M_H^4 - 12D_{21}E_2M_W^2M_H^2)c_\theta^2M_H^2$$
$$+ \Bigg[(3D_{13}E_7M_H^6 + 32D_{44}E_3M_t^2M_W^2M_H^2 + 9E_4M_W^2)c_\theta^2$$
$$+ 9(-E_6M_W^2 + F_0)M_W^2M_H^2\Bigg]s_\theta^2\Bigg\}c_\theta^{-1}s_\theta^{-1}M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,4} = \frac{1}{384}\Bigg[3(E_{10} + F_1)M_W^4M_H^2 - (-E_9 + 9E_{11}F_1M_W^2M_H^2)c_\theta^2\Bigg]c_\theta^{-2}M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,5} = -\frac{1}{96}\Bigg[3(-E_6 + F_1)M_W^4M_H^2 - (-E_{12} + 3E_{13}F_1M_W^2M_H^2)c_\theta^2\Bigg]c_\theta^{-2}M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,9} = \frac{1}{32}(-E_{14} + 12F_1M_t^2)s_\theta$$

$$\mathcal{H}_{a,10} = \frac{1}{192}\Bigg[6(4D_{19}E_{15} - E_{14}c_\theta^2)c_\theta^2 + D_{19}E_{16}M_W^2 + 12F_2\Bigg]c_\theta^{-3}$$

$$\mathcal{H}_{a,11} = -\frac{1}{16}E_{17}s_\theta M_t^2M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,12} = -\frac{1}{16}E_{17}c_\theta M_t^2M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,15} = -\frac{1}{16}(2E_{18} + 3F_1M_H^2)M_t^2M_H^{-2}$$

$$\mathcal{H}_{a,16} = \frac{1}{192}(E_{19}c_\theta^2 + 12E_{20}M_W^4M_H^2)c_\theta^{-2}M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,17} = \frac{1}{192}(E_{19}c_\theta^2 + 12E_{20}M_W^4M_H^2)c_\theta^{-2}M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,18} = \frac{1}{192}(E_{19}c_\theta^2 + 12E_{20}M_W^4 M_H^2)c_\theta^{-2}M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,19} = -\frac{1}{32}\left[2E_{23}M_W^4 M_H^2 - (-E_{21} + 6E_{22}F_1 M_W^2 M_H^2)c_\theta^2\right]c_\theta^{-2}M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,20} = \frac{1}{32}(-E_{24} + 2E_{25}F_1 M_W^2 M_H^2)M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,21} = \frac{1}{32}(-E_{24} + 2E_{25}F_1 M_W^2 M_H^2)M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,22} = \frac{1}{32}(-E_{24} + 2E_{25}F_1 M_W^2 M_H^2)M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,28} = \frac{1}{96}(E_{19}c_\theta^2 + 12E_{20}M_W^4 M_H^2)c_\theta^{-2}M_W^{-2}M_H^{-2}$$

$$\mathcal{H}_{a,35} = \frac{1}{16}(-E_{26} + 2F_1)M_H^2$$

$$\mathcal{H}_{a,36} = -\frac{1}{16}(-E_{26} + 2F_1)M_H^2$$

$$\mathcal{H}_{a,43} = \frac{1}{32}(-E_{27} + 2E_{28}F_1)M_t M_b^{-1} + \frac{3}{8}(E_{29}M_{t,g}M_{b,g} + F_1\sum_g M_{t,g}M_{b,g})$$

$$\mathcal{H}_{a,50} = \frac{1}{32}(-E_{26} + 2F_1)M_H^2 M_b^{-1}$$

$$\mathcal{H}_{a,51} = -\frac{1}{32}(E_{30} + E_{31}F_1)M_t M_b^{-1}$$

$\mathcal{H}_{b,i}$ **functions**

$$\begin{aligned}\mathcal{H}_{b,1} = -\frac{1}{576}\Big\{ &-2\Big[2E_{48}F_4M_W^2 - (72D_{24}E_{41}c_\theta^4M_H^2 + D_{27}E_{42}c_\theta^2M_H^2 \\ &- 4D_{32}E_{43}c_\theta^2M_H^4 - D_{36}E_{44}M_W^2 - 2D_{36}E_{45}c_\theta^2M_H^2 + E_{47}c_\theta^4 - 9E_{49}c_\theta^2M_H^2 \\ &- 27E_{50}M_W^2M_H^2 + F_5c_\theta^2M_H^2)c_\theta^2\Big]s_\theta^2 \\ &+ \Big[(48D_2E_{34}c_\theta^2 + 12D_4E_{35}c_\theta^2 + 4D_6E_{36}c_\theta^2 + 4D_8E_{37}c_\theta^2 - 6D_{10}E_{38}c_\theta^2 \\ &- 12D_{15}E_{39}c_\theta^2 - 36D_{22}E_2c_\theta^2 - 3D_{48}E_{46} + 72E_1c_\theta^2)c_\theta^2 + 3D_{16}E_{40} \\ &+ D_{49}E_6\Big]c_\theta^2M_H^2\Big\}c_\theta^{-6}M_WM_H^{-2}\end{aligned}$$

$$\begin{aligned}\mathcal{H}_{b,2} = \frac{1}{576}\Big\{ &4E_{48}F_6 - 24(D_{13}E_{43}M_H^2 - 30E_{41}c_\theta^2)c_\theta^6s_\theta^2M_H^2 \\ &+ \Big[(48D_2E_{34}c_\theta^2M_H^2 + 12D_4E_{35}c_\theta^2M_H^2 + 4D_6E_{36}c_\theta^2M_H^2 + 4D_8E_{37}c_\theta^2M_H^2 \\ &+ 8D_8E_{51}M_H^2 - 6D_{10}E_{38}c_\theta^2M_H^2 + 9D_{17}E_{40}M_H^2 - 18D_{26}E_{52}M_H^2 \\ &- 6D_{28}E_{42}c_\theta^2M_H^2 - 6D_{46}E_{54}M_W^2M_H^2 - 3D_{52}E_{46}c_\theta^2M_H^2 + 3D_{53}E_6M_H^2 \\ &+ 18D_{54}E_{55}M_H^2 - 4D_{55}E_{45}M_H^2 + 72E_1c_\theta^2M_H^2 - 6E_{56}c_\theta^4 - 2F_7M_H^2)c_\theta^2 \\ &- 2(4D_{39}E_{53} - 27D_{45}E_{50}M_H^2)M_W^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_WM_H^{-2}\end{aligned}$$

$$\begin{aligned}\mathcal{H}_{b,3} = -\frac{1}{576}\Big\{ &\Big[(24D_3E_{34} + 6D_5E_{35} + 2D_7E_{36} + 2D_9E_{37} - 3D_{11}E_{38} \\ &+ 36D_{13}E_1 - 36D_{23}E_2c_\theta^2 - 3D_{58}E_{46} - 12E_{39}c_\theta^4)c_\theta^2 \\ &+ 3(D_{18}E_{40} + D_{59}E_6)\Big]c_\theta^2M_W^2M_H^2\end{aligned}$$

$$
\begin{aligned}
&- 2\Big[(24D_{19}E_{15}c_\theta^4M_H^2 - 4D_{19}E_{58}c_\theta^2M_W^2M_H^2 + 4D_{19}E_{59}M_W^4 \\
&- 72D_{25}E_{41}c_\theta^4M_W^2M_H^2 - D_{29}E_{42}c_\theta^2M_W^2M_H^2 + 4D_{33}E_{43}c_\theta^2M_W^2M_H^4 \\
&+ D_{40}E_{60}M_W^4 + 6E_{45}F_9c_\theta^2M_W^2 + 9E_{49}c_\theta^2M_W^2M_H^2 + 27E_{50}M_W^4M_H^2 \\
&- 9E_{61}c_\theta^4M_W^2 - F_{10}c_\theta^2M_W^2M_H^2)c_\theta^2 \\
&- 2(2D_{19}E_{57}M_W^2 - E_{48}F_8)M_W^4\Big]s_\theta^2\Big\}c_\theta^{-5}s_\theta^{-1}M_W^{-1}M_H^{-2}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{H}_{b,4} = -\frac{1}{2304}\Big\{&4E_{48}F_4M_W^2 + \Big[3D_{10}E_{40}M_H^2 + 8D_{36}E_{53}M_W^2 - 2D_{47}E_{54}M_W^2M_H^2 \\
&- D_{64}E_6M_H^2 - 54E_{50}M_W^2M_H^2 - (144D_{15}E_{41}c_\theta^2M_H^2 + 2D_{30}E_{42}M_H^2 \\
&- 8D_{34}E_{43}M_H^4 + 3D_{63}E_{46}M_H^2 - 4D_{65}E_{45}M_H^2 + 6E_{62}c_\theta^2 - 18E_{63}M_H^2 \\
&- 2F_{11}M_H^2)c_\theta^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_WM_H^{-2}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{H}_{b,5} = -\frac{1}{576}\Big\{&4E_{48}F_{12}M_W^2 - 8(D_{35}E_{43}M_H^2 - 18E_{41}c_\theta^2)c_\theta^4s_\theta^2M_H^2 \\
&+ \Big[2D_8E_{44}M_W^2 + 3D_{10}E_{40}M_H^2 + D_{69}E_6M_H^2 - 162E_{50}M_W^2M_H^2 \\
&+ (4D_8E_{45}M_H^2 + 2D_{31}E_{42}M_H^2 + 3D_{68}E_{46}M_H^2 + 6E_{62}c_\theta^2 - 18E_{64}M_H^2 \\
&+ 2F_{13}M_H^2)c_\theta^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_WM_H^{-2}
\end{aligned}
$$

$$
\mathcal{H}_{b,6} = -\frac{9}{8}E_{65}M_WM_H^2
$$

$$
\mathcal{H}_{b,9} = -\frac{1}{64}(-E_{66} + 12F_1)s_\theta M_W
$$

$$
\mathcal{H}_{b,10} = \frac{1}{192}\Big\{2D_{19}E_{48}M_W^2 + \Big[3E_{68}c_\theta^4 - (2D_{19}E_{67} + F_{14})\Big]c_\theta^2\Big\}c_\theta^{-5}M_W
$$

$$\mathcal{H}_{b,11} = -\frac{1}{32} E_{69} s_\theta M_t^2 M_W M_H^{-2}$$

$$\mathcal{H}_{b,12} = -\frac{1}{32} E_{69} c_\theta M_t^2 M_W M_H^{-2}$$

$$\begin{aligned}\mathcal{H}_{b,14} = \frac{1}{576}\Big\{ & -8(D_{35}E_{70} - 18E_{41}c_\theta^2)c_\theta^2 s_\theta^2 \\ & - \Big[6E_{71}c_\theta^2 - (2D_{31}E_{42} + 3D_{74}E_{46} + 18E_{52} - 2F_{15})\Big]c_\theta^2 + 2D_8E_{54}M_W^2 \\ & + 3D_{10}E_{40} + D_{69}E_6 - 54E_{50}M_W^2\Big\}c_\theta^{-4}M_W\end{aligned}$$

$$\mathcal{H}_{b,15} = -\frac{1}{16} E_{72} M_t^2 M_W M_H^{-2}$$

$$\begin{aligned}\mathcal{H}_{b,16} = \frac{1}{576}\Big\{ & 4E_{48}F_{12}M_W^2 - 8(D_{35}E_{43}M_H^2 - 18E_{41}c_\theta^2)c_\theta^4 s_\theta^2 M_H^2 \\ & + \Big[2D_8E_{44}M_W^2 + 3D_{10}E_{40}M_H^2 + D_{69}E_6M_H^2 - 54E_{50}M_W^2M_H^2 \\ & + (4D_8E_{45}M_H^2 + 2D_{31}E_{42}M_H^2 + 3D_{74}E_{46}M_H^2 - 18E_{49}M_H^2 + 18E_{73}c_\theta^2 \\ & + 2F_{16}M_H^2)c_\theta^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_W M_H^{-2}\end{aligned}$$

$$\begin{aligned}\mathcal{H}_{b,17} = \frac{1}{576}\Big\{ & 4E_{48}F_{12}M_W^2 - 8(D_{35}E_{43}M_H^2 - 18E_{41}c_\theta^2)c_\theta^4 s_\theta^2 M_H^2 \\ & + \Big[2D_8E_{44}M_W^2 + 3D_{10}E_{40}M_H^2 + D_{69}E_6M_H^2 - 54E_{50}M_W^2M_H^2 \\ & + (4D_8E_{45}M_H^2 + 2D_{31}E_{42}M_H^2 + 3D_{74}E_{46}M_H^2 - 18E_{49}M_H^2 + 18E_{73}c_\theta^2 \\ & + 2F_{16}M_H^2)c_\theta^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_W M_H^{-2}\end{aligned}$$

$$\mathcal{H}_{b,18} = \frac{1}{576}\Big\{4E_{48}F_{12}M_W^2 - 8(D_{35}E_{43}M_H^2 - 18E_{41}c_\theta^2)c_\theta^4 s_\theta^2 M_H^2 + \Big[2D_8E_{44}M_W^2 + 3D_{10}E_{40}M_H^2 + D_{69}E_6M_H^2 - 54E_{50}M_W^2M_H^2 + (4D_8E_{45}M_H^2 + 2D_{31}E_{42}M_H^2 + 3D_{74}E_{46}M_H^2 - 18E_{49}M_H^2 + 18E_{73}c_\theta^2 + 2F_{16}M_H^2)c_\theta^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_WM_H^{-2}$$

$$\mathcal{H}_{b,19} = -\frac{1}{64}\Big\{16E_{48}M_W^4 + \Big[4E_{44}M_W^2 - (4E_{74}M_H^2 - E_{75}c_\theta^2 + F_{17}M_H^2)c_\theta^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_WM_H^{-2}$$

$$\mathcal{H}_{b,20} = -\frac{1}{192}\Big\{16D_{19}E_{48}M_W^4 - \Big[4D_{19}E_{77}M_W^2 - (4D_{19}E_{76}M_H^2 + 3E_{75}c_\theta^2 - F_{18}M_H^2)c_\theta^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_WM_H^{-2}$$

$$\mathcal{H}_{b,21} = \frac{1}{64}(-E_{78} + 12F_1M_H^2)M_WM_H^{-2}$$

$$\mathcal{H}_{b,22} = \frac{1}{64}(-E_{78} + 12F_1M_H^2)M_WM_H^{-2}$$

$$\mathcal{H}_{b,28} = \frac{1}{288}\Big\{4E_{48}F_{12}M_W^2 - 8(D_{35}E_{43}M_H^2 - 18E_{41}c_\theta^2)c_\theta^4 s_\theta^2 M_H^2 + \Big[2D_8E_{44}M_W^2 + 3D_{10}E_{40}M_H^2 + D_{69}E_6M_H^2 - 54E_{50}M_W^2M_H^2 + (4D_8E_{45}M_H^2 + 2D_{31}E_{42}M_H^2 + 3D_{74}E_{46}M_H^2 - 18E_{49}M_H^2 + 6E_{79}c_\theta^2 - 2F_{19}M_H^2)c_\theta^2\Big]c_\theta^2\Big\}c_\theta^{-6}M_WM_H^{-2}$$

$$\mathcal{H}_{b,42} = -\frac{1}{16}\left[3(D_{19}E_{40} - E_6) + (-E_{80} + 15F_1)c_\theta^2\right]c_\theta^{-2}M_W$$

$$\mathcal{H}_{b,43} = -\frac{1}{64}(-E_{81} + 2F_1)M_t M_W M_b^{-1}$$

$\mathcal{H}_c$ **function**

$$\mathcal{H}_c = -\frac{1}{192}\left[E_{32}c_\theta^2 - 6(-E_{33}M_W^2 + F_3)M_W^2\right]c_\theta^{-2}M_W^{-2}$$

D.2.4 $\mathcal{M}_{QCD}$ terms

$$\mathcal{M}_{QCD}^{(1,4)} = -\frac{1}{6}\mathcal{M}_{LO}^{(0,4)}$$

$$\mathcal{M}_{QCD}^{(1,6)} = -\frac{1}{6}\mathcal{M}_{LO}^{(0,6)} - \frac{1}{3}W_{49}\mathcal{M}_{LO}^{(0,4)}$$

D.3 HAA

D.3.1 Notations

The renormalized amplitude can be cast in the following form

$$\mathcal{M}_{HAA} = ig\left[\frac{g_6}{\sqrt{2}}\mathcal{M}_{LO}^{(0,6)} + \frac{\sqrt{2}G_F}{\pi^2}\left(\mathcal{M}^{(1,4)} + \frac{g_6}{\sqrt{2}}\mathcal{M}^{(1,6)}\right)\right]. \tag{D.11}$$

It is convenient express the one-loop amplitude as

$$\mathcal{M}^{(1,4)} = -\frac{1}{6}s_\theta^2 M_W^2 \mathcal{M}_{fact} \tag{D.12}$$

$$\mathcal{M}^{(1,6)} = \mathcal{M}_{fact}\mathcal{K} + \mathcal{M}_{nfact} \tag{D.13}$$

where $\mathcal{M}^{(1,6)}$ has been decomposed in a factorizable amplitude $\mathcal{M}_{fact}$, which derives from the SM contribution, and in a non-factorizable one denoted by $\mathcal{M}_{nfact}$ [26]. It is worth noting that the non-factorizable terms depend on LG coefficients: working in a PTG scenario, the LG coefficients do not give any contribution, simplifying the outcome of numerical implementation. However, we have considered in our results the full scenario PTG + LG, giving a complete result up to one-loop.

As usual, the upper index in D.11 indicate the order in perturbation theory and the mass dimension of the amplitude.

Loop scalar functions

We have used the shorthand notation

$$CX(p^2; m_1, m_2) = \mathrm{Re}X(p^2; m_1, m_2) + i\ \mathrm{Im}X(p^2; m_1, m_2), \tag{D.14}$$

where $X = \left\{B_{0f}\right\}$ and

$$B_{0p}(p^2; m_1, m_2) = \frac{d}{dp^2}B_0(p^2; m_1, m_2). \tag{D.15}$$

The sum over the fermionic generations is indicated by $\sum_{gen}$.

$$a_{0f}(M_t) = a_{0f}(1) \qquad a_{0f}(M_b) = a_{0f}(2) \qquad a_{0f}(M_W) = a_{0f}(3) \qquad a_{0f}(M_Z) = a_{0f}(4)$$

$$a_{0f}(M_H) = a_{0f}(5) \qquad \sum_{gen} a_{0f}(M_{l,g}) = a_{0f}(6) \qquad \sum_{gen} a_{0f}(M_{t,g}) = a_{0f}(7) \qquad \sum_{gen} a_{0f}(M_{b,g}) = a_{0f}(8)$$

$$B_{0f}(-M_Z^2; M_l, M_l) = B_{0f}(1)$$
$$B_{0f}(-M_Z^2; M_t, M_t) = B_{0f}(2)$$
$$B_{0f}(-M_Z^2; M_b, M_b) = B_{0f}(3)$$
$$B_{0f}(-M_Z^2; 0, 0) = B_{0f}(4)$$
$$\sum_{gen} B_{0f}(-M_Z^2; M_{l,g}, M_{l,g}) = B_{0f}(5)$$
$$\sum_{gen} B_{0f}(-M_Z^2; M_{t,g}, M_{t,g}) = B_{0f}(6)$$
$$\sum_{gen} B_{0f}(-M_Z^2; M_{b,g}, M_{b,g}) = B_{0f}(7)$$
$$B_{0f}(-M_Z^2; M_W, M_W) = B_{0f}(8)$$
$$B_{0f}(-M_Z^2; M_H, M_Z) = B_{0f}(9)$$
$$B_{0f}(-M_W^2; 0, M_l) = B_{0f}(10)$$
$$B_{0f}(-M_W^2; M_t, M_b) = B_{0f}(11)$$
$$\sum_{gen} B_{0f}(-M_W^2; 0, M_{l,g}) = B_{0f}(11)$$
$$\sum_{gen} B_{0f}(-M_W^2; M_{t,g}, M_{b,g}) = B_{0f}(12)$$
$$B_{0f}(-M_W^2; M_W, M_Z) = B_{0f}(13)$$
$$B_{0f}(-M_W^2; M_W, M_H) = B_{0f}(14)$$
$$B_{0f}(-M_W^2; 0, M_W) = B_{0f}(15)$$

$$CB_{0f}(-M_Z^2; M_l, M_l) = CB_{0f}(1)$$
$$CB_{0f}(-M_Z^2; M_t, M_t) = CB_{0f}(2)$$
$$CB_{0f}(-M_Z^2; M_b, M_b) = CB_{0f}(3)$$
$$CB_{0f}(-M_Z^2; 0, 0) = CB_{0f}(4)$$
$$\sum_{gen} CB_{0f}(-M_Z^2; M_{l,g}, M_{l,g}) = CB_{0f}(5)$$
$$\sum_{gen} CB_{0f}(-M_Z^2; M_{t,g}, M_{t,g}) = CB_{0f}(6)$$
$$\sum_{gen} CB_{0f}(-M_Z^2; M_{b,g}, M_{b,g}) = CB_{0f}(7)$$
$$CB_{0f}(-M_Z^2; M_W, M_W) = CB_{0f}(8)$$
$$CB_{0f}(-M_Z^2; M_H, M_Z) = CB_{0f}(9)$$
$$CB_{0f}(-M_W^2; 0, M_l) = CB_{0f}(10)$$
$$CB_{0f}(-M_W^2; M_t, M_b) = CB_{0f}(11)$$
$$\sum_{gen} CB_{0f}(-M_W^2; 0, M_{l,g}) = CB_{0f}(11)$$
$$\sum_{gen} CB_{0f}(-M_W^2; M_{t,g}, M_{b,g}) = CB_{0f}(12)$$
$$CB_{0f}(-M_W^2; M_W, M_Z) = CB_{0f}(13)$$
$$CB_{0f}(-M_W^2; M_W, M_H) = CB_{0f}(14)$$
$$CB_{0f}(-M_W^2; 0, M_W) = CB_{0f}(15)$$

$$CB_{0f}(-M_l^2; M_Z, M_l) = CB_{0f}(16)$$
$$CB_{0f}(-M_l^2; 0, M_W) = CB_{0f}(18)$$
$$B_{0f}(-M_H^2; M_b, M_b) = B_{0f}(17)$$
$$B_{0f}(-M_H^2; M_Z, M_Z) = B_{0f}(19)$$
$$B_{0f}(-M_W^2; 0, 0) = B_{0f}(21)$$
$$CB_{0f}(-M_H^2; M_b, M_b) = CB_{0f}(17)$$
$$CB_{0f}(-M_H^2; M_Z, M_Z) = CB_{0f}(19)$$
$$B_{0p}(-M_H^2; M_t, M_t) = B_{0p}(14)$$
$$B_{0p}(-M_H^2; M_W, M_W) = B_{0p}(16)$$
$$B_{0p}(-M_H^2; M_H, M_H) = B_{0p}(28)$$

$$CB_{0f}(-M_l^2; M_H, M_l) = CB_{0f}(17)$$
$$B_{0f}(-M_H^2; M_t, M_t) = B_{0f}(16)$$
$$B_{0f}(-M_H^2; M_W, M_W) = B_{0f}(18)$$
$$B_{0f}(-M_H^2; M_H, M_H) = B_{0f}(20)$$
$$CB_{0f}(-M_H^2; M_t, M_t) = CB_{0f}(16)$$
$$CB_{0f}(-M_H^2; M_W, M_W) = CB_{0f}(18)$$
$$CB_{0f}(-M_H^2; M_H, M_H) = CB_{0f}(20)$$
$$B_{0p}(-M_H^2; M_b, M_b) = B_{0p}(15)$$
$$B_{0p}(-M_H^2; M_Z, M_Z) = B_{0p}(17)$$

$$C_0(-M_H^2,0,0;M_t,M_t,M_t) = C_0(4) \qquad C_0(-M_H^2,0,0;M_b,M_b,M_b) = C_0(5)$$
$$C_0(-M_H^2,0,0;M_W,M_W,M_W) = C_0(6)$$

$\mathbf{G}_i$ functions

$$G_0 = -3M_t^4 + 2M_W^4 - 3M_W^2M_b^2 + 6M_t^2M_b^2 - 3M_t^2M_W^2 \qquad G_1 = M_H^4 + 12M_W^4 - 4M_W^2M_H^2$$
$$G_2 = M_H^2 - 2M_W^2 \qquad G_3 = M_H^2 - 4M_t^2$$
$$G_4 = 2M_H^4 - 6M_W^4 - 3M_W^2M_H^2 \qquad G_5 = M_b^2 + 2M_W^2 - M_t^2$$
$$G_6 = 2M_W^2 + M_t^2 \qquad G_7 = M_H^2 - 3M_W^2$$
$$G_8 = M_b^2 + 4M_t^2 \qquad G_9 = M_H^2 - 6M_b^2 - 6M_t^2$$
$$G_{10} = M_H^2 - 6M_W^2 \qquad G_{11} = 5M_H^2 + 4M_b^2 - 6M_W^2 + 16M_t^2$$

$\mathbf{H}_i$ functions

$$H_0 = 99 + 212s_\theta^4 - 48s_\theta^6 - 264s_\theta^2 \qquad H_1 = 1 + 4s_\theta^4 - 4s_\theta^2$$
$$H_2 = 1 - s_\theta^2 \qquad H_3 = 1 - 2s_\theta^2$$
$$H_4 = 4 - 3s_\theta^2 \qquad H_5 = 2 - 3s_\theta^2$$
$$H_6 = 4 - s_\theta^2 \qquad H_7 = 8 + 3s_\theta^2$$
$$H_8 = 2 - s_\theta^2 \qquad H_9 = 4 + 3s_\theta^2$$
$$H_{10} = 67 - 168s_\theta^4 + 96s_\theta^6 + 6s_\theta^2 \qquad H_{11} = 18 + 24s_\theta^4 - 43s_\theta^2$$
$$H_{12} = 3 + 4s_\theta^4 - 4s_\theta^2 \qquad H_{13} = 14 - 4s_\theta^4 + 72s_\theta^6 - 79s_\theta^2$$
$$H_{14} = 155 + 120s_\theta^4 + 180s_\theta^6 - 452s_\theta^2 \qquad H_{15} = 1 - 6s_\theta^2$$

$\mathbf{I}_i$ functions

$$I_0 = G_3H_4 \qquad I_1 = G_3H_2 \qquad I_2 = G_3H_5$$
$$I_3 = G_3 \qquad I_4 = H_4 \qquad I_5 = H_2$$
$$I_6 = H_5 \qquad I_7 = 6H_6M_W^2 - H_7M_H^2 \qquad I_8 = G_4H_2$$
$$I_9 = 6H_8M_W^2 - H_9M_H^2$$

$$\begin{aligned} I_{10} = {} & 4B_{0f}(11)G_0 + 24B_{0f}(12)M_W^4 + 2B_{0f}(14)G_1 + 36B_{0f}(16)M_t^2M_W^2 + 36B_{0f}(17)M_W^2M_b^2 \\ & + 12B_{0f}(18)G_2H_1M_W^2 + 6B_{0f}(19)M_W^2M_H^2 + 18B_{0f}(20)M_W^2M_H^2 + 8B_{0f}(21)M_W^4 + 36B_{0p}(14)G_3M_t^2M_W^2 \\ & + 36B_{0p}(15)M_W^2M_b^2M_H^2 + 6B_{0p}(16)G_1M_W^2 + 3B_{0p}(17)M_W^2M_H^4 + 27B_{0p}(28)M_W^2M_H^4 + 12a_{0f}(1)G_5M_t^2 \\ & + 12a_{0f}(2)G_6M_b^2 - 2a_{0f}(3)M_W^2M_H^2 + 2a_{0f}(5)G_7M_H^2 - 4G_9M_W^2 + 12H_{12}M_W^2L_RM_H^2 \end{aligned}$$

$$
\begin{aligned}
I_{11} &= B_{0f}(13)H_0 - 18B_{0p}(17)M_W^2 + a_{0f}(4)H_{11} \\
I_{12} &= 9B_{0f}(15)M_H^2 - 6a_{0f}(6)M_H^2 - 8a_{0f}(7)M_H^2 - 2a_{0f}(8)M_H^2 + 6G_8H_4 - 27H_6M_W^2 \\
I_{13} &= 18B_{0f}(19) + 18B_{0p}(17)M_H^2 + 3a_{0f}(3)H_{10} - 6H_{13}L_R + 2H_{14} \\
I_{14} &= B_{0f}(18)G_2H_2M_H^2 + 4a_{0f}(3)H_2M_W^2M_H^2 + G_{10}H_2L_RM_H^2 + G_{11}H_2M_W^2 \\
I_{15} &= 5 + 4a_{0f}(3) \\
I_{16} &= 3B_{0f}(18)G_2H_3M_H^2 + 8G_8H_5M_W^2 + 3H_3L_RM_H^4 - 36H_8M_W^4 - 6H_{15}M_W^2L_RM_H^2 \\
I_{17} &= B_{0f}(17) + a_{0f}(2) \\
I_{18} &= B_{0f}(16) + a_{0f}(1) \\
I_{19} &= 3 - 3L_R - 2B_{0f}(18) + a_{0f}(3) \\
I_{20} &= H_8
\end{aligned}
$$

$\mathbf{J}_i$ functions

$$
\begin{aligned}
J_0 &= 12C_0(4)I_0M_t^2 + 3C_0(5)I_4M_b^2M_H^2 + 9C_0(6)I_7M_W^2 + I_{12} \\
J_1 &= C_0(6) \\
J_2 &= I_{10} \\
J_3 &= I_{11} \\
J_4 &= I_{13} \\
J_5 &= 8C_0(4)I_1M_t^2M_W^2 + 2C_0(5)I_5M_W^2M_b^2M_H^2 - 2C_0(6)I_8M_W^2 + I_{14} \\
J_6 &= 16C_0(4)I_2M_t^2M_W^2 + 4C_0(5)I_6M_W^2M_b^2M_H^2 + 12C_0(6)I_9M_W^4 + I_{16} \\
J_7 &= 4C_0(6)M_H^2 - I_{15} \\
J_8 &= I_{17} \\
J_9 &= 2C_0(4)M_t^2 - I_{18} \\
J_{10} &= C_0(5)M_H^2 + 2 \\
J_{11} &= C_0(4)I_3 + 2 \\
J_{12} &= 2C_0(6)M_W^2 + I_{19} \\
J_{13} &= I_{20}
\end{aligned}
$$

D.3.2 $\mathcal{M}_{LO}^{(0,6)}$ terms

$$\mathcal{M}_{LO}^{(0,6)} = 2M_H^2M_W^{-1}W_1$$

D.3.3 $\mathcal{M}^{(1,4)}$ terms

$$\mathcal{M}^{(1,4)} = -\frac{1}{6}s_\theta^2M_W^2\mathcal{M}_{fact}$$

where we have defined

$$\mathcal{M}_{fact} = \Big[3M_W M_H^2 + 18M_W^3 - 4M_b^2 M_W - 16M_t^2 M_W - 8C_0(4)M_t^2 M_W M_H^2 + 32C_0(4)M_t^4 M_W - 2C_0(5)M_b^2 M_W M_H^2 + 18C_0(6)M_W^3 M_H^2 - 36C_0(6)M_W^5\Big] M_W^{-2}$$

D.3.4 $\mathcal{M}^{(1,6)}$ terms

$$\mathcal{M}^{(1,6)} = \mathcal{M}_{fact}\sum_i W_i\mathcal{K}_i + \sum_i W_i\mathcal{M}_{nfact,i}$$

where the functions $\mathcal{K}$ and $\mathcal{M}_{nfact}$ defined in Eq.(D.13) are decomposed in terms of Wilson coefficients.

Factorizable tems

$$\mathcal{K}_4 = \frac{1}{24}J_{13}M_W^2 \qquad \mathcal{K}_5 = -\frac{1}{6}s_\theta^2 M_W^2 \qquad \mathcal{K}_{16} = \frac{1}{6}s_\theta^2 M_W^2$$
$$\mathcal{K}_{17} = \frac{1}{6}s_\theta^2 M_W^2 \qquad \mathcal{K}_{18} = \frac{1}{6}s_\theta^2 M_W^2 \qquad \mathcal{K}_{28} = \frac{1}{3}s_\theta^2 M_W^2$$

Non-factorizable terms

$$\mathcal{M}_{nfact,1} = \frac{1}{9}J_0 s_\theta^2 M_W + 2J_1 s_\theta^4 M_W M_H^4 - \frac{1}{96}J_2 M_W^{-3} M_H^2 + \frac{1}{48}J_3 c_\theta^{-4} M_W M_H^2 + \frac{1}{144}J_4 c_\theta^{-2} M_W M_H^2$$

$$\mathcal{M}_{nfact,2} = -\frac{1}{2}J_5 s_\theta^2 M_W^{-1} \qquad \mathcal{M}_{nfact,3} = \frac{1}{12}J_6 c_\theta s_\theta M_W^{-1} + \frac{1}{2}J_7 c_\theta s_\theta^3 M_W M_H^2$$
$$\mathcal{M}_{nfact,9} = -\frac{1}{4}J_8 s_\theta M_b^2 M_W^{-1} M_H^2 \qquad \mathcal{M}_{nfact,11} = -\frac{1}{2}J_9 s_\theta M_t^2 M_W^{-1} M_H^2$$
$$\mathcal{M}_{nfact,14} = -\frac{1}{3}J_{10} s_\theta^2 M_b^2 M_W \qquad \mathcal{M}_{nfact,15} = \frac{4}{3}J_{11} s_\theta^2 M_t^2 M_W$$
$$\mathcal{M}_{nfact,42} = -\frac{3}{2}J_{12} s_\theta^2 M_W M_H^2$$

D.4 Anomalous magnetic moment of the muon

D.4.1 Notations

The (g-2)$_\mu$ of the muon can be decomposed as

$$a_\mu = \frac{g_6}{\sqrt{2}} a_\mu^{(0,6)} + a_\mu^{(QED)} + \frac{\sqrt{2} G_F M_\mu^2}{\pi^2} \left[a_\mu^{(1,4)} + \frac{g_6}{\sqrt{2}}\, a_\mu^{(1,6)} \right],$$

where the term $a_\mu^{(0,6)}$ derives from the SMEFT part of the vertex $A\mu^+\mu^-$ at $\mathcal{O}(gg_6)$. The terms $a_\mu^{(QED)}$ is the well known Schwinger term.

Loop scalar functions

$$a_{0f}(M_t) = a_{0f}(1)$$

$$B_{0f}(-M_Z^2; M_t, M_t) = B_{0f}(2) \qquad B_{0f}(-M_Z^2; M_W, M_W) = B_{0f}(8) \qquad B_{0f}(-M_Z^2; M_H, M_Z) = B_{0f}(9)$$
$$B_{0f}(-M_W^2; M_t, 0) = B_{0f}(11) \qquad B_{0f}(-M_W^2; M_W, M_Z) = B_{0f}(13) \qquad B_{0f}(-M_W^2; M_W, M_H) = B_{0f}(14)$$
$$B_{0f}(-M_W^2; 0, M_W) = B_{0f}(15)$$

$\mathbf{K}_i$ functions

$$v_f = 1 - 2\frac{Q_f}{I_f^3} s_\theta^2,$$

where $Q_l = -1$, $Q_b = -\frac{1}{3}$ and $Q_t = \frac{2}{3}$; I_f^3 is the third component of isospin.

$$K_0 = 3 + 11v_l^2 + 34v_l \qquad K_1 = 6 + v_l \qquad K_2 = 33 - 5v_l^2 - 4v_l$$
$$K_3 = 50 + 3v_l \qquad K_4 = 3 + v_l^2 + 4v_l \qquad K_5 = 3 + 3v_l^2 + 10v_l$$
$$K_6 = 9 + v_l^2 + 6v_l \qquad K_7 = 3 + v_l \qquad K_8 = 49 - 116v_l^2 - 329v_l$$
$$K_9 = 57 + 2v_l^2 + 25v_l \qquad K_{10} = 1 + v_l \qquad K_{11} = 21 + 13v_l$$
$$K_{12} = 23 + 2v_l \qquad K_{13} = 10 - v_l^2 + 3v_l \qquad K_{14} = 3 - v_l^2 - 5v_l$$
$$K_{15} = 5 - v_l \qquad K_{16} = 83 + 21v_l^2 + 96v_l \qquad K_{17} = 159 + 5v_l$$
$$K_{18} = 495 - 29v_l^2 + 110v_l \qquad K_{19} = 5 + 2v_l$$

$\mathbf{L}_i$ functions

We have use the shorthand notation:

$$L_m = \ln \frac{M_\mu^2}{M_W^2} \qquad L_h = \frac{M_Z^2}{M_W^2} \qquad L_h = \ln \frac{M_H^2}{M_W^2} \qquad L_R = \ln \frac{\mu_R^2}{M_W^2}$$

$$L_0 = L_R \qquad L_1 = 3 - 2L_m \qquad L_2 = 3 - 2L_m + 2L_h$$
$$L_3 = L_h \qquad L_4 = Lz \qquad L_5 = 1 - L_h$$
$$L_6 = 2 - 3L_h \qquad L_7 = 13 + 84L_z \qquad L_8 = 62 - 8L_z - 132L_m + 141L_R$$
$$L_9 = 1 - 2L_z + 2L_R \qquad L_{10} = 3 - 4L_z + 4L_R$$

$\mathbf{R}_i$ functions

$$R_0 = 24M_W^2 + 32M_W^2 c_\theta^2 + 39M_t^2 + 16K_1 M_t^2 c_\theta^2 \qquad R_1 = K_0 M_W^2 - 2K_2 M_t^2$$
$$R_2 = 66 + K_3 c_\theta^2 \qquad R_3 = 2M_H^2 + M_H^2 c_\theta^2 - 4M_W^2$$
$$R_4 = 8K_4 M_W^2 M_H^2 - K_5 M_H^4 - 12K_6 M_W^4 \qquad R_5 = -M_t^4 + 2M_W^4 - M_t^2 M_W^2$$
$$R_6 = -K_7 M_t^4 + 2K_7 M_W^4 - K_7 M_t^2 M_W^2 \qquad R_7 = 11 + 4c_\theta^2$$
$$R_8 = M_H^4 + 12M_W^4 - 4M_W^2 M_H^2 \qquad R_9 = K_7 M_H^4 + 12K_7 M_W^4 - 4K_7 M_W^2 M_H^2$$
$$R_{10} = 6K_7 L_0 + K_9 \qquad R_{11} = 6K_{10} L_0 M_H^2 + K_{11} M_H^2 + 8L_2 M_W^2$$
$$R_{12} = K_{12} + 6L_0 \qquad R_{13} = K_7 L_1 + 6L_3$$
$$R_{14} = 6K_4 L_4 M_H^2 + 6K_{16} L_4 M_W^2 - K_{18} M_W^2$$

$$R_{15} = 32K_{14} M_t^2 M_W^2 c_\theta^2 - 3K_{15} L_5 c_\theta^2 M_H^4 + 6K_{17} L_0 M_W^4 - 432L_0 M_t^2 M_W^2 c_\theta^2 - 12L_4 M_W^2 s_\theta^2 M_H^2 + 24L_6 M_W^2 M_H^2 c_\theta^2 - 8L_7 s_\theta^2 M_W^4 + 16L_8 s_\theta^2 c_\theta^2 M_W^4$$

$\mathbf{T}_i$ functions

$$T_0 = 27s_\theta^2 M_t^2 + 4K_{14} c_\theta^2 M_W^2$$
$$T_1 = 8R_{12} c_\theta^4 s_\theta^2 M_H^2 + 6L_2 K_7 M_W^2 - 3(6L_0 K_{10} M_H^2 + 8L_2 M_W^2 + K_{11} M_H^2) c_\theta^2$$
$$T_2 = R_{13} - 4L_2 c_\theta^2$$
$$T_3 = 3L_9 - 2L_{10} c_\theta^2$$
$$T_4 = 33K_0 + 64(66 + K_3 c_\theta^2) c_\theta^2 s_\theta^4$$
$$T_5 = K_8 - 144(11 + 4c_\theta^2) c_\theta^2 s_\theta^4$$
$$T_6 = 16(2M_H^2 - 4M_W^2 + c_\theta^2 M_H^2) c_\theta^2 s_\theta^4 M_H^2 - (8K_4 M_W^2 M_H^2 - K_5 M_H^4 - 12K_6 M_W^4)$$
$$T_7 = M_H^4 - 4M_W^2 M_H^2 + 12M_W^4$$
$$T_8 = 16(24M_W^2 + 39M_t^2 + 32c_\theta^2 M_W^2 + 16K_1 c_\theta^2 M_t^2) c_\theta^2 s_\theta^4 + 3(K_0 M_W^2 - 2K_2 M_t^2)$$
$$T_9 = 2M_W^4 - M_t^2 M_W^2 - M_t^4$$
$$T_{10} = -4(-432L_0 c_\theta^2 M_t^2 M_W^2 + 6L_0 K_{17} M_W^4 - 12L_4 s_\theta^2 M_W^2 M_H^2 - 3L_5 K_{15} c_\theta^2 M_H^4 + 24L_6 c_\theta^2 M_W^2 M_H^2 - 8L_7 s_\theta^2 M_W^4 + 16L_8 c_\theta^2 s_\theta^2 M_W^4 + 32K_{14} c_\theta^2 M_t^2 M_W^2) c_\theta^2 s_\theta^2 + (6L_4 K_4 M_H^2 + 6L_4 K_{16} M_W^2 - K_{18} M_W^2) M_W^2$$
$$T_{11} = K_7 M_H^4 - 4K_7 M_W^2 M_H^2 + 12K_7 M_W^4$$
$$T_{12} = 2K_7 M_W^4 - K_7 M_t^2 M_W^2 - K_7 M_t^4$$

D.4.2 SMEFT-LO terms

As we have said before, there is a term with no corresponding in the SM which contributes at LO:

$$a_\mu^{(0,6)} = \frac{1}{2s_\theta} \frac{M_\mu^2}{M_W^2} W_7,$$

given by the LO term in $A\mu^+\mu^-$ vertex.

D.4.3 QED terms

The Schwinger term results to be

$$a_\mu^{(QED)} = \frac{\alpha}{2\pi},$$

where α is the fine-structure constant.

D.4.4 SM-EW terms

The well known SM result can be cast in the form

$$a_\mu^{(1,4)} = \frac{1}{48}(5 + v_l^2), \quad v_l = 1 - 4s_\theta^2$$

D.4.5 SMEFT-EW terms

We have introduced the following shorthand notation:

$$a_\mu^{(1,6)} = \sum_i W_i \mathcal{A}_i$$

where the functions $\mathcal{A}_i$ are defined as

$$\mathcal{A}_1 = -\frac{1}{96} R_{10}$$

$$\mathcal{A}_2 = \frac{1}{96} R_{10}$$

$$\begin{aligned}\mathcal{A}_3 = &\frac{1}{192}(T_1 + 12 L_3 v_l c_\theta^2 M_H^2) c_\theta^{-3} s_\theta^{-1} M_H^{-2} \\ &- \frac{1}{32}(T_2 + 2 L_4 v_l) c_\theta^{-3} s_\theta^{-1} M_W^2 (M_H^2 - M_Z^2)^{-1}\end{aligned}$$

$$\mathcal{A}_4 = \frac{1}{192} K_{13} s_\theta^{-2}$$

$$\begin{aligned}\mathcal{A}_7 = -\frac{1}{9216}\Big[&2B_{0f}(2)T_8M_W^2 - 3B_{0f}(8)T_4M_W^4 + 3B_{0f}(9)T_6 \\ &- 72B_{0f}(11)(T_{12}v_l + 12T_9c_\theta^2s_\theta^4) - 12B_{0f}(13)T_5M_W^4 \\ &- 12B_{0f}(14)(T_{11}v_l + 12T_7c_\theta^2s_\theta^4) + 576B_{0f}(14)K_7c_\theta^4s_\theta^2M_W^4 \\ &+ 8a_{0f}(1)(4T_0c_\theta^2s_\theta^2 + 9K_7v_lM_t^2)M_t^2 - T_{10}\Big]c_\theta^{-4}s_\theta^{-3}M_W^{-4}\end{aligned}$$

$$\mathcal{A}_8 = \frac{1}{16}T_3c_\theta^{-1}$$

$$\mathcal{A}_{16} = -\frac{5}{24}$$

$$\mathcal{A}_{17} = \frac{1}{24}K_{19}$$

$$\mathcal{A}_{18} = \frac{5}{24}$$

$$\mathcal{A}_{27} = -\frac{1}{2}\left(1 + \frac{1}{M_\mu}\sum_{gen\neq\mu} M_{l,g}\right)$$

$$\mathcal{A}_{42} = \frac{3}{16}$$

D.5 Mass of the W boson

D.5.1 Notations

Starting from the LO expression

$$M_W^2 = M_Z^2 c_\theta^2, \tag{D.16}$$

we express the cosine of weak mixing angle as $c_\theta^2 = 1 - s_\theta^2$. After the expansion in terms of barred quantities $\bar{c}_\theta/\bar{s}_\theta$, we can collect the full result in the following form

$$\frac{M_W^2}{M_Z^2} = \bar{c}_\theta^2 + \frac{g_6}{\sqrt{2}}\Delta_{M_W}^{(0,6)} + \frac{\alpha}{\pi}\left[\Delta_{M_W}^{(1,4)} + \frac{g_6}{\sqrt{2}}\Delta_{M_W}^{(1,6)}\right]. \tag{D.17}$$

To give a more readable notation and solution, we will denote the barred quantity as

$$s = \bar{s}_\theta, \quad c = \bar{c}_\theta \tag{D.18}$$

Loop scalar functions

We have introduced the notation

$$B_{0p}(p^2; m_1, m_2) = \frac{d}{dp^2} B_0(p^2; m_1, m_2). \tag{D.19}$$

The quantity M_τ indicates the mass of the τ lepton.

$$a_{0f}(cM_Z) = a_{0f}(1) \qquad a_{0f}(M_Z) = a_{0f}(2) \qquad a_{0f}(M_H) = a_{0f}(3)$$
$$a_{0f}(M_\tau) = a_{0f}(4) \qquad a_{0f}(M_t) = a_{0f}(5) \qquad a_{0f}(M_b) = a_{0f}(6)$$

$$B_{0f}(-M_Z^2; cM_Z, cM_Z) = B_{0f}(1) \qquad B_{0f}(-M_Z^2; M_t, M_t) = B_{0f}(2)$$
$$B_{0f}(-M_Z^2; M_b, M_b) = B_{0f}(3) \qquad B_{0f}(-M_Z^2; M_H, M_Z) = B_{0f}(4)$$
$$B_{0f}(-M_Z^2; M_\tau, M_\tau) = B_{0f}(5) \qquad B_{0f}(-M_Z^2; 0, 0) = B_{0f}(6)$$
$$B_{0f}(-c^2 M_Z^2; cM_Z, M_Z) = B_{0f}(7) \qquad B_{0f}(-c^2 M_Z^2; cM_Z, M_H) = B_{0f}(8)$$
$$B_{0f}(-c^2 M_Z^2; M_t, M_b) = B_{0f}(9) \qquad B_{0f}(-c^2 M_Z^2; 0, cM_Z) = B_{0f}(10)$$
$$B_{0f}(-c^2 M_Z^2; 0, M_\tau) = B_{0f}(11)$$

$$B_{0p}(0; cM_Z, M_Z) = B_{0p}(1) \qquad B_{0p}(0; cM_Z, M_H) = B_{0p}(2)$$
$$B_{0p}(0; M_t, M_b) = B_{0p}(3)$$

$\mathbf{U}_i$ functions

$$U_0 = 33 - 32s^2$$
$$U_1 = 66 - 247s^2 + 312s^4 - 180s^6 + 48s^8$$
$$U_2 = 21 - 32s^2 + 28s^4 - 16s^6$$
$$U_3 = 6 - 7s^2 + 4s^4$$
$$U_4 = 33 - 95s^2 + 164s^4 - 132s^6 + 48s^8$$
$$U_5 = 51 - 128s^2 + 132s^4 - 48s^6$$
$$U_6 = 51 - 74s^2 + 24s^4$$
$$U_7 = 99 - 264s^2 + 212s^4 - 48s^6$$
$$U_8 = 3 - 4s^2$$
$$U_9 = 3 - 88s^2 + 132s^4 - 48s^6$$
$$U_{10} = 99 - 162s^2 + 64s^4$$
$$U_{11} = 9 - 52s^2 + 32s^4$$
$$U_{12} = 9 + 26s^2 - 16s^4$$
$$U_{13} = 61 - 136s^2 + 64s^4$$
$$U_{14} = 3 + 8s^2 - 64s^4$$
$$U_{15} = 9 - 6s^2 - 16s^4$$
$$U_{16} = 9 + 12s^2 + 32s^4$$
$$U_{17} = 17 - 68s^2 + 32s^4$$
$$U_{18} = 2 + 7s^2 - 4s^4$$
$$U_{19} = 9 - 10s^2 + 8s^4$$
$$U_{20} = 9 + 5s^2 - 4s^4$$
$$U_{21} = 19 - 28s^2 + 16s^4$$
$$U_{22} = 3 + 4s^2 + 16s^4$$
$$U_{23} = 6 - s^2 - 4s^4$$
$$U_{24} = 9 - 3s^2 - 4s^4$$
$$U_{25} = 9 + 6s^2 + 8s^4$$
$$U_{26} = 3 - s^2$$
$$U_{27} = 4 - s^2$$
$$U_{28} = 1 - 14s^2 + 8s^4$$
$$U_{29} = 1 + 7s^2 - 4s^4$$
$$U_{30} = 15 - 36s^2 + 16s^4$$
$$U_{31} = 1 - s^2 - 4s^4$$
$$U_{32} = 1 + 2s^2 + 8s^4$$
$$U_{33} = 1 + 4s^2 - 16s^4$$
$$U_{34} = 3 - 9s^2 + 4s^4$$
$$U_{35} = 33 - 116s^2 + 132s^4 - 48s^6$$
$$U_{36} = 33 - 44s^2 - 12s^4 + 16s^6$$
$$U_{37} = 33 - 8s^2 - 84s^4 + 48s^6$$
$$U_{38} = 2 - s^2$$
$$U_{39} = 7 - 2s^2$$
$$U_{40} = 9 - 8s^2$$
$$U_{41} = 9 - 17s^2 + 8s^4$$
$$U_{42} = 31 - 78s^2 + 66s^4 - 24s^6$$
$$U_{43} = 1 + 9s^2 - 24s^4$$
$$U_{44} = 8 - 31s^2 + 24s^4$$
$$U_{45} = 3 - 11s^2$$
$$U_{46} = 1 - 28s^2 + 16s^4$$
$$U_{47} = 1 + 14s^2 - 8s^4$$
$$U_{48} = 3 - 2s^2$$
$$U_{49} = 4 - 5s^2$$
$$U_{50} = 4 - 3s^2$$
$$U_{51} = 5 - 4s^2$$
$$U_{52} = 8 - 7s^2$$
$$U_{53} = 9 + 10s^2 - 8s^4$$
$$U_{54} = 9 + 52s^2 - 32s^4$$
$$U_{55} = 13 - 36s^2 + 16s^4$$
$$U_{56} = 43 - 136s^2 + 64s^4$$
$$U_{57} = 43 - 40s^2$$
$$U_{58} = 97 - 20s^2 - 64s^4$$
$$U_{59} = 196 + 9s^2 - 76s^4 - 260s^6 + 144s^8$$
$$U_{60} = 10 - 7s^2 + 4s^4 - 14s^6 + 8s^8$$
$$U_{61} = 19 + 8s^2$$
$$U_{62} = 29 - 17s^2 - 24s^4$$
$$U_{63} = 60 - 116s^2 + 23s^4 + 24s^6$$
$$U_{64} = 26 - 11s^2$$
$$U_{65} = 1 - 2s^2 - 8s^4$$
$$U_{66} = 1 + s^2 - 4s^4$$
$$U_{67} = 1 + 2s^2$$

$U_{68} = 3 + 2s^2$

$U_{69} = 3 + 4s^2$

$U_{70} = 3 + 4s^2 - 16s^4$

$U_{71} = 9 - 12s^2 - 32s^4$

$U_{72} = 9 - 6s^2 - 8s^4$

$U_{73} = 11 - 8s^2$

$U_{74} = 15 - 4s^2 - 16s^4$

$U_{75} = 15 + 4s^2 - 32s^4$

$U_{76} = 21 + 8s^2 - 64s^4$

$U_{77} = 137 - 308s^2 + 28s^4 + 144s^6$

$U_{78} = 165 - 100s^2 - 64s^4$

$U_{79} = 27 - 32s^2 + 12s^4$

$U_{80} = 57 - 106s^2 + 136s^4 - 132s^6 + 48s^8$

$U_{81} = 9 + 14s^2 - 48s^4$

$U_{82} = 53 - 84s^2 - 10s^4 + 44s^6$

$U_{83} = 3 - 28s^2 + 22s^4$

$U_{84} = 9 - 10s^2$

$U_{85} = 1 + s^2$

$U_{86} = 2 + s^2$

$U_{87} = 4 + s^2$

$U_{88} = 5 - 2s^2$

$U_{89} = 9 - 4s^2$

$U_{90} = 43 - 218s^2 + 160s^4$

$U_{91} = 98 + 73s^2 - 124s^4 - 116s^6 + 144s^8$

$U_{92} = 255 - 188s^2 + 8s^4$

$U_{93} = 7 + 21s^2 - 45s^4 + 12s^6$

$U_{94} = 15 - 96s^2 + 128s^4 - 48s^6$

$U_{95} = 14 - 39s^2 + 24s^4$

$U_{96} = 15 - 25s^2 + 8s^4$

$U_{97} = 61 - 160s^2 + 88s^4$

$U_{98} = 98 - 327s^2 + 500s^4 - 404s^6 + 144s^8$

$U_{99} = 5s^(- 1) - 49s + 44s^3$

$U_{100} = 1 - 3s^2$

$U_{101} = 11 - 16s^2$

$U_{102} = 1 - 4s^2$

$U_{103} = 5 - 8s^2$

$U_{104} = 7 - 8s^2$

$U_{105} = 25 - 32s^2$

$U_{106} = 3 - 8s^2$

$U_{107} = 13 - 12s^2$

$U_{108} = 15 + 5s^2 - 33s^4 + 12s^6$

$U_{109} = 31 - 14s^2 - 16s^4$

$U_{110} = 17 - 42s^2 + 24s^4$

$U_{111} = 60 - 103s^2 + 32s^4$

$U_{112} = 55 - 48s^2 - 16s^4$

$U_{113} = 103 - 108s^2 + 44s^4 - 48s^6$

$\mathbf{V}_i$ functions

$V_0 = 2B_{0p}(3)M_b^2 - B_{0p}(3)M_t^2$

$V_1 = B_{0f}(9) - a_{0f}(6)$

$V_2 = B_{0f}(9) + a_{0f}(5)$

$V_3 = 2B_{0f}(9) + a_{0f}(5) + a_{0f}(6)$

$V_4 = B_{0f}(9) + a_{0f}(6)$

$V_5 = 2 - 8B_{0f}(3) + B_{0f}(9) + 2a_{0f}(6)$

$V_6 = 2 - B_{0f}(9) + 2a_{0f}(5)$

$V_7 = 2 - 8B_{0f}(3) - 2a_{0f}(5) + 3a_{0f}(6)$

$V_8 = 2 - 2B_{0p}(3)M_b^2 + B_{0p}(3)M_t^2 + 3a_{0f}(5)$

$V_9 = 2 - 8B_{0f}(2) + B_{0f}(9) + 2a_{0f}(5)$

$V_{10} = 2 - B_{0f}(9) + 2a_{0f}(6)$

$V_{11} = 2 - 8B_{0f}(2) + 2B_{0p}(3)M_b^2 - B_{0p}(3)M_t^2 + a_{0f}(5)$

$V_{12} = 2 + 2a_{0f}(5) + a_{0f}(6)$

$V_{13} = 1 + 2B_{0f}(2) - B_{0f}(9) + a_{0f}(5)$

$V_{14} = 1 + 2B_{0f}(3) - B_{0f}(9) + a_{0f}(6)$

$V_{15} = 1 + 2B_{0f}(5) - B_{0f}(11) + a_{0f}(4)$

$V_{16} = 1 + 2B_{0f}(2) - 2B_{0p}(3)M_b^2 + B_{0p}(3)M_t^2 + 2a_{0f}(5)$

$V_{17} = 1 + 2B_{0f}(3) + 2a_{0f}(5)$

$V_{18} = 1 + 4B_{0f}(5) + 4a_{0f}(4)$

$\mathbf{X}_i$ functions

$$X_0 = 1 - B_{0f}(2)$$
$$X_1 = 1 + B_{0f}(2) + a_{0f}(5)$$
$$X_2 = 1 - B_{0f}(3)$$
$$X_3 = 1 + B_{0f}(3) + a_{0f}(6)$$
$$X_4 = 1 - B_{0f}(5)$$
$$X_5 = 1 + B_{0f}(5) + a_{0f}(4)$$
$$X_6 = 1 + a_{0f}(4)$$
$$X_7 = 1 + a_{0f}(5)$$
$$X_8 = M_\tau^2 - 12a_{0f}(5)M_b^2 + 6V_0M_t^2$$
$$X_9 = B_{0f}(2) + a_{0f}(5)$$
$$X_{10} = B_{0f}(4) + a_{0f}(3)$$
$$X_{11} = 2B_{0f}(4) + B_{0f}(6)$$
$$X_{12} = 4B_{0f}(4) - 16B_{0f}(8) - a_{0f}(1) + a_{0f}(2) - 15a_{0f}(3)$$
$$X_{13} = 4B_{0f}(4) - 8a_{0f}(1) + a_{0f}(2)$$
$$X_{14} = B_{0f}(5) + a_{0f}(4)$$
$$X_{15} = 9B_{0f}(8) + 3B_{0f}(9) + B_{0f}(11)$$
$$X_{16} = 3B_{0f}(9)M_t^2 + B_{0f}(11)M_\tau^2 + 3V_1M_b^2$$
$$X_{17} = a_{0f}(1) - a_{0f}(3)$$
$$X_{18} = 3a_{0f}(4) + 4a_{0f}(5) + a_{0f}(6)$$
$$X_{19} = V_2M_t^2 - V_3M_b^2$$
$$X_{20} = 1 + a_{0f}(1)$$
$$X_{21} = 16B_{0f}(8) + a_{0f}(1) - a_{0f}(2)$$
$$X_{22} = 8a_{0f}(1) - a_{0f}(2)$$
$$X_{23} = 4B_{0f}(4) - 16B_{0f}(8) - a_{0f}(1) + a_{0f}(2)$$
$$X_{24} = 1 + a_{0f}(6)$$
$$X_{25} = 3 - 12a_{0f}(4) - 16a_{0f}(5) - 4a_{0f}(6)$$
$$X_{26} = 4 + 4B_{0f}(4) + 4B_{0f}(8) + a_{0f}(1) + a_{0f}(2) + 15a_{0f}(3)$$
$$X_{27} = 4 + 4B_{0f}(4) + 8a_{0f}(1) + a_{0f}(2) + 4a_{0f}(3)$$
$$X_{28} = 2B_{0f}(4) - B_{0f}(6)$$
$$X_{29} = 4B_{0f}(4) - 4B_{0f}(8) - a_{0f}(1) + a_{0f}(2)$$
$$X_{30} = 4B_{0f}(4) - 8a_{0f}(1) + a_{0f}(2) + 11a_{0f}(3)$$
$$X_{31} = 6B_{0f}(4) + a_{0f}(2)$$
$$X_{32} = 6B_{0f}(8) + a_{0f}(1)$$
$$X_{33} = B_{0f}(9) + a_{0f}(5)$$
$$X_{34} = V_2M_t^2 - V_4M_b^2$$
$$X_{35} = 2 - 8B_{0f}(5) + B_{0f}(11) + 2a_{0f}(4)$$
$$X_{36} = 5 - 16B_{0f}(5) + 2a_{0f}(4)$$
$$X_{37} = 1 - B_{0f}(11)$$
$$X_{38} = 1 - 3B_{0f}(6)$$
$$X_{39} = 1 + 2a_{0f}(4)$$
$$X_{40} = 2 - B_{0f}(11) + 2a_{0f}(4)$$
$$X_{41} = V_5M_b^2 - V_6M_t^2$$
$$X_{42} = V_7M_b^2 - V_8M_t^2$$
$$X_{43} = 1 - B_{0f}(9)$$
$$X_{44} = B_{0f}(9)M_b^2 - V_6M_t^2$$
$$X_{45} = 2a_{0f}(5)M_b^2 + V_8M_t^2$$
$$X_{46} = V_9M_t^2 - V_{10}M_b^2$$
$$X_{47} = V_{11}M_t^2 - V_{12}M_b^2$$
$$X_{48} = B_{0f}(9)M_t^2 - V_{10}M_b^2$$
$$X_{49} = -V_0M_t^2 + V_{12}M_b^2$$
$$X_{50} = B_{0f}(1) - B_{0f}(7)$$
$$X_{51} = a_{0f}(5) - a_{0f}(6)$$
$$X_{52} = 3B_{0f}(2) + 3B_{0f}(3) + 12B_{0f}(4) + B_{0f}(5) + 6B_{0f}(6)$$
$$X_{53} = 3B_{0f}(8) + 3B_{0f}(9) + B_{0f}(11)$$
$$X_{54} = 3V_{13}M_t^2 + 3V_{14}M_b^2 + V_{15}M_\tau^2$$
$$X_{55} = 6V_{16}M_t^2 + 6V_{17}M_b^2 + V_{18}M_\tau^2$$

$\mathbf{Y}_i$ functions

We have used the shorthand notation

$$z_b = c^2 - s^2, \quad \text{with } s = \bar{s}_\theta \text{ and } c = \bar{c}_\theta. \tag{D.20}$$

$$
\begin{aligned}
Y_0 &= B_{0p}(2)M_H^2 \\
Y_1 &= z_b \\
Y_2 &= 12c^2 - B_{0f}(2)U_{13} - B_{0f}(3)U_{21} - 3B_{0f}(5)U_{30} + 6B_{0p}(2)c^2M_H^2 \\
Y_3 &= 3B_{0f}(1)U_0 - 3B_{0f}(1)U_1z_b^{-1} - B_{0f}(2)U_{11} - B_{0f}(3)U_{19} - 3B_{0f}(5)U_{28} - 144B_{0f}(10)U_{38}c^2 \\
&\quad - 6a_{0f}(1)U_{42}c^2z_b^{-1} + 3a_{0f}(2)U_{43}z_b^{-1} - U_{58} - U_{59}z_b^{-1} \\
Y_4 &= B_{0f}(7)U_{35} - a_{0f}(1)U_{41} + a_{0f}(2)U_{44} \\
Y_5 &= 4B_{0f}(3)U_{18} - U_{46} \\
Y_6 &= 2B_{0f}(3)U_{20} + U_{53} \\
Y_7 &= 4B_{0f}(4)U_{27}z_b^{-1} - B_{0p}(2)c^2M_H^2z_b^{-1} + 15a_{0f}(3) + a_{0f}(3)U_{64}z_b^{-1} \\
Y_8 &= 9B_{0f}(1)U_0 - B_{0f}(2)U_{16} - B_{0f}(3)U_{25} - 3B_{0f}(5)U_{32} - 18B_{0f}(6) - 18a_{0f}(1)U_{60}z_b^{-1} \\
&\quad - 3a_{0f}(1)U_{61} - U_{78} \\
Y_9 &= 9B_{0f}(1)U_2c^2z_b^{-1} + B_{0f}(2)U_{14}c^2z_b^{-1} - B_{0f}(3)U_{22}c^2z_b^{-1} \\
&\quad - 36B_{0f}(4)U_{26}z_b^{-1} + 3B_{0f}(5)U_{33}c^2z_b^{-1} - 18B_{0f}(6)c^2z_b^{-1} + 144B_{0f}(10)c^4 \\
&\quad + 6B_{0p}(2)c^4M_H^2z_b^{-1} - U_{77}c^2z_b^{-1} \\
Y_{10} &= 12B_{0f}(4)U_{26} + 3B_{0f}(7)U_{36} + a_{0f}(2)U_{62} \\
Y_{11} &= 4B_{0f}(3)U_{23} + U_{74} \\
Y_{12} &= 2B_{0f}(3)U_{24} + U_{72} \\
Y_{13} &= 18B_{0f}(1)U_3 + 3B_{0f}(1)U_4z_b^{-1} + B_{0f}(2)U_{11} + B_{0f}(3)U_{19} + 3B_{0f}(5)U_{28} + 144B_{0f}(10)c^4 \\
&\quad + 6a_{0f}(1)U_{79} + 3a_{0f}(1)U_{80}z_b^{-1} + U_{91}z_b^{-1} + U_{92} \\
Y_{14} &= B_{0f}(2)U_{13} + B_{0f}(3)U_{21} + 3B_{0f}(5)U_{30} - 6B_{0p}(2)c^2M_H^2 - 3U_{85}c^2 \\
Y_{15} &= 2B_{0f}(7)U_{37} + a_{0f}(2)U_{81} \\
Y_{16} &= s^{-1}z_b \\
Y_{17} &= U_{90}s^{-1} \\
Y_{18} &= c \\
Y_{19} &= s \\
Y_{20} &= B_{0f}(8) + a_{0f}(3)U_{38} \\
Y_{21} &= 3B_{0f}(1)U_5c^2z_b^{-1} - 3B_{0f}(1)U_6z_b + 3B_{0f}(1)U_7 + B_{0f}(2)U_{11} + B_{0f}(3)U_{19} + 3B_{0f}(5)U_{28} \\
&\quad - 36B_{0f}(8)c^2 + 144B_{0f}(10)c^4 - 24a_{0f}(1)U_{50}c^2z_b - 12a_{0f}(1)U_{93}c^2z_b^{-1} + 3a_{0f}(1)U_{94} \\
&\quad + 12a_{0f}(2)U_{96}z_b^{-1} - U_{97} + U_{98}z_b^{-1} \\
Y_{22} &= B_{0f}(2)U_{13} + B_{0f}(3)U_{21} + 3B_{0f}(5)U_{30} - 6B_{0p}(2)c^2M_H^2 \\
Y_{23} &= B_{0f}(7)U_7 + a_{0f}(2)U_{95} \\
Y_{24} &= U_{99}s^{-1} \\
Y_{25} &= 2B_{0p}(2)M_H^2 - 9a_{0f}(1) \\
Y_{26} &= U_{67}s^{-1} \\
Y_{27} &= U_{48}s^{-1} \\
Y_{28} &= U_{68}s^{-1} \\
Y_{29} &= 2c^2 - B_{0f}(5) \\
Y_{30} &= 1 - 2a_{0f}(4)U_{101} - 4U_{103}
\end{aligned}
$$

$$Y_{31} = 2c^2 - B_{0f}(3)$$
$$Y_{32} = a_{0f}(6)U_{101} + 2U_{104}$$
$$Y_{33} = 2c^2 - B_{0f}(2)$$
$$Y_{34} = a_{0f}(5)U_{105} + 2U_{101}$$
$$Y_{35} = 1 + 3B_{0f}(1)U_8 s^2 z_b^{-1} + 12a_{0f}(1)c^2 s^2 z_b^{-1} + U_{107} s^2 z_b^{-1}$$
$$Y_{36} = B_{0f}(1)U_9 z_b^{-1} + 48B_{0f}(10)c^2 - 2B_{0p}(2)c^2 M_H^2 z_b^{-1} + U_{113} z_b^{-1}$$
$$Y_{37} = B_{0f}(1)U_{10} - 4a_{0f}(1)U_{108}c^2 z_b^{-1} - a_{0f}(1)U_{109} + a_{0f}(2)U_{111} z_b^{-1} - U_{112}$$
$$Y_{38} = B_{0f}(7)U_7 + a_{0f}(2)U_{110}$$

$\mathbf{Z}_i$ functions

$$Z_0 = 2B_{0f}(2)U_{12} - U_{56} s^2 z_b^{-1}$$
$$Z_1 = B_{0f}(4)c^2 - B_{0f}(8) - a_{0f}(3)s^2$$
$$Z_2 = 2B_{0f}(5)U_{29} - U_{55} s^2 z_b^{-1}$$
$$Z_3 = B_{0p}(1)s^4 + B_{0p}(2)c^4$$
$$Z_4 = a_{0f}(3)U_{45} + Y_0 s^2$$
$$Z_5 = U_{48} + U_{50} z_b^{-1}$$
$$Z_6 = U_{49} z_b^{-1} + Y_1$$
$$Z_7 = U_{51} + U_{52} z_b^{-1}$$
$$Z_8 = Y_2 c^2 s^2 z_b^{-1} + Y_3 c^2 + 3Y_4$$
$$Z_9 = Y_5 s^2 z_b^{-1} + Y_6$$
$$Z_{10} = 3s^2 z_b^{-1} + U_{67}$$
$$Z_{11} = 5s^2 z_b^{-1} + U_{68}$$
$$Z_{12} = 7s^2 z_b^{-1} + U_{69}$$
$$Z_{13} = 2B_{0f}(2)U_{15} + U_{76} s^2 z_b^{-1}$$
$$Z_{14} = 4B_{0f}(4)U_{27} + Y_7 s^2$$
$$Z_{15} = 2B_{0f}(5)U_{31} + U_{70} s^2 z_b^{-1}$$
$$Z_{16} = 9B_{0p}(1)s^4 + B_{0p}(2)c^4$$
$$Z_{17} = 3a_{0f}(2)U_{63} z_b^{-1} + Y_8 c^2 + Y_9 s^2 - 3Y_{10}$$
$$Z_{18} = Y_{11} s^2 z_b^{-1} + Y_{12}$$
$$Z_{19} = B_{0p}(1)U_{39} s^4 + B_{0p}(2)c^4$$
$$Z_{20} = 3a_{0f}(2)U_{82} z_b^{-1} - 2Y_{13}c^2 - 2Y_{14}c^2 s^2 z_b^{-1} - 3Y_{15}$$
$$Z_{21} = a_{0f}(3)U_{83} z_b^{-1} + 3a_{0f}(3)U_{84} + 2Y_0 c^2 s^2 z_b^{-1}$$
$$Z_{22} = U_{38} z_b a^{-1} + U_{48}$$
$$Z_{23} = U_{86} z_b^{-1} + U_{88}$$
$$Z_{24} = U_{87} z_b^{-1} + U_{89}$$
$$Z_{25} = Y_{16}$$
$$Z_{26} = Y_{17}$$
$$Z_{27} = Y_{18}$$
$$Z_{28} = Y_{19}$$
$$Z_{29} = B_{0f}(2)U_{12} - B_{0f}(2)U_{17} s^2 z_b^{-1}$$
$$Z_{30} = 2B_{0f}(3)U_{18} s^2 z_b^{-1} + B_{0f}(3)U_{20}$$
$$Z_{31} = B_{0f}(4)c^2 + Y_{20}$$
$$Z_{32} = B_{0f}(5)U_{29} - 2B_{0f}(5)U_{34} s^2 z_b^{-1}$$
$$Z_{33} = B_{0p}(1)U_{40} s^4 - B_{0p}(2)c^4$$
$$Z_{34} = U_{11} + U_{13} s^2 z_b^{-1}$$
$$Z_{35} = U_{19} + U_{21} s^2 z_b^{-1}$$
$$Z_{36} = U_{28} + U_{30} s^2 z_b^{-1}$$
$$Z_{37} = Y_{21} c^2 + Y_{22} c^2 s^2 z_b^{-1} - 3Y_{23}$$
$$Z_{38} = Y_{24}$$
$$Z_{39} = 4 - 2U_{100} z_b^{-1} + 3Y_{25} c^2 z_b^{-1}$$
$$Z_{40} = Y_{26}$$
$$Z_{41} = B_{0f}(5) + B_{0f}(5) s^2 z_b^{-1}$$
$$Z_{42} = Y_{27}$$
$$Z_{43} = B_{0f}(3) + B_{0f}(3) s^2 z_b^{-1}$$
$$Z_{44} = Y_{28}$$
$$Z_{45} = B_{0f}(2) + B_{0f}(2) s^2 z_b^{-1}$$
$$Z_{46} = B_{0f}(5) - B_{0f}(11)c^2 - Y_{29} s^2 z_b^{-1}$$
$$Z_{47} = 2B_{0f}(11) + Y_{30} s^2 z_b^{-1}$$
$$Z_{48} = s^2 - 3B_{0f}(6) + B_{0f}(11)c^2$$
$$Z_{49} = B_{0f}(3) - B_{0f}(9)c^2 - Y_{31} s^2 z_b^{-1}$$
$$Z_{50} = Y_{32}$$
$$Z_{51} = B_{0f}(2) - B_{0f}(9)c^2 - Y_{33} s^2 z_b^{-1}$$
$$Z_{52} = Y_{34}$$
$$Z_{53} = 3a_{0f}(1)c^2 - 3a_{0f}(2) - Y_{35} s^2$$
$$Z_{54} = B_{0f}(9) - a_{0f}(5) s^2 z_b^{-1}$$
$$Z_{55} = B_{0p}(1)U_{40} s^4 + B_{0p}(2)c^4$$
$$Z_{56} = Y_{36} c^2 s^2 - Y_{37} c^2 + Y_{38}$$

D.5.2 SM terms

As usual we have introduced

$$v_f = 1 - 2\frac{Q_f}{I_f^3}s_\theta^2, \tag{D.21}$$

where $Q_l = -1$, $Q_b = -\frac{1}{3}$ and $Q_t = \frac{2}{3}$; I_f^3 is the third component of isospin. Furthermore, we have used the shorthand notation

$$z_b = c^2 - s^2, \quad \text{with } s = \bar{s}_\theta \text{ and } c = \bar{c}_\theta. \tag{D.22}$$

$$\begin{aligned}
\Delta_{M_W}^{(1,4)} = \frac{1}{48}\Big[&12\delta_G^{(4)}c^2M_Z^2 + 6X_{19}M_t^2 - X_{29}c^2M_Z^2M_H^2 - 4X_{53}c^4M_Z^4 \\
&- 2X_{54}c^2M_Z^2 + Z_1M_H^4 + Z_{56}M_Z^4\Big]c^{-2}s^{-2}M_Z^{-4} \\
&+ \frac{1}{144}\Big[3B_{0p}(2)s^2M_Z^2M_H^4 - 36\delta_G^{(4)}s^2M_Z^2 - 9X_0c^2M_Z^4v_t^2 + 18X_1c^2M_t^2M_Z^2v_t^2 \\
&- 9X_2c^2M_Z^4v_b^2 + 18X_3c^2M_b^2M_Z^2v_b^2 - 3X_4v_l^2c^2M_Z^4 + 6X_5v_l^2c^2M_Z^2M_\tau^2 \\
&+ 3X_{10}s^2M_H^4 + 24X_{17}s^2M_Z^2M_H^4(c^2M_Z^2 - M_H^2)^{-1} - 4X_{25}c^2s^4M_Z^4 \\
&- 3X_{30}s^2M_Z^2M_H^2 + 3X_{52}c^2M_Z^4 - 3X_{55}s^2M_Z^2 + 3Z_{55}s^2M_Z^6\Big]s^{-2}z_b^{-1}M_Z^{-4}
\end{aligned}$$

D.5.3 SMEFT terms

LO terms

We decompose the LO as:

$$\Delta_{M_W}^{(0,6)} = \sum_i W_i\Delta_{i,M_W}^{(0,6)}$$

where the functions $\Delta_{i,M_W}^{(0,6)}$ are defined as

$$\begin{aligned}
&\Delta_{1,M_W}^{(0,6)} = 4c^4s^2z_b^{-1} \qquad \Delta_{2,M_W}^{(0,6)} = -4c^4s^2z_b^{-1} \qquad \Delta_{3,M_W}^{(0,6)} = -2Z_{28}c^3s^4z_b^{-1} \\
&\Delta_{4,M_W}^{(0,6)} = -\frac{1}{2}c^4s^4z_b^{-1} \quad \Delta_{16,M_W}^{(0,6)} = -c^2s^2z_b^{-1} \qquad \Delta_{17,M_W}^{(0,6)} = -c^2s^2z_b^{-1} \\
&\Delta_{18,M_W}^{(0,6)} = -c^2s^2z_b^{-1} \qquad \Delta_{28,M_W}^{(0,6)} = -2c^2s^2z_b^{-1}
\end{aligned}$$

NLO terms

The NLO terms are decomposed as

$$\Delta_{M_W}^{(1,6)} = \sum_i W_i \Delta_{i,M_W}^{(1,6)}.$$

We also introduce the quantity

$$L_R = \ln \frac{\mu_R^2}{M_W^2}$$

The functions $\Delta_{i,M_W}^{(1,6)}$ are defined by

$$\begin{aligned}
\Delta_{1,M_W}^{(1,6)} &= \frac{1}{72}\Big[9c^4 M_Z^2 M_H^2 L_R - 6U_{47}X_6 c^2 s^2 M_Z^2 M_\tau^2 - 2U_{54}X_7 c^2 s^2 M_t^2 M_Z^2 \\
&+ 3U_{57}c^4 M_Z^4 L_R + 3X_0 Z_7 c^2 s^2 M_Z^4 v_t - 6X_1 Z_7 c^2 s^2 M_t^2 M_Z^2 v_t + 3X_2 Z_6 c^2 s^2 M_Z^4 v_b \\
&- 6X_3 Z_6 c^2 s^2 M_b^2 M_Z^2 v_b + 3X_4 Z_5 v_l c^2 s^2 M_Z^4 - 6X_5 Z_5 v_l c^2 s^2 M_Z^2 M_\tau^2 \\
&- 3X_{12} c^2 s^2 M_Z^2 M_H^2 - 12X_{15} c^4 s^2 M_Z^4 + 6X_{16} c^2 s^2 M_Z^2 + 18X_{19} s^2 M_t^2 \\
&- 2Z_0 c^2 s^2 M_t^2 M_Z^2 + 3Z_1 s^2 M_H^4 - 6Z_2 c^2 s^2 M_Z^2 M_\tau^2 - Z_8 s^2 M_Z^4 \\
&- 2Z_9 c^2 s^2 M_b^2 M_Z^2\Big] c^{-2} s^{-2} M_Z^{-4} \\
&+ \frac{1}{72}\Big[4U_{17}X_9 s^2 M_t^2 M_Z^2 + 24U_{34}X_{14} s^2 M_Z^2 M_\tau^2 + 16U_{38}X_{18} c^2 s^2 M_Z^4 + 3X_8 s^2 M_Z^2 \\
&+ 3X_{10} s^2 M_H^4 + 18X_{11} c^2 M_Z^4 - 3X_{13} s^2 M_Z^2 M_H^2 + 24X_{17} s^2 M_Z^2 M_H^4 (c^2 M_Z^2 - M_H^2)^{-1} \\
&+ 3Z_3 s^2 M_Z^6 + 3Z_4 M_Z^2 M_H^2\Big] z_b^{-1} M_Z^{-4}
\end{aligned}$$

$$\begin{aligned}
\Delta_{2,M_W}^{(1,6)} &= -\frac{1}{72}\Big[24U_{66}X_{14} s^2 M_Z^2 M_\tau^2 + 4U_{75}X_9 s^2 M_t^2 M_Z^2 - 3X_8 s^2 M_Z^2 - 3X_{10} s^2 M_H^4 \\
&- 24X_{17} s^2 M_Z^2 M_H^4 (c^2 M_Z^2 - M_H^2)^{-1} + 16X_{18} c^2 s^4 M_Z^4 \\
&- 3X_{22} s^2 M_Z^2 M_H^2 - 3Z_{16} s^2 M_Z^6\Big] c^2 s^{-2} z_b^{-1} M_Z^{-4}
\end{aligned}$$

$$
\begin{aligned}
&- \frac{1}{72}\Big[9c^2M_Z^2M_H^2L_R + 6U_{65}X_6c^2M_Z^2M_\tau^2 + 2U_{71}X_7c^2M_t^2M_Z^2 \\
&+ 15U_{73}c^2M_Z^4L_R + 3X_0Z_{12}c^2M_Z^4v_t - 6X_1Z_{12}c^2M_t^2M_Z^2v_t + 3X_2Z_{11}c^2M_Z^4v_b \\
&- 6X_3Z_{11}c^2M_b^2M_Z^2v_b + 3X_4Z_{10}v_lc^2M_Z^4 - 6X_5Z_{10}v_lc^2M_Z^2M_\tau^2 + 12X_{15}c^4M_Z^4 \\
&- 6X_{16}c^2M_Z^2 - 18X_{19}M_t^2 - 3X_{21}c^2M_Z^2M_H^2 - 3Z_1M_H^4 + 2Z_{13}c^2M_t^2M_Z^2 \\
&+ 3Z_{14}M_Z^2M_H^2 + 6Z_{15}c^2M_Z^2M_\tau^2 + Z_{17}M_Z^4 + 2Z_{18}c^2M_b^2M_Z^2\Big]s^{-2}M_Z^{-4}
\end{aligned}
$$

$$
\begin{aligned}
\Delta^{(1,6)}_{3,M_W} &= \frac{1}{72}\Big[4U_{17}X_9s^2M_t^2M_Z^2 + 24U_{34}X_{14}s^2M_Z^2M_\tau^2 + 3X_8s^2M_Z^2 + 3X_{10}s^2M_H^4 \\
&+ 18X_{11}c^2M_Z^4 - 3X_{13}s^2M_Z^2M_H^2 + 24X_{17}s^2M_Z^2M_H^4(c^2M_Z^2 - M_H^2)^{-1} \\
&+ 16X_{18}Z_{27}c^3s^2M_Z^4 + 3Z_{19}s^2M_Z^6\Big]cs^{-1}z_b^{-1}M_Z^{-4} \\
&- \frac{1}{144}\Big[12U_{47}X_6c^2sM_Z^2M_\tau^2 + 4U_{54}X_7c^2sM_t^2M_Z^2 - 6X_0Z_{24}c^2sM_Z^4v_t \\
&+ 12X_1Z_{24}c^2sM_t^2M_Z^2v_t - 6X_2Z_{22}c^2sM_Z^4v_b + 12X_3Z_{22}c^2sM_b^2M_Z^2v_b \\
&- 6X_4Z_{23}v_lc^2sM_Z^4 + 12X_5Z_{23}v_lc^2sM_Z^2M_\tau^2 + 24X_{15}c^4sM_Z^4 \\
&- 12X_{16}c^2sM_Z^2 - 36X_{19}sM_t^2 + 6X_{23}c^2sM_Z^2M_H^2 + 4Z_0c^2sM_t^2M_Z^2 \\
&- 6Z_1sM_H^4 + 12Z_2c^2sM_Z^2M_\tau^2 + 4Z_9c^2sM_b^2M_Z^2 + Z_{20}sM_Z^4 - 3Z_{21}sM_Z^2M_H^2 \\
&- 9Z_{25}c^2M_Z^2M_H^2L_R - 3Z_{26}c^2M_Z^4L_R\Big]c^{-1}s^{-2}M_Z^{-4}
\end{aligned}
$$

$$
\begin{aligned}
\Delta^{(1,6)}_{4,M_W} &= -\frac{1}{288}\Big[3B_{0p}(2)s^2M_Z^2M_H^4 - 3X_{10}s^2M_H^4 + 24X_{17}s^2M_Z^2M_H^4(c^2M_Z^2 - M_H^2)^{-1} \\
&- 4X_{25}c^4s^2M_Z^4 + 3X_{27}s^2M_Z^2M_H^2 - 18X_{28}c^2M_Z^4 - 3Z_{33}s^2M_Z^6\Big]s^{-2}z_b^{-1}M_Z^{-4} \\
&- \frac{1}{288}\Big[3X_0Z_{24}c^2M_Z^4v_t - 6X_1Z_{24}c^2M_t^2M_Z^2v_t + 3X_2Z_{22}c^2M_Z^4v_b \\
&- 6X_3Z_{22}c^2M_b^2M_Z^2v_b + 3X_4Z_{23}v_lc^2M_Z^4 - 6X_5Z_{23}v_lc^2M_Z^2M_\tau^2
\end{aligned}
$$

$$+ 6X_6 Z_{36} c^2 M_Z^2 M_\tau^2 + 2X_7 Z_{34} c^2 M_t^2 M_Z^2 + 2X_{24} Z_{35} c^2 M_b^2 M_Z^2$$
$$+ 3X_{26} c^2 M_Z^2 M_H^2 - 4Z_{29} c^2 M_t^2 M_Z^2 - 4Z_{30} c^2 M_b^2 M_Z^2 - 3Z_{31} M_H^4$$
$$- 12Z_{32} c^2 M_Z^2 M_\tau^2 + Z_{37} M_Z^4 + 3Z_{38} c^2 M_Z^4 L_R \Big] c^{-2} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{5,M_W} = \frac{1}{24}\Big[B_{0p}(2) s^2 M_Z^2 M_H^4 + B_{0p}(2) c^4 s^2 M_Z^6 + X_{10} s^2 M_H^4$$
$$+ 8X_{17} s^2 M_Z^2 M_H^4 (c^2 M_Z^2 - M_H^2)^{-1} - X_{30} s^2 M_Z^2 M_H^2$$
$$+ 2X_{31} c^2 M_Z^4 \Big] s^{-2} z_b^{-1} M_Z^{-4}$$
$$- \frac{1}{72}\Big[3c^4 M_Z^4 L_R + 3X_{29} c^2 M_Z^2 M_H^2 + 6X_{32} c^4 M_Z^4 - 3Z_1 M_H^4$$
$$+ Z_{39} c^2 s^2 M_Z^4 \Big] c^{-2} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{7,M_W} = \frac{1}{2} X_6 M_\tau^2 s z_b^{-1} M_Z^{-2} - \frac{1}{16}\Big[2B_{0f}(11) s M_\tau^2 - Z_{40} L_R M_\tau^2\Big] s^{-2} M_Z^{-2}$$

$$\Delta^{(1,6)}_{8,M_W} = \frac{1}{8}\Big[c^2 L_R M_\tau^2 - B_{0f}(11) c^2 M_\tau^2 + Z_{41} v_l M_\tau^2\Big] c^{-1} s^{-2} M_Z^{-2}$$

$$\Delta^{(1,6)}_{9,M_W} = \frac{1}{2} X_{24} M_b^2 s z_b^{-1} M_Z^{-2}$$
$$- \frac{1}{16}\Big[6B_{0f}(9) c^2 s M_b^2 M_Z^2 + 6X_{33} s M_t^2 M_b^2 - Z_{42} c^2 M_b^2 M_Z^2 L_R\Big] c^{-2} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{10,M_W} = -\frac{1}{8}\Big[c^2 M_b^2 M_Z^2 L_R + 3B_{0f}(9) c^2 M_b^2 M_Z^2 + 3X_{33} M_t^2 M_b^2$$
$$- 3Z_{43} M_b^2 M_Z^2 v_b \Big] c^{-1} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{11,M_W} = -X_7 M_t^2 s z_b^{-1} M_Z^{-2} \frac{1}{16}\Big[6B_{0f}(9)c^2 s M_t^2 M_Z^2 - 6X_{34} s M_t^2 - Z_{44} c^2 M_t^2 M_Z^2 L_R\Big] c^{-2} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{12,M_W} = -\frac{1}{8}\Big[c^2 M_t^2 M_Z^2 L_R - 3B_{0f}(9)c^2 M_t^2 M_Z^2 + 3X_{34} M_t^2 + 3Z_{45} M_t^2 M_Z^2 v_t\Big] c^{-1} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{16,M_W} = \frac{1}{48} X_{36} z_b^{-1} M_\tau^2 M_Z^{-2} - \frac{1}{24}\Big[6c^2 M_Z^2 L_R - X_{35} M_\tau^2 - 2Z_{46} M_Z^2\Big] s^{-2} M_Z^{-2}$$

$$\Delta^{(1,6)}_{17,M_W} = -\frac{1}{12}\Big[X_4 v_l M_Z^2 - 2X_5 v_l M_\tau^2\Big] c^2 s^{-2} z_b^{-1} M_Z^{-2} + \frac{1}{48}\Big[12c^2 M_Z^2 L_R - 4U_{102} X_6 M_\tau^2 + 4X_{37} c^2 M_Z^2 + Z_{47} M_\tau^2\Big] s^{-2} M_Z^{-2}$$

$$\Delta^{(1,6)}_{18,M_W} = -\frac{1}{48}\Big[4X_{38} M_Z^2 + 3X_{39} M_\tau^2\Big] z_b^{-1} M_Z^{-2} - \frac{1}{24}\Big[6c^2 M_Z^2 L_R + X_{40} M_\tau^2 + 2Z_{48} M_Z^2\Big] s^{-2} M_Z^{-2}$$

$$\Delta^{(1,6)}_{19,M_W} = \frac{1}{8} X_{42} z_b^{-1} M_Z^{-2} - \frac{1}{8}\Big[6c^4 M_Z^4 L_R - X_{19} M_t^2 - X_{41} c^2 M_Z^2 - 2Z_{49} c^2 M_Z^4\Big] c^{-2} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{20,M_W} = -\frac{1}{24}\Big[6X_2 c^2 M_Z^2 v_b - 12X_3 c^2 M_b^2 v_b + 3X_{45} s^2 + Z_{50} s^2 M_b^2\Big] s^{-2} z_b^{-1} M_Z^{-2} - \frac{1}{24}\Big[6c^4 M_Z^4 L_R + 2U_8 X_{24} c^2 M_b^2 M_Z^2 - 3X_{19} M_t^2 - 6X_{43} c^4 M_Z^4 - 3X_{44} c^2 M_Z^2\Big] c^{-2} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{21,M_W} = \frac{1}{8} X_{47} z_b^{-1} M_Z^{-2} - \frac{1}{8}\Big[6c^4 M_Z^4 L_R - X_{19} M_t^2 - X_{46} c^2 M_Z^2 - 2Z_{51} c^2 M_Z^4\Big] c^{-2} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{22,M_W} = -\frac{1}{24}\Big[6X_0 c^2 M_Z^2 v_t - 12X_1 c^2 M_t^2 v_t + 3X_{49} s^2 + Z_{52} s^2 M_t^2\Big] s^{-2} z_b^{-1} M_Z^{-2} + \frac{1}{24}\Big[6c^4 M_Z^4 L_R - 2U_{106} X_7 c^2 M_t^2 M_Z^2 + 3X_{19} M_t^2 + 6X_{43} c^4 M_Z^4 + 3X_{48} c^2 M_Z^2\Big] c^{-2} s^{-2} M_Z^{-4}$$

$$\Delta^{(1,6)}_{42,M_W} = \frac{3}{4} L_R c^2 z_b^{-1} - \frac{1}{4}\Big[3U_8 X_{50} - Z_{53}\Big] s^{-2}$$

$$\Delta^{(1,6)}_{43,M_W} = -\frac{3}{4} X_{51} M_t M_b^3 (M_t^2 - M_b^2)^{-1} z_b^{-1} M_Z^{-2} + \frac{3}{4} Z_{54} M_t M_b s^{-2} M_Z^{-2}$$

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